



SDS Sheets

**Custom Electric, Inc.
52 Main Street
Manchester CT 06042**

Last Revision: August 2026



The Safety Data Sheets for the below Hazardous Chemicals follow this list as noted. The SDS sheets are arranged in the order below.

<u>Manufacturer</u>	<u>Chemical Name</u>	<u>SDS Date</u>
3M	Fire Barrier CP-25WB+	1/29/2026
3M	WL Wire Pulling Lubricant	1/19/2026
3M	Scotchcast Resin 4 Kit	5/3/2021
3M	82-A Series Inline Resin Splicing Kit	8/12/2022
3M	Heat Shrink tubing FP-301	4/30/2021
Aervoe	Cold Galvanize Coating	11/12/2025
Amer. Polywater	FO Fiber Optic Cleaner	3/4/2022
Amer. Polywater	HydraSol Cable Gel Remover	10/14/2025
Burndy	Penetrox Oxide Compound	2/8/2023
CRC	Zinc-It Instant Cold Galvanize	6/5/2018
CRC	Wasp & Hornet Killer	1/2/2020
CRC	Spray Paint- Red	11/10/2016
CRC	Spray Paint- Orange	11/11/2016
CRC	Spray Paint- Yellow	11/11/2016
CRC	RTC Silicone Sealant	9/29/2021
Eaton Crouse-Hinds	Chico Sealing Compound	Jun-24
Eaton Crouse-Hinds	Chico X Fiber	Apr-24
Erico	Caldweld Plus	12/19/2019
GE	Clear Silicone Sealant	5/19/2023
Hilti	FS-One Firestop Sealant	10/30/2024
Hilti	GC-22 Gas Canister	7/29/2024
Hilti	CFS-P Fire Protection	12/16/2024
Ideal	Clear-Glide Wire Pulling Lubricant	4/9/2023
Ideal	Multi-Purpose Wipes	6/6/2023
Milwaukee	Rechargeable Lithium-Ion Batteries	Jan-18
Owens Corning	Bonded Mineral Wool	6/5/2023
Ridgid	Nu-Clear Thread Cutting Oil	1/14/2026
Sakrete	Non-Shrink Construction Grout	2/18/2022
STI	Intumescent Firestop Plug	1/16/2023
STI	Smoke 'N' Sound Sealant	1/17/2023
Thomas & Betts	Carlton Low-VOC "Quick-Set" Solvent Cement	3/26/2025
Thomas & Betts	Carlton Low-VOC Solvent Cement for PVC	3/26/2025
Thomas & Betts	Carlton Low-VOC Solvent Cement for PVC	3/26/2025

SDS Sheets Index

****Note:** All SDS sheets are verified to be the most up-to-date version.

1) 3M

- a) Fire Barrier CP-25WB+
- b) WL Wire Pulling Lubricant
- c) Scotchcast Resin 4 Kit
- d) 82-A Series Inline Resin Splicing Kit
- e) Heat Shrink Tubing FP-301

2) Aervoe

- a) Cold Galvanize Coating

3) American Polywater Corporation

- a) FO Fiber Optic Cleaner
- b) HydraSol Cable Gel Remover

4) Burndy

- a) Penetrox Oxide Inhibiting Compound

5) CRC Industries

- a) Zinc-It Instant Cold Galvanize
- b) Wasp & Hornet Killer
- c) Spray paint- Red
- d) Spray paint- Orange
- e) Spray paint- Yellow
- f) RTV Silicone Sealant

6) Eaton Crouse-Hinds

- a) Chico Sealing Compound
- b) Chico X Fiber

7) Erico

- a) CADWELD Plus Welding Material

8) **GE**

- a) Clear Silicone Sealant

9) **Hilti**

- a) FS-One Firestop Sealant
- b) GC-22 Gas Canister
- c) CFS-P Fire Protection

10) **Ideal**

- a) Clear-Glide Wire Pulling Lubricant
- b) Multi-Purpose Wipes

11) **Milwaukee**

- a) Rechargeable Lithium-Ion Batteries

12) **Owens Corning**

- a) Bonded Mineral Wool

13) **Ridgid**

- a) Nu-Clear Thread Cutting Oil

14) **Sakrete**

- a) Non-Shrink Construction Grout

15) **STI**

- a) Intumescent Firestop Plug
- b) Smoke 'N' Sound Sealant

16) **Thomas & Betts**

- a) Carlon All-Weather Low VOC "Quick-Set" Solvent Cement for ENT & PVC
- b) Carlon Low-VOC Solvent Cement for PVC
- c) Carlon Low-VOC Solvent Cement for PVC



Safety Data Sheet

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Supersedes Date: 01/20/25

SECTION 1: Identification

1.1. Product identifier

3M Brand Fire Barrier CP-25WB+

Product Identification Numbers

70-0091-7202-7, 70-0091-9188-6, 98-0400-5380-7, 98-0400-5381-5, 98-0400-5382-3, 98-0400-5383-1, 98-0400-5406-0, 98-0400-5456-5, 98-0400-5562-0, 98-0400-5573-7, 98-0400-5610-7, 98-0400-5629-7, 98-0441-1101-5, 7100006311, 7000006379, 7000059394, 7000145569, 7100025518, 7000006383, 7010353050, 7100137423, 7100271914, 7100330628, 7100377147

1.2. Recommended use and restrictions on use

Recommended use

Fire Protection, Industrial use

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Industrial Specialties Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Serious Eye Damage/Irritation: Category 2A.

Skin Sensitizer: Category 1.

Germ Cell Mutagenicity: Category 2.

Reproductive Toxicity: Category 2.

2.2. Label elements

Signal word

Warning

Symbols

Exclamation mark |Health Hazard |

Pictograms**Hazard Statements**

Causes serious eye irritation.

May cause an allergic skin reaction.

Suspected of causing genetic defects.

Suspected of damaging fertility or the unborn child.

Precautionary statements**General:**

Keep out of reach of children.

Prevention:

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Avoid breathing vapors.

Wash exposed skin thoroughly after handling.

Contaminated work clothing should not be allowed out of the workplace.

Wear protective gloves and eye protection.

Response:

IF ON SKIN: Wash with plenty of soap and water.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do.

Continue rinsing.

IF exposed or concerned: Get medical attention.

If eye irritation persists or if skin irritation or rash occurs: Get medical attention.

Take off contaminated clothing and wash it before reuse.

Storage:

Store locked up.

Disposal:

Dispose of contents and container in accordance with applicable local, regional, national, and international regulations.

3% of the mixture consists of ingredients of unknown acute oral toxicity.

3% of the mixture consists of ingredients of unknown acute dermal toxicity.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
Water	7732-18-5	10 - 35
Zinc Borate 2335	138265-88-0	10 - 30 Trade Secret *
Polymer (NJTS Reg. No. 04499600-7270)	Trade Secret*	10 - 30
2-Ethylhexyldiphenyl Phosphate	1241-94-7	3 - 7 Trade Secret *
Sodium Silicate	1344-09-8	3 - 7 Trade Secret *
Iron Oxide	1309-37-1	1 - 3
Polyethylene Glycol	25322-68-3	1 - 3
Continuous Filament Glass Fiber	None	1 - 3

Fiberglass	None	1 - 3
TRIPHENYL PHOSPHATE	115-86-6	< 0.3
Polyether Polyol	68815-56-5	< 0.2
Quartz Silica	14808-60-7	< 0.2

NJTS or NJTSRN: New Jersey Trade Secret Registry Number.

*The specific chemical identity and/or exact percentage (concentration) of this composition has been withheld as a trade secret.

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation:

Remove person to fresh air. If you feel unwell, get medical attention.

Skin Contact:

Immediately wash with soap and water. Remove contaminated clothing and wash before reuse. If signs/symptoms develop, get medical attention.

Eye Contact:

Immediately flush with large amounts of water. Remove contact lenses if easy to do. Continue rinsing. Get medical attention.

If Swallowed:

Rinse mouth. If you feel unwell, get medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Allergic skin reaction (redness, swelling, blistering, and itching).

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

Non-combustible. Use a fire fighting agent suitable for surrounding fire.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products

Substance

Carbon monoxide
Carbon dioxide
Oxides of Phosphorus

Condition

During Combustion
During Combustion
During Combustion

5.3. Special protective actions for fire-fighters

Wear full protective clothing, including helmet, self-contained, positive pressure or pressure demand breathing apparatus, bunker coat and pants, bands around arms, waist and legs, face mask, and protective covering for exposed areas of the head.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Evacuate area. Ventilate the area with fresh air. Use personal protective equipment based on the results of an exposure

assessment. Refer to Section 8 for PPE recommendations. If anticipated exposure resulting from an accidental release exceeds the protective capabilities of the PPE listed in Section 8, or are unknown, select PPE that offers an appropriate level of protection. Consider the physical and chemical hazards of the material when doing so. Examples of PPE ensembles for emergency response could include wearing bunker gear for a release of flammable material; wearing chemical protective clothing if the spilled material is a corrosive, a sensitizer, a significant dermal irritant, or can be absorbed through the skin; or donning a positive pressure supplied-air respirator for chemicals with inhalation hazards. For information regarding physical and health hazards, refer to sections 2 and 11 of the SDS.

6.2. Environmental precautions

Avoid release to the environment.

6.3. Methods and material for containment and cleaning up

Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Clean up residue. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Keep out of reach of children. Do not handle until all safety precautions have been read and understood. Avoid breathing dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wash contaminated clothing before reuse. Use personal protective equipment (gloves, respirators, etc.) as required.

7.2. Conditions for safe storage including any incompatibilities

Keep cool. Store away from heat. Store away from areas where product may come into contact with food or pharmaceuticals. Store in a dry place.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Occupational exposure limits

If a component is disclosed in section 3 but does not appear in the table below, an occupational exposure limit is not available for the component.

Ingredient	C.A.S. No.	Agency	Limit type	Additional Comments
TRIPHENYL PHOSPHATE	115-86-6	ACGIH	TWA:3 mg/m3	A4: Not class. as human carcin
TRIPHENYL PHOSPHATE	115-86-6	OSHA	TWA:3 mg/m3	
Iron Oxide	1309-37-1	ACGIH	TWA(respirable fraction):5 mg/m3	A4: Not class. as human carcin
Iron Oxide	1309-37-1	OSHA	TWA(as fume):10 mg/m3	
Borate compounds, inorganic, inhalable fraction	138265-88-0	ACGIH	TWA(inhalable fraction):2 mg/m3;STEL(inhalable fraction):6 mg/m3	A4: Not class. as human carcin
Quartz Silica	14808-60-7	OSHA	TWA Table Z-1(respirable):0.05 mg/m3;TWA Table Z-3(respirable):0.1 mg/m3;TWA concentration(respirable):0.1 mg/m3(2.4 millions of particles/cu. ft.)	
Silica, crystalline, respirable	14808-60-7	ACGIH	TWA(respirable	A2: Suspected human

fraction			fraction):0.025 mg/m3	carcin.
Polyethylene Glycol	25322-68-3	AIHA	TWA:10 mg/m3	

ACGIH : American Conference of Governmental Industrial Hygienists

AIHA : American Industrial Hygiene Association

CMRG : Chemical Manufacturer's Recommended Guidelines

OSHA : United States Department of Labor - Occupational Safety and Health Administration

TWA: Time-Weighted-Average

STEL: Short Term Exposure Limit

CEIL: Ceiling

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Select and use eye/face protection to prevent contact based on the results of an exposure assessment. The following eye/face protection(s) are recommended:

Full Face Shield

Indirect Vented Goggles

Skin/hand protection

Select and use gloves and/or protective clothing approved to relevant local standards to prevent skin contact based on the results of an exposure assessment. Selection should be based on use factors such as exposure levels, concentration of the substance or mixture, frequency and duration, physical challenges such as temperature extremes, and other use conditions. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible gloves/protective clothing. Note: Nitrile gloves may be worn over polymer laminate gloves to improve dexterity.

Gloves made from the following material(s) are recommended: Polymer laminate

If this product is used in a manner that presents a higher potential for exposure (e.g., spraying, high splash potential, etc.), then use of a protective apron may be necessary. See recommended glove material(s) for determining appropriate apron material(s). If a glove material is not available as an apron, polymer laminate is a suitable option.

Respiratory protection

An exposure assessment may be needed to decide if a respirator is required. If a respirator is needed, use respirators as part of a full respiratory protection program. Based on the results of the exposure assessment, select from the following respirator type(s) to reduce inhalation exposure:

Half facepiece or full facepiece air-purifying respirator suitable for particulates

Half facepiece or full facepiece supplied-air respirator

For questions about suitability for a specific application, consult with your respirator manufacturer.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Solid
Specific Physical Form:	Paste
Color	Red
Odor	Odorless

Odor threshold	No Data Available
pH	7.5 - 8
Melting point/Freezing point	No Data Available
Boiling point/Initial boiling point/Boiling range	100 °C
Flash Point	No flash point
Evaporation rate	0.33 [Ref Std:BUOAC=1]
Flammability	Not Applicable
Flammable Limits(LEL)	Not Applicable
Flammable Limits(UEL)	Not Applicable
Vapor Pressure	17.5 mmHg [@ 20 °C]
Relative Vapor Density	No Data Available
Density	No Data Available
Relative Density	1.35 [Ref Std:WATER=1]
Water solubility	Complete
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	No Data Available
Autoignition temperature	Not Applicable
Decomposition temperature	No Data Available
Kinematic Viscosity	No Data Available
Volatile Organic Compounds	<=0.5 % weight [Test Method:tested per EPA method 24]
Percent volatile	No Data Available
VOC Less H2O & Exempt Solvents	<=6 g/l [Test Method:tested per EPA method 24]
Molecular weight	No Data Available

Particle Characteristics	Not Applicable
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SECTION 10: Stability and reactivity

10.1. Reactivity

This material is considered to be non reactive under normal use conditions.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Substance

Condition

None known.

Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient

classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

Skin Contact:

Mild Skin Irritation: Signs/symptoms may include localized redness, swelling, itching, and dryness. Allergic Skin Reaction (non-photo induced): Signs/symptoms may include redness, swelling, blistering, and itching.

Eye Contact:

Severe Eye Irritation: Signs/symptoms may include significant redness, swelling, pain, tearing, cloudy appearance of the cornea, and impaired vision.

Ingestion:

Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

May cause additional health effects (see below).

Additional Health Effects:

Reproductive/Developmental Toxicity:

Contains a chemical or chemicals which can cause birth defects or other reproductive harm.

Genotoxicity:

Genotoxicity and Mutagenicity: May interact with genetic material and possibly alter gene expression.

Carcinogenicity:

<u>Ingredient</u>	<u>CAS No.</u>	<u>Class Description</u>	<u>Regulation</u>
Silica, Crystalline (Respirable Size)	14808-60-7	Known To Be Human Carcinogen.	National Toxicology Program Carcinogens
Silica dust, crystalline, in the form of quartz or cristobalite	14808-60-7	Grp. 1: Carcinogenic to humans	International Agency for Research on Cancer

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

<u>Name</u>	<u>Route</u>	<u>Species</u>	<u>Value</u>
Overall product	Dermal		No data available; calculated ATE >5,000 mg/kg
Overall product	Ingestion		No data available; calculated ATE >5,000 mg/kg
Zinc Borate 2335	Dermal	Rabbit	LD50 > 5,000 mg/kg
Zinc Borate 2335	Inhalation-Dust/Mist	Rat	LC50 > 4.95 mg/l
Zinc Borate 2335	Ingestion	Rat	LD50 > 5,000 mg/kg
Polymer (NJTS Reg. No. 04499600-7270)	Dermal		LD50 estimated to be > 5,000 mg/kg
Polymer (NJTS Reg. No. 04499600-7270)	Ingestion	Rat	LD50 > 2,000 mg/kg

Sodium Silicate	Dermal	Rabbit	LD50 > 4,640 mg/kg
Sodium Silicate	Ingestion	Rat	LD50 500 mg/kg
2-Ethylhexyldiphenyl Phosphate	Dermal	Rabbit	LD50 > 7,940 mg/kg
2-Ethylhexyldiphenyl Phosphate	Inhalation-Dust/Mist (4 hours)	Rat	LC50 > 4.8 mg/l
2-Ethylhexyldiphenyl Phosphate	Ingestion	Rat	LD50 > 15,800 mg/kg
Iron Oxide	Dermal	Not available	LD50 3,100 mg/kg
Iron Oxide	Ingestion	Not available	LD50 3,700 mg/kg
Polyethylene Glycol	Dermal	Rabbit	LD50 > 20,000 mg/kg
Polyethylene Glycol	Ingestion	Rat	LD50 32,770 mg/kg
TRIPHENYL PHOSPHATE	Dermal	Rabbit	LD50 > 10,000 mg/kg
TRIPHENYL PHOSPHATE	Ingestion	Rat	LD50 > 20,000 mg/kg
Polyether Polyol	Ingestion	Mouse	LD50 > 540 mg/kg
Quartz Silica	Dermal		LD50 estimated to be > 5,000 mg/kg
Quartz Silica	Ingestion		LD50 estimated to be > 5,000 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
Zinc Borate 2335	Rabbit	No significant irritation
Polymer (NJTS Reg. No. 04499600-7270)	Rabbit	Minimal irritation
Sodium Silicate	Rabbit	Corrosive
2-Ethylhexyldiphenyl Phosphate	Rabbit	Minimal irritation
Iron Oxide	Rabbit	No significant irritation
Polyethylene Glycol	Rabbit	Minimal irritation
TRIPHENYL PHOSPHATE	Rabbit	No significant irritation
Polyether Polyol	In vitro data	Corrosive
Quartz Silica	Professional judgement	No significant irritation

Serious Eye Damage/Irritation

Name	Species	Value
Zinc Borate 2335	Rabbit	Severe irritant
Polymer (NJTS Reg. No. 04499600-7270)	Professional judgement	Mild irritant
Sodium Silicate	In vitro data	Corrosive
2-Ethylhexyldiphenyl Phosphate	Rabbit	Mild irritant
Iron Oxide	Rabbit	No significant irritation
Polyethylene Glycol	Rabbit	Mild irritant
TRIPHENYL PHOSPHATE	Rabbit	Mild irritant
Polyether Polyol	In vitro data	Corrosive

Skin Sensitization

Name	Species	Value
Zinc Borate 2335	Guinea pig	Not classified
Sodium Silicate	Mouse	Not classified
2-Ethylhexyldiphenyl Phosphate	Human	Not classified
Iron Oxide	Human	Not classified
Polyethylene Glycol	Guinea pig	Not classified

TRIPHENYL PHOSPHATE	Human	Not classified
Polyether Polyol	In vitro data	Sensitizing

Respiratory Sensitization

For the component/components, either no data are currently available or the data are not sufficient for classification.

Germ Cell Mutagenicity

Name	Route	Value
Zinc Borate 2335	In Vitro	Some positive data exist, but the data are not sufficient for classification
Zinc Borate 2335	In vivo	Mutagenic
Sodium Silicate	In Vitro	Not mutagenic
Sodium Silicate	In vivo	Not mutagenic
2-Ethylhexyldiphenyl Phosphate	In Vitro	Not mutagenic
2-Ethylhexyldiphenyl Phosphate	In vivo	Not mutagenic
Iron Oxide	In Vitro	Not mutagenic
Polyethylene Glycol	In Vitro	Not mutagenic
Polyethylene Glycol	In vivo	Not mutagenic
TRIPHENYL PHOSPHATE	In Vitro	Not mutagenic
Polyether Polyol	In Vitro	Not mutagenic
Quartz Silica	In Vitro	Some positive data exist, but the data are not sufficient for classification
Quartz Silica	In vivo	Some positive data exist, but the data are not sufficient for classification

Carcinogenicity

Name	Route	Species	Value
Iron Oxide	Inhalation	Human	Some positive data exist, but the data are not sufficient for classification
Polyethylene Glycol	Ingestion	Rat	Not carcinogenic
Quartz Silica	Inhalation	Human and animal	Carcinogenic

Reproductive Toxicity

Reproductive and/or Developmental Effects

Name	Route	Value	Species	Test Result	Exposure Duration
Zinc Borate 2335	Ingestion	Toxic to male reproduction	Rat	NOAEL 100 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	Toxic to development	Rat	LOAEL 100 mg/kg/day	during gestation
Sodium Silicate	Ingestion	Not classified for development	Mouse	NOAEL 200 mg/kg/day	during gestation
2-Ethylhexyldiphenyl Phosphate	Ingestion	Not classified for male reproduction	Rat	NOAEL 890 mg/kg/day	2 generation
2-Ethylhexyldiphenyl Phosphate	Ingestion	Toxic to female reproduction	Rat	NOAEL 160 mg/kg/day	2 generation
2-Ethylhexyldiphenyl Phosphate	Ingestion	Toxic to development	Rat	NOAEL 160 mg/kg/day	2 generation
Polyethylene Glycol	Ingestion	Not classified for female reproduction	Rat	NOAEL 1,125 mg/kg/day	during gestation
Polyethylene Glycol	Ingestion	Not classified for male reproduction	Rat	NOAEL 5699 +/- 1341 mg/kg/day	5 days
Polyethylene Glycol	Not Specified	Not classified for reproduction and/or development		NOEL N/A	
Polyethylene Glycol	Ingestion	Not classified for development	Mouse	NOAEL 562 mg/animal/day	during gestation

TRIPHENYL PHOSPHATE	Ingestion	Not classified for female reproduction	Rat	NOAEL 690 mg/kg/day	premating & during gestation
TRIPHENYL PHOSPHATE	Ingestion	Not classified for male reproduction	Rat	NOAEL 690 mg/kg/day	91 days
TRIPHENYL PHOSPHATE	Ingestion	Toxic to development	Rat	NOAEL 77 mg/kg/day	1 generation

Target Organ(s)

Specific Target Organ Toxicity - single exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
Zinc Borate 2335	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification	similar health hazards	NOAEL Not available	
Sodium Silicate	Inhalation	respiratory irritation	May cause respiratory irritation	official classification	NOAEL Not available	
Polyethylene Glycol	Inhalation	respiratory irritation	Not classified	Rat	NOAEL 1.008 mg/l	2 weeks
Polyether Polyol	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification	similar health hazards	NOAEL not available	

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
Zinc Borate 2335	Inhalation	immune system	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	respiratory system	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	heart	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	endocrine system	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	hematopoietic system	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	liver	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	nervous system	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Inhalation	kidney and/or bladder	Not classified	Rat	NOAEL 0.15 mg/l	2 weeks
Zinc Borate 2335	Ingestion	endocrine system	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	liver	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	kidney and/or bladder	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	heart	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	skin	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	bone, teeth, nails, and/or hair	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	hematopoietic system	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	immune system	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	nervous system	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	eyes	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Zinc Borate 2335	Ingestion	respiratory system	Not classified	Rat	NOAEL 375	92 days

					mg/kg/day	
Zinc Borate 2335	Ingestion	vascular system	Not classified	Rat	NOAEL 375 mg/kg/day	92 days
Sodium Silicate	Ingestion	kidney and/or bladder	Some positive data exist, but the data are not sufficient for classification	Dog	LOAEL 2,400 mg/kg/day	4 weeks
Sodium Silicate	Ingestion	endocrine system	Not classified	Rat	NOAEL 804 mg/kg/day	3 months
Sodium Silicate	Ingestion	blood	Not classified	Rat	NOAEL 804 mg/kg/day	3 months
Sodium Silicate	Ingestion	heart	Not classified	Rat	NOAEL 1,259 mg/kg/day	8 weeks
Sodium Silicate	Ingestion	liver	Not classified	Rat	NOAEL 1,259 mg/kg/day	8 weeks
2-Ethylhexyldiphenyl Phosphate	Ingestion	heart	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	endocrine system	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	gastrointestinal tract	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	hematopoietic system	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	liver	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	nervous system	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
2-Ethylhexyldiphenyl Phosphate	Ingestion	kidney and/or bladder	Not classified	Rat	NOAEL 463 mg/kg/day	90 days
Iron Oxide	Inhalation	pulmonary fibrosis	Not classified	Human	NOAEL Not available	occupational exposure
Iron Oxide	Inhalation	pneumoconiosis	Not classified	Human	NOAEL Not available	occupational exposure
Polyethylene Glycol	Inhalation	respiratory system	Not classified	Rat	NOAEL 1,008 mg/l	2 weeks
Polyethylene Glycol	Ingestion	kidney and/or bladder	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
Polyethylene Glycol	Ingestion	heart	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
Polyethylene Glycol	Ingestion	endocrine system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
Polyethylene Glycol	Ingestion	hematopoietic system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
Polyethylene Glycol	Ingestion	liver	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
Polyethylene Glycol	Ingestion	nervous system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
TRIPHENYL PHOSPHATE	Dermal	endocrine system	Not classified	Rabbit	NOAEL 1,000 mg/kg/day	3 weeks
TRIPHENYL PHOSPHATE	Dermal	hematopoietic system	Not classified	Rabbit	NOAEL 1,000 mg/kg/day	3 weeks
TRIPHENYL PHOSPHATE	Dermal	liver	Not classified	Rabbit	NOAEL 1,000 mg/kg/day	3 weeks
TRIPHENYL PHOSPHATE	Dermal	nervous system	Not classified	Rabbit	NOAEL 1,000 mg/kg/day	3 weeks
TRIPHENYL	Dermal	kidney and/or	Not classified	Rabbit	NOAEL	3 weeks

PHOSPHATE		bladder			1,000 mg/kg/day	
TRIPHENYL PHOSPHATE	Ingestion	endocrine system	Not classified	Rat	NOAEL 583 mg/kg/day	90 days
TRIPHENYL PHOSPHATE	Ingestion	liver	Not classified	Rat	NOAEL 583 mg/kg/day	90 days
TRIPHENYL PHOSPHATE	Ingestion	immune system	Not classified	Rat	NOAEL 700 mg/kg/day	120 days
TRIPHENYL PHOSPHATE	Ingestion	gastrointestinal tract	Not classified	Rat	NOAEL 583 mg/kg/day	90 days
TRIPHENYL PHOSPHATE	Ingestion	nervous system	Not classified	Chicken	NOAEL 10,000 mg/kg/day	42 days
Quartz Silica	Inhalation	silicosis	Causes damage to organs through prolonged or repeated exposure	Human	NOAEL Not available	occupational exposure

Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Test Organism	Test Type	Result
Water flea, Daphnia magna	48 hours Aquatic Toxicity - Acute	27 mg/l
Green algae, Pseudokirchneriella subcapitata	72 hours Aquatic Toxicity - Chronic	2.6 mg/l

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Dispose of waste product in a permitted industrial waste facility. Empty drums/barrels/containers used for transporting and handling hazardous chemicals (chemical substances/mixtures/preparations classified as Hazardous as per applicable regulations) shall be considered, stored, treated & disposed of as hazardous wastes unless otherwise defined by applicable waste regulations. Consult with the respective regulating authorities to determine the available treatment and disposal facilities.

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information**15.1. US Federal Regulations**

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:**Physical Hazards**

Not Applicable.

Health Hazards

Germ cell mutagenicity

Reproductive toxicity

Respiratory or Skin Sensitization

Serious eye damage or eye irritation

Section 313 Toxic Chemicals subject to the reporting requirements of that section and 40 CFR part 372 (EPCRA):

<u>Ingredient</u>	<u>C.A.S. No</u>	<u>% by Wt</u>
Zinc Borate 2335	138265-88-0	Trade Secret 10 - 30

This material contains a chemical which requires export notification under TSCA Section 12[b]:

<u>Ingredient (Category if applicable)</u>	<u>C.A.S. No</u>	<u>Regulation</u>	<u>Status</u>
TRIPHENYL PHOSPHATE	115-86-6	Toxic Substances Control Act (TSCA) 4 Test Rule Chemicals	Applicable
TRIPHENYL PHOSPHATE	115-86-6	Toxic Substances Control Act (TSCA) 5 SNUR or Consent Order Chemicals	Proposed

This material contains a chemical subject to a proposed EPA Significant New Use Rule (TSCA Section 5)

<u>Ingredient (Category if applicable)</u>	<u>C.A.S. No</u>	<u>Reference</u>
TRIPHENYL PHOSPHATE	115-86-6	40 CFR 721.11778

15.2. State Regulations**15.3. Chemical Inventories**

The components of this material are in compliance with the provisions of Australia National Industrial Chemical Notification and Assessment Scheme (NICNAS). Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the new substance notification requirements of CEPA.

The components of this material are in compliance with the China "Measures on Environmental Management of New Chemical Substance". Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of the Korean Toxic Chemical Control Law. Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

15.4. International Regulations

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 16: Other information

NFPA Hazard Classification**Health:** 2 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

Document Group: 09-5451-1**Version Number:** 30.00**Issue Date:** 01/29/26**Supersedes Date:** 01/20/25

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Safety Data Sheet

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Document Group: 06-5049-9
Issue Date: 01/19/26

Version Number: 13.02
Supersedes Date: 04/30/21

SECTION 1: Identification

1.1. Product identifier

3M™ Wire Pulling Lubricant - WL Series (WL-QT, WL-1, WL-5)

Product Identification Numbers

ID Number	UPC	ID Number	UPC
80-6107-3662-3	00-54007-50622-5	80-6107-3663-1	00-54007-50623-2
80-6107-3664-9	00-54007-50624-9	80-6107-3665-6	00-54007-50625-6
80-6107-3931-2	00-54007-41313-4	80-6108-3578-9	00-54007-41925-9
80-6114-9092-3			

7000058273, 7100027866, 7000031543, 7000058274

1.2. Recommended use and restrictions on use

Recommended use

lubricant, wire pulling, LUBRICANT, WIRE PULLING

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Electrical Markets Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Not classified as hazardous according to OSHA Hazard Communication Standard, 29 CFR 1910.1200.

2.2. Label elements

Signal word

Not applicable.

Symbols

Not applicable

Pictograms

Not applicable

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
WATER	7732-18-5	90 - 99
GLYCOLS, POLYPROPYLENE	25322-69-4	0.1 - 2
POLYETHYLENE GLYCOL	25322-68-3	0.1 - 2
SODIUM POLYACRYLATE	9003-04-7	0.1 - 1

SECTION 4: First aid measures**4.1. Description of first aid measures****Inhalation:**

No need for first aid is anticipated. If symptoms develop, remove the affected person to fresh air. Get medical attention.

Skin Contact:

Wash with soap and water. If signs/symptoms develop, get medical attention.

Eye Contact:

Flush with large amounts of water. Remove contact lenses if easy to do. Continue rinsing. If signs/symptoms persist, get medical attention.

If Swallowed:

Rinse mouth. If you feel unwell, get medical attention.

4.2. Most important symptoms and effects, both acute and delayed

No critical symptoms or effects. See Section 11.1, information on toxicological effects.

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures**5.1. Suitable extinguishing media**

Material will not burn. Non-combustible. Use a fire fighting agent suitable for surrounding fire. Use a fire fighting agent suitable for the surrounding fire.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products**Substance**

Carbon monoxide

Carbon dioxide

Condition

During Combustion

During Combustion

5.3. Special protective actions for fire-fighters

No special protective actions for fire-fighters are anticipated.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment based on the results of an exposure assessment. Refer to Section 8 for PPE recommendations. If anticipated exposure resulting from an accidental release exceeds the protective capabilities of the PPE listed in Section 8, or are unknown, select PPE that offers an appropriate level of protection. Consider the physical and chemical hazards of the material when doing so. Examples of PPE ensembles for emergency response could include wearing bunker gear for a release of flammable material; wearing chemical protective clothing if the spilled material is a corrosive, a sensitizer, a significant dermal irritant, or can be absorbed through the skin; or donning a positive pressure supplied-air respirator for chemicals with inhalation hazards. For information regarding physical and health hazards, refer to sections 2 and 11 of the SDS. Evacuate area. Ventilate the area with fresh air.

6.2. Environmental precautions

Avoid release to the environment. For larger spills, cover drains and build dikes to prevent entry into sewer systems or bodies of water.

6.3. Methods and material for containment and cleaning up

Working from around the edges of the spill inward, cover with bentonite, vermiculite, or commercially available inorganic absorbent material. Mix in sufficient absorbent until it appears dry. Remember, adding an absorbent material does not remove a physical, health, or environmental hazard. Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Clean up residue with water. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

For industrial/occupational use only. Not for consumer sale or use. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Avoid release to the environment.

7.2. Conditions for safe storage including any incompatibilities

No special storage requirements.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Occupational exposure limits

If a component is disclosed in section 3 but does not appear in the table below, an occupational exposure limit is not available for the component.

Ingredient	C.A.S. No.	Agency	Limit type	Additional Comments
POLYETHYLENE GLYCOL	25322-68-3	AIHA	TWA:10 mg/m3	
GLYCOLS, POLYPROPYLENE	25322-69-4	AIHA	TWA(as aerosol):10 mg/m3	

ACGIH : American Conference of Governmental Industrial Hygienists

AIHA : American Industrial Hygiene Association

CMRG : Chemical Manufacturer's Recommended Guidelines

OSHA : United States Department of Labor - Occupational Safety and Health Administration

TWA: Time-Weighted-Average

STEL: Short Term Exposure Limit

CEIL: Ceiling

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure

Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Under normal use conditions, eye exposure is not expected to be significant enough to require eye protection.

Skin/hand protection

No protective gloves required.

Respiratory protection

Under normal use conditions, airborne exposures are not expected to be significant enough to require respiratory protection.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Liquid
Specific Physical Form:	Gel
Color	White
Odor	Odorless
Odor threshold	No Data Available
pH	6.5 - 8.5
Melting point/Freezing point	Not Applicable
Boiling point/Initial boiling point/Boiling range	100 °C
Flash Point	No flash point
Evaporation rate	No Data Available
Flammability	Not Applicable
Flammable Limits(LEL)	Not Applicable
Flammable Limits(UEL)	Not Applicable
Vapor Pressure	18 mmHg [@ 68.0000000000 °F] [Details:@20C MITS data]
Relative Vapor Density	0.9 - 1.1 [Ref Std: AIR=1]
Density	Not Applicable
Relative Density	1.01 [Ref Std: WATER=1]
Water solubility	Complete
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	Not Applicable
Autoignition temperature	Not Applicable
Decomposition temperature	No Data Available
Kinematic Viscosity	111,386 mm2/sec
Volatile Organic Compounds	0 lb/gal
Percent volatile	0 %
VOC Less H2O & Exempt Solvents	0 g/l
Average particle size	Not Applicable
Bulk density	Not Applicable
Molecular weight	Not Applicable
Softening point	Not Applicable

Particle Characteristics	Not Applicable
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SECTION 10: Stability and reactivity

10.1. Reactivity

This material is considered to be non reactive under normal use conditions.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

None known.

Not Applicable

10.6. Hazardous decomposition products

<u>Substance</u>	<u>Condition</u>
None known.	

Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

No known health effects.

Skin Contact:

Contact with the skin during product use is not expected to result in significant irritation.

Eye Contact:

Contact with the eyes during product use is not expected to result in significant irritation.

Ingestion:

Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

Name	Route	Species	Value
Overall product	Ingestion		No data available; calculated ATE >5,000 mg/kg
GLYCOLS, POLYPROPYLENE	Dermal	Rabbit	LD50 > 10,000 mg/kg
POLYETHYLENE GLYCOL	Dermal	Rabbit	LD50 > 20,000 mg/kg
GLYCOLS, POLYPROPYLENE	Ingestion	Rat	LD50 > 1,000 mg/kg
POLYETHYLENE GLYCOL	Ingestion	Rat	LD50 32,770 mg/kg
SODIUM POLYACRYLATE	Dermal	Rabbit	LD50 > 5,000 mg/kg
SODIUM POLYACRYLATE	Ingestion	Rat	LD50 > 2,000 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
GLYCOLS, POLYPROPYLENE	Not available	No significant irritation
POLYETHYLENE GLYCOL	Rabbit	Minimal irritation
SODIUM POLYACRYLATE	Rabbit	No significant irritation

Serious Eye Damage/Irritation

Name	Species	Value
GLYCOLS, POLYPROPYLENE	Not available	Mild irritant
POLYETHYLENE GLYCOL	Rabbit	Mild irritant
SODIUM POLYACRYLATE	Rabbit	No significant irritation

Skin Sensitization

Name	Species	Value
GLYCOLS, POLYPROPYLENE	Human and animal	Not classified
POLYETHYLENE GLYCOL	Guinea pig	Not classified

Respiratory Sensitization

For the component/components, either no data are currently available or the data are not sufficient for classification.

Germ Cell Mutagenicity

Name	Route	Value
GLYCOLS, POLYPROPYLENE	In Vitro	Not mutagenic
POLYETHYLENE GLYCOL	In Vitro	Not mutagenic
POLYETHYLENE GLYCOL	In vivo	Not mutagenic

Carcinogenicity

Name	Route	Species	Value
POLYETHYLENE GLYCOL	Ingestion	Rat	Not carcinogenic

Reproductive Toxicity**Reproductive and/or Developmental Effects**

Name	Route	Value	Species	Test Result	Exposure Duration
POLYETHYLENE GLYCOL	Ingestion	Not classified for female reproduction	Rat	NOAEL 1,125 mg/kg/day	during gestation
POLYETHYLENE GLYCOL	Ingestion	Not classified for male reproduction	Rat	NOAEL 5699 +/- 1341 mg/kg/day	5 days

POLYETHYLENE GLYCOL	Not Specified	Not classified for reproduction and/or development		NOEL N/A	
POLYETHYLENE GLYCOL	Ingestion	Not classified for development	Mouse	NOAEL 562 mg/animal/day	during gestation

Target Organ(s)**Specific Target Organ Toxicity - single exposure**

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
POLYETHYLENE GLYCOL	Inhalation	respiratory irritation	Not classified	Rat	NOAEL 1.008 mg/l	2 weeks

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
POLYETHYLENE GLYCOL	Inhalation	respiratory system	Not classified	Rat	NOAEL 1.008 mg/l	2 weeks
POLYETHYLENE GLYCOL	Ingestion	kidney and/or bladder	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
POLYETHYLENE GLYCOL	Ingestion	heart	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
POLYETHYLENE GLYCOL	Ingestion	endocrine system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
POLYETHYLENE GLYCOL	Ingestion	hematopoietic system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
POLYETHYLENE GLYCOL	Ingestion	liver	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks
POLYETHYLENE GLYCOL	Ingestion	nervous system	Not classified	Rat	NOAEL 5,640 mg/kg/day	13 weeks

Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Prior to disposal, consult all applicable authorities and regulations to insure proper classification. Dispose of waste product in a permitted industrial waste facility. Empty and clean product containers may be disposed as non-hazardous waste. Consult your specific regulations and service providers to determine available options and requirements.

EPA Hazardous Waste Number (RCRA): Not regulated

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information

15.1. US Federal Regulations

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:

Physical Hazards

Not Applicable.

Health Hazards

Not Applicable.

15.2. State Regulations

Contact 3M for more information.

15.3. Chemical Inventories

The components of this material are in compliance with the provisions of Australia National Industrial Chemical Notification and Assessment Scheme (NICNAS). Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the new substance notification requirements of CEPA.

The components of this material are in compliance with the China "Measures on Environmental Management of New Chemical Substance". Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of the Korean Toxic Chemical Control Law. Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of Japan Chemical Substance Control Law. Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of Philippines RA 6969 requirements. Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

Contact 3M for more information.

15.4. International Regulations

Contact 3M for more information.

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 16: Other information**NFPA Hazard Classification****Health:** 0 **Flammability:** 0 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

Document Group: 06-5049-9**Version Number:** 13.02**Issue Date:** 01/19/26**Supersedes Date:** 04/30/21

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Issue Date:	05/03/21	Supersedes Date:	12/16/20

Product identifier

3M™ Scotchcast™ Electrical Insulating Resin 4N, Part A and B

ID Number	UPC	ID Number	UPC
80-6116-1677-4		80-6116-1678-2	
80-6116-1679-0		80-6116-1680-8	
80-6116-1681-6		80-6116-1682-4	
80-6116-1683-2		80-6116-2766-4	000-51128-61352-8

7100127062, 7100127061, 7100123070, 7100165596, 7100165597, 4100026994, 7100165598, 7100180522

Recommended use

Electrical

Supplier's details

MANUFACTURER:	3M
DIVISION:	Electrical Markets Division
ADDRESS:	3M Center, St. Paul, MN 55144-1000, USA
Telephone:	1-888-3M HELPS (1-888-364-3577)

Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

This product is a kit or a multipart product which consists of multiple, independently packaged components. A Safety Data Sheet (SDS), Article Information Sheet (AIS), or Article Information Letter (AIL) for each of these components is included. Please do not separate the component documents from this cover page. The document numbers for components of this product are:

24-9848-3, 35-7972-9

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Document Group: 24-9848-3
Issue Date: 02/19/20

Version Number: 3.02
Supersedes Date: 11/08/19

SECTION 1: Identification

1.1. Product identifier

3M™ Scotchcast™ Electrical Insulating Resin 4N, Part A and 3M™ Scotchcast™ Electrical Insulating Resin 4, Part A

1.2. Recommended use and restrictions on use

Recommended use

Electrical, Part A of Resin 4 & Resin 4N

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Electrical Markets Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Serious Eye Damage/Irritation: Category 2B.

Skin Sensitizer: Category 1A.

2.2. Label elements

Signal word

Warning

Symbols

Exclamation mark |

Pictograms



Hazard Statements

Causes eye irritation.
May cause an allergic skin reaction.

Precautionary Statements**Prevention:**

Avoid breathing dust/fume/gas/mist/vapors/spray.
Wear protective gloves.
Wash thoroughly after handling.
Contaminated work clothing must not be allowed out of the workplace.

Response:

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
If eye irritation persists: Get medical advice/attention.
IF ON SKIN: Wash with plenty of soap and water.
If skin irritation or rash occurs: Get medical advice/attention.
Wash contaminated clothing before reuse.

Disposal:

Dispose of contents/container in accordance with applicable local/regional/national/international regulations.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	25085-99-8	80 - 100 Trade Secret *
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	68609-97-2	< 20 Trade Secret *

*The specific chemical identity and/or exact percentage (concentration) of this composition has been withheld as a trade secret.

SECTION 4: First aid measures**4.1. Description of first aid measures****Inhalation:**

Remove person to fresh air. If you feel unwell, get medical attention.

Skin Contact:

Immediately wash with soap and water. Remove contaminated clothing and wash before reuse. If signs/symptoms develop, get medical attention.

Eye Contact:

Flush with large amounts of water. Remove contact lenses if easy to do. Continue rinsing. If signs/symptoms persist, get medical attention.

If Swallowed:

Rinse mouth. If you feel unwell, get medical attention.

4.2. Most important symptoms and effects, both acute and delayed

See Section 11.1. Information on toxicological effects.

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

In case of fire: Use a fire fighting agent suitable for ordinary combustible material such as water or foam to extinguish.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products

<u>Substance</u>	<u>Condition</u>
Carbon monoxide	During Combustion
Carbon dioxide	During Combustion
Toxic Vapor, Gas, Particulate	During Combustion

5.3. Special protective actions for fire-fighters

Wear full protective clothing, including helmet, self-contained, positive pressure or pressure demand breathing apparatus, bunker coat and pants, bands around arms, waist and legs, face mask, and protective covering for exposed areas of the head.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Evacuate area. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Refer to other sections of this SDS for information regarding physical and health hazards, respiratory protection, ventilation, and personal protective equipment.

6.2. Environmental precautions

Avoid release to the environment. For larger spills, cover drains and build dikes to prevent entry into sewer systems or bodies of water.

6.3. Methods and material for containment and cleaning up

Contain spill. Working from around the edges of the spill inward, cover with bentonite, vermiculite, or commercially available inorganic absorbent material. Mix in sufficient absorbent until it appears dry. Remember, adding an absorbent material does not remove a physical, health, or environmental hazard. Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Clean up residue with an appropriate solvent selected by a qualified and authorized person. Ventilate the area with fresh air. Read and follow safety precautions on the solvent label and SDS. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Avoid breathing dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wash contaminated clothing before reuse.

7.2. Conditions for safe storage including any incompatibilities

No special storage requirements.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Occupational exposure limits

No occupational exposure limit values exist for any of the components listed in Section 3 of this SDS.

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Select and use eye/face protection to prevent contact based on the results of an exposure assessment. The following eye/face protection(s) are recommended:

Indirect Vented Goggles

Skin/hand protection

Select and use gloves and/or protective clothing approved to relevant local standards to prevent skin contact based on the results of an exposure assessment. Selection should be based on use factors such as exposure levels, concentration of the substance or mixture, frequency and duration, physical challenges such as temperature extremes, and other use conditions. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible gloves/protective clothing. Note: Nitrile gloves may be worn over polymer laminate gloves to improve dexterity.

Gloves made from the following material(s) are recommended: Polymer laminate

If this product is used in a manner that presents a higher potential for exposure (eg. spraying, high splash potential etc.), then use of protective coveralls may be necessary. Select and use body protection to prevent contact based on the results of an exposure assessment. The following protective clothing material(s) are recommended: Apron - polymer laminate

Respiratory protection

An exposure assessment may be needed to decide if a respirator is required. If a respirator is needed, use respirators as part of a full respiratory protection program. Based on the results of the exposure assessment, select from the following respirator type(s) to reduce inhalation exposure:

Half facepiece or full facepiece air-purifying respirator suitable for organic vapors and particulates

For questions about suitability for a specific application, consult with your respirator manufacturer.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state

Liquid

Color

Amber

Specific Physical Form:

Resin

Odor

Epoxy

Odor threshold

No Data Available

pH

No Data Available

Melting point

No Data Available

Boiling Point	>= 201 °F
Flash Point	>= 201 °F [Test Method: Closed Cup]
Evaporation rate	No Data Available
Flammability (solid, gas)	Not Applicable
Flammable Limits(LEL)	No Data Available
Flammable Limits(UEL)	No Data Available
Vapor Pressure	<= 27 psia [@ 131 °F]
Vapor Density	No Data Available
Density	1.16 g/ml
Specific Gravity	1.16 [Ref Std: WATER=1]
Solubility in Water	Negligible
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	No Data Available
Autoignition temperature	No Data Available
Decomposition temperature	No Data Available
Viscosity	3,000 centipoise - 5,000 centipoise
Average particle size	No Data Available
Bulk density	No Data Available
Hazardous Air Pollutants	No Data Available
Molecular weight	No Data Available
Volatile Organic Compounds	No Data Available
Percent volatile	Negligible
Softening point	No Data Available
VOC Less H2O & Exempt Solvents	No Data Available

SECTION 10: Stability and reactivity

10.1. Reactivity

This material is considered to be non reactive under normal use conditions.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Substance

None known.

Condition

Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient

classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

Skin Contact:

Mild Skin Irritation: Signs/symptoms may include localized redness, swelling, itching, and dryness. Allergic Skin Reaction (non-photo induced): Signs/symptoms may include redness, swelling, blistering, and itching.

Eye Contact:

Moderate Eye Irritation: Signs/symptoms may include redness, swelling, pain, tearing, and blurred or hazy vision.

Ingestion:

Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

Name	Route	Species	Value
Overall product	Ingestion		No data available; calculated ATE >5,000 mg/kg
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Rat	LD50 > 1,600 mg/kg
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Rat	LD50 > 1,000 mg/kg
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	Rabbit	LD50 > 4,000 mg/kg
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Ingestion	Rat	LD50 17,100 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Rabbit	Mild irritant
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Rabbit	Mild irritant

Serious Eye Damage/Irritation

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Rabbit	Moderate irritant
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Rabbit	Mild irritant

Skin Sensitization

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Human and animal	Sensitizing
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Guinea	Sensitizing

	pig	
--	-----	--

Respiratory Sensitization

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Human	Not classified

Germ Cell Mutagenicity

Name	Route	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	In vivo	Not mutagenic
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	In Vitro	Some positive data exist, but the data are not sufficient for classification
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	In vivo	Not mutagenic
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	In Vitro	Some positive data exist, but the data are not sufficient for classification

Carcinogenicity

Name	Route	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Mouse	Some positive data exist, but the data are not sufficient for classification

Reproductive Toxicity**Reproductive and/or Developmental Effects**

Name	Route	Value	Species	Test Result	Exposure Duration
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for female reproduction	Rat	NOAEL 750 mg/kg/day	2 generation
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for male reproduction	Rat	NOAEL 750 mg/kg/day	2 generation
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Not classified for development	Rabbit	NOAEL 300 mg/kg/day	during organogenesis
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for development	Rat	NOAEL 750 mg/kg/day	2 generation
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	Not classified for development	Rat	NOAEL 200 mg/kg/day	during organogenesis

Target Organ(s)**Specific Target Organ Toxicity - single exposure**

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	heart blood liver nervous system kidney and/or bladder	Not classified	Rabbit	NOAEL 4,000 mg/kg	24 hours

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	liver	Not classified	Rat	NOAEL 1,000 mg/kg/day	2 years
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	nervous system	Not classified	Rat	NOAEL 1,000 mg/kg/day	13 weeks
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	auditory system heart endocrine system hematopoietic	Not classified	Rat	NOAEL 1,000 mg/kg/day	28 days

		system liver eyes kidney and/or bladder				
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL] DERIVATIVES	Dermal	nervous system respiratory system	Not classified	Rat	NOAEL 100 mg/kg/day	14 weeks
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL] DERIVATIVES	Dermal	blood liver eyes kidney and/or bladder	Not classified	Rat	NOAEL 100 mg/kg/day	13 weeks

Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Incinerate in a permitted waste incineration facility. Proper destruction may require the use of additional fuel during incineration processes. As a disposal alternative, utilize an acceptable permitted waste disposal facility. Empty drums/barrels/containers used for transporting and handling hazardous chemicals (chemical substances/mixtures/preparations classified as Hazardous as per applicable regulations) shall be considered, stored, treated & disposed of as hazardous wastes unless otherwise defined by applicable waste regulations. Consult with the respective regulating authorities to determine the available treatment and disposal facilities.

EPA Hazardous Waste Number (RCRA): Not regulated

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information**15.1. US Federal Regulations**

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:**Physical Hazards**

Not applicable

Health Hazards

Respiratory or Skin Sensitization

Serious eye damage or eye irritation

15.2. State Regulations

Contact 3M for more information.

15.3. Chemical Inventories

The components of this material are in compliance with the provisions of Australia National Industrial Chemical Notification and Assessment Scheme (NICNAS). Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the new substance notification requirements of CEPA.

The components of this material are in compliance with the China "Measures on Environmental Management of New Chemical Substance". Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of the Korean Toxic Chemical Control Law. Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of Philippines RA 6969 requirements. Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

Contact 3M for more information.

15.4. International Regulations

Contact 3M for more information.

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.
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SECTION 16: Other information**NFPA Hazard Classification****Health:** 2 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

HMIS Hazard Classification**Health:** 2 **Flammability:** 1 **Physical Hazard:** 0 **Personal Protection:** X - See PPE section.

Hazardous Material Identification System (HMIS® IV) hazard ratings are designed to inform employees of chemical hazards in the workplace. These ratings are based on the inherent properties of the material under expected conditions of normal use and are not intended for use in emergency situations. HMIS® IV ratings are to be used with a fully implemented HMIS® IV program. HMIS® is a registered mark of the American Coatings Association (ACA).

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Issue Date: 02/19/20**Supersedes Date:** 11/08/19

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SECTION 1: Identification

1.1. Product identifier

3M™ Scotchcast™ Electrical Insulating Resin 4N, Part B

1.2. Recommended use and restrictions on use

Recommended use

Electrical, Part B of Resin 4N

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Electrical Markets Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Acute Toxicity (oral): Category 4.
Acute Toxicity (dermal): Category 4.
Serious Eye Damage/Irritation: Category 1.
Skin Corrosion/Irritation: Category 1B.
Skin Sensitizer: Category 1A.
Reproductive Toxicity: Category 2.
Carcinogenicity: Category 1B.
Specific Target Organ Toxicity (repeated exposure): Category 1.

2.2. Label elements

Signal word

Danger

Symbols

Corrosion | Exclamation mark | Health Hazard |

Pictograms

**Hazard Statements**

Harmful if swallowed.

Harmful in contact with skin.

Causes severe skin burns and eye damage.

May cause an allergic skin reaction.

Suspected of damaging fertility or the unborn child.

May cause cancer.

Causes damage to organs through prolonged or repeated exposure:
respiratory system |

Precautionary Statements**Prevention:**

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Do not breathe dust/fume/gas/mist/vapors/spray.

Wear protective gloves, protective clothing, and eye/face protection.

Do not eat, drink or smoke when using this product.

Wash thoroughly after handling.

Contaminated work clothing must not be allowed out of the workplace.

Response:

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Immediately call a POISON CENTER or doctor/physician.

If skin irritation or rash occurs: Get medical advice/attention.

Wash contaminated clothing before reuse.

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

Storage:

Store locked up.

Disposal:

Dispose of contents/container in accordance with applicable local/regional/national/international regulations.

2.3. Hazards not otherwise classified

May cause chemical gastrointestinal burns.

Supplemental Information:

Persons previously sensitized to amines may develop a cross-sensitization reaction to certain other amines.

18% of the mixture consists of ingredients of unknown acute oral toxicity.

36% of the mixture consists of ingredients of unknown acute dermal toxicity.

100% of the mixture consists of ingredients of unknown acute inhalation toxicity.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
STYRENATED PHENOL	Trade Secret*	25 - 70 Trade Secret *
N-AMINOETHYLPIPERAZINE	140-31-8	5 - 22 Trade Secret *
HEAVY NAPHTHENIC DISTILLATE SOLVENT PETROLEUM EXTRACTS	64742-11-6	5 - 20 Trade Secret *
Alkyl Acids, Reaction Products With Triethylenetetramine	Trade Secret*	5 - 17 Trade Secret *
Alykl Acids, Reaction Products With TETA And DGEBA	Trade Secret*	4 - 10 Trade Secret *
Reaction product of cycloaliphatic amine with aromatic epoxy resin	Trade Secret*	1 - 8 Trade Secret *
Thermal cracked residuum (petroleum)	64741-80-6	1 - 7 Trade Secret *
PETROLEUM DISTILLATES	Trade Secret*	1 - 7 Trade Secret *
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	90-72-2	1 - 5 Trade Secret *
TRIETHYLENETETRAMINE	112-24-3	<= 2 Trade Secret *
BIS[(DIMETHYLAMINO)METHYL]PHENOL	71074-89-0	<= 1 Trade Secret *
CARBON BLACK	1333-86-4	< 1 Trade Secret *

*The specific chemical identity and/or exact percentage (concentration) of this composition has been withheld as a trade secret.

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation:

Remove person to fresh air. If you feel unwell, get medical attention.

Skin Contact:

Immediately flush with large amounts of water for at least 15 minutes. Remove contaminated clothing. Get immediate medical attention. Wash clothing before reuse.

Eye Contact:

Immediately flush with large amounts of water for at least 15 minutes. Remove contact lenses if easy to do. Continue rinsing. Immediately get medical attention.

If Swallowed:

Rinse mouth. Do not induce vomiting. Get immediate medical attention.

4.2. Most important symptoms and effects, both acute and delayed

See Section 11.1. Information on toxicological effects.

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

Use a fire fighting agent suitable for the surrounding fire.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products**Substance**

Amine Compounds
Carbon monoxide
Carbon dioxide
Oxides of Nitrogen

Condition

During Combustion
During Combustion
During Combustion
During Combustion

5.3. Special protective actions for fire-fighters

No special protective actions for fire-fighters are anticipated.

SECTION 6: Accidental release measures**6.1. Personal precautions, protective equipment and emergency procedures**

Evacuate area. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Refer to other sections of this SDS for information regarding physical and health hazards, respiratory protection, ventilation, and personal protective equipment.

6.2. Environmental precautions

Avoid release to the environment. For larger spills, cover drains and build dikes to prevent entry into sewer systems or bodies of water.

6.3. Methods and material for containment and cleaning up

Contain spill. For large spills, if necessary, get assistance from professional spill clean up team. For small spills, carefully neutralize spill by adding appropriate dilute acid such as vinegar. Work slowly to avoid boiling or spattering. Continue to add neutralizing agent until reaction stops. Let cool before collecting. Or use a commercially available caustic (alkaline or basic) spill clean-up kit. Follow kit directions exactly. Working from around the edges of the spill inward, cover with bentonite, vermiculite, or commercially available inorganic absorbent material. Mix in sufficient absorbent until it appears dry. Remember, adding an absorbent material does not remove a physical, health, or environmental hazard. Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage**7.1. Precautions for safe handling**

Do not handle until all safety precautions have been read and understood. Do not breathe dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wash contaminated clothing before reuse. Use personal protective equipment (gloves, respirators, etc.) as required.

7.2. Conditions for safe storage including any incompatibilities

Store away from acids.

SECTION 8: Exposure controls/personal protection**8.1. Control parameters****Occupational exposure limits**

If a component is disclosed in section 3 but does not appear in the table below, an occupational exposure limit is not available for the component.

Ingredient	C.A.S. No.	Agency	Limit type	Additional Comments
TRIETHYLENETETRAMINE	112-24-3	AIHA	TWA:6 mg/m3(1 ppm)	SKIN
CARBON BLACK	1333-86-4	ACGIH	TWA(inhalable fraction):3	A3: Confirmed animal

			mg/m3	carcin.
CARBON BLACK	1333-86-4	OSHA	TWA:3.5 mg/m3	

ACGIH : American Conference of Governmental Industrial Hygienists

AIHA : American Industrial Hygiene Association

CMRG : Chemical Manufacturer's Recommended Guidelines

OSHA : United States Department of Labor - Occupational Safety and Health Administration

TWA: Time-Weighted-Average

STEL: Short Term Exposure Limit

CEIL: Ceiling

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Select and use eye/face protection to prevent contact based on the results of an exposure assessment. The following eye/face protection(s) are recommended:

Full Face Shield

Indirect Vented Goggles

Skin/hand protection

Select and use gloves and/or protective clothing approved to relevant local standards to prevent skin contact based on the results of an exposure assessment. Selection should be based on use factors such as exposure levels, concentration of the substance or mixture, frequency and duration, physical challenges such as temperature extremes, and other use conditions. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible gloves/protective clothing.

Gloves made from the following material(s) are recommended: Butyl Rubber

If this product is used in a manner that presents a higher potential for exposure (eg. spraying, high splash potential etc.), then use of protective coveralls may be necessary. Select and use body protection to prevent contact based on the results of an exposure assessment. The following protective clothing material(s) are recommended: Apron – Butyl rubber

Respiratory protection

An exposure assessment may be needed to decide if a respirator is required. If a respirator is needed, use respirators as part of a full respiratory protection program. Based on the results of the exposure assessment, select from the following respirator type(s) to reduce inhalation exposure:

Half facepiece or full facepiece air-purifying respirator suitable for organic vapors and particulates

For questions about suitability for a specific application, consult with your respirator manufacturer.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state

Liquid

Color

Black

Specific Physical Form:

Resin

Odor

Amine

Odor threshold

No Data Available

pH	10 - 12
Melting point	No Data Available
Boiling Point	607 °F
Flash Point	No flash point
Evaporation rate	No Data Available
Flammability (solid, gas)	Not Applicable
Flammable Limits(LEL)	No Data Available
Flammable Limits(UEL)	No Data Available
Vapor Pressure	4.0 mmHg
Vapor Density	No Data Available
Density	1.03 g/ml
Specific Gravity	1.03 [Ref Std: WATER=1]
Solubility In Water	660 ppm [@ 77 °F]
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	No Data Available
Autoignition temperature	No Data Available
Decomposition temperature	No Data Available
Viscosity	3,000 centipoise - 4,500 centipoise [@ 25 °C]
Average particle size	No Data Available
Bulk density	No Data Available
Molecular weight	Not Applicable
Percent volatile	3 - 5 %

SECTION 10: Stability and reactivity

10.1. Reactivity

This material may be reactive with certain agents under certain conditions - see the remaining headings in this section.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

Strong acids

No Data Available

10.6. Hazardous decomposition products

<u>Substance</u>	<u>Condition</u>
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None known.	
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Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

May be harmful if inhaled. Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

May cause additional health effects (see below).

Skin Contact:

Harmful in contact with skin. Corrosive (Skin Burns): Signs/symptoms may include localized redness, swelling, itching, intense pain, blistering, ulceration, and tissue destruction.

Allergic Skin Reaction (non-photo induced): Signs/symptoms may include redness, swelling, blistering, and itching.

Eye Contact:

Corrosive (Eye Burns): Signs/symptoms may include cloudy appearance of the cornea, chemical burns, severe pain, tearing, ulcerations, significantly impaired vision or complete loss of vision.

Ingestion:

Harmful if swallowed. Gastrointestinal Corrosion: Signs/symptoms may include severe mouth, throat and abdominal pain; nausea; vomiting; and diarrhea; blood in the feces and/or vomitus may also be seen.

May cause additional health effects (see below).

Additional Health Effects:

Prolonged or repeated exposure may cause target organ effects:

Respiratory Effects: Signs/symptoms may include cough, shortness of breath, chest tightness, wheezing, increased heart rate, bluish colored skin (cyanosis), sputum production, changes in lung function tests, and/or respiratory failure.

Reproductive/Developmental Toxicity:

Contains a chemical or chemicals which can cause birth defects or other reproductive harm.

Carcinogenicity:

Contains a chemical or chemicals which can cause cancer.

Ingredient	CAS No.	Class Description	Regulation
FUEL OILS, RES HEAVY	64741-80-6	Grp. 2B: Possible human carc.	International Agency for Research on Cancer
PETROLEUM DISTILLATES	Trade Secret	Grp. 2B: Possible human carc.	International Agency for Research on Cancer
CARBON BLACK	1333-86-4	Grp. 2B: Possible human carc.	International Agency for Research on Cancer

Additional Information:

Persons previously sensitized to amines may develop a cross-sensitization reaction to certain other amines.

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

Name	Route	Species	Value
Overall product	Dermal		No data available; calculated ATE1,000 - 2,000 mg/kg

Overall product	Inhalation-Dust/Mist(4 hr)		No data available; calculated ATE5 - 12.5 mg/l
Overall product	Ingestion		No data available; calculated ATE300 - 2,000 mg/kg
STYRENATED PHENOL	Dermal	Rat	LD50 > 2,000 mg/kg
STYRENATED PHENOL	Ingestion	Rat	LD50 > 2,000 mg/kg
N-AMINOETHYLPIPERAZINE	Dermal	Rabbit	LD50 865 mg/kg
N-AMINOETHYLPIPERAZINE	Ingestion	Rat	LD50 1,470 mg/kg
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Dermal	Rat	LD50 1,280 mg/kg
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Ingestion	Rat	LD50 1,000 mg/kg
TRIETHYLENETETRAMINE	Dermal	Rabbit	LD50 550 mg/kg
TRIETHYLENETETRAMINE	Ingestion	Rat	LD50 2,500 mg/kg
BIS[(DIMETHYLAMINO)METHYL]PHENOL	Ingestion		LD50 estimated to be 300 - 2,000 mg/kg
CARBON BLACK	Dermal	Rabbit	LD50 > 3,000 mg/kg
CARBON BLACK	Ingestion	Rat	LD50 > 8,000 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
STYRENATED PHENOL	Rabbit	No significant irritation
N-AMINOETHYLPIPERAZINE	Rabbit	Corrosive
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Rabbit	Corrosive
TRIETHYLENETETRAMINE	Rabbit	Corrosive
BIS[(DIMETHYLAMINO)METHYL]PHENOL	similar compounds	Corrosive
CARBON BLACK	Rabbit	No significant irritation

Serious Eye Damage/Irritation

Name	Species	Value
STYRENATED PHENOL	Rabbit	Mild irritant
N-AMINOETHYLPIPERAZINE	Rabbit	Corrosive
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Rabbit	Corrosive
TRIETHYLENETETRAMINE	Rabbit	Corrosive
BIS[(DIMETHYLAMINO)METHYL]PHENOL	similar compounds	Corrosive
CARBON BLACK	Rabbit	No significant irritation

Skin Sensitization

Name	Species	Value
STYRENATED PHENOL	Mouse	Sensitizing
N-AMINOETHYLPIPERAZINE	Guinea pig	Sensitizing
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Guinea pig	Not classified
TRIETHYLENETETRAMINE	Guinea pig	Sensitizing

Respiratory Sensitization

For the component/components, either no data are currently available or the data are not sufficient for classification.

Germ Cell Mutagenicity

Name	Route	Value
N-AMINOETHYLPIPERAZINE	In vivo	Not mutagenic
N-AMINOETHYLPIPERAZINE	In Vitro	Some positive data exist, but the data are not sufficient for classification
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	In Vitro	Not mutagenic
CARBON BLACK	In Vitro	Not mutagenic

CARBON BLACK	In vivo	Some positive data exist, but the data are not sufficient for classification
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Carcinogenicity

Name	Route	Species	Value
CARBON BLACK	Dermal	Mouse	Not carcinogenic
CARBON BLACK	Ingestion	Mouse	Not carcinogenic
CARBON BLACK	Inhalation	Rat	Carcinogenic

Reproductive Toxicity**Reproductive and/or Developmental Effects**

Name	Route	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Ingestion	Not classified for female reproduction	Rat	NOAEL 598 mg/kg/day	premating & during gestation
N-AMINOETHYLPIPERAZINE	Ingestion	Not classified for male reproduction	Rat	NOAEL 409 mg/kg/day	32 days
N-AMINOETHYLPIPERAZINE	Ingestion	Toxic to development	Rabbit	NOAEL 75 mg/kg/day	during gestation

Target Organ(s)**Specific Target Organ Toxicity - single exposure**

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification		NOAEL Not available	
TRIS(2,4,6-DIMETHYLAMINOMETHYL)PHENOL	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification		NOAEL Not available	

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Dermal	skin	Not classified	Rat	NOAEL 100 mg/kg/day	29 days
N-AMINOETHYLPIPERAZINE	Dermal	hematopoietic system nervous system kidney and/or bladder	Not classified	Rat	NOAEL 1,000 mg/kg/day	29 days
N-AMINOETHYLPIPERAZINE	Inhalation	respiratory system	Causes damage to organs through prolonged or repeated exposure	Rat	NOAEL 0.2 mg/m3	13 weeks
N-AMINOETHYLPIPERAZINE	Inhalation	hematopoietic system eyes kidney and/or bladder	Not classified	Rat	NOAEL 53.8 mg/m3	13 weeks
N-AMINOETHYLPIPERAZINE	Ingestion	heart endocrine system hematopoietic system liver nervous system kidney and/or bladder	Not classified	Rat	NOAEL 598 mg/kg/day	28 days
TRIS(2,4,6-DIMETHYLAMINOMETHYL)PHENOL	Dermal	skin liver nervous system auditory system hematopoietic system eyes	Not classified	Rat	NOAEL 125 mg/kg/day	28 days
CARBON BLACK	Inhalation	pneumoconiosis	Not classified	Human	NOAEL Not	occupational

					available	exposure
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Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Dispose of waste product in a permitted industrial waste facility. As a disposal alternative, incinerate in a permitted waste incineration facility. Proper destruction may require the use of additional fuel during incineration processes. Empty drums/barrels/containers used for transporting and handling hazardous chemicals (chemical substances/mixtures/preparations classified as Hazardous as per applicable regulations) shall be considered, stored, treated & disposed of as hazardous wastes unless otherwise defined by applicable waste regulations. Consult with the respective regulating authorities to determine the available treatment and disposal facilities.

EPA Hazardous Waste Number (RCRA): Not regulated

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information**15.1. US Federal Regulations**

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:**Physical Hazards**

Not applicable

Health Hazards

Acute toxicity

Carcinogenicity

Hazard Not Otherwise Classified (HNOC)

Reproductive toxicity

Respiratory or Skin Sensitization

Serious eye damage or eye irritation
Skin Corrosion or Irritation
Specific target organ toxicity (single or repeated exposure)

15.2. State Regulations

Contact 3M for more information.

California Proposition 65

<u>Ingredient</u>	<u>C.A.S. No.</u>	<u>Listing</u>
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RESIDUAL (HEAVY) FUEL OILS

Trade Secret

Carcinogen

15.3. Chemical Inventories

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

Contact 3M for more information.

15.4. International Regulations

Contact 3M for more information.

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.
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SECTION 16: Other information

NFPA Hazard Classification

Health: 3 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

HMIS Hazard Classification

Health: *3 **Flammability:** 1 **Physical Hazard:** 0 **Personal Protection:** X - See PPE section.

Hazardous Material Identification System (HMIS® IV) hazard ratings are designed to inform employees of chemical hazards in the workplace. These ratings are based on the inherent properties of the material under expected conditions of normal use and are not intended for use in emergency situations. HMIS® IV ratings are to be used with a fully implemented HMIS® IV program. HMIS® is a registered mark of the American Coatings Association (ACA).

Document Group:	35-7972-9
Issue Date:	11/04/20

Version Number:	1.05
Supersedes Date:	01/15/19

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Document Group:	36-4098-4	Version Number:	2.02
Issue Date:	08/12/22	Supersedes Date:	05/03/21

Product identifier

3M™ Scotchcast™ Inline Resin Power Cable Splice Kits (82-AN, 82-A1N, 82-A2N, 82-A3N), with 3M™ Scotchcast™ Resin 4N

ID Number(s):

80-6116-1668-3, 80-6116-1671-7, 80-6116-1672-5, 80-6116-1673-3

7100123072, 7100123073, 7100123110, 7100117646

Recommended use

Electrical

Supplier's details

MANUFACTURER:	3M
DIVISION:	Electrical Markets Division
ADDRESS:	3M Center, St. Paul, MN 55144-1000, USA
Telephone:	1-888-3M HELPS (1-888-364-3577)

Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

This product is a kit or a multipart product which consists of multiple, independently packaged components. A Safety Data Sheet (SDS), Article Information Sheet (AIS), or Article Information Letter (AIL) for each of these components is included. Please do not separate the component documents from this cover page. The document numbers for components of this product are:

24-9848-3, 35-7972-9

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Document Group: 24-9848-3
Issue Date: 02/19/20

Version Number: 3.02
Supersedes Date: 11/08/19

SECTION 1: Identification

1.1. Product identifier

3M™ Scotchcast™ Electrical Insulating Resin 4N, Part A and 3M™ Scotchcast™ Electrical Insulating Resin 4, Part A

1.2. Recommended use and restrictions on use

Recommended use

Electrical, Part A of Resin 4 & Resin 4N

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Electrical Markets Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Serious Eye Damage/Irritation: Category 2B.

Skin Sensitizer: Category 1A.

2.2. Label elements

Signal word

Warning

Symbols

Exclamation mark |

Pictograms



Hazard Statements

Causes eye irritation.
May cause an allergic skin reaction.

Precautionary Statements**Prevention:**

Avoid breathing dust/fume/gas/mist/vapors/spray.
Wear protective gloves.
Wash thoroughly after handling.
Contaminated work clothing must not be allowed out of the workplace.

Response:

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
If eye irritation persists: Get medical advice/attention.
IF ON SKIN: Wash with plenty of soap and water.
If skin irritation or rash occurs: Get medical advice/attention.
Wash contaminated clothing before reuse.

Disposal:

Dispose of contents/container in accordance with applicable local/regional/national/international regulations.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	25085-99-8	80 - 100 Trade Secret *
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	68609-97-2	< 20 Trade Secret *

*The specific chemical identity and/or exact percentage (concentration) of this composition has been withheld as a trade secret.

SECTION 4: First aid measures**4.1. Description of first aid measures****Inhalation:**

Remove person to fresh air. If you feel unwell, get medical attention.

Skin Contact:

Immediately wash with soap and water. Remove contaminated clothing and wash before reuse. If signs/symptoms develop, get medical attention.

Eye Contact:

Flush with large amounts of water. Remove contact lenses if easy to do. Continue rinsing. If signs/symptoms persist, get medical attention.

If Swallowed:

Rinse mouth. If you feel unwell, get medical attention.

4.2. Most important symptoms and effects, both acute and delayed

See Section 11.1. Information on toxicological effects.

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

In case of fire: Use a fire fighting agent suitable for ordinary combustible material such as water or foam to extinguish.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products

<u>Substance</u>	<u>Condition</u>
Carbon monoxide	During Combustion
Carbon dioxide	During Combustion
Toxic Vapor, Gas, Particulate	During Combustion

5.3. Special protective actions for fire-fighters

Wear full protective clothing, including helmet, self-contained, positive pressure or pressure demand breathing apparatus, bunker coat and pants, bands around arms, waist and legs, face mask, and protective covering for exposed areas of the head.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

Evacuate area. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Refer to other sections of this SDS for information regarding physical and health hazards, respiratory protection, ventilation, and personal protective equipment.

6.2. Environmental precautions

Avoid release to the environment. For larger spills, cover drains and build dikes to prevent entry into sewer systems or bodies of water.

6.3. Methods and material for containment and cleaning up

Contain spill. Working from around the edges of the spill inward, cover with bentonite, vermiculite, or commercially available inorganic absorbent material. Mix in sufficient absorbent until it appears dry. Remember, adding an absorbent material does not remove a physical, health, or environmental hazard. Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Clean up residue with an appropriate solvent selected by a qualified and authorized person. Ventilate the area with fresh air. Read and follow safety precautions on the solvent label and SDS. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Avoid breathing dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wash contaminated clothing before reuse.

7.2. Conditions for safe storage including any incompatibilities

No special storage requirements.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Occupational exposure limits

No occupational exposure limit values exist for any of the components listed in Section 3 of this SDS.

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Select and use eye/face protection to prevent contact based on the results of an exposure assessment. The following eye/face protection(s) are recommended:

Indirect Vented Goggles

Skin/hand protection

Select and use gloves and/or protective clothing approved to relevant local standards to prevent skin contact based on the results of an exposure assessment. Selection should be based on use factors such as exposure levels, concentration of the substance or mixture, frequency and duration, physical challenges such as temperature extremes, and other use conditions. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible gloves/protective clothing. Note: Nitrile gloves may be worn over polymer laminate gloves to improve dexterity.

Gloves made from the following material(s) are recommended: Polymer laminate

If this product is used in a manner that presents a higher potential for exposure (eg. spraying, high splash potential etc.), then use of protective coveralls may be necessary. Select and use body protection to prevent contact based on the results of an exposure assessment. The following protective clothing material(s) are recommended: Apron - polymer laminate

Respiratory protection

An exposure assessment may be needed to decide if a respirator is required. If a respirator is needed, use respirators as part of a full respiratory protection program. Based on the results of the exposure assessment, select from the following respirator type(s) to reduce inhalation exposure:

Half facepiece or full facepiece air-purifying respirator suitable for organic vapors and particulates

For questions about suitability for a specific application, consult with your respirator manufacturer.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state

Liquid

Color

Amber

Specific Physical Form:

Resin

Odor

Epoxy

Odor threshold

No Data Available

pH

No Data Available

Melting point

No Data Available

Boiling Point	>= 201 °F
Flash Point	>= 201 °F [Test Method: Closed Cup]
Evaporation rate	No Data Available
Flammability (solid, gas)	Not Applicable
Flammable Limits(LEL)	No Data Available
Flammable Limits(UEL)	No Data Available
Vapor Pressure	<= 27 psia [@ 131 °F]
Vapor Density	No Data Available
Density	1.16 g/ml
Specific Gravity	1.16 [Ref Std: WATER=1]
Solubility in Water	Negligible
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	No Data Available
Autoignition temperature	No Data Available
Decomposition temperature	No Data Available
Viscosity	3,000 centipoise - 5,000 centipoise
Average particle size	No Data Available
Bulk density	No Data Available
Hazardous Air Pollutants	No Data Available
Molecular weight	No Data Available
Volatile Organic Compounds	No Data Available
Percent volatile	Negligible
Softening point	No Data Available
VOC Less H2O & Exempt Solvents	No Data Available

SECTION 10: Stability and reactivity

10.1. Reactivity

This material is considered to be non reactive under normal use conditions.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

None known.

10.6. Hazardous decomposition products

Substance

None known.

Condition

Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient

classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

Skin Contact:

Mild Skin Irritation: Signs/symptoms may include localized redness, swelling, itching, and dryness. Allergic Skin Reaction (non-photo induced): Signs/symptoms may include redness, swelling, blistering, and itching.

Eye Contact:

Moderate Eye Irritation: Signs/symptoms may include redness, swelling, pain, tearing, and blurred or hazy vision.

Ingestion:

Gastrointestinal Irritation: Signs/symptoms may include abdominal pain, stomach upset, nausea, vomiting and diarrhea.

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

Name	Route	Species	Value
Overall product	Ingestion		No data available; calculated ATE >5,000 mg/kg
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Rat	LD50 > 1,600 mg/kg
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Rat	LD50 > 1,000 mg/kg
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	Rabbit	LD50 > 4,000 mg/kg
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Ingestion	Rat	LD50 17,100 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Rabbit	Mild irritant
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Rabbit	Mild irritant

Serious Eye Damage/Irritation

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Rabbit	Moderate irritant
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Rabbit	Mild irritant

Skin Sensitization

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Human and animal	Sensitizing
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Guinea	Sensitizing

	pig	
--	-----	--

Respiratory Sensitization

Name	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Human	Not classified

Germ Cell Mutagenicity

Name	Route	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	In vivo	Not mutagenic
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	In Vitro	Some positive data exist, but the data are not sufficient for classification
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	In vivo	Not mutagenic
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	In Vitro	Some positive data exist, but the data are not sufficient for classification

Carcinogenicity

Name	Route	Species	Value
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Mouse	Some positive data exist, but the data are not sufficient for classification

Reproductive Toxicity**Reproductive and/or Developmental Effects**

Name	Route	Value	Species	Test Result	Exposure Duration
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for female reproduction	Rat	NOAEL 750 mg/kg/day	2 generation
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for male reproduction	Rat	NOAEL 750 mg/kg/day	2 generation
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	Not classified for development	Rabbit	NOAEL 300 mg/kg/day	during organogenesis
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	Not classified for development	Rat	NOAEL 750 mg/kg/day	2 generation
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	Not classified for development	Rat	NOAEL 200 mg/kg/day	during organogenesis

Target Organ(s)**Specific Target Organ Toxicity - single exposure**

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL]DERIVATIVES	Dermal	heart blood liver nervous system kidney and/or bladder	Not classified	Rabbit	NOAEL 4,000 mg/kg	24 hours

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	liver	Not classified	Rat	NOAEL 1,000 mg/kg/day	2 years
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Dermal	nervous system	Not classified	Rat	NOAEL 1,000 mg/kg/day	13 weeks
2,2-Bis(p-hydroxyphenyl)propane diglycidyl ether polymer	Ingestion	auditory system heart endocrine system hematopoietic	Not classified	Rat	NOAEL 1,000 mg/kg/day	28 days

		system liver eyes kidney and/or bladder				
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL] DERIVATIVES	Dermal	nervous system respiratory system	Not classified	Rat	NOAEL 100 mg/kg/day	14 weeks
OXIRANE, MONO[(C12-14-ALKYLOXY)METHYL] DERIVATIVES	Dermal	blood liver eyes kidney and/or bladder	Not classified	Rat	NOAEL 100 mg/kg/day	13 weeks

Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Incinerate in a permitted waste incineration facility. Proper destruction may require the use of additional fuel during incineration processes. As a disposal alternative, utilize an acceptable permitted waste disposal facility. Empty drums/barrels/containers used for transporting and handling hazardous chemicals (chemical substances/mixtures/preparations classified as Hazardous as per applicable regulations) shall be considered, stored, treated & disposed of as hazardous wastes unless otherwise defined by applicable waste regulations. Consult with the respective regulating authorities to determine the available treatment and disposal facilities.

EPA Hazardous Waste Number (RCRA): Not regulated

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information**15.1. US Federal Regulations**

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:**Physical Hazards**

Not applicable

Health Hazards

Respiratory or Skin Sensitization

Serious eye damage or eye irritation

15.2. State Regulations

Contact 3M for more information.

15.3. Chemical Inventories

The components of this material are in compliance with the provisions of Australia National Industrial Chemical Notification and Assessment Scheme (NICNAS). Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the new substance notification requirements of CEPA.

The components of this material are in compliance with the China "Measures on Environmental Management of New Chemical Substance". Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of the Korean Toxic Chemical Control Law. Certain restrictions may apply. Contact the selling division for additional information.

The components of this material are in compliance with the provisions of Philippines RA 6969 requirements. Certain restrictions may apply. Contact the selling division for additional information.

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

Contact 3M for more information.

15.4. International Regulations

Contact 3M for more information.

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.
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SECTION 16: Other information**NFPA Hazard Classification****Health:** 2 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

HMIS Hazard Classification**Health:** 2 **Flammability:** 1 **Physical Hazard:** 0 **Personal Protection:** X - See PPE section.

Hazardous Material Identification System (HMIS® IV) hazard ratings are designed to inform employees of chemical hazards in the workplace. These ratings are based on the inherent properties of the material under expected conditions of normal use and are not intended for use in emergency situations. HMIS® IV ratings are to be used with a fully implemented HMIS® IV program. HMIS® is a registered mark of the American Coatings Association (ACA).

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Issue Date: 02/19/20**Supersedes Date:** 11/08/19

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Safety Data Sheet

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Document Group: 35-7972-9
Issue Date: 11/04/20

Version Number: 1.05
Supersedes Date: 01/15/19

SECTION 1: Identification

1.1. Product identifier

3M™ Scotchcast™ Electrical Insulating Resin 4N, Part B

1.2. Recommended use and restrictions on use

Recommended use

Electrical, Part B of Resin 4N

1.3. Supplier's details

MANUFACTURER: 3M
DIVISION: Electrical Markets Division
ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA
Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

2.1. Hazard classification

Acute Toxicity (oral): Category 4.
Acute Toxicity (dermal): Category 4.
Serious Eye Damage/Irritation: Category 1.
Skin Corrosion/Irritation: Category 1B.
Skin Sensitizer: Category 1A.
Reproductive Toxicity: Category 2.
Carcinogenicity: Category 1B.
Specific Target Organ Toxicity (repeated exposure): Category 1.

2.2. Label elements

Signal word

Danger

Symbols

Corrosion | Exclamation mark | Health Hazard |

Pictograms

**Hazard Statements**

Harmful if swallowed.

Harmful in contact with skin.

Causes severe skin burns and eye damage.

May cause an allergic skin reaction.

Suspected of damaging fertility or the unborn child.

May cause cancer.

Causes damage to organs through prolonged or repeated exposure:
respiratory system |

Precautionary Statements**Prevention:**

Obtain special instructions before use.

Do not handle until all safety precautions have been read and understood.

Do not breathe dust/fume/gas/mist/vapors/spray.

Wear protective gloves, protective clothing, and eye/face protection.

Do not eat, drink or smoke when using this product.

Wash thoroughly after handling.

Contaminated work clothing must not be allowed out of the workplace.

Response:

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Immediately call a POISON CENTER or doctor/physician.

If skin irritation or rash occurs: Get medical advice/attention.

Wash contaminated clothing before reuse.

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

Storage:

Store locked up.

Disposal:

Dispose of contents/container in accordance with applicable local/regional/national/international regulations.

2.3. Hazards not otherwise classified

May cause chemical gastrointestinal burns.

Supplemental Information:

Persons previously sensitized to amines may develop a cross-sensitization reaction to certain other amines.

18% of the mixture consists of ingredients of unknown acute oral toxicity.

36% of the mixture consists of ingredients of unknown acute dermal toxicity.

100% of the mixture consists of ingredients of unknown acute inhalation toxicity.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
STYRENATED PHENOL	Trade Secret*	25 - 70 Trade Secret *
N-AMINOETHYLPIPERAZINE	140-31-8	5 - 22 Trade Secret *
HEAVY NAPHTHENIC DISTILLATE SOLVENT PETROLEUM EXTRACTS	64742-11-6	5 - 20 Trade Secret *
Alkyl Acids, Reaction Products With Triethylenetetramine	Trade Secret*	5 - 17 Trade Secret *
Alykl Acids, Reaction Products With TETA And DGEBA	Trade Secret*	4 - 10 Trade Secret *
Reaction product of cycloaliphatic amine with aromatic epoxy resin	Trade Secret*	1 - 8 Trade Secret *
Thermal cracked residuum (petroleum)	64741-80-6	1 - 7 Trade Secret *
PETROLEUM DISTILLATES	Trade Secret*	1 - 7 Trade Secret *
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	90-72-2	1 - 5 Trade Secret *
TRIETHYLENETETRAMINE	112-24-3	<= 2 Trade Secret *
BIS[(DIMETHYLAMINO)METHYL]PHENOL	71074-89-0	<= 1 Trade Secret *
CARBON BLACK	1333-86-4	< 1 Trade Secret *

*The specific chemical identity and/or exact percentage (concentration) of this composition has been withheld as a trade secret.

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation:

Remove person to fresh air. If you feel unwell, get medical attention.

Skin Contact:

Immediately flush with large amounts of water for at least 15 minutes. Remove contaminated clothing. Get immediate medical attention. Wash clothing before reuse.

Eye Contact:

Immediately flush with large amounts of water for at least 15 minutes. Remove contact lenses if easy to do. Continue rinsing. Immediately get medical attention.

If Swallowed:

Rinse mouth. Do not induce vomiting. Get immediate medical attention.

4.2. Most important symptoms and effects, both acute and delayed

See Section 11.1. Information on toxicological effects.

4.3. Indication of any immediate medical attention and special treatment required

Not applicable

SECTION 5: Fire-fighting measures

5.1. Suitable extinguishing media

Use a fire fighting agent suitable for the surrounding fire.

5.2. Special hazards arising from the substance or mixture

None inherent in this product.

Hazardous Decomposition or By-Products**Substance**

Amine Compounds
Carbon monoxide
Carbon dioxide
Oxides of Nitrogen

Condition

During Combustion
During Combustion
During Combustion
During Combustion

5.3. Special protective actions for fire-fighters

No special protective actions for fire-fighters are anticipated.

SECTION 6: Accidental release measures**6.1. Personal precautions, protective equipment and emergency procedures**

Evacuate area. Ventilate the area with fresh air. For large spill, or spills in confined spaces, provide mechanical ventilation to disperse or exhaust vapors, in accordance with good industrial hygiene practice. Refer to other sections of this SDS for information regarding physical and health hazards, respiratory protection, ventilation, and personal protective equipment.

6.2. Environmental precautions

Avoid release to the environment. For larger spills, cover drains and build dikes to prevent entry into sewer systems or bodies of water.

6.3. Methods and material for containment and cleaning up

Contain spill. For large spills, if necessary, get assistance from professional spill clean up team. For small spills, carefully neutralize spill by adding appropriate dilute acid such as vinegar. Work slowly to avoid boiling or spattering. Continue to add neutralizing agent until reaction stops. Let cool before collecting. Or use a commercially available caustic (alkaline or basic) spill clean-up kit. Follow kit directions exactly. Working from around the edges of the spill inward, cover with bentonite, vermiculite, or commercially available inorganic absorbent material. Mix in sufficient absorbent until it appears dry. Remember, adding an absorbent material does not remove a physical, health, or environmental hazard. Collect as much of the spilled material as possible. Place in a closed container approved for transportation by appropriate authorities. Seal the container. Dispose of collected material as soon as possible in accordance with applicable local/regional/national/international regulations.

SECTION 7: Handling and storage**7.1. Precautions for safe handling**

Do not handle until all safety precautions have been read and understood. Do not breathe dust/fume/gas/mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wash contaminated clothing before reuse. Use personal protective equipment (gloves, respirators, etc.) as required.

7.2. Conditions for safe storage including any incompatibilities

Store away from acids.

SECTION 8: Exposure controls/personal protection**8.1. Control parameters****Occupational exposure limits**

If a component is disclosed in section 3 but does not appear in the table below, an occupational exposure limit is not available for the component.

Ingredient	C.A.S. No.	Agency	Limit type	Additional Comments
TRIETHYLENETETRAMINE	112-24-3	AIHA	TWA:6 mg/m3(1 ppm)	SKIN
CARBON BLACK	1333-86-4	ACGIH	TWA(inhalable fraction):3	A3: Confirmed animal

			mg/m3	carcin.
CARBON BLACK	1333-86-4	OSHA	TWA:3.5 mg/m3	

ACGIH : American Conference of Governmental Industrial Hygienists

AIHA : American Industrial Hygiene Association

CMRG : Chemical Manufacturer's Recommended Guidelines

OSHA : United States Department of Labor - Occupational Safety and Health Administration

TWA: Time-Weighted-Average

STEL: Short Term Exposure Limit

CEIL: Ceiling

8.2. Exposure controls

8.2.1. Engineering controls

Use general dilution ventilation and/or local exhaust ventilation to control airborne exposures to below relevant Exposure Limits and/or control dust/fume/gas/mist/vapors/spray. If ventilation is not adequate, use respiratory protection equipment.

8.2.2. Personal protective equipment (PPE)

Eye/face protection

Select and use eye/face protection to prevent contact based on the results of an exposure assessment. The following eye/face protection(s) are recommended:

Full Face Shield

Indirect Vented Goggles

Skin/hand protection

Select and use gloves and/or protective clothing approved to relevant local standards to prevent skin contact based on the results of an exposure assessment. Selection should be based on use factors such as exposure levels, concentration of the substance or mixture, frequency and duration, physical challenges such as temperature extremes, and other use conditions. Consult with your glove and/or protective clothing manufacturer for selection of appropriate compatible gloves/protective clothing.

Gloves made from the following material(s) are recommended: Butyl Rubber

If this product is used in a manner that presents a higher potential for exposure (eg. spraying, high splash potential etc.), then use of protective coveralls may be necessary. Select and use body protection to prevent contact based on the results of an exposure assessment. The following protective clothing material(s) are recommended: Apron – Butyl rubber

Respiratory protection

An exposure assessment may be needed to decide if a respirator is required. If a respirator is needed, use respirators as part of a full respiratory protection program. Based on the results of the exposure assessment, select from the following respirator type(s) to reduce inhalation exposure:

Half facepiece or full facepiece air-purifying respirator suitable for organic vapors and particulates

For questions about suitability for a specific application, consult with your respirator manufacturer.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Appearance

Physical state

Liquid

Color

Black

Specific Physical Form:

Resin

Odor

Amine

Odor threshold

No Data Available

pH	10 - 12
Melting point	No Data Available
Boiling Point	607 °F
Flash Point	No flash point
Evaporation rate	No Data Available
Flammability (solid, gas)	Not Applicable
Flammable Limits(LEL)	No Data Available
Flammable Limits(UEL)	No Data Available
Vapor Pressure	4.0 mmHg
Vapor Density	No Data Available
Density	1.03 g/ml
Specific Gravity	1.03 [Ref Std: WATER=1]
Solubility In Water	660 ppm [@ 77 °F]
Solubility- non-water	No Data Available
Partition coefficient: n-octanol/ water	No Data Available
Autoignition temperature	No Data Available
Decomposition temperature	No Data Available
Viscosity	3,000 centipoise - 4,500 centipoise [@ 25 °C]
Average particle size	No Data Available
Bulk density	No Data Available
Molecular weight	Not Applicable
Percent volatile	3 - 5 %

SECTION 10: Stability and reactivity

10.1. Reactivity

This material may be reactive with certain agents under certain conditions - see the remaining headings in this section.

10.2. Chemical stability

Stable.

10.3. Possibility of hazardous reactions

Hazardous polymerization will not occur.

10.4. Conditions to avoid

None known.

10.5. Incompatible materials

Strong acids

No Data Available

10.6. Hazardous decomposition products

<u>Substance</u>	<u>Condition</u>
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None known.	
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Refer to section 5.2 for hazardous decomposition products during combustion.

SECTION 11: Toxicological information

The information below may not be consistent with the material classification in Section 2 if specific ingredient classifications are mandated by a competent authority. In addition, toxicological data on ingredients may not be reflected in the material classification and/or the signs and symptoms of exposure, because an ingredient may be present below the threshold for labeling, an ingredient may not be available for exposure, or the data may not be relevant to the material as a whole.

11.1. Information on Toxicological effects

Signs and Symptoms of Exposure

Based on test data and/or information on the components, this material may produce the following health effects:

Inhalation:

May be harmful if inhaled. Respiratory Tract Irritation: Signs/symptoms may include cough, sneezing, nasal discharge, headache, hoarseness, and nose and throat pain.

May cause additional health effects (see below).

Skin Contact:

Harmful in contact with skin. Corrosive (Skin Burns): Signs/symptoms may include localized redness, swelling, itching, intense pain, blistering, ulceration, and tissue destruction.

Allergic Skin Reaction (non-photo induced): Signs/symptoms may include redness, swelling, blistering, and itching.

Eye Contact:

Corrosive (Eye Burns): Signs/symptoms may include cloudy appearance of the cornea, chemical burns, severe pain, tearing, ulcerations, significantly impaired vision or complete loss of vision.

Ingestion:

Harmful if swallowed. Gastrointestinal Corrosion: Signs/symptoms may include severe mouth, throat and abdominal pain; nausea; vomiting; and diarrhea; blood in the feces and/or vomitus may also be seen.

May cause additional health effects (see below).

Additional Health Effects:

Prolonged or repeated exposure may cause target organ effects:

Respiratory Effects: Signs/symptoms may include cough, shortness of breath, chest tightness, wheezing, increased heart rate, bluish colored skin (cyanosis), sputum production, changes in lung function tests, and/or respiratory failure.

Reproductive/Developmental Toxicity:

Contains a chemical or chemicals which can cause birth defects or other reproductive harm.

Carcinogenicity:

Contains a chemical or chemicals which can cause cancer.

Ingredient	CAS No.	Class Description	Regulation
FUEL OILS, RES HEAVY	64741-80-6	Grp. 2B: Possible human carc.	International Agency for Research on Cancer
PETROLEUM DISTILLATES	Trade Secret	Grp. 2B: Possible human carc.	International Agency for Research on Cancer
CARBON BLACK	1333-86-4	Grp. 2B: Possible human carc.	International Agency for Research on Cancer

Additional Information:

Persons previously sensitized to amines may develop a cross-sensitization reaction to certain other amines.

Toxicological Data

If a component is disclosed in section 3 but does not appear in a table below, either no data are available for that endpoint or the data are not sufficient for classification.

Acute Toxicity

Name	Route	Species	Value
Overall product	Dermal		No data available; calculated ATE1,000 - 2,000 mg/kg

Overall product	Inhalation-Dust/Mist(4 hr)		No data available; calculated ATE5 - 12.5 mg/l
Overall product	Ingestion		No data available; calculated ATE300 - 2,000 mg/kg
STYRENATED PHENOL	Dermal	Rat	LD50 > 2,000 mg/kg
STYRENATED PHENOL	Ingestion	Rat	LD50 > 2,000 mg/kg
N-AMINOETHYLPIPERAZINE	Dermal	Rabbit	LD50 865 mg/kg
N-AMINOETHYLPIPERAZINE	Ingestion	Rat	LD50 1,470 mg/kg
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Dermal	Rat	LD50 1,280 mg/kg
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Ingestion	Rat	LD50 1,000 mg/kg
TRIETHYLENETETRAMINE	Dermal	Rabbit	LD50 550 mg/kg
TRIETHYLENETETRAMINE	Ingestion	Rat	LD50 2,500 mg/kg
BIS[(DIMETHYLAMINO)METHYL]PHENOL	Ingestion		LD50 estimated to be 300 - 2,000 mg/kg
CARBON BLACK	Dermal	Rabbit	LD50 > 3,000 mg/kg
CARBON BLACK	Ingestion	Rat	LD50 > 8,000 mg/kg

ATE = acute toxicity estimate

Skin Corrosion/Irritation

Name	Species	Value
STYRENATED PHENOL	Rabbit	No significant irritation
N-AMINOETHYLPIPERAZINE	Rabbit	Corrosive
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Rabbit	Corrosive
TRIETHYLENETETRAMINE	Rabbit	Corrosive
BIS[(DIMETHYLAMINO)METHYL]PHENOL	similar compounds	Corrosive
CARBON BLACK	Rabbit	No significant irritation

Serious Eye Damage/Irritation

Name	Species	Value
STYRENATED PHENOL	Rabbit	Mild irritant
N-AMINOETHYLPIPERAZINE	Rabbit	Corrosive
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Rabbit	Corrosive
TRIETHYLENETETRAMINE	Rabbit	Corrosive
BIS[(DIMETHYLAMINO)METHYL]PHENOL	similar compounds	Corrosive
CARBON BLACK	Rabbit	No significant irritation

Skin Sensitization

Name	Species	Value
STYRENATED PHENOL	Mouse	Sensitizing
N-AMINOETHYLPIPERAZINE	Guinea pig	Sensitizing
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	Guinea pig	Not classified
TRIETHYLENETETRAMINE	Guinea pig	Sensitizing

Respiratory Sensitization

For the component/components, either no data are currently available or the data are not sufficient for classification.

Germ Cell Mutagenicity

Name	Route	Value
N-AMINOETHYLPIPERAZINE	In vivo	Not mutagenic
N-AMINOETHYLPIPERAZINE	In Vitro	Some positive data exist, but the data are not sufficient for classification
TRIS(2,4,6-DIMETHYLAMINOMONOMETHYL)PHENOL	In Vitro	Not mutagenic
CARBON BLACK	In Vitro	Not mutagenic

CARBON BLACK	In vivo	Some positive data exist, but the data are not sufficient for classification
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Carcinogenicity

Name	Route	Species	Value
CARBON BLACK	Dermal	Mouse	Not carcinogenic
CARBON BLACK	Ingestion	Mouse	Not carcinogenic
CARBON BLACK	Inhalation	Rat	Carcinogenic

Reproductive Toxicity**Reproductive and/or Developmental Effects**

Name	Route	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Ingestion	Not classified for female reproduction	Rat	NOAEL 598 mg/kg/day	prematuring & during gestation
N-AMINOETHYLPIPERAZINE	Ingestion	Not classified for male reproduction	Rat	NOAEL 409 mg/kg/day	32 days
N-AMINOETHYLPIPERAZINE	Ingestion	Toxic to development	Rabbit	NOAEL 75 mg/kg/day	during gestation

Target Organ(s)**Specific Target Organ Toxicity - single exposure**

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification		NOAEL Not available	
TRIS(2,4,6-DIMETHYLAMINOMETHYL)PHENOL	Inhalation	respiratory irritation	Some positive data exist, but the data are not sufficient for classification		NOAEL Not available	

Specific Target Organ Toxicity - repeated exposure

Name	Route	Target Organ(s)	Value	Species	Test Result	Exposure Duration
N-AMINOETHYLPIPERAZINE	Dermal	skin	Not classified	Rat	NOAEL 100 mg/kg/day	29 days
N-AMINOETHYLPIPERAZINE	Dermal	hematopoietic system nervous system kidney and/or bladder	Not classified	Rat	NOAEL 1,000 mg/kg/day	29 days
N-AMINOETHYLPIPERAZINE	Inhalation	respiratory system	Causes damage to organs through prolonged or repeated exposure	Rat	NOAEL 0.2 mg/m3	13 weeks
N-AMINOETHYLPIPERAZINE	Inhalation	hematopoietic system eyes kidney and/or bladder	Not classified	Rat	NOAEL 53.8 mg/m3	13 weeks
N-AMINOETHYLPIPERAZINE	Ingestion	heart endocrine system hematopoietic system liver nervous system kidney and/or bladder	Not classified	Rat	NOAEL 598 mg/kg/day	28 days
TRIS(2,4,6-DIMETHYLAMINOMETHYL)PHENOL	Dermal	skin liver nervous system auditory system hematopoietic system eyes	Not classified	Rat	NOAEL 125 mg/kg/day	28 days
CARBON BLACK	Inhalation	pneumoconiosis	Not classified	Human	NOAEL Not	occupational

					available	exposure
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Aspiration Hazard

For the component/components, either no data are currently available or the data are not sufficient for classification.

Please contact the address or phone number listed on the first page of the SDS for additional toxicological information on this material and/or its components.

SECTION 12: Ecological information**Ecotoxicological information**

Please contact the address or phone number listed on the first page of the SDS for additional ecotoxicological information on this material and/or its components.

Chemical fate information

Please contact the address or phone number listed on the first page of the SDS for additional chemical fate information on this material and/or its components.

SECTION 13: Disposal considerations**13.1. Disposal methods**

Dispose of contents/ container in accordance with the local/regional/national/international regulations.

Dispose of waste product in a permitted industrial waste facility. As a disposal alternative, incinerate in a permitted waste incineration facility. Proper destruction may require the use of additional fuel during incineration processes. Empty drums/barrels/containers used for transporting and handling hazardous chemicals (chemical substances/mixtures/preparations classified as Hazardous as per applicable regulations) shall be considered, stored, treated & disposed of as hazardous wastes unless otherwise defined by applicable waste regulations. Consult with the respective regulating authorities to determine the available treatment and disposal facilities.

EPA Hazardous Waste Number (RCRA): Not regulated

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information**15.1. US Federal Regulations**

Contact 3M for more information.

EPCRA 311/312 Hazard Classifications:**Physical Hazards**

Not applicable

Health Hazards

Acute toxicity

Carcinogenicity

Hazard Not Otherwise Classified (HNOC)

Reproductive toxicity

Respiratory or Skin Sensitization

Serious eye damage or eye irritation
Skin Corrosion or Irritation
Specific target organ toxicity (single or repeated exposure)

15.2. State Regulations

Contact 3M for more information.

California Proposition 65

<u>Ingredient</u>	<u>C.A.S. No.</u>	<u>Listing</u>
RESIDUAL (HEAVY) FUEL OILS	Trade Secret	Carcinogen

15.3. Chemical Inventories

The components of this product are in compliance with the chemical notification requirements of TSCA. All required components of this product are listed on the active portion of the TSCA Inventory.

Contact 3M for more information.

15.4. International Regulations

Contact 3M for more information.

This SDS has been prepared to meet the U.S. OSHA Hazard Communication Standard, 29 CFR 1910.1200.
--

SECTION 16: Other information

NFPA Hazard Classification

Health: 3 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

HMIS Hazard Classification

Health: *3 **Flammability:** 1 **Physical Hazard:** 0 **Personal Protection:** X - See PPE section.

Hazardous Material Identification System (HMIS® IV) hazard ratings are designed to inform employees of chemical hazards in the workplace. These ratings are based on the inherent properties of the material under expected conditions of normal use and are not intended for use in emergency situations. HMIS® IV ratings are to be used with a fully implemented HMIS® IV program. HMIS® is a registered mark of the American Coatings Association (ACA).

Document Group:	35-7972-9	Version Number:	1.05
Issue Date:	11/04/20	Supersedes Date:	01/15/19

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Article Information Sheet

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This Article Information Sheet is provided as a courtesy in response to a customer request. A Safety Data Sheet (SDS) has not been prepared for these product(s) because they are articles. Articles are not subject to the Occupational Safety and Health Administration's Hazard Communication Standard (29 CFR 1910.1200(b)(6)(v)). As defined in this standard: "Article" means a manufactured item other than a fluid or particle: (i) which is formed to a specific shape or design during manufacture; (ii) which has end use function(s) dependent in whole or in part upon its shape or design during end use; and (iii) which under normal conditions of use does not release more than very small quantities, e.g., minute or trace amounts of a hazardous chemical, and does not pose a physical hazard or health risk to employees.

Document Group: 06-2459-3

Issue Date: 04/30/21

Version Number: 2.01

Supersedes Date: 05/12/17

SECTION 1: Identification

1.1. Product identifier

3M™ Heat Shrink Tubing FP-301, Black (1/4, 3/16, 1/8, 3/32 Inch Nominal Sizes Only)

1.2. Recommended use and restrictions on use

Recommended use

Electrical, Electrical insulation for wires/cables

1.3. Supplier's details

MANUFACTURER: 3M

DIVISION: Electrical Markets Division

ADDRESS: 3M Center, St. Paul, MN 55144-1000, USA

Telephone: 1-888-3M HELPS (1-888-364-3577)

1.4. Emergency telephone number

1-800-364-3577 or (651) 737-6501 (24 hours)

SECTION 2: Hazard identification

This product is exempt from hazard classification according to OSHA Hazard Communication Standard, 29 CFR 1910.1200.

SECTION 3: Composition/information on ingredients

Ingredient	C.A.S. No.	% by Wt
Flame Retardant Polyolefin Tubing	Mixture	100

SECTION 4: First aid measures

4.1. Description of first aid measures

Inhalation:

No need for first aid is anticipated.

Skin Contact:

No need for first aid is anticipated.

Eye Contact:

No need for first aid is anticipated.

If Swallowed:

No need for first aid is anticipated.

SECTION 5: Fire-fighting measures

In case of fire: Use a carbon dioxide or dry chemical extinguisher to extinguish.

SECTION 6: Accidental release measures**6.1. Personal precautions, protective equipment and emergency procedures**

Not applicable.

6.2. Environmental precautions

Not applicable.

6.3. Methods and material for containment and cleaning up

Not applicable.

SECTION 7: Handling and storage**7.1. Precautions for safe handling**

This product is considered to be an article which does not release or otherwise result in exposure to a hazardous chemical under normal use conditions.

7.2. Conditions for safe storage including any incompatibilities

No special storage requirements.

SECTION 8: Exposure controls/personal protection

This product is considered to be an article which does not release or otherwise result in exposure to a hazardous chemical under normal use conditions. No engineering controls or personal protective equipment (PPE) are necessary.

SECTION 9: Physical and chemical properties**9.1. Information on basic physical and chemical properties****Appearance****Physical state**

Solid

Color

Black

Odor

Odorless

Odor threshold*No Data Available***pH***Not Applicable*

Melting point	<i>No Data Available</i>
Boiling Point	<i>No Data Available</i>
Flash Point	<i>No Data Available</i>
Evaporation rate	<i>Not Applicable</i>
Flammability (solid, gas)	<i>Not Classified</i>
Flammable Limits(LEL)	<i>No Data Available</i>
Flammable Limits(UEL)	<i>No Data Available</i>
Vapor Pressure	<i>Not Applicable</i>
Vapor Density	<i>Not Applicable</i>
Density	<i>No Data Available</i>
Specific Gravity	<i>No Data Available</i>
Solubility In Water	<i>Not Applicable</i>
Solubility- non-water	<i>No Data Available</i>
Partition coefficient: n-octanol/ water	<i>No Data Available</i>
Autoignition temperature	<i>No Data Available</i>
Decomposition temperature	<i>No Data Available</i>
Viscosity	<i>Not Applicable</i>
Volatile Organic Compounds	<i>Not Applicable</i>
Percent volatile	<i>Not Applicable</i>
VOC Less H2O & Exempt Solvents	<i>Not Applicable</i>

SECTION 10: Stability and reactivity

This material is considered to be non reactive under normal use conditions.

SECTION 11: Toxicological information

Inhalation:

No health effects are expected

Skin Contact:

No health effects are expected

Eye Contact:

No health effects are expected

Ingestion:

No health effects are expected

Additional Information:

This product, when used under reasonable conditions and in accordance with the directions for use, should not present a health hazard. However, use or processing of the product in a manner not in accordance with the product's directions for use may affect the performance of the product and may present potential health and safety hazards.

SECTION 12: Ecological information

This article is expected to present a low environmental risk either because use and disposal are unlikely to result in a significant release of components to the environment or because those components that may be released are expected to have insignificant environmental impact.

SECTION 13: Disposal considerations

Dispose of contents/container in accordance with the local/regional/national/international regulations.

SECTION 14: Transport Information

For Transport Information, please visit <http://3M.com/Transportinfo> or call 1-800-364-3577 or 651-737-6501.

SECTION 15: Regulatory information

Chemical Inventories

This product is an article as defined by TSCA regulations, and is exempt from TSCA Inventory requirements.

SECTION 16: Other information

NFPA Hazard Classification

Health: 0 **Flammability:** 1 **Instability:** 0 **Special Hazards:** None

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel to address the hazards that are presented by short-term, acute exposure to a material under conditions of fire, spill, or similar emergencies. Hazard ratings are primarily based on the inherent physical and toxic properties of the material but also include the toxic properties of combustion or decomposition products that are known to be generated in significant quantities.

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Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25 Version no.: 06 Supersedes: (3/27/25)

1.) Identification of the Mixture and of the Company

Product identifier: **Crown Cold Galvanize Coating 93% Zinc Rich - Aerosol**

Product name:
7007 Cold Galvanize Coating 93% Zinc Rich

Relevant identified uses of the substance: Apply directly to metal or galvanized surfaces that are free of loose rust, heavy mill scale, old paint, grease, moisture, and other contaminants.

Uses advised against: Do not apply at temperatures below 40°F (4°C), or if rain is imminent within 6 hours of application

CAS No:	Not Applicable (mixture)
EC No:	Not Applicable (mixture)
Index No:	Not Applicable (mixture)
Manufacturer/Supplier:	Aervoe LLC
Street address/P.O. Box:	1100 Mark Circle
Country ID/Postcode/Place:	Gardnerville, Nevada 89410
Telephone number:	1-775-782-0100
e-mail:	mailbox@aervoe.com
National contact:	Aervoe LLC
For Product Information:	1-800-227-0196
Emergency telephone number:	1-800-424-9300 (CHEMTREC – 24 hrs)

2. Hazards identification

Classifications

Physical Hazards: Flammable Aerosol - Category 1

Health Hazards: Asp. Tox. 1
 Eye Irrit. 2
 Skin Irrit. 2
 STOT SE 3

Environmental Hazards: Aquatic Chronic 1

Labeling

Signal Word: Danger

Hazard Statements: H222 – Extremely Flammable Aerosol
 H229 – Pressurized container: may burst if heated
 H304 – May be fatal if swallowed and enters airways
 H315 – Causes skin irritation.
 H319 – Causes serious eye irritation.
 H336 – May cause drowsiness or dizziness.
 H410 – Very toxic to aquatic life with long lasting effects.



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25

Version no.: 06

Supersedes: (3/27/25)

Precautionary Statements:

- P101 - If medical advice is needed, have product container or label at hand
- P102 - Keep out of reach of children
- P103 - Read label before use
- P210 - Keep away from heat/sparks/open flames/hot surfaces - no smoking
- P211 - Do not spray on an open flame or other ignition source
- P251 - Pressurized container: Do not pierce or burn, even after use
- P261 - Avoid breathing dust/fume/gas/mist/vapours/spray
- P262 - Do not get in eyes, on skin, or on clothing
- P264 - Wash hands thoroughly after handling
- P280 - Wear protective gloves/eye protection/face protection

P303+P361+P353 - If on skin or hair, remove/takeoff immediately all contaminated clothing. Rinse skin with water/shower.

P410+P412 - Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F

P501 - Dispose of contents/container in accordance with local/regional/national/international regulation



Symbols/Pictograms:

3. Composition / Information on Ingredients

Composition

Chemical	Synonyms	CAS Number	EINECS Number	Weight Percent	Hazard Category	H-Code
Hydrocarbon Propellant	LPG	68476-86-8	270-705-8	10-30%	Press. Gas Flam. Gas 1	H220 H229
Aliphatic Petroleum Distillates	Solvent Naphtha	64742-89-8	265-192-2	10-30%	Flam Liq. 2 Skin Irr. 2 Asp. Tox. 1 STOT SE 3 Aquatic Tox. 2	H224 H304 H315 H336 H411
Zinc Powder	Zinc Dust	7440-66-6	231-175-3	30-60%	Aquatic Acute 1 Aquatic Chronic 1	H400 H410
Methyl Ethyl Ketone	M.E.K.	78-93-3	201-159-0	7-13%	Flam. Liq. 2 Eye Irrit. 2 STOT SE 3	H225 H319 H336

Other Product Information

Chemical Identity: Mixture

4.) First Aid Measures



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25

Version no.: 06

Supersedes: (3/27/25)

General Advice:	If symptoms persist, always call a doctor.
Inhalation First Aid:	Remove victim to fresh air and provide oxygen if breathing is difficult. If not breathing, give artificial respiration, preferably mouth to mouth. Get medical attention immediately.
Skin Contact First Aid:	Wash with soap and water. Remove contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse.
Eye Contact First Aid:	If contact with eyes, immediately flush eyes with plenty of water for at least 15 minutes, while holding eyelids open. Get medical attention immediately.
Ingestion First Aid:	If swallowed, wash out mouth with water provided the person is conscious. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.
Most Important Symptoms/Effects:	Exposure may cause slight irritation to the skin, eyes, and respiratory tract. Excessive exposure may cause central nervous system effects.

5. Fire Fighting Measures

Flammable Properties:	Aerosol
Auto Ignition Temperature:	Not Available
Suitable extinguishing media:	Carbon dioxide, dry chemical, water spray.
Unsuitable extinguishing media:	None known
Special hazards arising from the substance or mixture:	None known
Hazardous combustion products:	Carbon dioxide, Carbon monoxide
Fire & Explosion Hazards:	Closed Containers may rupture due to the buildup of pressure from extreme temperatures.
Precautions for fire-fighters:	Use water spray to cool containers exposed to heat or fire to prevent pressure build up. In the event of a fire, wear full protective clothing and NIOSH- approved self-contained breathing apparatus with full face piece operated in the pressure demand or other positive pressure mode.

6. Accidental Release Measures

PERSONAL PRECAUTIONARY MEASURES:

- 1) Follow personal protective equipment recommendations found in section 8.
- 2) Maintain adequate ventilation.

SPILL CLEAN-UP PROCEDURES:

- 1.) Evacuate unprotected personnel from the area.
- 2.) Remove sources of ignition if safe to do so.
- 3.) Pickup spilled materials using non-sparking tools and place in an appropriate container for disposal.
- 4.) Contain spill to prevent material from entering sewage or ground water systems.
- 5.) Always dispose of waste materials in accordance with all EU, National and Local Regulations.



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25

Version no.: 06

Supersedes: (3/27/25)

7. Handling and Storage

Handling: Flammable Aerosol, use in a well ventilated area.
Do not use near sources of ignition.
Do not to eat, drink and smoke while working with this material.
Wash hands after use.

Conditions for safe storage, including any incompatibilities:

Store out of direct sunlight.
Storage Temperature: 32° to 120°F (0° to 49°C).
No known incompatibilities.

8. Exposure Controls / Personal Protection

Appropriate engineering controls:

Ensure adequate ventilation. A system of local and/or general exhaust is recommended to keep employee exposures below the Airborne Exposure Limits.

Keep away from sources of ignition.

Take precautionary measures against static discharge.

Personal Protection:

Eye & face protection devices such as safety glasses, safety goggles or face shield are recommended.

Skin protection

Wear the appropriate protective clothing, including boots, gloves, lab coat, apron or coveralls, as appropriate, to prevent skin contact.

Respiratory protection:

Use only in an adequately ventilated area. For unknown vapor concentrations use a positive-pressure, pressure-demand, self-contained breathing apparatus (SCBA).

Hazardous Ingredient	CAS Number	ACGIH TLV (TWA)	ACGIH TLV (STEL)	OSHA PEL (TWA)	OSHA PEL (STEL)
Hydrocarbon Propellant	68476-86-8	N/A	N/A	N/A	N/A
Zinc Powder	7440-66-6	N/A	N/A	N/A	N/A
Aliphatic Petroleum Distillates	64742-89-8	N/A	N/A	N/A	N/A
Methyl Ethyl Ketone	78-93-3	200PPM	300PPM	200PPM	N/A

***Values are based on the 2019 Guide to Occupational Exposure Values by ACGIH**

9. Information on Basic Physical and Chemical Properties

Appearance: Metallic gray	Odor: Ketone Odor
Odor Threshold: N/AV	pH: Not Applicable (solvent Base)
Melting Point: N/AV	Freezing Point: N/AV
Initial Boiling Point: N/AV	Boiling Point Range: N/AV



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25

Version no.: 06

Supersedes: (3/27/25)

Flash Point: <0° F (-18° C)	Evaporation Rate: Faster than n-Butyl Acetate
Flammability Solid/Gas: Flammable gas	LEL: 0.9% UEL: 11.5%
Vapor Pressure: N/AV	Vapor Density: Heavier Than Air
Relative Density: N/AV	Solubility: Negligible
Partition Coefficient: n-octanol/ water: N/AV	Auto-ignition Temperature: N/AV
Decomposition Temperature: N/AV	Viscosity: N/AV
Explosive Properties: N/AV	Oxidizing Properties: N/AV

10. Stability & Reactivity

Possibility of hazardous reactions: Hazardous polymerization will not occur under normal conditions

Chemical stability: Stable under normal conditions

Conditions to avoid: Heat and ignition sources

Incompatible materials: Strong Oxidizing Agents

Hazardous decomposition products: Will not occur

11. Toxicological Information

Reports have associated repeated and prolonged overexposure to solvents with permanent brain and nervous system damage. Repeated overexposure can also damage kidneys, lungs, liver, heart and blood

Routes of exposure: Eyes, skin, ingestion, and/or inhalation

Acute toxicological data: N/AV

Eye irritation data: Eye Irrit. 2

Skin irritation/sensitization/absorption data: Skin irrit. 2

Reproductive toxicity data: N/AV

Mutagenicity data: N/AV

Symptoms associated with physical contact: N/AV

Acute/chronic effects from short/long term exposure:

Irritating to skin. Prolonged/repeated contact may cause defatting of the skin which can lead to dermatitis. Not expected to be a skin sensitizer.

Known reportable carcinogens via the following agencies:

NTP: N/AV

IARC: N/AV



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25

Version no.: 06

Supersedes: (3/27/25)

OSHA:

N/AV

12. Ecological Information

Ecotoxicity: **No Data Available**

Persistence and degradability: **No Data Available**

Bioaccumulative potential: **No Data Available**

Mobility in soil: **No Data Available**

Results of PBT and vPvB assessment: **No Data Available**

Other adverse effects: **No Data Available**

13. Disposal Considerations

Waste Disposal: Dispose of material in accordance with EU, national and local requirements. For proper disposal of used material, an assessment must be completed to determine the proper and permissible waste management options permitted under applicable rules, regulations and/or laws governing your location.

Product / Packaging disposal: Dispose of packaging in accordance with federal, state and local requirements, regulations and/or laws governing your location.

14. Transportation Information

US DOT

UN Number	Proper Shipping Name	Hazard Class	Packing Group	Marine Pollutant	Special Provisions
UN1950	Aerosols	2.1	Not Applicable	Not Applicable	Reference 49 CFR 172.101

IMDG

UN Number	Proper Shipping Name	Hazard Class	Packing Group	Marine Pollutant	Special Provisions
UN1950	Aerosols	2.1	Not Applicable	Not Applicable	Reference IMDG code part 3

IATA:

UN Number	Proper Shipping Name	Hazard Class	Packing Group	Marine Pollutant	Special Provisions
UN1950	Aerosols, Flammable	2.1	Not Applicable	Not Applicable	Reference IATA Dangerous Goods Regulation

15. Regulatory Information

Workplace classification:

This product is considered hazardous under the OSHA Hazard Communication Standard (29 CFR



Safety Data Sheet (SDS)

Date Prepared/Revised: 11/12/25 Version no.: 06 Supersedes: (3/27/25)

1910.1200). The Occupational Safety and Health Administration's interpretation of the product's hazard to workers.

SARA Title 3:

Section 311/312 Categorizations (40 CFR 372): This product is a hazardous chemical under 29 CFR 1910.1200, and is categorized as an immediate and delayed health, and flammability physical hazard. Superfund Amendment and Reauthorization Act (SARA) category. SARA requires reporting any spill of any hazardous substance.

TSCA status: All chemicals in this product are listed, or are exempt from listing, on the TSCA Inventory.

WHMIS: This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the (M)SDS contains all of the information required by the CPR.

16. Other Information

This SDS has been completed in accordance with GHS Rev04 (2011): U.S OSHA, CMA, ANSI, Canadian WHMIS standards, and European Directives.

Date of Preparation/Revision: 11/12/25

Supersedes: 3/27/25

To the best of our knowledge, the information contained herein is believed to be accurate. However, the above data does not imply any guarantee or warranty of any kind, expressed or implied. The final determination of the suitability of any material is the sole responsibility of the user. All materials made present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee these are the only hazards existing.

SAFETY DATA SHEET

1. Identification of the substance/mixture and of the company

1.1 Product identifier

Product Name:
Type FO™ Fiber Optic Cleaner/Degreaser

Product ID numbers: FO-XXX (Where XXX is the package code.)

1.2 Relevant identified uses of the mixture and uses advised against

Identified uses: Fiber and precision cleaning

List of advices against: Not applicable.

1.3 Details of the supplier of the safety data sheet

Supplier/Manufacturer:

American Polywater Corporation

11222 - 60th Street North

Stillwater, MN 55082 USA

Tel: 1-651-430-2270

Email: sds@polywater.com

1.4 Emergency telephone numbers

INFOTRAC: 1-800-535-5053 (USA) 1-352-323-3500 (INT'L)

2. Hazards Identification

2.1 Classification of the substance or mixture

Classification according to USA OSHA 29 CFR 1910.1200 (2012) and Canada HPR (SOR/2015-17; WHMIS 2015).

Flam Liq 2 H225

Eye Irrit. 2A H319

STOT SE 3 H336

2.2 Label elements

Contains: Isopropanol



Pictograms:

Signal word: Danger

Hazard Statements:

H225 Highly flammable liquid and vapor

H319 Causes serious eye irritation

H336 May cause drowsiness or dizziness

Precautionary Statements:

P210 Keep away from sparks, flames and hot surfaces. No smoking.

P233 Keep container tightly closed.

P240 Ground, bond container and receiving equipment.

P242 Use only non-sparking tools.

P243	Take precautionary measures against static discharge.
P261	Avoid breathing vapor.
P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P303 + P361 + P353	IF ON SKIN: Take off immediately all contaminated clothing. Rinse skin with water.
P304 + P340	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337 + P313	If eye irritation persists: Get medical attention.
P370 + P378	In case of fire: Use water fog, foam, dry chemical or carbon dioxide for extinction.
P403 + P235	Store in a well-ventilated place. Keep cool.
P501	Dispose of contents/container in accordance with local and national regulations.

2.3 Other hazards: No information available.

3. Composition/Information on Ingredients

<u>Component</u>	<u>CAS #</u>	<u>EC #</u>	<u>Wt. %</u>
Isopropanol	67-63-0	200-661-7	100

4. First Aid Measures

4.1 Description of first aid measures

Eye Contact:	If eye irritation from exposure to vapors develops, move to fresh air. Flush eyes with clean water. If irritation persists, seek medical attention. For direct eye contact, flush with large quantity of water for 15 minutes. Seek medical attention.
Skin Contact:	Remove contaminated clothing; flush skin thoroughly with water. If irritation occurs, seek medical attention.
Inhalation (Breathing):	If irritation of nose or throat develops, move to fresh air. If irritation persists, seek medical attention. If breathing is difficult, provide oxygen. If not breathing, give artificial respiration. Seek immediate medical attention.
Ingestion (Swallowing):	Do not induce vomiting or give anything by mouth unless directed to do so by medical personnel. Get medical attention if symptoms appear.

4.2 Most important symptoms and effects, both acute and delayed

Refer to Section 11 for more information.

4.3 Indication of immediate medical attention and special treatment needed.

Causes serious eye irritation.

5. Firefighting Measures

5.1 Extinguishing media:

Carbon dioxide, water fog, dry chemical or foam.

5.2 Special hazards arising from the substance or mixture

Hazardous decomposition and by-products:

Burning generates carbon monoxide, carbon dioxide.

5.3 Advice for firefighters

Wear appropriate, protective clothing, including self-contained, positive pressure or pressure-demand breathing apparatus. Sealed container can build up pressure when exposed to high heat. Use water spray to cool fire exposed containers.

6. Accidental Release Measures

6.1 Personal precautions, protective equipment and emergency procedures:

Keep away from heat/sparks/open flames/hot surfaces. No smoking. For a spill in a confined space, provide mechanical ventilation to disperse or exhaust vapors. For emergency responders: use respiratory protection: half-face or full-face respirator with filter(s) for organic vapor for spills in a confined space. Chemical goggles are recommended if splashes or contact with eyes is possible. For small spills: normal antistatic work clothes are usually adequate.

6.2 Environmental precautions:

Avoid release to the environment. Dyke the spill to prevent entry into waterways, sewers, basements or confined areas.

6.3 Methods materials for containment and cleaning up:

Absorb spill with sand or absorbents. Collect as much of the spilled material as possible using non-sparking tools and transfer to a container. Seal the container. Remember, adding an absorbent material does not change the toxicity or flammability hazard.

6.4 Reference to other sections:

Refer to Sections 4, 5, 8, and 13 for more information.

7. Handling and Storage**7.1 Precautions for safe handling**

Keep away from heat/sparks/open flames/hot surfaces. No smoking. Avoid breathing vapors or spray. Do not get in eyes, on skin, or on clothing. Do not eat, drink or smoke when using this product. Wash thoroughly after handling. Wash contaminated clothing before reuse. Use only outdoors or in a well-ventilated area. For industrial or professional use only.

7.2 Conditions for safe storage, including incompatibilities

Store in a well-ventilated place. Keep container tightly closed. Keep cool. Store away from acids and oxidizing agents.

7.3 Specific end uses

See technical data sheet on this product for further information.

8. Exposure Controls / Personal Protection**8.1 Control parameters****Exposure limits and recommendations:****Isopropanol, 2-propanol (67-63-0)**

Country/Source	Long-term exposure limit – 8 hr. TWA	Short-term exposure limit – 15 min
USA, OSHA NIOSH	980 mg/m ³ , 400 ppm	1,225 mg/m ³ , 500 ppm
USA, ACGIH	200 ppm	400 ppm
British Columbia	200 ppm	400 ppm
Alberta	492 mg/m ³ , 200 ppm	984 mg/m ³ , 400 ppm
Quebec	985 mg/m ³ , 400 ppm	1,230 mg/m ³ , 500 ppm
Saskatchewan*	200 ppm	400 ppm

* Manitoba, Newfoundland and Labrador, Nova Scotia, and Prince Edward Island are all based on the current ACGIH TLVs. New Brunswick is based on an older version ACGIH. Nunavut and Northwest Territories are based heavily on current ACGIH TLVs.

8.2 Exposure controls**Respiratory protection:**

Normal ventilation is adequate. If exposure exceeds recommended limits, respirator protection is recommended. Use a respirator or gas mask with cartridges for organic vapors (NIOSH-approved) or use supplied air equipment.

Protective gloves:

For repeated or prolonged skin contact, the use of impermeable gloves is recommended to prevent drying and possible irritation.

Suggested Material: Nitrile Rubber
Suggested Thickness: For short term contact (<15 minutes), splashes use 0.2 mm. For full contact use 0.4 mm

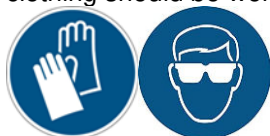
Nitrile, minimum 0.38 mm thickness or comparable protective barrier material with a high performance level for continuous contact use conditions, permeation breakthrough minimum 480 minutes in accordance with CEN standards EN 420 and EN 374.

Eye protection:

Safety glasses recommended.

Other protective equipment:

It is suggested that a source of clean water be available in work area for flushing eyes and skin. Impervious clothing should be worn as needed.



9. Physical and Chemical

9.1 Information of basic physical and chemical properties

Appearance:	Clear, colorless liquid; typical alcohol odor.
Odor threshold:	22 ppm
pH:	Does not apply
Freezing point:	-130°F / -90°C
Boiling point:	180°F / 82°C
Flash point:	55°F / 13°C (TCC)
Evaporation rate:	1.7 (n-butyl acetate = 1)
Flammability (solid, gas):	Not applicable to liquids
Upper/lower flammability or explosive limits:	LEL: 2% UEL: 12.7%
Vapor pressure:	4.4 kPa @20°C
Vapor density (Air = 1):	2.07 (Air = 1)
Specific gravity (H₂O = 1):	0.79
Solubility in water:	Complete
Coefficient of Water/Oil Distribution:	0.1 This product is equally soluble in oil and water.
Auto-ignition temperature:	Not available
Decomposition temperature:	Not available
Viscosity:	Not available

9.2 Other Information

Volatiles (Weight %):	100%
VOC Content:	790 g/l

10. Stability and Reactivity

10.1 Reactivity:

See remaining headings in Section 10.

10.2 Chemical stability:

Stable

10.3 Possibility of hazardous reactions:

None known.

10.4 Conditions to avoid:

Avoid heat, flame, and sparks.

10.5 Incompatible materials :

Strong oxidizing agents.

10.6 Hazardous decomposition products:

Carbon dioxide, carbon monoxide.

11. Toxicological Information

11.1 Information on toxicological effects:**Acute toxicity****Eye contact:**

Direct eye contact may cause eye irritation. This irritation is minimal and expected to be transient.

Skin contact:

Prolonged or repeated skin exposure can remove oils, causing redness, drying and cracking. Persons with pre-existing skin disorders may be more susceptible to skin irritation from this material.

Irritation and Sensitization Potential:

Product may be irritating to skin and eyes. It is not a sensitizer.

Inhalation (Breathing):

Concentrated solvent vapors may cause irritation of the nose and throat. Prolonged exposure to excessively high vapor concentrations can result in central nervous system depression (e.g., drowsiness, dizziness, loss of coordination, and fatigue).

Ingestion:

Ingestion of large quantities may cause irritation of the digestive tract, nervous system depression (e.g., drowsiness, dizziness, loss of coordination, and fatigue).

Toxicity to Animals:

Isopropyl Alcohol	LD ₅₀ (oral rat) 5000 mg/kg
	LD ₅₀ (dermal rabbit) 12800 mg/kg
	LC ₅₀ (inhl rat) 12000, 8 hours

Chronic Exposure:

Reproductive Toxicity: Not classified as a reproductive system toxin.

Mutagenicity: Not classified as a mutagen.

Teratogenicity: Not classified as teratogenic or embryotoxic.

Specific Target Organ Toxicity (STOT) No end point data.

Toxicologically Synergistic Products: Not available.

Carcinogenic Status: This substance has not been identified as a carcinogen or probable carcinogen by NTP, IARC, or OSHA, nor have any of its components.

12. Ecological Information

12.1 Toxicity:

Ecotoxicity: No information available.

Aquatic Toxicity:

Fish (acute)	96 h LC ₅₀ Fathead Minnow > 1000 µl/l 48 h LC ₅₀ Golden Orfe 8970 - 9280 mg/l
Aquatic crustacea (acute)	96 h LC ₅₀ Daphnid > 1000 µl/l

12.2 Persistence and degradability: No information available

12.3 Bioaccumulation potential: No information available

12.4 Mobility in soil: No information available

12.5 Results of PBT and vPvB**Assessment:**

This product is not, nor does it contain a substance that is a PBT or vPvB.

12.6 Other adverse effects:

None known.

13. Disposal Considerations

Dispose of product in accordance with National and Local Regulations.

14. Transport Information**US DOT Domestic Ground Transportation:**

Isopropyl Alcohol LTD QTY, (packages FO-2LP, FO-8LF, FO-16, FO-16C, FO-32)

Isopropyl Alcohol, (packages FO-128, FO-128C)

UN Number:

1219

UN Proper shipping name:

Isopropyl Alcohol LTD QTY, (packages FO-2LP, FO-8LF, FO-16, FO-16C, FO-32)

Isopropyl Alcohol, (packages FO-128, FO-128C)

Transport hazard class(es):

Class 3

Packing group:

II

Environmental hazards:

None known

Special precautions:

None known

ICAO/IATA-DGR:

Isopropyl Alcohol LTD QTY, (packages FO-2LP, FO-8LF, FO-16, FO-16C, FO-32)

Isopropyl Alcohol, (packages FO-128, FO-128C)

IMDG:

Isopropyl Alcohol LTD QTY, (packages FO-2LP, FO-8LF, FO-16, FO-16C, FO-32)

Isopropyl Alcohol, (packages FO-128, FO-128C)

15. Regulatory Information**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture****USA Federal and State**

All components are listed on the TSCA inventory.

Hazard Categories for SARA Section 311/312 Reporting

Acute
Yes

Chronic
No

Fire
Yes

Pressure
No

Reactive
No

Components**CERCLA/SARA Sec 302****Hazardous Substance RQ****EHS TPQ****SARA Sec. 313****Toxic Release**

Components are not affected by these Superfund regulations.

NFPA Ratings:

Health: 1

Fire: 3

Reactivity: 0

National Fire Protection Association (NFPA) hazard ratings are designed for use by emergency response personnel during spill, fire or similar emergencies. Hazard ratings are based on physical and toxic properties of combustion or decomposition.

California Proposition 65

This product does not contain any chemicals known to the State of California to cause cancer, birth defects, or any other reproductive harm or has been assessed to be below OEHHA Safe Harbor exposure levels required for labeling.

European Union

Product complies with the communication requirements of REACH Regulation (EC) No. 1907/2006. All components are listed on the European Inventory of Existing Chemical Substances (EINECS). Contains no substance on the

REACH candidate list $\geq 0.1\%$ SCL. Does not contain notified substances from the ELINCS List, Directive 92/32/EEC. Contains no REACH substances with Annex XVII restrictions.

Canada

All components are listed on the DSL inventory.

This product has been classified according to the hazard criteria of the CPR and the SDS contains all the information required by the CPR.

Australia

All components are listed on the AICS.

Hazardous according to criteria of NOHSC Australia.

15.2 Chemical Safety Assessment

No chemical safety assessment has been carried out for the mixture by the supplier.

16. Other Information**Abbreviations and acronyms:**

OSHA = Occupational Safety and Health Administration

CLP = Classification, Labeling and Packaging Regulation

STOT = Specific Target Organ Toxicity

LD₅₀ = Median Lethal Dose

DNEL = Derived No Effect Level

ACGIH = American Conference of Governmental Industrial Hygienists

TSCA = Toxic Substances Control Act (USA)

DSL = Domestic Substances List (Canada)

AICS = Australian Inventory of Chemical Substances

Mixture classification according to Regulation (EC) No 1272/2008:

H225 Highly flammable liquid and vapor
H319 Causes serious eye irritation
H336 May cause drowsiness or dizziness

Classification Procedure

Calculation method.
Calculation method.
Calculation method.

Revision Date: March 4, 2022

Revision Number: 8 NA

Supersedes: September 21, 2018

Other: Not Applicable

Indication of Changes: Section 8 updated; added PPE pictograms. Written in accordance with the provisions of OSHA 1910.1200 App D (2012) and Canada HPR (SOR/2015-17) (WHMIS 2015). (GHS format)

The information and recommendations contained herein are believed to be reliable. However, the supplier makes no warranties, express or implied, concerning the use of this product. The buyer must determine conditions of safe usage and assumes all risk and liability in handling this product.

Section 1. Identification

Product identifier

Product Identity HydraSol® Cable Gel Remover Saturated Wipe or Towel Package

Other means of identification HS-1, HS-1C, HS-1M, HS-1D42, HS-D72, HS-D72C

Relevant identified uses of the substance or mixture and uses advised against

See Technical Data Sheet.

Details of the supplier of the safety data sheet

Company Name American Polywater Corporation
11222 - 60th Street North
Stillwater, MN 55082 USA

Emergency
24 hour Emergency Telephone No. INFOTRAC: 1-800-535-5053 (USA)
1-352-323-3500 (INT'L)

Customer Service: Tel: 1-651-430-2270
Email: sds@polywater.com

Section 2. Hazard(s) identification

Classification of the substance or mixture under US OSHA's Hazard Communication Standard (1910.1200) revised 2024 and Canadian Hazardous Products Regulations (SOR/2015-17) (GHS revision 7)

Flammable Liquid 4;H227	Combustible Liquid.
Skin corrosion/irritation category 2;H315	Causes skin irritation.
Serious eye damage / eye irritation, category 2A;H319	Causes serious eye irritation.
Skin sensitizer category 1;H317	May cause an allergic skin reaction.

Label elements



Warning

H227 Combustible liquid.
H315 Causes skin irritation.
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.

[Prevention]

P210 Keep away from heat, sparks, open flames, and other ignition sources - No smoking.
P260 Do not breathe dust, fume, mist, vapors or spray.
P261 Avoid breathing dust, fume, gas, mist, vapors, spray.
P262 Do not get in eyes, on skin, or on clothing.
P264 Wash thoroughly after handling.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves, eye protection, and face protection.

[Response]

P301+310 IF SWALLOWED: Immediately call a POISON CENTER, doctor or physician.
P302+352 IF ON SKIN: Wash with plenty of soap and water.
P305+351+338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P331 Do NOT induce vomiting.
P333+313 If skin irritation or a rash occurs: Get medical advice or attention.
P337+313 If eye irritation persists: Get medical advice or attention.
P362+364 Take off contaminated clothing and wash it before reuse.
P370+378 In case of fire: Use extinguishing media listed in section 5 of SDS for extinction.
P391 Collect spillage.

[Storage]

P403+235 Store in a well ventilated place. Keep cool.

[Disposal]

P501 Dispose of contents or container in accordance with local and national regulations.

Other hazards

This product contains no PBT/vPvB/vPvM chemicals.

This product contains no endocrine disrupting chemicals.

Does not contain component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS) per the Organisation for Economic Co-operation and Development (OECD) list of Per- and Polyfluoroalkyl Substances (PFASs).

Section 3. Composition/information on ingredients

This product contains the following substances that present a hazard within the meaning of US OSHA's Hazard Communication Standard (1910.1200) revised 2024 and Canadian Hazardous Products Regulations (SOR/2015-17) (GHS revision 7)

Ingredient/Chemical Designations	Weight %	GHS Classification	Notes
Hydrocarbon Solvent CAS Number: 64742-48-9 Synonyms: Hydrotreated heavy naphtha (petroleum), Naphtha (petroleum), hydrotreated heavy	10 - 30	Aspiration hazard, category 1;H304	-----
Citrus Terpenes CAS Number: 5989-27-5 Synonyms: (R)-p-mentha-1,8-diene, Cyclohexene, 1-methyl-4-(1-methylethenyl)-, (R)-, D-Limonene	10 - 30	Flammable Liquid, category 3;H226 Skin corrosion/irritation category 2;H315 Skin sensitizer category 1;H317 Aquatic toxicity (acute), category 1;H400 Aquatic toxicity (chronic), category 1;H410 Aspiration hazard, category 1;H304	-----
Sorbitan monooleate CAS Number: 1338-43-8 Synonyms: Sorbitan oleate	1 - 5	Not Classified	-----
Alcohols, C12-15, ethoxylated CAS Number: 68131-39-5 Synonyms: Ethoxylated Alcohols	0.5 - 1.5	Serious eye damage / eye irritation, category 1;H318 Aquatic toxicity (acute), category 1;H400 Acute toxicity(oral), category 4;H302	-----

The actual concentration or concentration range is withheld as a trade secret.

*PBT/vPvB - PBT, vPvM or vPvB-substance.

The full texts of the phrases are shown in Section 16.

Section 4. First aid measures

Description of first aid measures

- General** In all cases of doubt, or when symptoms persist, seek medical attention.
Never give anything by mouth to an unconscious person.
- Inhalation** Remove to fresh air, keep patient warm and at rest. If breathing is irregular or stopped, give artificial respiration. If unconscious, place in the recovery position and obtain immediate medical attention. Give nothing by mouth.
- Eyes** Irrigate copiously with clean water for at least 15 minutes, holding the eyelids apart and seek medical attention.
- Skin** Remove contaminated clothing. Wash skin thoroughly with soap and water or use a recognized skin cleanser.
- Ingestion** If swallowed obtain immediate medical attention. Keep at rest. Do NOT induce vomiting.

Most important symptoms and effects, both acute and delayed

Overview POTENTIAL HEALTH EFFECTS

Eye Contact: May cause tearing, stinging, redness, irritation, and burns.

Inhalation: Irritating to respiratory tract. Prolonged or repeated breathing of very high vapor concentrations cause euphoria, excitation, and dizziness, headaches, nausea, and vomiting, abdominal pain, fatigue, muscular weakness. Aspiration into the lungs can cause CNS (central nervous system) and subsequent aspiration into the lungs can cause pulmonary edema and chemical pneumonia depression. Chronic overexposure in high concentrations may produce CNS depression.

Ingestion: Irritation of the mouth, esophagus, and stomach can develop following ingestion. Symptoms include burning of the mouth, sore throat, vomiting, nausea, dizziness, loss of consciousness. Due to its light viscosity, there is danger of aspiration into the lungs during vomiting. Aspiration can result in severe lung damage or death.

Skin Contact: Prolonged or repeated skin contact may cause moderate to severe irritation including itching and redness of the skin, defatting, and/or dermatitis. This product can also be absorbed through the skin and produce CNS symptoms. Single prolonged exposure is not likely to result in the product being absorbed through the skin in harmful amounts.

Signs And Symptoms Of Exposure: Eye irritation, respiratory irritation, drying and cracking of skin, dizziness, fatigue, headache, unconsciousness or asphyxiation. Chronic effects of ingestion and subsequent aspiration into the lungs can cause pneumatocele (lung cavity) formation and chronic lung dysfunction. Repeated breathing of vapors can cause effects to liver and kidneys.

Treat symptomatically. Exposure to solvent vapor concentrations from the component solvents in excess of the stated occupational exposure limits may

result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Symptoms include headache, nausea, dizziness, fatigue, muscular weakness, drowsiness and in extreme cases, loss of consciousness.

Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin resulting in dryness, irritation and possible non-allergic contact dermatitis. Solvents may also be absorbed through the skin. Splashes of liquid in the eyes may cause irritation and soreness with possible reversible damage. See section 2 for further details.

Eyes Causes serious eye irritation.
Skin May cause an allergic skin reaction. Causes skin irritation.

Section 5. Fire-fighting measures

Extinguishing media

Use dry chemicals, carbon dioxide foam, water fog, or inert gas (nitrogen) for small fires. For large fires use foam, water fog, or water spray. Water fog and spray are effective in cooling containers and adjacent structures but might cause frothing and/or not achieve extinguishment. A water jet may be used to cool the container's external walls to prevent pressure build-up, auto ignition, or explosion. NEVER use a water jet directly on the fire. Product will float and can be re-ignited on surface of water.

Special hazards arising from the substance or mixture

Hazardous decomposition: High temperatures and fires may produce such toxic substances as carbon monoxide and carbon dioxide.
Keep away from heat, sparks, open flames, and other ignition sources - No smoking.
Do not breathe dust, fume, mist, vapors or spray.
Avoid breathing dust, fume, gas, mist, vapors, spray.
Do not get in eyes, on skin, or on clothing.

Advice for fire-fighters

As with all fires, wear positive pressure, self-contained breathing apparatus, (SCBA) with a full face piece and protective clothing. Persons without respiratory protection should leave area. Wear SCBA during clean-up immediately after fire. No smoking.

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Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

Put on appropriate personal protective equipment (see section 8).
Use good personal hygiene practices. Wash hands before eating, drinking, smoking or using toilet. Promptly remove soiled clothing and wash thoroughly before reuse.

Environmental precautions

Do not allow spills to enter drains or waterways.

Methods and material for containment and cleaning up

Ventilate the area and avoid breathing vapors. Take the personal protective measures listed in section 8.
Contain and absorb spillage with non-combustible materials e.g. sand, earth, and vermiculite. Place in closed containers outside buildings and dispose of according to the Waste Regulations.
Eliminate ignition sources. Soak up with noncombustible absorbent material. Remove absorbent material for proper disposal.

Section 7. Handling and storage

Precautions for safe handling

Handle containers carefully to prevent damage and spillage.
Store in accordance with the National Fire Protection Association's publication NFPA 30, Flammable and Combustible Liquids Code. 29 CFR 1910.106 applies to the handling, storage, and use of flammable and combustible liquids.
See section 2 for further details. - [Prevention]

Conditions for safe storage, including any incompatibilities

Store in a cool dry area, away from heat, sparks and open flame.
Keep containers sealed when not in use.
Store out of direct sunlight.

Incompatible materials: Strong acids, alkalis, and oxidizers such as liquid chlorine, halogens, hydrogen peroxide, oxygen.

Other Precautions: All labeled precautions must be observed when handling, storing and transporting empty containers due to product residues. Do not reuse containers. Empty containers may contain material residues which can ignite with explosive force. Cutting or welding of empty containers can cause fire, explosion, or release fumes from residues. Keep containers closed and drum bungs in place. Dispose of in a licensed facility.

See section 2 for further details. - [Storage]

Specific end use(s)

No available information

Section 8. Exposure controls / personal protection

Control parameters

Exposure Limits

CAS No.	Ingredient	Source	Value
1338-43-8	Sorbitan monooleate	ACGIH	No Established Limit
		OSHA	No Established Limit
		NIOSH	No Established Limit
		Alberta	No Established Limit
		British Columbia	No Established Limit
		Manitoba	No Established Limit
		New Brunswick	No Established Limit
		Newfoundland and Labrador	No Established Limit
		Nova Scotia	No Established Limit
		Northwest Territories	No Established Limit
		Nunavut	No Established Limit
		Ontario	No Established Limit
		Prince Edward Island	No Established Limit
		Quebec	No Established Limit
		Saskatchewan	No Established Limit
		Yukon	No Established Limit
5989-27-5	Citrus Terpenes	ACGIH	No Established Limit
		OSHA	No Established Limit
		NIOSH	No Established Limit
		Alberta	No Established Limit
		British Columbia	No Established Limit
		Manitoba	No Established Limit
		New Brunswick	No Established Limit
		Newfoundland and Labrador	No Established Limit
		Nova Scotia	No Established Limit
		Northwest Territories	No Established Limit
		Nunavut	No Established Limit
		Ontario	No Established Limit

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CAS No.	Ingredient	Source	Value
		Prince Edward Island	No Established Limit
		Quebec	No Established Limit
		Saskatchewan	No Established Limit
		Yukon	No Established Limit
64742-48-9	Hydrocarbon Solvent	ACGIH	No Established Limit
		OSHA	No Established Limit
		NIOSH	No Established Limit
		Alberta	No Established Limit
		British Columbia	No Established Limit
		Manitoba	No Established Limit
		New Brunswick	No Established Limit
		Newfoundland and Labrador	No Established Limit
		Nova Scotia	No Established Limit
		Northwest Territories	No Established Limit
		Nunavut	No Established Limit
		Ontario	No Established Limit
		Prince Edward Island	No Established Limit
		Quebec	No Established Limit
		Saskatchewan	No Established Limit
		Yukon	No Established Limit
68131-39-5	Alcohols, C12-15, ethoxylated	ACGIH	No Established Limit
		OSHA	No Established Limit
		NIOSH	No Established Limit
		Alberta	No Established Limit
		British Columbia	No Established Limit
		Manitoba	No Established Limit
		New Brunswick	No Established Limit
		Newfoundland and Labrador	No Established Limit
		Nova Scotia	No Established Limit
		Northwest Territories	No Established Limit
		Nunavut	No Established Limit
		Ontario	No Established Limit
		Prince Edward Island	No Established Limit
		Quebec	No Established Limit
		Saskatchewan	No Established Limit
		Yukon	No Established Limit

Exposure

controls

Respiratory

If workers are exposed to concentrations above the exposure limit they must use the appropriate, certified respirators.

Eyes

Protective safety glasses recommended

Skin

Avoid skin contact. Wear nitrile or similar chemical resistant gloves to keep skin contact to a minimum.

Refer to the manufacturer's recommendations regarding the suitability of any gloves used.

Engineering Controls

Provide adequate ventilation. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations of particulates and any

vapor below occupational exposure limits suitable respiratory protection must be worn.

Other Work Practices Use good personal hygiene practices. Wash hands before eating, drinking, smoking or using toilet. Promptly remove soiled clothing and wash thoroughly before reuse.

See section 2 for further details.

Section 9. Physical and chemical properties

Information on basic physical and chemical properties

Physical State	Liquid
Color	Milky white emulsion solution
Odor	Citrus
Odor threshold	No available information
Melting point / freezing point	No available information
Initial boiling point and boiling range	212°F (100°C) Initial
Flammability (solid, gas)	Not Applicable
Upper/lower flammability or explosive limits	Lower Explosive Limit: No available information Upper Explosive Limit: No available information
Flash Point	155°F (68°C), Closed Cup (PMCC)
Auto-ignition temperature	No available information
Decomposition temperature	No available information
pH	7.0
Viscosity	No available information
Solubility in Water	No available information
Partition coefficient n-octanol/water (Log Kow)	No available information
Vapor pressure (Pa)	10.5 mm Hg @ 20°C
Relative Density	0.91
Vapor Density	No available information
Evaporation rate (Ether = 1)	<0.06 (n-butyl acetate = 1)
VOC Content	375 g/l
Oxidising properties	No available information
Explosive properties	No available information
% Volatile	>97%

Other information

No other relevant information.

Section 10. Stability and reactivity

Reactivity

Hazardous Polymerization will not occur.

Chemical stability

Stable under normal circumstances.

Possibility of hazardous reactions

No available information

Conditions to avoid

Excessive heat and open flame.

Incompatible materials

Strong acids, alkalis, and oxidizers such as liquid chlorine, halogens, hydrogen peroxide, oxygen.

Hazardous decomposition products

High temperatures and fires may produce such toxic substances as carbon monoxide and carbon dioxide.

Section 11. Toxicological information

Acute toxicity

Exposure to solvent vapor concentrations from the component solvents in excess of the stated occupational exposure limits may result in adverse health effects such as mucous membrane and respiratory system irritation and adverse effects on the kidneys, liver and central nervous system. Symptoms include headache, nausea, dizziness, fatigue, muscular weakness, drowsiness and in extreme cases, loss of consciousness.

Repeated or prolonged contact with the preparation may cause removal of natural fat from the skin resulting in dryness, irritation and possible non-allergic contact dermatitis. Solvents may also be absorbed through the skin. Splashes of liquid in the eyes may cause irritation and soreness with possible reversible damage.

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Note: When no route specific LD50 data is available for an acute toxin, the converted acute toxicity point estimate was used in the calculation of the product's ATE (Acute Toxicity Estimate).

Ingredient	Oral LD50, mg/kg	Skin LD50, mg/kg	Inhalation Vapor LC50, mg/L/4hr	Inhalation Dust/Mist LC50, mg/L/4hr	Inhalation Gas LC50, ppm
Hydrocarbon Solvent - (64742-48-9)	> 5,000.00, Rat - Category: NA	>2,000.00, Rabbit - Category: 5	No data available.	No data available.	No data available.
Citrus Terpenes - (5989-27-5)	>2,000.00, Rat - Category: 5	> 5,000.00, Rabbit - Category: NA	No data available.	No data available.	No data available.
Sorbitan monooleate - (1338-43-8)	No data available.	No data available.	No data available.	No data available.	No data available.
Alcohols, C12-15, ethoxylated - (68131-39-5)	> 5,000.00, Rat - Category: NA	>2,000.00, Rat - Category: 5	No data available.	No data available.	No data available.

Carcinogen Data

CAS No.	Ingredient	Source	Value
1338-43-8	Sorbitan monooleate	IARC	No
		ACGIH	No Established Limit
5989-27-5	Citrus Terpenes	IARC	Group 3
		ACGIH	No Established Limit
64742-48-9	Hydrocarbon Solvent	IARC	No
		ACGIH	No Established Limit
68131-39-5	Alcohols, C12-15, ethoxylated	IARC	No
		ACGIH	No Established Limit

Classification	Category	Hazard Description
Acute toxicity (oral)	---	Not Applicable
Acute toxicity (dermal)	---	Not Applicable
Acute toxicity (inhalation)	---	Not Applicable
Skin corrosion/irritation	2	Causes skin irritation.
Serious eye damage/irritation	2A	Causes serious eye irritation.
Respiratory sensitization	---	Not Applicable
Skin sensitization	1	May cause an allergic skin reaction.
Germ cell mutagenicity	---	Not Applicable
Carcinogenicity	---	Not Applicable

Reproductive toxicity	---	Not Applicable
STOT-single exposure	---	Not Applicable
STOT-repeated exposure	---	Not Applicable
Aspiration hazard	---	Not Applicable

Possible routes of entry:

Inhalation, ingestion, skin contact, and skin absorption.

Symptoms and effects, both acute and delayed:

POTENTIAL HEALTH EFFECTS

Eye Contact: May cause tearing, stinging, redness, irritation, and burns.

Inhalation: Irritating to respiratory tract. Prolonged or repeated breathing of very high vapor concentrations cause euphoria, excitation, and dizziness, headaches, nausea, and vomiting, abdominal pain, fatigue, muscular weakness. Aspiration into the lungs can cause CNS (central nervous system) and subsequent aspiration into the lungs can cause pulmonary edema and chemical pneumonia depression. Chronic overexposure in high concentrations may produce CNS depression.

Ingestion: Irritation of the mouth, esophagus, and stomach can develop following ingestion. Symptoms include burning of the mouth, sore throat, vomiting, nausea, dizziness, loss of consciousness. Due to its light viscosity, there is danger of aspiration into the lungs during vomiting. Aspiration can result in severe lung damage or death.

Skin Contact: Prolonged or repeated skin contact may cause moderate to severe irritation including itching and redness of the skin, defatting, and/or dermatitis. This product can also be absorbed through the skin and produce CNS symptoms. Single prolonged exposure is not likely to result in the product being absorbed through the skin in harmful amounts.

Signs And Symptoms Of Exposure: Eye irritation, respiratory irritation, drying and cracking of skin, dizziness, fatigue, headache, unconsciousness or asphyxiation. Chronic effects of ingestion and subsequent aspiration into the lungs can cause pneumatocele (lung cavity) formation and chronic lung dysfunction. Repeated breathing of vapors can cause effects to liver and kidneys.

Treat symptomatically.

Eyes Causes serious eye irritation.

Skin May cause an allergic skin reaction. Causes skin irritation.

Section 12. Ecological information

Toxicity

Toxic to aquatic life with long lasting effects.

Aquatic Ecotoxicity

Ingredient	96 hr LC50 fish, mg/L	48 hr EC50 crustacea, mg/L	ErC50 algae, mg/L
Hydrocarbon Solvent - (64742-48-9)	18.00, Oncorhynchus mykiss	4.50, Daphnia magna	3.10, Pseudokirchneriella subcapitata
Citrus Terpenes - (5989-27-5)	0.72, Pimephales promelas	0.31, Daphnia magna	0.32, Pseudokirchneriella subcapitata
Sorbitan monooleate - (1338-43-8)	No data available.	No data available.	No data available.
Alcohols, C12-15, ethoxylated - (68131-39-5)	3.10, Scophthalmus maximus	0.23, Daphnia magna	3.20, Skeletonema costatum

Persistence and degradability

There is no data available on the preparation itself.

Bioaccumulative potential

No available information

Mobility in soil

No available information

Results of PBT and vPvB assessment

This product contains no PBT/vPvB/vPvM chemicals.

Other adverse effects

No available information

Section 13. Disposal considerations

Waste treatment methods

Observe all federal, provincial and local regulations when disposing of this substance.

Section 14. Transport information

Not regulated for packages under 5L (1.3 gallons) or 5.0 kg (11 lbs).

Classification Method: Classified as per Part 2, Sections 2.1-2.8 of the Transportation of Dangerous Goods Regulations.

DOT (Domestic Surface Transportation)

UN number	NA1993
UN proper shipping name	Combustible liquid, n.o.s., (CHEMICAL NAME)
Transport hazard class(es)	Combustible liquid
Sub Class	Not Applicable
Packing group	III

TDG (Domestic Surface Transportation)

UN number	Not Regulated
UN proper shipping name	Not Regulated
Transport hazard class(es)	Combustible liquid
Sub Class	Not Applicable
Packing group	Not Applicable

IMO / IMDG (Ocean Transportation)

UN number	Not Regulated
UN proper shipping name	Not Regulated
Transport hazard class(es)	Not Applicable
Sub Class	Not Applicable
Packing group	Not Applicable

ICAO/IATA

UN number	Not Regulated
UN proper shipping name	Not Regulated
Transport hazard class(es)	Not Applicable
Sub Class	Not Applicable
Packing group	Not Applicable

Environmental hazards

IMDG Marine Pollutant: Yes; (Citrus Terpenes)

Special precautions for user

No available information

Section 15. Regulatory information

**Regulatory
Overview**

The regulatory data in Section 15 is not intended to be all-inclusive, only selected regulations are represented.

This product has been classified in accordance with US OSHA's Hazard Communication Standard (1910.1200) revised 2024 and Canadian Hazardous Products Regulations (SOR/2015-17 amended 2022-12-15) (GHS revision 7) and the SDS contains all of the information required by those regulations.

All components of this material are either listed or exempt from listing on the TSCA Inventory.

All components of this material are either listed or exempt from listing on the DSL or NDSL Inventories.

EPCRA 302 Extremely Hazardous:

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

EPCRA 313 Toxic Chemicals:

To the best of our knowledge, there are no chemicals at levels which require reporting under this statute.

Proposition 65 - Carcinogens (>0.0%):

Benzene

Cumene

Ethylbenzene

Naphthalene

Proposition 65 - Developmental Toxins (>0.0%):

Benzene

Toluene

Proposition 65 - Female Repro Toxins (>0.0%):

Toluene

Proposition 65 - Male Repro Toxins (>0.0%):

Benzene

Proposition 65 Label Warning:



WARNING: This product can expose you to chemicals including [Benzene, Cumene, Ethylbenzene , Naphthalene], which are known to the State of California to cause cancer, and [Benzene, Toluene], which are known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

Section 16. Other information

Revision Date

14 October, 2025

The information and recommendations contained herein are based upon data believed to be correct. However, no guarantee or warranty of any kind, expressed or implied, is made with respect to the information contained herein. We accept no responsibility and disclaim all liability for any harmful effects which may be caused by exposure to our products. Customers/users of this product must comply with all applicable health and safety laws, regulations, and orders.

The full text of the phrases appearing in section 3 is:

H226 Flammable liquid and vapor.

H302 Harmful if swallowed.

H304 May be fatal if swallowed and enters airways.

H315 Causes skin irritation.

H317 May cause an allergic skin reaction.

H318 Causes serious eye damage.

H400 Very toxic to aquatic life.

H410 Very toxic to aquatic life with long lasting effects.

End of Document

Product Name: PENETROX™ A-13 OXIDE INHIBITING COMPOUND

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SAFETY DATA SHEET

SECTION 1 PRODUCT AND COMPANY IDENTIFICATION

PRODUCT

Product Name: PENETROX™ A-13 OXIDE INHIBITING COMPOUND
Product Description: Oxide inhibiting compound with evenly distributed zinc particles
Intended Use: Corrosion inhibitor

COMPANY IDENTIFICATION

Supplier: BURNDY LLC
47 East Industrial Park Drive
Manchester, NH 03109USA

24 Hour Emergency (INFOTRAC) (800) 535-5053 (*US and Canada*)
(352) 323-3500 (*International*)
Burndy Informational Number (603) 647-5000

SECTION 2 HAZARDS IDENTIFICATION

CLASSIFICATION

Health	Environmental	Physical
No Classifiable hazards	No Classifiable hazards	No Classifiable hazards

LABELLING

Symbols: Not Applicable	
Signal Word: Not Applicable	
Hazard Statements Not Applicable	Precautionary Statements Not Applicable

Hazard not otherwise classified: Not applicable

SECTION 3 COMPOSITION / INFORMATION ON INGREDIENTS

MIXTURES

Name	CAS #	Wt. Percentage*
Zinc Oxide	1314-13-2	1 - 5

* 95-99% material composition inclusive of inert and non-hazardous filler withheld as trade secret in accordance with paragraph 1910.1200(i)(1).

SECTION 4 FIRST AID MEASURES

DESCRIPTION OF THE FIRST AID MEASURE

Eye: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. This compound contains abrasive particles. If eye irritation persists: Get medical advice/attention

Skin: If skin irritation occurs: Wash skin with plenty of water. Obtain medical attention if irritation persists.

Inhalation: Not a normal route of exposure. If symptoms develop, remove to fresh air. Get medical attention if condition worsens.

Ingestion: Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person. Get medical advice/attention if you feel unwell.

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MOST IMPORTANT SYMPTOMS AND EFFECTS, BOTH ACUTE AND DELAYED

Eye: May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling.

Skin: May cause skin irritation. Symptoms may include redness, drying, and cracking of the skin.

Inhalation: Not a normal route of exposure.

Ingestion: May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.

INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENTS NEEDED

Symptoms may be delayed. In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

SECTION 5 FIRE FIGHTING MEASURES

EXTINGUISHING MEDIA

Suitable extinguishing media: Dry chemical. Foam. Carbon dioxide.

Unsuitable extinguishing media: Do not use water jet.

SPECIAL HAZARDS ARISING FROM THE CHEMICAL

Fire hazard: Products of combustion may include and are not limited to: oxides of carbon.

SPECIAL PROTECTIVE EQUIPMENT AND PRECAUTIONS FOR FIRE FIGHTERS

Keep upwind of fire. Wear full firefighting turn-out gear (full Bunker gear) and respiratory protection (SCBA).

SECTION 6 ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.

For non-emergency personnel: No additional information available

For emergency responders: No additional information available

Environmental precautions: Prevent entry to sewers and public waters.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING - UP

Methods for Containment: Absorb and/or contain spill with inert material (sand, vermiculite or other appropriate material), then place in suitable container. Do not flush into surface water or sewer system. Wear recommended personal protective equipment.

Methods for Cleaning-Up: Sweep or shovel spills into appropriate container for disposal. For further information refer to section 8: "Exposure controls/personal protection".

SECTION 7 HANDLING AND STORAGE

PRECAUTIONS FOR SAFE HANDLING

Handling: Avoid contact with skin and eyes. Avoid breathing dust/fume/gas/mist/vapors/spray. Do not swallow. Handle and open container with care. When using do not eat, drink or smoke.

Hygiene Measures: Wash contaminated clothing before reuse. Always wash hands after handling the product.

Storage: Keep out of the reach of children. Keep container tightly closed.

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SECTION 8 EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits				
Ingredient	OSHA-PEL	ACGIH	IDLH	NIOSH
Zinc oxide	TWA: 5 mg/m ³ (fume) 15 mg/m ³ (total dust) 5 mg/m ³ (respirable fraction) Regulatory reference: OSHA Annotated Table Z-1	TWA: 2 mg/m ³ (respirable particulate matter) STEL: 10 mg/m ³ (respirable particulate matter) TLV® Basis: Metal fume fever Regulatory reference - ACGIH 2021	500 mg/m ³	TWA: 5 mg/m ³ (dust and fume) STEL: 10 mg/m ³ (fume) Ceiling: 15 mg/m ³ (dust)

EXPOSURE CONTROLS

Engineering Controls: Ensure good ventilation of the workstation.

Environmental Controls: Avoid release to the environment

INDIVIDUAL PROTECTIVE MEASURES

Eye/Face Protection: Safety glasses or goggles are recommended when using product.

Skin Protection:

Hand Protection: None required.

Body Protection: Wear suitable protective clothing

Respiratory Protection: None required

Other Information: Handle according to established industrial hygiene and safety practices. Do not eat, smoke or drink where material is handled, processed or stored. Wash hands carefully before eating or smoking.

SECTION 9 PHYSICAL/CHEMICAL PROPERTIES

Physical State:	Solid.
Appearance:	Paste. Thick grease
Color:	Gray
Odor:	Not available
Odor Threshold:	Not available
pH:	Not available
Melting Point/Freezing Point:	Not available.
Boiling Point:	Not available
Flash Point:	>260 °C (>500 °F)
Relative Evaporation Rate (butyl acetate=1):	Not available
Flammability:	Not flammable.
Vapor Pressure:	Not available.
Relative vapor density at 20 °C:	Not available
Relative Density:	Not available
Solubility:	Not available
Partition coefficient n-octanol/water:	Not available
Auto-ignition Temperature:	Not available
Decomposition Temperature:	Not available
Viscosity, Kinematic:	Not available
Viscosity, Dynamic:	Not available
Explosion Limits:	Not available
Explosive Properties:	Not available
Oxidizing Properties:	Not available

SECTION 10 STABILITY AND REACTIVITY

Reactivity: No dangerous reaction known under conditions of normal use

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Chemical Stability: Stable under normal conditions
Possibility Of Hazardous Reactions: No dangerous reaction known under conditions of normal use
Conditions To Avoid: Heat
Incompatible Materials: None known
Hazardous Decomposition Products: May include, and are not limited to: oxides of carbon

SECTION 11 TOXICOLOGICAL INFORMATION

Acute toxicity (oral): Not classified
Acute toxicity (dermal): Not classified
Acute toxicity (inhalation): Not classified

Ingredient	LC50	LD50
Zinc oxide	Inhalation: > 5700 mg/m ³ (Exposure time: 4 h), rat	Oral:>5000 mg/kg, rat Dermal: > 2000 mg/kg body weight Animal: rat, Guideline: OECD Guideline 402 (Acute Dermal Toxicity)

Skin Corrosion/Irritation: Not classified
Serious Eye Damage/Irritation: Not classified
Respiratory Or Skin Sensitization: Not classified
Germ Cell Mutagenicity: Not classified
Carcinogenicity: Not classified
Reproductive Toxicity: Not classified
STOT-Single Exposure: Not classified
STOT-Repeated Exposure: Not classified
Aspiration Hazard: Not classified
Viscosity, Kinematic: No data available
Symptoms/Effects After Inhalation: Not a normal route of exposure.
Symptoms/Effects After Skin Contact: May cause skin irritation. Symptoms may include redness, drying, and cracking of the skin.
Symptoms/Effects After Eye Contact: May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling.
Symptoms/Effects After Ingestion: May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea

SECTION 12 ECOLOGICAL INFORMATION

General: Very toxic to aquatic life with long lasting effects.

Ingredient	LC50
Zinc oxide	Fish, 1.55 mg/l (Exposure time: 96 h - Species: Danio rerio [static])

Persistence And Degradability: Not established
Bioaccumulative Potential: Not established
Mobility in Soil: Not available
Other Information: No other effects known

SECTION 13 DISPOSAL CONSIDERATIONS

Disposal Method: This material must be disposed of in accordance with all local, state, provincial, and federal regulations.

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SECTION 14 TRANSPORTATION

Regulatory Information	UN Number	Proper Shipping Name	HazardClass	PackingGroup	Label(s)
US DOT	UN3077	Environmentally hazardous substances, solid, n.o.s. (Zinc, Zinc Oxide)	9	III	

Other Information: No supplementary information available.

Special Transport Precautions: Do not handle until all safety precautions have been read and understood

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code: Not applicable

SECTION 15 REGULATORY INFORMATION

Us Federal Regulations: All components of this product are listed, or excluded from listing, on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory.

International Regulations: No additional information available

US State Regulations: California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm

SECTION 16 OTHER INFORMATION

THIS SAFETY DATA SHEET CONTAINS THE FOLLOWING REVISIONS:

Revision Date	Description	Sections Affected
06/02/2011	MSDS Version written	1-11
07/21/2011	Updated to GHS criteria, additional sections added.	1-16
08/01/2011	Updated	15
11/08/2012	Updated	1, 3, 8, 9, 11, 12, 15
04/02/2013	Updated	11
07/05/2013	Version number update	-
07/25/2014	Updates	3,15
09/16/2015	Updates	1-16
10/28/2015	Updates	1-16
11/21/2016	Updates	8
10/20/2017	Updates	1-16
8/30/2018	Updates	15
07/18/2019	Updates	1-16
09/13/2019	Updates	1-16
02/08/2023	Updates	1-16

This SDS provides a good faith representation of information believed to be accurate as of the last revision date. This document does not create any express or implied product warranties. Since conditions of use are beyond the control of Burndy LLC, all risks associated with product use are assumed by the user.



SAFETY DATA SHEET

1. Identification

Product identifier Zinc-It® Instant Cold Galvanize

Other means of identification

Product Code No. 18412 (Item# 1005240)

Recommended use Coating

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Manufactured or sold by:

Company name CRC Industries, Inc.

Address 885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical Assistance 800-521-3168

Customer Service 800-272-4620

**24-Hour Emergency
(CHEMTREC)** 800-424-9300 (US)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards	Flammable aerosols	Category 1
	Gases under pressure	Liquefied gas
Health hazards	Skin corrosion/irritation	Category 2
	Serious eye damage/eye irritation	Category 2
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Aspiration hazard	Category 1
Environmental hazards	Hazardous to the aquatic environment, acute hazard	Category 1
	Hazardous to the aquatic environment, long-term hazard	Category 1
OSHA defined hazards	Not classified.	

Label elements



Signal word Danger

Hazard statement Extremely flammable aerosol. Contains gas under pressure; may explode if heated. May be fatal if swallowed and enters airways. Causes skin irritation. Causes serious eye irritation. May cause drowsiness or dizziness. Very toxic to aquatic life. Very toxic to aquatic life with long lasting effects.

Precautionary statement

Prevention Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not spray on an open flame or other ignition source. Pressurized container: Do not pierce or burn, even after use. Do not apply while equipment is energized. Extinguish all flames, pilot lights, and heaters. Vapors will accumulate readily and may ignite. Use only with adequate ventilation; maintain ventilation during use and until all vapors are gone. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Avoid breathing mist or vapor. Wash thoroughly after handling. Wear eye protection/face protection. Wear protective gloves. Avoid release to the environment.

Response	If swallowed: Immediately call a poison center/doctor. Do NOT induce vomiting. If on skin: Wash with plenty of water. If skin irritation occurs: Get medical advice/attention. Take off contaminated clothing and wash before reuse. If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center/doctor if you feel unwell. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. Collect spillage.
Storage	Store in a well-ventilated place. Store locked up. Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F. Exposure to high temperature may cause can to burst.
Disposal	Dispose of contents/container in accordance with local/regional/national regulations.
Hazard(s) not otherwise classified (HNOC)	None known.
Supplemental information	None.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
zinc		7440-66-6	30 - 60
liquefied petroleum gas		68476-86-8	10 - 30
solvent naphtha (petroleum), light aliph.		64742-89-8	10 - 30
methyl ethyl ketone		78-93-3	5 - 10

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
Skin contact	Remove contaminated clothing. Wash with plenty of soap and water. If skin irritation occurs: Get medical advice/attention. Wash contaminated clothing before reuse.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	Call a physician or poison control center immediately. Rinse mouth. Do not induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs.
Most important symptoms/effects, acute and delayed	Aspiration may cause pulmonary edema and pneumonitis. May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Skin irritation. May cause redness and pain.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.
General information	Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Water spray. Dry chemical powder. Carbon dioxide (CO2).
Unsuitable extinguishing media	None known.
Specific hazards arising from the chemical	Contents under pressure. Pressurized container may rupture when exposed to heat or flame. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Fire-fighting equipment/instructions	In case of fire: Stop leak if safe to do so. Move containers from fire area if you can do so without risk. Containers should be cooled with water to prevent vapor pressure build up. Use standard firefighting procedures and consider the hazards of other involved materials. In the event of fire and/or explosion do not breathe fumes.
General fire hazards	Extremely flammable aerosol. Contents under pressure. Pressurized container may rupture when exposed to heat or flame.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Many gases are heavier than air and will spread along ground and collect in low or confined areas (sewers, basements, tanks). Wear appropriate protective equipment and clothing during clean-up. Avoid breathing mist or vapor. Emergency personnel need self-contained breathing equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. This product is miscible in water. This material is classified as a water pollutant under the Clean Water Act and should be prevented from contaminating soil or from entering sewage and drainage systems which lead to waterways. Stop the flow of material, if this is without risk. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid release to the environment. Inform appropriate managerial or supervisory personnel of all environmental releases. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage

Precautions for safe handling	Pressurized container: Do not pierce or burn, even after use. Do not use if spray button is missing or defective. Do not spray on a naked flame or any other incandescent material. Do not smoke while using or until sprayed surface is thoroughly dry. Do not cut, weld, solder, drill, grind, or expose containers to heat, flame, sparks, or other sources of ignition. Use caution around energized equipment. The metal container will conduct electricity if it contacts a live source. This may result in injury to the user from electrical shock and/or flash fire. Avoid breathing mist or vapor. Avoid contact with eyes, skin, and clothing. Avoid prolonged exposure. Use only in well-ventilated areas. Wear appropriate personal protective equipment. Wash hands thoroughly after handling. Avoid release to the environment. Observe good industrial hygiene practices. For product usage instructions, see the product label.
Conditions for safe storage, including any incompatibilities	Level 1 Aerosol. Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C/122 °F. Do not puncture, incinerate or crush. Do not handle or store near an open flame, heat or other sources of ignition. This material can accumulate static charge which may cause spark and become an ignition source. Store in a well-ventilated place. Stored containers should be periodically checked for general condition and leakage. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
methyl ethyl ketone (CAS 78-93-3)	PEL	590 mg/m3
solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)	PEL	200 ppm
		400 mg/m3
		100 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
methyl ethyl ketone (CAS 78-93-3)	STEL	300 ppm
	TWA	200 ppm

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
methyl ethyl ketone (CAS 78-93-3)	STEL	885 mg/m3

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)	TWA	300 ppm
		590 mg/m3
		200 ppm
	TWA	400 mg/m3
		100 ppm

Biological limit values**ACGIH Biological Exposure Indices**

Components	Value	Determinant	Specimen	Sampling Time
methyl ethyl ketone (CAS 78-93-3)	2 mg/l	MEK	Urine	*

* - For sampling details, please see the source document.

Appropriate engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station. Eye wash fountain and emergency showers are recommended.

Individual protection measures, such as personal protective equipment

Eye/face protection Wear safety glasses with side shields (or goggles).

Skin protection

Hand protection Wear protective gloves such as: Nitrile.

Other Wear appropriate chemical resistant clothing.

Respiratory protection If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.

Thermal hazards Wear appropriate thermal protective clothing, when necessary.

General hygiene considerations

When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties**Appearance**

Physical state Liquid.

Form Aerosol.

Color Gray.

Odor Ketone.

Odor threshold Not available.

pH Not available.

Melting point/freezing point Not available.

Initial boiling point and boiling range 95 °F (35 °C) estimated

Flash point < 0 °F (< -17.8 °C)

Evaporation rate Moderate.

Flammability (solid, gas) Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%) 0.9 % estimated

Flammability limit - upper (%) 11.5 % estimated

Vapor pressure 965.7 hPa estimated

Vapor density > 1 (air = 1)

Relative density	1.38
Solubility(ies)	
Solubility (water)	Negligible.
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	550 °F (287.8 °C) estimated
Decomposition temperature	Not available.
Viscosity	Not available.
Percent volatile	52.6 % estimated

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	No dangerous reaction known under conditions of normal use.
Conditions to avoid	Heat, flames and sparks. Contact with incompatible materials.
Incompatible materials	Strong oxidizing agents.
Hazardous decomposition products	Carbon oxides.

11. Toxicological information

Information on likely routes of exposure

Inhalation	May cause drowsiness and dizziness. Headache. Nausea, vomiting. Prolonged inhalation may be harmful.
Skin contact	Causes skin irritation.
Eye contact	Causes serious eye irritation.
Ingestion	Droplets of the product aspirated into the lungs through ingestion or vomiting may cause a serious chemical pneumonia.

Symptoms related to the physical, chemical and toxicological characteristics	Aspiration may cause pulmonary edema and pneumonitis. May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Skin irritation. May cause redness and pain.
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Information on toxicological effects

Acute toxicity	May be fatal if swallowed and enters airways.
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Components	Species	Test Results
methyl ethyl ketone (CAS 78-93-3)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 8000 mg/kg
Inhalation		
LC50	Rat	11700 ppm, 4 Hours
Oral		
LD50	Rat	2300 - 3500 mg/kg
solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 2000 mg/kg
Oral		
LD50	Rat	> 3000 mg/kg
zinc (CAS 7440-66-6)		
<u>Acute</u>		
Oral		
LD50	Rat	> 2000 mg/kg
Skin corrosion/irritation	Causes skin irritation.	

Serious eye damage/eye irritation	Causes serious eye irritation.
Respiratory or skin sensitization	
Respiratory sensitization	Not a respiratory sensitizer.
Skin sensitization	This product is not expected to cause skin sensitization.
Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
Carcinogenicity	Not classifiable as to carcinogenicity to humans.
IARC Monographs. Overall Evaluation of Carcinogenicity	Not listed.
OSHA Specifically Regulated Substances (29 CFR 1910.1001-1052)	Not regulated.
US. National Toxicology Program (NTP) Report on Carcinogens	Not listed.
Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity - single exposure	May cause drowsiness and dizziness.
Specific target organ toxicity - repeated exposure	Not classified.
Aspiration hazard	May be fatal if swallowed and enters airways.
Chronic effects	Prolonged inhalation may be harmful.

12. Ecological information

Ecotoxicity	Very toxic to aquatic life with long lasting effects.		
Components	Species		Test Results
methyl ethyl ketone (CAS 78-93-3)			
Aquatic			
Crustacea	EC50	Water flea (Daphnia magna)	4025 - 6440 mg/l, 48 hours
Acute			
Fish	LC50	Fathead minnow (Pimephales promelas)	2993 mg/l, 96 hours
solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)			
Aquatic			
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	8.8 mg/l, 96 hours
			8.8 mg/l, 96 hours
Acute			
Crustacea	EC50	Water flea (Daphnia magna)	1.5 mg/l, 48 hours
zinc (CAS 7440-66-6)			
Aquatic			
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	0.56 mg/l, 96 hours
Acute			
Crustacea	EC50	Water flea (Daphnia magna)	0.068 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	0.56 mg/l, 96 hours
			0.482 mg/l, 96 hours
Persistence and degradability	No data is available on the degradability of any ingredients in the mixture.		
Bioaccumulative potential			
Partition coefficient n-octanol / water (log Kow)			
methyl ethyl ketone	0.29		
Mobility in soil	No data available.		
Other adverse effects	No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.		

13. Disposal considerations

Hazardous waste code	D001: Waste Flammable material with a flash point <140 F
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.
Disposal instructions	If discarded, this product is considered a RCRA ignitable waste, D001. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Contents under pressure. Do not puncture, incinerate or crush. Do not allow this material to drain into sewers/water supplies. Do not contaminate ponds, waterways or ditches with chemical or used container. Dispose in accordance with all applicable regulations.

14. Transport information

DOT

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Label(s)	2.1
Packing group	Not applicable.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Special provisions	N82
Packaging exceptions	306
Packaging non bulk	304
Packaging bulk	None

IATA

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
ERG Code	10L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	UN1950
UN proper shipping name	AEROSOLS, Limited Quantity
Transport hazard class(es)	
Class	2
Subsidiary risk	-
Packing group	Not applicable.
Environmental hazards	
Marine pollutant	No.
EmS	F-D, S-U
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

15. Regulatory information

US federal regulations	This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.
TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)	
Not regulated.	
SARA 304 Emergency release notification	
Not regulated.	
OSHA Specifically Regulated Substances (29 CFR 1910.1001-1052)	
Not regulated.	

US EPCRA (SARA Title III) Section 313 - Toxic Chemical: Listed substance

zinc (CAS 7440-66-6)

CERCLA Hazardous Substance List (40 CFR 302.4)

methyl ethyl ketone (CAS 78-93-3)

Listed.

zinc (CAS 7440-66-6)

Listed.

CERCLA Hazardous Substances: Reportable quantity

methyl ethyl ketone (CAS 78-93-3)

5000 LBS

zinc (CAS 7440-66-6)

1000 LBS

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Other federal regulations**Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List**

Not regulated.

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

Not regulated.

Safe Drinking Water Act (SDWA)

Not regulated.

Food and Drug

Not regulated.

Administration (FDA)**Drug Enforcement Administration (DEA). List 2, Essential Chemicals (21 CFR 1310.02(b) and 1310.04(f)(2) and Chemical Code Number**

methyl ethyl ketone (CAS 78-93-3)

6714

Drug Enforcement Administration (DEA). List 1 & 2 Exempt Chemical Mixtures (21 CFR 1310.12(c))

methyl ethyl ketone (CAS 78-93-3)

35 %WV

DEA Exempt Chemical Mixtures Code Number

methyl ethyl ketone (CAS 78-93-3)

6714

FEMA Priority Substances Respiratory Health and Safety in the Flavor Manufacturing Workplace

methyl ethyl ketone (CAS 78-93-3)

Low priority

Superfund Amendments and Reauthorization Act of 1986 (SARA)**Classified hazard categories**

Flammable (gases, aerosols, liquids, or solids)

Gas under pressure

Acute toxicity (any route of exposure)

Skin corrosion or irritation

Serious eye damage or eye irritation

Specific target organ toxicity (single or repeated exposure)

Aspiration hazard

SARA 302 Extremely hazardous substance

Not listed.

SARA 311/312 Hazardous chemical

Yes

SARA 313 (TRI reporting)

Chemical name	CAS number	% by wt.
zinc	7440-66-6	30 - 60

US state regulations**US. New Jersey Worker and Community Right-to-Know Act**

methyl ethyl ketone (CAS 78-93-3)

solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)

zinc (CAS 7440-66-6)

US. Massachusetts RTK - Substance List

methyl ethyl ketone (CAS 78-93-3)

solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)

zinc (CAS 7440-66-6)

US. Pennsylvania Worker and Community Right-to-Know Law

methyl ethyl ketone (CAS 78-93-3)

solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)

zinc (CAS 7440-66-6)

US. Rhode Island RTK

methyl ethyl ketone (CAS 78-93-3)

solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)
zinc (CAS 7440-66-6)

California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 2016 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins. For more information go to www.P65Warnings.ca.gov.

US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))

liquefied petroleum gas (CAS 68476-86-8)
methyl ethyl ketone (CAS 78-93-3)
solvent naphtha (petroleum), light aliph. (CAS 64742-89-8)
zinc (CAS 7440-66-6)

Volatile organic compounds (VOC) regulations

EPA

Aerosol coatings (40 CFR 59, Subpt. E) Compliant

State

Aerosol coatings This product is regulated as a Metallic Coating. This product is compliant for sale in all 50 states.
Maximum incremental reactivity (MIR) 0.5

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	No
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
Taiwan	Taiwan Toxic Chemical Substances (TCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date	08-28-2015
Revision date	06-05-2018
Prepared by	Allison Yoon
Version #	04
Disclaimer	The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..
Revision information	This document has undergone significant changes and should be reviewed in its entirety.



SAFETY DATA SHEET

1. Identification

Product identifier Wasp & Hornet Killer Plus™ - 14 oz

Other means of identification

Product Code No. 14010 (Item# 1004768)

Product registration number 55809-3

Registration number EPA: 55809-3

Recommended use Wasp and hornet insecticide

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Manufactured or sold by:

Company name CRC Industries, Inc.

Address 885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical Assistance 800-521-3168

Customer Service 800-272-4620

24-Hour Emergency (CHEMTREC) 800-424-9300 (US)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards Flammable aerosols Category 1
Gases under pressure Compressed gas

Health hazards Aspiration hazard Category 1

Environmental hazards Hazardous to the aquatic environment, acute hazard Category 1

Hazardous to the aquatic environment, long-term hazard Category 1

OSHA defined hazards Not classified.

Label elements



Signal word Danger

Hazard statement Extremely flammable aerosol. Contains gas under pressure; may explode if heated. May be fatal if swallowed and enters airways. Very toxic to aquatic life. Very toxic to aquatic life with long lasting effects.

Precautionary statement

Prevention

Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not spray on an open flame or other ignition source. Pressurized container: Do not pierce or burn, even after use. Do not apply while equipment is energized. Extinguish all flames, pilot lights, and heaters. Vapors will accumulate readily and may ignite. Use only with adequate ventilation; maintain ventilation during use and until all vapors are gone. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Avoid release to the environment.

Response If swallowed: Immediately call a poison center/doctor. Do NOT induce vomiting. Collect spillage.

Storage	Store in a well-ventilated place. Store locked up. Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F. Exposure to high temperature may cause can to burst.
Disposal	Dispose of contents/container in accordance with local/regional/national/international regulations.
Hazard(s) not otherwise classified (HNOC)	None known.
Supplemental information	None.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
distillates (petroleum), hydrotreated light		64742-47-8	90 - 100
carbon dioxide		124-38-9	1 - 3
tetramethrin		7696-12-0	0.2
d-phenothrin		26002-80-2	0.125

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Move to fresh air. Call a physician if symptoms develop or persist.
Skin contact	Take off immediately all contaminated clothing. Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	Call a physician or poison control center immediately. Rinse mouth. Do not induce vomiting. If vomiting occurs, keep head low so that stomach content doesn't get into the lungs.
Most important symptoms/effects, acute and delayed	Aspiration may cause pulmonary edema and pneumonitis. Headache. Nausea, vomiting. Diarrhea.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.
General information	Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Water fog. Alcohol resistant foam. Dry chemical powder. Dry chemicals. Carbon dioxide (CO2).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Contents under pressure. Pressurized container may rupture when exposed to heat or flame. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Fire fighting equipment/instructions	In case of fire: Stop leak if safe to do so. Move containers from fire area if you can do so without risk. Cool containers exposed to heat with water spray and remove container, if no risk is involved. Containers should be cooled with water to prevent vapor pressure build up.
Specific methods	Use standard firefighting procedures and consider the hazards of other involved materials. In the event of fire and/or explosion do not breathe fumes.
General fire hazards	Extremely flammable aerosol. Contents under pressure. Pressurized container may rupture when exposed to heat or flame. Will burn if involved in a fire.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Many gases are heavier than air and will spread along ground and collect in low or confined areas (sewers, basements, tanks). Wear appropriate protective equipment and clothing during clean-up. Emergency personnel need self-contained breathing equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
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Methods and materials for containment and cleaning up

Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. This product is miscible in water. Prevent product from entering drains. Stop the flow of material, if this is without risk. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. For waste disposal, see section 13 of the SDS.

Environmental precautions

Avoid release to the environment. Inform appropriate managerial or supervisory personnel of all environmental releases. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage**Precautions for safe handling**

Pressurized container: Do not pierce or burn, even after use. Do not use if spray button is missing or defective. Do not spray on a naked flame or any other incandescent material. Do not smoke while using or until sprayed surface is thoroughly dry. Do not cut, weld, solder, drill, grind, or expose containers to heat, flame, sparks, or other sources of ignition. Use caution around energized equipment. The metal container will conduct electricity if it contacts a live source. This may result in injury to the user from electrical shock and/or flash fire. Avoid prolonged or repeated contact with skin. Avoid prolonged exposure. Use only in well-ventilated areas. Wear appropriate personal protective equipment. Wash hands thoroughly after handling. Avoid release to the environment. Observe good industrial hygiene practices. For product usage instructions, see the product label.

Conditions for safe storage, including any incompatibilities

Level 3 Aerosol.

Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C/122 °F. Do not puncture, incinerate or crush. Do not handle or store near an open flame, heat or other sources of ignition. This material can accumulate static charge which may cause spark and become an ignition source. Store in tightly closed container. Store in a well-ventilated place. Stored containers should be periodically checked for general condition and leakage. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection**Occupational exposure limits**

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
carbon dioxide (CAS 124-38-9)	PEL	9000 mg/m3
		5000 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
carbon dioxide (CAS 124-38-9)	STEL	30000 ppm
	TWA	5000 ppm

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
carbon dioxide (CAS 124-38-9)	STEL	54000 mg/m3
		30000 ppm
		9000 mg/m3
	TWA	5000 ppm
		100 mg/m3
distillates (petroleum), hydrotreated light (CAS 64742-47-8)	TWA	

Biological limit values

No biological exposure limits noted for the ingredient(s).

Appropriate engineering controls

Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.

Individual protection measures, such as personal protective equipment**Eye/face protection**

Wear safety glasses with side shields (or goggles).

Skin protection	
Hand protection	Wear protective gloves such as: Neoprene. Nitrile.
Other	Wear suitable protective clothing.
Respiratory protection	If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.
Thermal hazards	Wear appropriate thermal protective clothing, when necessary.
General hygiene considerations	When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties

Appearance

Physical state	Liquid.
Form	Aerosol.
Color	Clear.
Odor	Petroleum.
Odor threshold	Not available.
pH	Not available.
Melting point/freezing point	Not available.
Initial boiling point and boiling range	347 °F (175 °C) estimated
Flash point	205 °F (96.1 °C) Pensky-Martens Closed Cup
Evaporation rate	Slow.
Flammability (solid, gas)	Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%)	0.5 % estimated
Flammability limit - upper (%)	6.5 % estimated
Vapor pressure	1832.6 hPa estimated
Vapor density	> 2 (air = 1)
Relative density	0.82 estimated
Solubility(ies)	
Solubility (water)	Negligible.
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	410 °F (210 °C) estimated
Decomposition temperature	Not available.
Viscosity	Not available.

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	No dangerous reaction known under conditions of normal use.
Conditions to avoid	Heat. Avoid temperatures exceeding the flash point. Contact with incompatible materials.
Incompatible materials	Strong oxidizing agents.
Hazardous decomposition products	Carbon oxides. Nitrogen oxides (NOx). Hydrocarbons.

11. Toxicological information

Information on likely routes of exposure

Inhalation	Prolonged inhalation may be harmful.
Skin contact	No adverse effects due to skin contact are expected.
Eye contact	Direct contact with eyes may cause temporary irritation.
Ingestion	Droplets of the product aspirated into the lungs through ingestion or vomiting may cause a serious chemical pneumonia.

Symptoms related to the physical, chemical and toxicological characteristics	Aspiration may cause pulmonary edema and pneumonitis. Headache. Nausea, vomiting. Diarrhea.
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Information on toxicological effects

Acute toxicity	May be fatal if swallowed and enters airways.
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Components	Species	Test Results
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d-phenothrin (CAS 26002-80-2)

Acute

Dermal

LD50	Rat	> 2000 mg/kg
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Inhalation

LC50	Rat	> 3.76 mg/l, 4 Hours
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Oral

LD50	Rat	> 500 mg/kg
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tetramethrin (CAS 7696-12-0)

Acute

Dermal

LD50	Rabbit	> 2000 mg/kg
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Oral

LD50	Rat	> 5000 mg/kg
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Skin corrosion/irritation	Prolonged skin contact may cause temporary irritation.
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Serious eye damage/eye irritation	Direct contact with eyes may cause temporary irritation.
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Respiratory or skin sensitization

Respiratory sensitization	Not a respiratory sensitizer.
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Skin sensitization	This product is not expected to cause skin sensitization.
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Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
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Carcinogenicity	Not classifiable as to carcinogenicity to humans.
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IARC Monographs. Overall Evaluation of Carcinogenicity

Not listed.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

US. National Toxicology Program (NTP) Report on Carcinogens

Not listed.

Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
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Specific target organ toxicity - single exposure	Not classified.
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Specific target organ toxicity - repeated exposure	Not classified.
--	-----------------

Aspiration hazard	May be fatal if swallowed and enters airways.
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Chronic effects	Prolonged inhalation may be harmful.
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12. Ecological information

Ecotoxicity	Very toxic to aquatic life with long lasting effects.
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Components	Species		Test Results
distillates (petroleum), hydrotreated light (CAS 64742-47-8)			
Aquatic			
Acute			
Crustacea	EC50	Water flea (Daphnia magna)	> 1000 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	> 1000 mg/l, 96 hours
tetramethrin (CAS 7696-12-0)			
Aquatic			
Acute			
Crustacea	EC50	Water flea (Daphnia magna)	0.11 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	0.0037 mg/l, 96 hours
Persistence and degradability	No data is available on the degradability of any ingredients in the mixture.		
Bioaccumulative potential			
Partition coefficient n-octanol / water (log Kow)			
tetramethrin			4.58
Mobility in soil	No data available.		
Other adverse effects	No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.		

13. Disposal considerations

Disposal instructions	If discarded, this product is considered a RCRA ignitable waste, D001. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Contents under pressure. Do not puncture, incinerate or crush. Do not allow this material to drain into sewers/water supplies. Do not contaminate ponds, waterways or ditches with chemical or used container. Dispose in accordance with all applicable regulations.
Hazardous waste code	D001: Waste Flammable material with a flash point <140 F
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

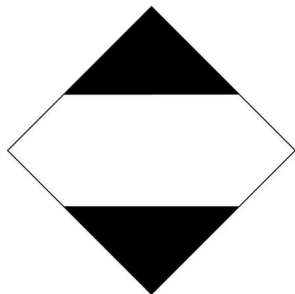
DOT

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Label(s)	2.1
Packing group	Not applicable.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Special provisions	N82
Packaging exceptions	306
Packaging non bulk	None
Packaging bulk	None

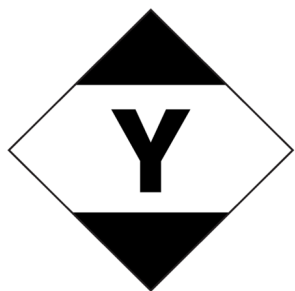
IATA

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
ERG Code	10L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.

Cargo aircraft only	Allowed with restrictions.
IMDG	
UN number	UN1950
UN proper shipping name	AEROSOLS, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
Environmental hazards	
Marine pollutant	Yes, but exempt from the regulations.
EmS	Not available.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
DOT; IMDG	



IATA



15. Regulatory information

US federal regulations This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.

TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)

Not regulated.

SARA 304 Emergency release notification

Not regulated.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

US EPCRA (SARA Title III) Section 313 - Toxic Chemical: Listed substance

d-phenothrin (CAS 26002-80-2)

tetramethrin (CAS 7696-12-0)

CERCLA Hazardous Substance List (40 CFR 302.4)

Not listed.

CERCLA Hazardous Substances: Reportable quantity

Not listed.

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Other federal regulations

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

Not regulated.

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

Not regulated.

Safe Drinking Water Act (SDWA)	Not regulated.
Food and Drug Administration (FDA)	Not regulated.
FIFRA Information	This chemical is a pesticide product registered by the United States Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals. The hazard information required on the pesticide label is reproduced below. The pesticide label also includes other important information, including directions for use.
Signal word	Warning.
Hazard statement	Harmful if absorbed through skin. Causes moderate eye irritation. This product is registered in all 50 United States and Puerto Rico. This product is not registered outside of the United States and Puerto Rico.

Superfund Amendments and Reauthorization Act of 1986 (SARA)

Classified hazard categories	Flammable (gases, aerosols, liquids, or solids) Gas under pressure Aspiration hazard
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SARA 302 Extremely hazardous substance

Not listed.

SARA 311/312 Hazardous chemical	Yes
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SARA 313 (TRI reporting)	Not regulated.
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US state regulations

US. New Jersey Worker and Community Right-to-Know Act

carbon dioxide (CAS 124-38-9)
d-phenothrin (CAS 26002-80-2)
tetramethrin (CAS 7696-12-0)

US. Massachusetts RTK - Substance List

carbon dioxide (CAS 124-38-9)

US. Pennsylvania Worker and Community Right-to-Know Law

carbon dioxide (CAS 124-38-9)
distillates (petroleum), hydrotreated light (CAS 64742-47-8)

US. Rhode Island RTK

carbon dioxide (CAS 124-38-9)
distillates (petroleum), hydrotreated light (CAS 64742-47-8)

California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 2016 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins. For more information go to www.P65Warnings.ca.gov.

US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))

distillates (petroleum), hydrotreated light (CAS 64742-47-8)

Volatile organic compounds (VOC) regulations

EPA

VOC content (40 CFR 51.100(s))	96.5 %
Consumer products (40 CFR 59, Subpt. C)	Not regulated

State

Consumer products	This product is regulated as a Wasp and Hornet Insecticide. This product is compliant for use in all 50 states.
VOC content (CA)	1.4 %
VOC content (OTC)	1.4 %

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	No
Canada	Domestic Substances List (DSL)	No
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	No
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	No
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	No
Korea	Existing Chemicals List (ECL)	No
New Zealand	New Zealand Inventory	No
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	No
Taiwan	Taiwan Chemical Substance Inventory (TCSI)	No
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date	01-02-2020
Prepared by	Allison Yoon
Version #	01
Further information	CRC # 431J/1002412
Disclaimer	The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..
Revision information	This document has undergone significant changes and should be reviewed in its entirety.



SAFETY DATA SHEET

1. Identification

Product identifier All Purpose Enamel Swift Red Spray Paint

Other means of identification

Product code 18000

Recommended use Coating

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Company name CRC Industries, Inc.

Address 885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical 800-521-3168

Assistance

Customer Service 800-272-4620

24-Hour Emergency 800-424-9300 (US)

(CHEMTREC) 703-527-3887 (International)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards	Flammable aerosols	Category 1
	Gases under pressure	Liquefied gas
Health hazards	Serious eye damage/eye irritation	Category 2A
	Carcinogenicity	Category 2
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Specific target organ toxicity, repeated exposure	Category 2
Environmental hazards	Not classified.	
OSHA defined hazards	Not classified.	
Label elements		



Signal word Danger

Hazard statement Extremely flammable aerosol. Contains gas under pressure; may explode if heated. Causes serious eye irritation. May cause drowsiness or dizziness. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.

Precautionary statement

Prevention

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not spray on an open flame or other ignition source. Pressurized container: Do not pierce or burn, even after use. Do not apply while equipment is energized. Extinguish all flames, pilot lights and heaters. Vapors will accumulate readily and may ignite. Use only with adequate ventilation; maintain ventilation during use and until all vapors are gone. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Do not breathe mist or vapor. Wear protective gloves/protective clothing/eye protection/face protection. Wash thoroughly after handling.

Response	If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center/doctor if you feel unwell. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention. If exposed or concerned: Get medical attention.
Storage	Store in a well-ventilated place. Store locked up. Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F. Exposure to high temperature may cause can to burst.
Disposal	Dispose of contents/container in accordance with local/regional/national regulations.
Hazard(s) not otherwise classified (HNOC)	None known.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
acetone		67-64-1	30 - 40
isobutyl acetate		110-19-0	10 - 20
propane		74-98-6	10 - 20
n-butane		106-97-8	5 - 10
ethylene glycol propyl ether		2807-30-9	1 - 3
methyl isobutyl ketone		108-10-1	1 - 3
methyl propyl ketone		107-87-9	1 - 3
propylene glycol methyl ether acetate		108-65-6	1 - 3

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
Skin contact	Rinse skin with water/shower. Get medical attention if irritation develops and persists.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	In the unlikely event of swallowing contact a physician or poison control center. Rinse mouth.
Most important symptoms/effects, acute and delayed	May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Prolonged exposure may cause chronic effects.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.
General information	IF exposed or concerned: Get medical advice/attention. Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Water fog. Alcohol resistant foam. Dry chemical powder. Carbon dioxide (CO2).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Contents under pressure. Pressurized container may rupture when exposed to heat or flame. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Fire-fighting equipment/instructions	In case of fire: Stop leak if safe to do so. Move containers from fire area if you can do so without risk. Containers should be cooled with water to prevent vapor pressure build up.
General fire hazards	Extremely flammable aerosol. Contents under pressure. Pressurized container may rupture when exposed to heat or flame.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Many gases are heavier than air and will spread along ground and collect in low or confined areas (sewers, basements, tanks). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Emergency personnel need self-contained breathing equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. Prevent product from entering drains. Stop the flow of material, if this is without risk. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Put material in suitable, covered, labeled containers. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid release to the environment. Inform appropriate managerial or supervisory personnel of all environmental releases. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage

Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Pressurized container: Do not pierce or burn, even after use. Do not use if spray button is missing or defective. Do not spray on a naked flame or any other incandescent material. Do not smoke while using or until sprayed surface is thoroughly dry. Do not cut, weld, solder, drill, grind, or expose containers to heat, flame, sparks, or other sources of ignition. Use caution around energized equipment. The metal container will conduct electricity if it contacts a live source. This may result in injury to the user from electrical shock and/or flash fire. Do not breathe mist or vapor. Avoid contact with eyes. Avoid prolonged exposure. Should be handled in closed systems, if possible. Use only in well-ventilated areas. Wear appropriate personal protective equipment. Avoid release to the environment. Observe good industrial hygiene practices. For product usage instructions, please see the product label.
Conditions for safe storage, including any incompatibilities	Level 3 Aerosol. Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C/122 °F. Do not puncture, incinerate or crush. Do not handle or store near an open flame, heat or other sources of ignition. This material can accumulate static charge which may cause spark and become an ignition source. Store in a well-ventilated place. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value
acetone (CAS 67-64-1)	PEL	2400 mg/m3 1000 ppm
isobutyl acetate (CAS 110-19-0)	PEL	700 mg/m3 150 ppm
methyl isobutyl ketone (CAS 108-10-1)	PEL	410 mg/m3 100 ppm
methyl propyl ketone (CAS 107-87-9)	PEL	700 mg/m3 200 ppm
propane (CAS 74-98-6)	PEL	1800 mg/m3 1000 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
acetone (CAS 67-64-1)	STEL	500 ppm
	TWA	250 ppm
isobutyl acetate (CAS 110-19-0)	STEL	150 ppm

US. ACGIH Threshold Limit Values

Components	Type	Value
methyl isobutyl ketone (CAS 108-10-1)	TWA	50 ppm
	STEL	75 ppm
methyl propyl ketone (CAS 107-87-9)	TWA	20 ppm
	STEL	150 ppm
n-butane (CAS 106-97-8)	STEL	1000 ppm

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
acetone (CAS 67-64-1)	TWA	590 mg/m3
		250 ppm
isobutyl acetate (CAS 110-19-0)	TWA	700 mg/m3
		150 ppm
methyl isobutyl ketone (CAS 108-10-1)	STEL	300 mg/m3
		75 ppm
	TWA	205 mg/m3
		50 ppm
methyl propyl ketone (CAS 107-87-9)	TWA	530 mg/m3
		150 ppm
n-butane (CAS 106-97-8)	TWA	1900 mg/m3
		800 ppm
propane (CAS 74-98-6)	TWA	1800 mg/m3
		1000 ppm

US. AIHA Workplace Environmental Exposure Level (WEEL) Guides

Components	Type	Value
propylene glycol methyl ether acetate (CAS 108-65-6)	TWA	50 ppm

Biological limit values**ACGIH Biological Exposure Indices**

Components	Value	Determinant	Specimen	Sampling Time
acetone (CAS 67-64-1)	25 mg/l	Acetone	Urine	*
methyl isobutyl ketone (CAS 108-10-1)	1 mg/l	Methyl isobutyl ketone	Urine	*

* - For sampling details, please see the source document.

Exposure guidelines**US - California OELs: Skin designation**

propylene glycol methyl ether acetate (CAS 108-65-6) Can be absorbed through the skin.

Appropriate engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station.

Individual protection measures, such as personal protective equipment

Eye/face protection Wear safety glasses with side shields (or goggles).

Skin protection

Hand protection Wear protective gloves such as: Nitrile. Butyl rubber.

Other Wear suitable protective clothing.

Respiratory protection If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.

Thermal hazards Wear appropriate thermal protective clothing, when necessary.

General hygiene considerations

Observe any medical surveillance requirements. When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties

Appearance

Physical state Liquid.

Form Aerosol.

Color Red.

Odor Aromatic.

Odor threshold Not available.

pH Not available.

Melting point/freezing point -145.8 °F (-98.8 °C) estimated

Initial boiling point and boiling range -166 °F (-110 °C)

Flash point -2.2 °F (-19 °C)

Evaporation rate Not available.

Flammability (solid, gas) Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%) 1.7 %

Flammability limit - upper (%) 10.9 %

Vapor pressure 2251.4 hPa estimated

Vapor density > 1 (air = 1)

Relative density 0.77 - 0.85

Solubility (water) Not available.

Partition coefficient (n-octanol/water) Not available.

Auto-ignition temperature 689 °F (365 °C)

Decomposition temperature Not available.

Viscosity (kinematic) Not available.

Percent volatile 82 % estimated

10. Stability and reactivity

Reactivity The product is stable and non-reactive under normal conditions of use, storage and transport.

Chemical stability Material is stable under normal conditions.

Possibility of hazardous reactions No dangerous reaction known under conditions of normal use.

Conditions to avoid Heat, flames and sparks. Contact with incompatible materials.

Incompatible materials Acids. Strong oxidizing agents. Nitrates. Fluorine. Chlorine.

Hazardous decomposition products No hazardous decomposition products are known.

11. Toxicological information

Information on likely routes of exposure

Inhalation May cause damage to organs through prolonged or repeated exposure by inhalation. May cause drowsiness and dizziness. Headache. Nausea, vomiting.

Skin contact Prolonged skin contact may cause temporary irritation.

Eye contact Causes serious eye irritation.

Ingestion Health injuries are not known or expected under normal use.

Symptoms related to the physical, chemical and toxicological characteristics

May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision.

Information on toxicological effects

Acute toxicity

Not known.

Product

Species

Test Results

All Purpose Enamel Swift Red Spray Paint

Acute

Inhalation

LC50

Rat

7113.5137 mg/l, 4 hours

Components

Species

Test Results

acetone (CAS 67-64-1)

Acute

Dermal

LD50

Rabbit

20000 mg/kg

Inhalation

LC50

Rat

16000 ppm, 4 hours

Oral

LD50

Rat

5800 mg/kg

ethylene glycol propyl ether (CAS 2807-30-9)

Acute

Dermal

LD50

Rabbit

0.87 g/kg

Oral

LD50

Rat

4.45 g/kg

methyl isobutyl ketone (CAS 108-10-1)

Acute

Dermal

LD50

Rabbit

> 3 g/kg

Inhalation

LC50

Rat

8.2 mg/l, 4 Hours

Oral

LD50

Rat

2080 mg/kg

methyl propyl ketone (CAS 107-87-9)

Acute

Oral

LD50

Rat

3.73 g/kg

propane (CAS 74-98-6)

Acute

Dermal

LD50

Rabbit

> 5000 mg/kg

propylene glycol methyl ether acetate (CAS 108-65-6)

Acute

Oral

LD50

Rat

8500 mg/kg

* Estimates for product may be based on additional component data not shown.

Skin corrosion/irritation

Prolonged skin contact may cause temporary irritation.

Serious eye damage/eye irritation

Causes serious eye irritation.

Respiratory sensitization

Not a respiratory sensitizer.

Skin sensitization

This product is not expected to cause skin sensitization.

Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
Carcinogenicity	Suspected of causing cancer.
IARC Monographs. Overall Evaluation of Carcinogenicity	
methyl isobutyl ketone (CAS 108-10-1)	2B Possibly carcinogenic to humans.
US. National Toxicology Program (NTP) Report on Carcinogens	
Not listed.	
US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)	
Not regulated.	
Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity - single exposure	May cause drowsiness and dizziness.
Specific target organ toxicity - repeated exposure	May cause damage to organs through prolonged or repeated exposure.
Aspiration hazard	Not an aspiration hazard.
Chronic effects	May cause damage to organs through prolonged or repeated exposure. Prolonged inhalation may be harmful. Prolonged exposure may cause chronic effects.

12. Ecological information

Ecotoxicity

Components		Species	Test Results
acetone (CAS 67-64-1)			
Aquatic			
Crustacea	EC50	Water flea (Daphnia magna)	10294 - 17704 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	4740 - 6330 mg/l, 96 hours
methyl isobutyl ketone (CAS 108-10-1)			
Aquatic			
Fish	LC50	Fathead minnow (Pimephales promelas)	492 - 593 mg/l, 96 hours
methyl propyl ketone (CAS 107-87-9)			
Aquatic			
Fish	LC50	Fathead minnow (Pimephales promelas)	1190 - 1290 mg/l, 96 hours

* Estimates for product may be based on additional component data not shown.

Persistence and degradability No data is available on the degradability of this product.

Bioaccumulative potential

Partition coefficient n-octanol / water (log Kow)

acetone	-0.24
isobutyl acetate	1.78
methyl isobutyl ketone	1.31
methyl propyl ketone	0.91
n-butane	2.89
propane	2.36

Mobility in soil No data available.

Other adverse effects No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Disposal of waste from residues / unused products	If discarded, this product is considered a RCRA ignitable waste, D001. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Contents under pressure. Do not puncture, incinerate or crush. Do not allow this material to drain into sewers/water supplies. Do not contaminate ponds, waterways or ditches with chemical or used container. Dispose in accordance with all applicable regulations.
Hazardous waste code	D001: Waste Flammable material with a flash point <140 F
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

DOT

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Label(s)	2.1
Packing group	Not applicable.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Special provisions	N82
Packaging exceptions	306
Packaging non bulk	304
Packaging bulk	None

IATA

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
ERG Code	10L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	UN1950
UN proper shipping name	AEROSOLS, Limited Quantity
Transport hazard class(es)	
Class	2
Subsidiary risk	-
Packing group	Not applicable.
Environmental hazards	
Marine pollutant	No.
EmS	F-D, S-U
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

15. Regulatory information

US federal regulations	This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.
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TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)

Not regulated.

SARA 304 Emergency release notification

Not regulated.

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)

Not regulated.

US EPCRA (SARA Title III) Section 313 - Toxic Chemical: Listed substance

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

CERCLA Hazardous Substances: Reportable quantity

acetone (CAS 67-64-1)	5000 LBS
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isobutyl acetate (CAS 110-19-0)	5000 LBS
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methyl isobutyl ketone (CAS 108-10-1)	5000 LBS
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CERCLA Hazardous Substance List (40 CFR 302.4)

acetone (CAS 67-64-1)	Listed.
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ethylene glycol propyl ether (CAS 2807-30-9)	Listed.
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isobutyl acetate (CAS 110-19-0)	Listed.
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methyl isobutyl ketone (CAS 108-10-1) Listed.

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

n-butane (CAS 106-97-8)

propane (CAS 74-98-6)

Safe Drinking Water Act (SDWA) Not regulated.

Drug Enforcement Administration (DEA). List 2, Essential Chemicals (21 CFR 1310.02(b) and 1310.04(f)(2) and Chemical Code Number

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

Drug Enforcement Administration (DEA). List 1 & 2 Exempt Chemical Mixtures (21 CFR 1310.12(c))

acetone (CAS 67-64-1) 35 %WV

methyl isobutyl ketone (CAS 108-10-1) 35 %WV

DEA Exempt Chemical Mixtures Code Number

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

FEMA Priority Substances Respiratory Health and Safety in the Flavor Manufacturing Workplace

acetone (CAS 67-64-1) Low priority

isobutyl acetate (CAS 110-19-0) Low priority

methyl isobutyl ketone (CAS 108-10-1) Low priority

methyl propyl ketone (CAS 107-87-9) Low priority

Food and Drug Administration (FDA) Not regulated.

Superfund Amendments and Reauthorization Act of 1986 (SARA)

Section 311/312
Hazard categories Immediate Hazard - Yes
Delayed Hazard - Yes
Fire Hazard - Yes
Pressure Hazard - Yes
Reactivity Hazard - No

SARA 302 Extremely hazardous substance No

US state regulations

US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))

acetone (CAS 67-64-1)

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

n-butane (CAS 106-97-8)

US. New Jersey Worker and Community Right-to-Know Act

acetone (CAS 67-64-1)

ethylene glycol propyl ether (CAS 2807-30-9)

isobutyl acetate (CAS 110-19-0)

methyl isobutyl ketone (CAS 108-10-1)

methyl propyl ketone (CAS 107-87-9)

n-butane (CAS 106-97-8)

propane (CAS 74-98-6)

US. Massachusetts RTK - Substance List

acetone (CAS 67-64-1)

isobutyl acetate (CAS 110-19-0)

methyl isobutyl ketone (CAS 108-10-1)

methyl propyl ketone (CAS 107-87-9)

n-butane (CAS 106-97-8)

propane (CAS 74-98-6)

US. Pennsylvania Worker and Community Right-to-Know Law

acetone (CAS 67-64-1)

ethylbenzene (CAS 100-41-4)
ethylene glycol propyl ether (CAS 2807-30-9)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)

US. Rhode Island RTK

acetone (CAS 67-64-1)
ethylbenzene (CAS 100-41-4)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)

US. California Proposition 65

WARNING: This product contains a chemical known to the State of California to cause cancer and birth defects or other reproductive harm.

US - California Proposition 65 - CRT: Listed date/Carcinogenic substance

ethylbenzene (CAS 100-41-4)	Listed: June 11, 2004
methyl isobutyl ketone (CAS 108-10-1)	Listed: November 4, 2011

US - California Proposition 65 - CRT: Listed date/Developmental toxin

methyl isobutyl ketone (CAS 108-10-1)	Listed: March 28, 2014
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Volatile organic compounds (VOC) regulations

EPA

Aerosol coatings (40 CFR 59, Subpt. E) Compliant

State

Aerosol coatings This product is regulated as a Non-Flat Paint. This product is compliant for sale in all 50 states.
Maximum incremental reactivity (MIR) 0.71

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date 11-10-2016
Prepared by Allison Cho
Version # 01
Further information Not available.

HMIS® ratings

Health: 2*
Flammability: 4
Physical hazard: 1
Personal protection: B

NFPA ratings

Health: 2
Flammability: 4
Instability: 1

NFPA ratings**Disclaimer**

The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..

Revision Information

This document has undergone significant changes and should be reviewed in its entirety.



SAFETY DATA SHEET

1. Identification

Product identifier All Purpose Enamel Orange Spray Paint

Other means of identification

Product code 18004

Recommended use Coating

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Company name CRC Industries, Inc.

Address 885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical 800-521-3168

Assistance

Customer Service 800-272-4620

24-Hour Emergency 800-424-9300 (US)

(CHEMTREC) 703-527-3887 (International)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards	Flammable aerosols	Category 1
	Gases under pressure	Liquefied gas
Health hazards	Serious eye damage/eye irritation	Category 2A
	Carcinogenicity	Category 2
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Specific target organ toxicity, repeated exposure	Category 2
Environmental hazards	Not classified.	
OSHA defined hazards	Not classified.	
Label elements		

Signal word Danger

Hazard statement Extremely flammable aerosol. Contains gas under pressure; may explode if heated. Causes serious eye irritation. May cause drowsiness or dizziness. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.

Precautionary statement

Prevention

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not spray on an open flame or other ignition source. Pressurized container: Do not pierce or burn, even after use. Do not apply while equipment is energized. Extinguish all flames, pilot lights and heaters. Vapors will accumulate readily and may ignite. Use only with adequate ventilation; maintain ventilation during use and until all vapors are gone. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Do not breathe mist or vapor. Wear protective gloves/protective clothing/eye protection/face protection. Wash thoroughly after handling.

Response	If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center/doctor if you feel unwell. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention. If exposed or concerned: Get medical attention.
Storage	Store in a well-ventilated place. Store locked up. Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F. Exposure to high temperature may cause can to burst.
Disposal	Dispose of contents/container in accordance with local/regional/national regulations.
Hazard(s) not otherwise classified (HNOC)	None known.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
acetone		67-64-1	30 - 40
isobutyl acetate		110-19-0	10 - 20
propane		74-98-6	10 - 20
n-butane		106-97-8	5 - 10
ethylene glycol propyl ether		2807-30-9	1 - 3
methyl isobutyl ketone		108-10-1	1 - 3
methyl propyl ketone		107-87-9	1 - 3
propylene glycol methyl ether acetate		108-65-6	1 - 3
titanium dioxide		13463-67-7	< 1

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
Skin contact	Rinse skin with water/shower. Get medical attention if irritation develops and persists.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	In the unlikely event of swallowing contact a physician or poison control center. Rinse mouth.
Most important symptoms/effects, acute and delayed	May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Prolonged exposure may cause chronic effects.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.
General information	IF exposed or concerned: Get medical advice/attention. Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Water fog. Alcohol resistant foam. Dry chemical powder. Carbon dioxide (CO2).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Contents under pressure. Pressurized container may rupture when exposed to heat or flame. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Fire-fighting equipment/instructions	In case of fire: Stop leak if safe to do so. Move containers from fire area if you can do so without risk. Containers should be cooled with water to prevent vapor pressure build up.
General fire hazards	Extremely flammable aerosol. Contents under pressure. Pressurized container may rupture when exposed to heat or flame.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Many gases are heavier than air and will spread along ground and collect in low or confined areas (sewers, basements, tanks). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Emergency personnel need self-contained breathing equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. Prevent product from entering drains. Stop the flow of material, if this is without risk. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Put material in suitable, covered, labeled containers. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid release to the environment. Inform appropriate managerial or supervisory personnel of all environmental releases. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage

Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Pressurized container: Do not pierce or burn, even after use. Do not use if spray button is missing or defective. Do not spray on a naked flame or any other incandescent material. Do not smoke while using or until sprayed surface is thoroughly dry. Do not cut, weld, solder, drill, grind, or expose containers to heat, flame, sparks, or other sources of ignition. Use caution around energized equipment. The metal container will conduct electricity if it contacts a live source. This may result in injury to the user from electrical shock and/or flash fire. Do not breathe mist or vapor. Avoid contact with eyes. Avoid prolonged exposure. Should be handled in closed systems, if possible. Use only in well-ventilated areas. Wear appropriate personal protective equipment. Avoid release to the environment. Observe good industrial hygiene practices. For product usage instructions, please see the product label.
Conditions for safe storage, including any incompatibilities	Level 3 Aerosol. Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C/122 °F. Do not puncture, incinerate or crush. Do not handle or store near an open flame, heat or other sources of ignition. This material can accumulate static charge which may cause spark and become an ignition source. Store in a well-ventilated place. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value	Form
acetone (CAS 67-64-1)	PEL	2400 mg/m3 1000 ppm	
isobutyl acetate (CAS 110-19-0)	PEL	700 mg/m3 150 ppm	
methyl isobutyl ketone (CAS 108-10-1)	PEL	410 mg/m3 100 ppm	
methyl propyl ketone (CAS 107-87-9)	PEL	700 mg/m3 200 ppm	
propane (CAS 74-98-6)	PEL	1800 mg/m3 1000 ppm	
titanium dioxide (CAS 13463-67-7)	PEL	15 mg/m3	Total dust.

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value	Form
titanium dioxide (CAS 13463-67-7)	TWA	5 mg/m3	Respirable fraction.

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value	Form
		15 mg/m3	Total dust.
		50 mppcf	Total dust.
		15 mppcf	Respirable fraction.

US. ACGIH Threshold Limit Values

Components	Type	Value
acetone (CAS 67-64-1)	STEL	500 ppm
	TWA	250 ppm
isobutyl acetate (CAS 110-19-0)	STEL	150 ppm
	TWA	50 ppm
methyl isobutyl ketone (CAS 108-10-1)	STEL	75 ppm
	TWA	20 ppm
methyl propyl ketone (CAS 107-87-9)	STEL	150 ppm
n-butane (CAS 106-97-8)	STEL	1000 ppm
titanium dioxide (CAS 13463-67-7)	TWA	10 mg/m3

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
acetone (CAS 67-64-1)	TWA	590 mg/m3
		250 ppm
isobutyl acetate (CAS 110-19-0)	TWA	700 mg/m3
		150 ppm
methyl isobutyl ketone (CAS 108-10-1)	STEL	300 mg/m3
		75 ppm
	TWA	205 mg/m3
		50 ppm
methyl propyl ketone (CAS 107-87-9)	TWA	530 mg/m3
		150 ppm
n-butane (CAS 106-97-8)	TWA	1900 mg/m3
		800 ppm
propane (CAS 74-98-6)	TWA	1800 mg/m3
		1000 ppm

US. AIHA Workplace Environmental Exposure Level (WEEL) Guides

Components	Type	Value
propylene glycol methyl ether acetate (CAS 108-65-6)	TWA	50 ppm

Biological limit values
ACGIH Biological Exposure Indices

Components	Value	Determinant	Specimen	Sampling Time
acetone (CAS 67-64-1)	25 mg/l	Acetone	Urine	*
methyl isobutyl ketone (CAS 108-10-1)	1 mg/l	Methyl isobutyl ketone	Urine	*

* - For sampling details, please see the source document.

Exposure guidelines
US - California OELs: Skin designation

propylene glycol methyl ether acetate (CAS 108-65-6) Can be absorbed through the skin.

Appropriate engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station.

Individual protection measures, such as personal protective equipment

Eye/face protection	Wear safety glasses with side shields (or goggles).
Skin protection	
Hand protection	Wear protective gloves such as: Nitrile. Butyl rubber.
Other	Wear suitable protective clothing.
Respiratory protection	If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.
Thermal hazards	Wear appropriate thermal protective clothing, when necessary.
General hygiene considerations	Observe any medical surveillance requirements. When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties**Appearance**

Physical state	Liquid.
Form	Aerosol.
Color	Orange.
Odor	Aromatic.
Odor threshold	Not available.
pH	Not available.
Melting point/freezing point	Not available.
Initial boiling point and boiling range	-47.2 °F (-44 °C)
Flash point	-2.2 °F (-19 °C)
Evaporation rate	Not available.
Flammability (solid, gas)	Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%)	1.7 %
Flammability limit - upper (%)	10.9 %
Vapor pressure	2250.8 hPa estimated
Vapor density	> 1 (air = 1)
Relative density	0.77 - 0.85
Solubility (water)	Not available.
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	689 °F (365 °C)
Decomposition temperature	Not available.
Viscosity (kinematic)	Not available.
Percent volatile	82.3 % estimated

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	No dangerous reaction known under conditions of normal use.
Conditions to avoid	Heat, flames and sparks. Contact with incompatible materials.
Incompatible materials	Acids. Strong oxidizing agents. Nitrates. Fluorine. Chlorine.
Hazardous decomposition products	No hazardous decomposition products are known.

11. Toxicological information

Information on likely routes of exposure

Inhalation	May cause damage to organs through prolonged or repeated exposure by inhalation. May cause drowsiness and dizziness. Headache. Nausea, vomiting.
Skin contact	Prolonged skin contact may cause temporary irritation.
Eye contact	Causes serious eye irritation.
Ingestion	Health injuries are not known or expected under normal use.

Symptoms related to the physical, chemical and toxicological characteristics
May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision.

Information on toxicological effects

Acute toxicity Not known.

Components	Species	Test Results
acetone (CAS 67-64-1)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	20000 mg/kg
Inhalation		
LC50	Rat	16000 ppm, 4 hours
Oral		
LD50	Rat	5800 mg/kg
ethylene glycol propyl ether (CAS 2807-30-9)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	0.87 g/kg
Oral		
LD50	Rat	4.45 g/kg
methyl isobutyl ketone (CAS 108-10-1)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 3 g/kg
Inhalation		
LC50	Rat	8.2 mg/l, 4 Hours
Oral		
LD50	Rat	2080 mg/kg
methyl propyl ketone (CAS 107-87-9)		
<u>Acute</u>		
Oral		
LD50	Rat	3.73 g/kg
propane (CAS 74-98-6)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 5000 mg/kg
propylene glycol methyl ether acetate (CAS 108-65-6)		
<u>Acute</u>		
Oral		
LD50	Rat	8500 mg/kg
titanium dioxide (CAS 13463-67-7)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 10000 mg/kg

Components	Species	Test Results
Oral LD50	Rat	> 10000 mg/kg

* Estimates for product may be based on additional component data not shown.

Skin corrosion/irritation	Prolonged skin contact may cause temporary irritation.
Serious eye damage/eye irritation	Causes serious eye irritation.
Respiratory sensitization	Not a respiratory sensitizer.
Skin sensitization	This product is not expected to cause skin sensitization.
Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
Carcinogenicity	Suspected of causing cancer.

IARC Monographs. Overall Evaluation of Carcinogenicity

methyl isobutyl ketone (CAS 108-10-1)	2B Possibly carcinogenic to humans.
titanium dioxide (CAS 13463-67-7)	2B Possibly carcinogenic to humans.

US. National Toxicology Program (NTP) Report on Carcinogens

Not listed.

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)

Not regulated.

Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity - single exposure	May cause drowsiness and dizziness.
Specific target organ toxicity - repeated exposure	May cause damage to organs through prolonged or repeated exposure.
Aspiration hazard	Not an aspiration hazard.
Chronic effects	May cause damage to organs through prolonged or repeated exposure. Prolonged inhalation may be harmful. Prolonged exposure may cause chronic effects.

12. Ecological information

Ecotoxicity

Components	Species	Test Results
acetone (CAS 67-64-1)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia magna) 10294 - 17704 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss) 4740 - 6330 mg/l, 96 hours
methyl isobutyl ketone (CAS 108-10-1)		
Aquatic		
Fish	LC50	Fathead minnow (Pimephales promelas) 492 - 593 mg/l, 96 hours
methyl propyl ketone (CAS 107-87-9)		
Aquatic		
Fish	LC50	Fathead minnow (Pimephales promelas) 1190 - 1290 mg/l, 96 hours
titanium dioxide (CAS 13463-67-7)		
Aquatic		
Crustacea	EC50	Water flea (Daphnia magna) > 1000 mg/l, 48 hours
Acute		
Fish	LC50	Fathead minnow (Pimephales promelas) 1000 mg/l, 96 hours

* Estimates for product may be based on additional component data not shown.

Persistence and degradability	No data is available on the degradability of this product.
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Bioaccumulative potential

Partition coefficient n-octanol / water (log Kow)

acetone	-0.24
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Partition coefficient n-octanol / water (log Kow)

isobutyl acetate	1.78
methyl isobutyl ketone	1.31
methyl propyl ketone	0.91
n-butane	2.89
propane	2.36

Bioconcentration factor (BCF)

titanium dioxide	352
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Mobility in soil No data available.

Other adverse effects No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Disposal of waste from residues / unused products If discarded, this product is considered a RCRA ignitable waste, D001. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Contents under pressure. Do not puncture, incinerate or crush. Do not allow this material to drain into sewers/water supplies. Do not contaminate ponds, waterways or ditches with chemical or used container. Dispose in accordance with all applicable regulations.

Hazardous waste code D001: Waste Flammable material with a flash point <140 F

Contaminated packaging Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information
DOT

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Label(s)	2.1
Packing group	Not applicable.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Special provisions	N82
Packaging exceptions	306
Packaging non bulk	304
Packaging bulk	None

IATA

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
ERG Code	10L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	UN1950
UN proper shipping name	AEROSOLS, Limited Quantity
Transport hazard class(es)	
Class	2
Subsidiary risk	-
Packing group	Not applicable.
Environmental hazards	
Marine pollutant	No.
EmS	F-D, S-U
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

15. Regulatory information

US federal regulations

This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.

TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)

Not regulated.

SARA 304 Emergency release notification

Not regulated.

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)

Not regulated.

US EPCRA (SARA Title III) Section 313 - Toxic Chemical: Listed substance

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

CERCLA Hazardous Substances: Reportable quantity

acetone (CAS 67-64-1) 5000 LBS

isobutyl acetate (CAS 110-19-0) 5000 LBS

methyl isobutyl ketone (CAS 108-10-1) 5000 LBS

CERCLA Hazardous Substance List (40 CFR 302.4)

acetone (CAS 67-64-1) Listed.

ethylene glycol propyl ether (CAS 2807-30-9) Listed.

isobutyl acetate (CAS 110-19-0) Listed.

methyl isobutyl ketone (CAS 108-10-1) Listed.

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

n-butane (CAS 106-97-8)

propane (CAS 74-98-6)

Safe Drinking Water Act (SDWA)

Not regulated.

Drug Enforcement Administration (DEA). List 2, Essential Chemicals (21 CFR 1310.02(b) and 1310.04(f)(2) and Chemical Code Number

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

Drug Enforcement Administration (DEA). List 1 & 2 Exempt Chemical Mixtures (21 CFR 1310.12(c))

acetone (CAS 67-64-1) 35 %WV

methyl isobutyl ketone (CAS 108-10-1) 35 %WV

DEA Exempt Chemical Mixtures Code Number

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

FEMA Priority Substances Respiratory Health and Safety in the Flavor Manufacturing Workplace

acetone (CAS 67-64-1) Low priority

isobutyl acetate (CAS 110-19-0) Low priority

methyl isobutyl ketone (CAS 108-10-1) Low priority

methyl propyl ketone (CAS 107-87-9) Low priority

Food and Drug Administration (FDA)

Not regulated.

Superfund Amendments and Reauthorization Act of 1986 (SARA)

Section 311/312 Immediate Hazard - Yes

Hazard categories Delayed Hazard - Yes

Fire Hazard - Yes

Pressure Hazard - Yes

Reactivity Hazard - No

SARA 302 Extremely hazardous substance No

US state regulations

US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))

acetone (CAS 67-64-1)
ethylene glycol propyl ether (CAS 2807-30-9)
methyl isobutyl ketone (CAS 108-10-1)
n-butane (CAS 106-97-8)
titanium dioxide (CAS 13463-67-7)

US. New Jersey Worker and Community Right-to-Know Act

acetone (CAS 67-64-1)
ethylene glycol propyl ether (CAS 2807-30-9)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. Massachusetts RTK - Substance List

acetone (CAS 67-64-1)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)

US. Pennsylvania Worker and Community Right-to-Know Law

acetone (CAS 67-64-1)
ethylbenzene (CAS 100-41-4)
ethylene glycol propyl ether (CAS 2807-30-9)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. Rhode Island RTK

acetone (CAS 67-64-1)
ethylbenzene (CAS 100-41-4)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. California Proposition 65

WARNING: This product contains a chemical known to the State of California to cause cancer and birth defects or other reproductive harm.

US - California Proposition 65 - CRT: Listed date/Carcinogenic substance

ethylbenzene (CAS 100-41-4)	Listed: June 11, 2004
methyl isobutyl ketone (CAS 108-10-1)	Listed: November 4, 2011
titanium dioxide (CAS 13463-67-7)	Listed: September 2, 2011

US - California Proposition 65 - CRT: Listed date/Developmental toxin

methyl isobutyl ketone (CAS 108-10-1)	Listed: March 28, 2014
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Volatile organic compounds (VOC) regulations

EPA

Aerosol coatings (40 CFR 59, Subpt. E) Compliant

State

Aerosol coatings This product is regulated as a Non-Flat Paint. This product is compliant for sale in all 50 states.

Maximum incremental reactivity (MIR) 0.7

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date	11-11-2016
Prepared by	Allison Cho
Version #	01
Further information	Not available.
HMIS® ratings	Health: 2* Flammability: 4 Physical hazard: 1 Personal protection: B
NFPA ratings	Health: 2 Flammability: 4 Instability: 1

NFPA ratings



Disclaimer

The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..

Revision Information

This document has undergone significant changes and should be reviewed in its entirety.



SAFETY DATA SHEET

1. Identification

Product identifier All Purpose Enamel Yellow Spray Paint

Other means of identification

Product code 18001

Recommended use Coating

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Company name CRC Industries, Inc.

Address
885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical 800-521-3168

Assistance

Customer Service 800-272-4620

24-Hour Emergency 800-424-9300 (US)

(CHEMTREC) 703-527-3887 (International)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards	Flammable aerosols	Category 1
	Gases under pressure	Liquefied gas
Health hazards	Serious eye damage/eye irritation	Category 2A
	Carcinogenicity	Category 2
	Specific target organ toxicity, single exposure	Category 3 narcotic effects
	Specific target organ toxicity, repeated exposure	Category 2
Environmental hazards	Not classified.	
OSHA defined hazards	Not classified.	

Label elements



Signal word Danger

Hazard statement Extremely flammable aerosol. Contains gas under pressure; may explode if heated. Causes serious eye irritation. May cause drowsiness or dizziness. Suspected of causing cancer. May cause damage to organs through prolonged or repeated exposure.

Precautionary statement

Prevention

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Keep away from heat/sparks/open flames/hot surfaces. - No smoking. Do not spray on an open flame or other ignition source. Pressurized container: Do not pierce or burn, even after use. Do not apply while equipment is energized. Extinguish all flames, pilot lights and heaters. Vapors will accumulate readily and may ignite. Use only with adequate ventilation; maintain ventilation during use and until all vapors are gone. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Do not breathe mist or vapor. Wear protective gloves/protective clothing/eye protection/face protection. Wash thoroughly after handling.

Response	If inhaled: Remove person to fresh air and keep comfortable for breathing. Call a poison center/doctor if you feel unwell. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical attention. If exposed or concerned: Get medical attention.
Storage	Store in a well-ventilated place. Store locked up. Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F. Exposure to high temperature may cause can to burst.
Disposal	Dispose of contents/container in accordance with local/regional/national regulations.
Hazard(s) not otherwise classified (HNOC)	None known.

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
acetone		67-64-1	30 - 40
isobutyl acetate		110-19-0	10 - 20
propane		74-98-6	10 - 20
n-butane		106-97-8	5 - 10
titanium dioxide		13463-67-7	3 - 5
ethylene glycol propyl ether		2807-30-9	1 - 3
methyl isobutyl ketone		108-10-1	1 - 3
methyl propyl ketone		107-87-9	1 - 3

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
Skin contact	Rinse skin with water/shower. Get medical attention if irritation develops and persists.
Eye contact	Immediately flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention if irritation develops and persists.
Ingestion	In the unlikely event of swallowing contact a physician or poison control center. Rinse mouth.
Most important symptoms/effects, acute and delayed	May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision. Prolonged exposure may cause chronic effects.
Indication of immediate medical attention and special treatment needed	Provide general supportive measures and treat symptomatically. Keep victim under observation. Symptoms may be delayed.
General information	IF exposed or concerned: Get medical advice/attention. Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Water fog. Alcohol resistant foam. Dry chemical powder. Carbon dioxide (CO2).
Unsuitable extinguishing media	Do not use water jet as an extinguisher, as this will spread the fire.
Specific hazards arising from the chemical	Contents under pressure. Pressurized container may rupture when exposed to heat or flame. During fire, gases hazardous to health may be formed.
Special protective equipment and precautions for firefighters	Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.
Fire-fighting equipment/instructions	In case of fire: Stop leak if safe to do so. Move containers from fire area if you can do so without risk. Containers should be cooled with water to prevent vapor pressure build up.
General fire hazards	Extremely flammable aerosol. Contents under pressure. Pressurized container may rupture when exposed to heat or flame.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep unnecessary personnel away. Keep people away from and upwind of spill/leak. Keep out of low areas. Many gases are heavier than air and will spread along ground and collect in low or confined areas (sewers, basements, tanks). Wear appropriate protective equipment and clothing during clean-up. Do not breathe mist or vapor. Emergency personnel need self-contained breathing equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Ventilate closed spaces before entering them. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Eliminate all ignition sources (no smoking, flares, sparks, or flames in immediate area). Keep combustibles (wood, paper, oil, etc.) away from spilled material. Prevent product from entering drains. Stop the flow of material, if this is without risk. Wipe up with absorbent material (e.g. cloth, fleece). Clean surface thoroughly to remove residual contamination. Put material in suitable, covered, labeled containers. For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid release to the environment. Inform appropriate managerial or supervisory personnel of all environmental releases. Prevent further leakage or spillage if safe to do so. Avoid discharge into drains, water courses or onto the ground.

7. Handling and storage

Precautions for safe handling	Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Pressurized container: Do not pierce or burn, even after use. Do not use if spray button is missing or defective. Do not spray on a naked flame or any other incandescent material. Do not smoke while using or until sprayed surface is thoroughly dry. Do not cut, weld, solder, drill, grind, or expose containers to heat, flame, sparks, or other sources of ignition. Use caution around energized equipment. The metal container will conduct electricity if it contacts a live source. This may result in injury to the user from electrical shock and/or flash fire. Do not breathe mist or vapor. Avoid contact with eyes. Avoid prolonged exposure. Should be handled in closed systems, if possible. Use only in well-ventilated areas. Wear appropriate personal protective equipment. Avoid release to the environment. Observe good industrial hygiene practices. For product usage instructions, please see the product label.
Conditions for safe storage, including any incompatibilities	Level 3 Aerosol. Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C/122 °F. Do not puncture, incinerate or crush. Do not handle or store near an open flame, heat or other sources of ignition. This material can accumulate static charge which may cause spark and become an ignition source. Store in a well-ventilated place. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value	Form
acetone (CAS 67-64-1)	PEL	2400 mg/m3 1000 ppm	
isobutyl acetate (CAS 110-19-0)	PEL	700 mg/m3 150 ppm	
methyl isobutyl ketone (CAS 108-10-1)	PEL	410 mg/m3 100 ppm	
methyl propyl ketone (CAS 107-87-9)	PEL	700 mg/m3 200 ppm	
propane (CAS 74-98-6)	PEL	1800 mg/m3 1000 ppm	
titanium dioxide (CAS 13463-67-7)	PEL	15 mg/m3	Total dust.

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value	Form
titanium dioxide (CAS 13463-67-7)	TWA	5 mg/m3	Respirable fraction.

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value	Form
		15 mg/m3	Total dust.
		50 mppcf	Total dust.
		15 mppcf	Respirable fraction.

US. ACGIH Threshold Limit Values

Components	Type	Value
acetone (CAS 67-64-1)	STEL	500 ppm
	TWA	250 ppm
isobutyl acetate (CAS 110-19-0)	STEL	150 ppm
	TWA	50 ppm
methyl isobutyl ketone (CAS 108-10-1)	STEL	75 ppm
	TWA	20 ppm
methyl propyl ketone (CAS 107-87-9)	STEL	150 ppm
n-butane (CAS 106-97-8)	STEL	1000 ppm
titanium dioxide (CAS 13463-67-7)	TWA	10 mg/m3

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value
acetone (CAS 67-64-1)	TWA	590 mg/m3
		250 ppm
isobutyl acetate (CAS 110-19-0)	TWA	700 mg/m3
		150 ppm
methyl isobutyl ketone (CAS 108-10-1)	STEL	300 mg/m3
		75 ppm
	TWA	205 mg/m3
		50 ppm
methyl propyl ketone (CAS 107-87-9)	TWA	530 mg/m3
		150 ppm
n-butane (CAS 106-97-8)	TWA	1900 mg/m3
		800 ppm
propane (CAS 74-98-6)	TWA	1800 mg/m3
		1000 ppm

Biological limit values
ACGIH Biological Exposure Indices

Components	Value	Determinant	Specimen	Sampling Time
acetone (CAS 67-64-1)	25 mg/l	Acetone	Urine	*
methyl isobutyl ketone (CAS 108-10-1)	1 mg/l	Methyl isobutyl ketone	Urine	*

* - For sampling details, please see the source document.

Appropriate engineering controls

Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level. Provide eyewash station.

Individual protection measures, such as personal protective equipment

Eye/face protection Wear safety glasses with side shields (or goggles).

Skin protection

Hand protection Wear protective gloves such as: Nitrile. Butyl rubber.

Other Wear suitable protective clothing.

Respiratory protection	If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.
Thermal hazards	Wear appropriate thermal protective clothing, when necessary.
General hygiene considerations	Observe any medical surveillance requirements. When using do not smoke. Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties

Appearance

Physical state	Liquid.
Form	Aerosol.
Color	Yellow.
Odor	Aromatic.
Odor threshold	Not available.
pH	Not available.
Melting point/freezing point	Not available.
Initial boiling point and boiling range	-47.2 °F (-44 °C)
Flash point	-2.2 °F (-19 °C)
Evaporation rate	Not available.
Flammability (solid, gas)	Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%)	1.7 %
Flammability limit - upper (%)	10.9 %
Vapor pressure	2355.2 hPa estimated
Vapor density	> 1 (air = 1)
Relative density	0.77 - 0.85
Solubility (water)	Not available.
Partition coefficient (n-octanol/water)	Not available.
Auto-ignition temperature	689 °F (365 °C)
Decomposition temperature	Not available.
Viscosity (kinematic)	Not available.
Percent volatile	80.9 % estimated

10. Stability and reactivity

Reactivity	The product is stable and non-reactive under normal conditions of use, storage and transport.
Chemical stability	Material is stable under normal conditions.
Possibility of hazardous reactions	No dangerous reaction known under conditions of normal use.
Conditions to avoid	Heat, flames and sparks. Contact with incompatible materials.
Incompatible materials	Acids. Strong oxidizing agents. Nitrates. Fluorine. Chlorine.
Hazardous decomposition products	No hazardous decomposition products are known.

11. Toxicological information

Information on likely routes of exposure

Inhalation	May cause damage to organs through prolonged or repeated exposure by inhalation. May cause drowsiness and dizziness. Headache. Nausea, vomiting.
Skin contact	Prolonged skin contact may cause temporary irritation.

Eye contact	Causes serious eye irritation.
Ingestion	Health injuries are not known or expected under normal use.
Symptoms related to the physical, chemical and toxicological characteristics	May cause drowsiness and dizziness. Headache. Nausea, vomiting. Severe eye irritation. Symptoms may include stinging, tearing, redness, swelling, and blurred vision.

Information on toxicological effects

Acute toxicity Not known.

Components	Species	Test Results
acetone (CAS 67-64-1)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	20000 mg/kg
Inhalation		
LC50	Rat	16000 ppm, 4 hours
Oral		
LD50	Rat	5800 mg/kg
ethylene glycol propyl ether (CAS 2807-30-9)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	0.87 g/kg
Oral		
LD50	Rat	4.45 g/kg
methyl isobutyl ketone (CAS 108-10-1)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 3 g/kg
Inhalation		
LC50	Rat	8.2 mg/l, 4 Hours
Oral		
LD50	Rat	2080 mg/kg
methyl propyl ketone (CAS 107-87-9)		
<u>Acute</u>		
Oral		
LD50	Rat	3.73 g/kg
propane (CAS 74-98-6)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 5000 mg/kg
titanium dioxide (CAS 13463-67-7)		
<u>Acute</u>		
Dermal		
LD50	Rabbit	> 10000 mg/kg
Oral		
LD50	Rat	> 10000 mg/kg

* Estimates for product may be based on additional component data not shown.

Skin corrosion/irritation Prolonged skin contact may cause temporary irritation.

Serious eye damage/eye irritation Causes serious eye irritation.

Respiratory sensitization Not a respiratory sensitizer.

Skin sensitization This product is not expected to cause skin sensitization.

Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
Carcinogenicity	Suspected of causing cancer.
IARC Monographs. Overall Evaluation of Carcinogenicity	
methyl isobutyl ketone (CAS 108-10-1)	2B Possibly carcinogenic to humans.
titanium dioxide (CAS 13463-67-7)	2B Possibly carcinogenic to humans.
US. National Toxicology Program (NTP) Report on Carcinogens	
Not listed.	
US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)	
Not regulated.	
Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity - single exposure	May cause drowsiness and dizziness.
Specific target organ toxicity - repeated exposure	May cause damage to organs through prolonged or repeated exposure.
Aspiration hazard	Not an aspiration hazard.
Chronic effects	May cause damage to organs through prolonged or repeated exposure. Prolonged inhalation may be harmful. Prolonged exposure may cause chronic effects.

12. Ecological information

Ecotoxicity

Components		Species	Test Results
acetone (CAS 67-64-1)			
Aquatic			
Crustacea	EC50	Water flea (Daphnia magna)	10294 - 17704 mg/l, 48 hours
Fish	LC50	Rainbow trout,donaldson trout (Oncorhynchus mykiss)	4740 - 6330 mg/l, 96 hours
methyl isobutyl ketone (CAS 108-10-1)			
Aquatic			
Fish	LC50	Fathead minnow (Pimephales promelas)	492 - 593 mg/l, 96 hours
methyl propyl ketone (CAS 107-87-9)			
Aquatic			
Fish	LC50	Fathead minnow (Pimephales promelas)	1190 - 1290 mg/l, 96 hours
titanium dioxide (CAS 13463-67-7)			
Aquatic			
Crustacea	EC50	Water flea (Daphnia magna)	> 1000 mg/l, 48 hours
Acute			
Fish	LC50	Fathead minnow (Pimephales promelas)	1000 mg/l, 96 hours

* Estimates for product may be based on additional component data not shown.

Persistence and degradability No data is available on the degradability of this product.

Bioaccumulative potential

Partition coefficient n-octanol / water (log Kow)

acetone	-0.24
isobutyl acetate	1.78
methyl isobutyl ketone	1.31
methyl propyl ketone	0.91
n-butane	2.89
propane	2.36

Bioconcentration factor (BCF)

titanium dioxide	352
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Mobility in soil No data available.

Other adverse effects No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.

13. Disposal considerations

Disposal of waste from residues / unused products	If discarded, this product is considered a RCRA ignitable waste, D001. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Contents under pressure. Do not puncture, incinerate or crush. Do not allow this material to drain into sewers/water supplies. Do not contaminate ponds, waterways or ditches with chemical or used container. Dispose in accordance with all applicable regulations.
Hazardous waste code	D001: Waste Flammable material with a flash point <140 F
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

DOT

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Label(s)	2.1
Packing group	Not applicable.
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Special provisions	N82
Packaging exceptions	306
Packaging non bulk	304
Packaging bulk	None

IATA

UN number	UN1950
UN proper shipping name	Aerosols, flammable, Limited Quantity
Transport hazard class(es)	
Class	2.1
Subsidiary risk	-
Packing group	Not applicable.
ERG Code	10L
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.
Other information	
Passenger and cargo aircraft	Allowed with restrictions.
Cargo aircraft only	Allowed with restrictions.

IMDG

UN number	UN1950
UN proper shipping name	AEROSOLS, Limited Quantity
Transport hazard class(es)	
Class	2
Subsidiary risk	-
Packing group	Not applicable.
Environmental hazards	
Marine pollutant	No.
EmS	F-D, S-U
Special precautions for user	Read safety instructions, SDS and emergency procedures before handling.

15. Regulatory information

US federal regulations	This product is a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.
TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)	Not regulated.
SARA 304 Emergency release notification	Not regulated.
US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050)	Not regulated.

US EPCRA (SARA Title III) Section 313 - Toxic Chemical: Listed substance

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

CERCLA Hazardous Substances: Reportable quantity

acetone (CAS 67-64-1) 5000 LBS

isobutyl acetate (CAS 110-19-0) 5000 LBS

methyl isobutyl ketone (CAS 108-10-1) 5000 LBS

CERCLA Hazardous Substance List (40 CFR 302.4)

acetone (CAS 67-64-1) Listed.

ethylene glycol propyl ether (CAS 2807-30-9) Listed.

isobutyl acetate (CAS 110-19-0) Listed.

methyl isobutyl ketone (CAS 108-10-1) Listed.

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

n-butane (CAS 106-97-8)

propane (CAS 74-98-6)

Safe Drinking Water Act (SDWA) Not regulated.**Drug Enforcement Administration (DEA). List 2, Essential Chemicals (21 CFR 1310.02(b) and 1310.04(f)(2) and Chemical Code Number**

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

Drug Enforcement Administration (DEA). List 1 & 2 Exempt Chemical Mixtures (21 CFR 1310.12(c))

acetone (CAS 67-64-1) 35 %WV

methyl isobutyl ketone (CAS 108-10-1) 35 %WV

DEA Exempt Chemical Mixtures Code Number

acetone (CAS 67-64-1) 6532

methyl isobutyl ketone (CAS 108-10-1) 6715

FEMA Priority Substances Respiratory Health and Safety in the Flavor Manufacturing Workplace

acetone (CAS 67-64-1) Low priority

isobutyl acetate (CAS 110-19-0) Low priority

methyl isobutyl ketone (CAS 108-10-1) Low priority

methyl propyl ketone (CAS 107-87-9) Low priority

Food and Drug Administration (FDA) Not regulated.**Superfund Amendments and Reauthorization Act of 1986 (SARA)****Section 311/312** Immediate Hazard - Yes**Hazard categories** Delayed Hazard - Yes

Fire Hazard - Yes

Pressure Hazard - Yes

Reactivity Hazard - No

SARA 302 Extremely hazardous substance No**US state regulations****US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))**

acetone (CAS 67-64-1)

ethylene glycol propyl ether (CAS 2807-30-9)

methyl isobutyl ketone (CAS 108-10-1)

n-butane (CAS 106-97-8)

titanium dioxide (CAS 13463-67-7)

US. New Jersey Worker and Community Right-to-Know Act

acetone (CAS 67-64-1)

ethylene glycol propyl ether (CAS 2807-30-9)

isobutyl acetate (CAS 110-19-0)

methyl isobutyl ketone (CAS 108-10-1)

methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. Massachusetts RTK - Substance List

acetone (CAS 67-64-1)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. Pennsylvania Worker and Community Right-to-Know Law

acetone (CAS 67-64-1)
ethylbenzene (CAS 100-41-4)
ethylene glycol propyl ether (CAS 2807-30-9)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. Rhode Island RTK

acetone (CAS 67-64-1)
ethylbenzene (CAS 100-41-4)
isobutyl acetate (CAS 110-19-0)
methyl isobutyl ketone (CAS 108-10-1)
methyl propyl ketone (CAS 107-87-9)
n-butane (CAS 106-97-8)
propane (CAS 74-98-6)
titanium dioxide (CAS 13463-67-7)

US. California Proposition 65

WARNING: This product contains a chemical known to the State of California to cause cancer and birth defects or other reproductive harm.

US - California Proposition 65 - CRT: Listed date/Carcinogenic substance

ethylbenzene (CAS 100-41-4)	Listed: June 11, 2004
methyl isobutyl ketone (CAS 108-10-1)	Listed: November 4, 2011
titanium dioxide (CAS 13463-67-7)	Listed: September 2, 2011

US - California Proposition 65 - CRT: Listed date/Developmental toxin

methyl isobutyl ketone (CAS 108-10-1)	Listed: March 28, 2014
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Volatile organic compounds (VOC) regulations

EPA

Aerosol coatings (40 CFR 59, Subpt. E) Compliant

State

Aerosol coatings This product is regulated as a Non-Flat Paint. This product is compliant for sale in all 50 states.
Maximum incremental reactivity (MIR) 0.7

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	Yes
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes

Country(s) or region	Inventory name	On inventory (yes/no)*
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)
A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date	11-11-2016
Prepared by	Allison Cho
Version #	01
Further information	Not available.
HMIS® ratings	Health: 2* Flammability: 4 Physical hazard: 1 Personal protection: B
NFPA ratings	Health: 2 Flammability: 4 Instability: 1
NFPA ratings	
Disclaimer	The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..
Revision Information	This document has undergone significant changes and should be reviewed in its entirety.



SAFETY DATA SHEET

1. Identification

Product identifier RTV Silicone Adhesive & Sealant - Clear (cartridge) - 10.1 fl oz

Other means of identification

Product Code No. 14073 (Item# 1004802)

Recommended use Sealant and adhesive

Recommended restrictions None known.

Manufacturer/Importer/Supplier/Distributor information

Manufactured or sold by:

Company name CRC Industries, Inc.

Address 885 Louis Dr.
Warminster, PA 18974 US

Telephone

General Information 215-674-4300

Technical Assistance 800-521-3168

Customer Service 800-272-4620

**24-Hour Emergency
(CHEMTREC)** 800-424-9300 (US)

Website www.crcindustries.com

2. Hazard(s) identification

Physical hazards Not classified.

Health hazards Not classified.

Environmental hazards Not classified.

OSHA defined hazards Not classified.

Label elements

Hazard symbol None.

Signal word None.

Hazard statement The mixture does not meet the criteria for classification.

Precautionary statement

Prevention Use only outdoors or in a well-ventilated area. Open doors and windows or use other means to ensure a fresh air supply during use and while product is drying. If you experience any symptoms listed on this label, increase ventilation or leave the area. Observe good industrial hygiene practices.

Response Wash hands after handling.

Storage Store away from incompatible materials.

Disposal Dispose of contents/container in accordance with local/regional/national/international regulations.

**Hazard(s) not otherwise
classified (HNOC)** None known.

Supplemental information When heated to temperature above 300°F/150°C in the presence of air, product may form formaldehyde vapors. When exposed to water or humid air, product evolves acetic acid (HOAc).

3. Composition/information on ingredients

Mixtures

Chemical name	Common name and synonyms	CAS number	%
amorphous silica		7631-86-9	5 - 10
distillates (petroleum), hydrotreated middle		64742-46-7	5 - 10

Specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. First-aid measures

Inhalation	Move to fresh air. Call a physician if symptoms develop or persist.
Skin contact	Wash with water and soap as a precaution. Get medical attention if irritation develops and persists.
Eye contact	Flush eyes with water as a precaution. Get medical attention if irritation develops and persists.
Ingestion	Rinse mouth thoroughly. Get medical attention if symptoms occur.
Most important symptoms/effects, acute and delayed	Direct contact with eyes may cause temporary irritation.
Indication of immediate medical attention and special treatment needed	Treat symptomatically.
General information	Ensure that medical personnel are aware of the material(s) involved, and take precautions to protect themselves.

5. Fire-fighting measures

Suitable extinguishing media	Carbon dioxide (CO2). Dry chemicals. Alcohol resistant foam.
Unsuitable extinguishing media	Water.
Specific hazards arising from the chemical	During fire, gases hazardous to health may be formed. When heated to temperature above 300°F/150°C in the presence of air, product may form formaldehyde vapors. When exposed to water or humid air, product evolves acetic acid (HOAc).
Special protective equipment and precautions for firefighters	Self-contained breathing apparatus and full protective clothing must be worn in case of fire.
Fire-fighting equipment/instructions	Do not get water inside container. Use standard firefighting procedures and consider the hazards of other involved materials.
General fire hazards	No unusual fire or explosion hazards noted.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures	Keep people away from and upwind of spill/leak. Keep unnecessary personnel away. Wear appropriate protective equipment and clothing during clean-up. Ensure adequate ventilation. Local authorities should be advised if significant spillages cannot be contained. For personal protection, see section 8 of the SDS.
Methods and materials for containment and cleaning up	Do not get water on spilled substance or inside containers. Stop the flow of material, if this is without risk. Absorb in vermiculite, dry sand or earth and place into containers. Wipe up with absorbent material (e.g. cloth, fleece). For waste disposal, see section 13 of the SDS.
Environmental precautions	Avoid release to the environment. Avoid discharge into drains, water courses or onto the ground. Prevent further leakage or spillage if safe to do so. Contact local authorities in case of spillage to drain/aquatic environment.

7. Handling and storage

Precautions for safe handling	Wear appropriate personal protective equipment. When using, do not eat, drink or smoke. Wash hands thoroughly after handling. Observe good industrial hygiene practices. Provide adequate ventilation. For product usage instructions, see the product label.
Conditions for safe storage, including any incompatibilities	Store in a cool, dry place out of direct sunlight. Store in tightly closed container. Keep container dry. Store away from incompatible materials (see Section 10 of the SDS).

8. Exposure controls/personal protection

Occupational exposure limits

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit. At this time, the other constituents have no known exposure limits.

US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000)

Components	Type	Value	Form
distillates (petroleum), hydrotreated middle (CAS 64742-46-7)	PEL	5 mg/m3	Mist.

US. OSHA Table Z-3 (29 CFR 1910.1000)

Components	Type	Value
amorphous silica (CAS 7631-86-9)	TWA	0.8 mg/m3
		20 mppcf

US. ACGIH Threshold Limit Values

Components	Type	Value	Form
distillates (petroleum), hydrotreated middle (CAS 64742-46-7)	TWA	5 mg/m3	Inhalable fraction.

US. NIOSH: Pocket Guide to Chemical Hazards

Components	Type	Value	Form
amorphous silica (CAS 7631-86-9)	TWA	6 mg/m3	
distillates (petroleum), hydrotreated middle (CAS 64742-46-7)	STEL	10 mg/m3	Mist.
	TWA	5 mg/m3	Mist.

Biological limit values

No biological exposure limits noted for the ingredient(s).

Exposure guidelines

Occupational Exposure Limits are not relevant to the current physical form of the product.

Appropriate engineering controls

Good general ventilation should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.

Individual protection measures, such as personal protective equipment**Eye/face protection**

Wear safety glasses with side shields (or goggles).

Skin protection**Hand protection**

Wear protective gloves such as: Butyl rubber. Nitrile.

Other

Wear suitable protective clothing.

Respiratory protection

If engineering controls are not feasible or if exposure exceeds the applicable exposure limits, use a NIOSH-approved cartridge respirator with an organic vapor cartridge. Use a self-contained breathing apparatus in confined spaces and for emergencies. Air monitoring is needed to determine actual employee exposure levels.

Thermal hazards

Wear appropriate thermal protective clothing, when necessary.

General hygiene considerations

Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants.

9. Physical and chemical properties**Appearance****Physical state**

Solid.

Form

Paste.

Color

Colorless.

Odor

Acetic acid.

Odor threshold

Not available.

pH

Not available.

Melting point/freezing point

Not available.

Initial boiling point and boiling range

680 °F (360 °C) estimated

Flash point

> 212.0 °F (> 100.0 °C) Closed Cup

Evaporation rate

Not available.

Flammability (solid, gas)

Not available.

Upper/lower flammability or explosive limits

Flammability limit - lower (%)	2.9 % estimated
Flammability limit - upper (%)	10.3 % estimated

Vapor pressure 0.3 hPa estimated

Vapor density Not available.

Relative density 1.01

Solubility(ies)

Solubility (water) Not available.

Partition coefficient (n-octanol/water) Not available.

Auto-ignition temperature 500 °F (260 °C) estimated

Decomposition temperature Not available.

Viscosity Not available.

Percent volatile Not available.

10. Stability and reactivity

Reactivity The product is stable and non-reactive under normal conditions of use, storage and transport.

Chemical stability Material is stable under normal conditions.

Possibility of hazardous reactions No dangerous reaction known under conditions of normal use.

Conditions to avoid When heated to temperature above 300°F/150°C in the presence of air, product may form formaldehyde vapors. When exposed to water or humid air, product evolves acetic acid (HOAc). Contact with incompatible materials.

Incompatible materials Strong oxidizing agents.

Hazardous decomposition products Carbon oxides. Formaldehyde. Silicon oxides.

11. Toxicological information**Information on likely routes of exposure**

Inhalation Based on available data, the classification criteria are not met.

Skin contact Based on available data, the classification criteria are not met.

Eye contact Based on available data, the classification criteria are not met.

Ingestion Based on available data, the classification criteria are not met.

Symptoms related to the physical, chemical and toxicological characteristics Direct contact with eyes may cause temporary irritation.

Information on toxicological effects

Acute toxicity Not classified.

Components	Species	Test Results
------------	---------	--------------

amorphous silica (CAS 7631-86-9)

Acute**Dermal**

LD50	Rabbit	> 6000 mg/kg
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Oral

LD50	Rat	> 5000 mg/kg
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Skin corrosion/irritation Prolonged skin contact may cause temporary irritation.

Serious eye damage/eye irritation Direct contact with eyes may cause temporary irritation.

Respiratory or skin sensitization

Respiratory sensitization Not a respiratory sensitizer.

Skin sensitization This product is not expected to cause skin sensitization.

Germ cell mutagenicity	No data available to indicate product or any components present at greater than 0.1% are mutagenic or genotoxic.
Carcinogenicity	Not classifiable as to carcinogenicity to humans.
IARC Monographs. Overall Evaluation of Carcinogenicity	
amorphous silica (CAS 7631-86-9)	3 Not classifiable as to carcinogenicity to humans.
OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)	
Not listed.	
US. National Toxicology Program (NTP) Report on Carcinogens	
Not listed.	
Reproductive toxicity	This product is not expected to cause reproductive or developmental effects.
Specific target organ toxicity - single exposure	Not classified.
Specific target organ toxicity - repeated exposure	Not classified.
Aspiration hazard	Not an aspiration hazard.
Further information	This product has no known adverse effect on human health.

12. Ecological information

Ecotoxicity	The product is not classified as environmentally hazardous. However, this does not exclude the possibility that large or frequent spills can have a harmful or damaging effect on the environment.
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Components	Species		Test Results
amorphous silica (CAS 7631-86-9)			
Aquatic			
Acute			
Algae	EC50	Algae	> 10000 mg/l, 72 hours
Crustacea	EC50	Water flea (Daphnia magna)	> 1000 mg/l, 24 hours
Fish	LC0	Zebra fish	10000 mg/l, 96 hours
Persistence and degradability	No data is available on the degradability of any ingredients in the mixture.		
Bioaccumulative potential	No data available.		
Mobility in soil	No data available.		
Other adverse effects	No other adverse environmental effects (e.g. ozone depletion, photochemical ozone creation potential, endocrine disruption, global warming potential) are expected from this component.		

13. Disposal considerations

Disposal instructions	This product is not a RCRA hazardous waste (See 40 CFR Part 261.20 – 261.33). Empty containers may be recycled. Collect and reclaim or dispose in sealed containers at licensed waste disposal site. Dispose of contents/container in accordance with local/regional/national/international regulations.
Hazardous waste code	Not regulated.
Contaminated packaging	Since emptied containers may retain product residue, follow label warnings even after container is emptied. Empty containers should be taken to an approved waste handling site for recycling or disposal.

14. Transport information

DOT	Not regulated as dangerous goods.
IATA	Not regulated as dangerous goods.
IMDG	Not regulated as dangerous goods.

15. Regulatory information

US federal regulations	This product is not known to be a "Hazardous Chemical" as defined by the OSHA Hazard Communication Standard, 29 CFR 1910.1200.
TSCA Section 12(b) Export Notification (40 CFR 707, Subpt. D)	
Not regulated.	

SARA 304 Emergency release notification

Not regulated.

OSHA Specifically Regulated Substances (29 CFR 1910.1001-1053)

Not listed.

CERCLA Hazardous Substance List (40 CFR 302.4)

Not listed.

CERCLA Hazardous Substances: Reportable quantity

Not listed.

Spills or releases resulting in the loss of any ingredient at or above its RQ require immediate notification to the National Response Center (800-424-8802) and to your Local Emergency Planning Committee.

Other federal regulations**Clean Air Act (CAA) Section 112 Hazardous Air Pollutants (HAPs) List**

Not regulated.

Clean Air Act (CAA) Section 112(r) Accidental Release Prevention (40 CFR 68.130)

Not regulated.

Safe Drinking Water Act (SDWA) Not regulated.

Food and Drug Administration (FDA) Not regulated.

Superfund Amendments and Reauthorization Act of 1986 (SARA)**SARA 302 Extremely hazardous substance**

Not listed.

SARA 311/312 Hazardous chemical No

SARA 313 (TRI reporting)

Not regulated.

US state regulations**US. New Jersey Worker and Community Right-to-Know Act**

Not listed.

US. Massachusetts RTK - Substance List

amorphous silica (CAS 7631-86-9)

US. Pennsylvania Worker and Community Right-to-Know Law

amorphous silica (CAS 7631-86-9)

distillates (petroleum), hydrotreated middle (CAS 64742-46-7)

US. Rhode Island RTK

distillates (petroleum), hydrotreated middle (CAS 64742-46-7)

California Proposition 65

California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65): This material is not known to contain any chemicals currently listed as carcinogens or reproductive toxins. For more information go to www.P65Warnings.ca.gov.

US. California. Candidate Chemicals List. Safer Consumer Products Regulations (Cal. Code Regs, tit. 22, 69502.3, subd. (a))

distillates (petroleum), hydrotreated middle (CAS 64742-46-7)

Volatile organic compounds (VOC) regulations**EPA**

VOC content (40 CFR 51.100(s)) < 3 %

Consumer products (40 CFR 59, Subpt. C) Not regulated

State

Consumer products This product is regulated as a Sealant and Caulking Compound (chemically curing, non-aerosol). This product is compliant for use in all 50 states.

VOC content (CA) < 3 %

VOC content (OTC) < 3 %

International Inventories

Country(s) or region	Inventory name	On inventory (yes/no)*
Australia	Australian Inventory of Industrial Chemicals (AICIS)	Yes
Canada	Domestic Substances List (DSL)	Yes
Canada	Non-Domestic Substances List (NDSL)	No
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Europe	European Inventory of Existing Commercial Chemical Substances (EINECS)	No
Europe	European List of Notified Chemical Substances (ELINCS)	No
Japan	Inventory of Existing and New Chemical Substances (ENCS)	No
Korea	Existing Chemicals List (ECL)	Yes
New Zealand	New Zealand Inventory	Yes
Philippines	Philippine Inventory of Chemicals and Chemical Substances (PICCS)	Yes
Taiwan	Taiwan Chemical Substance Inventory (TCSI)	Yes
United States & Puerto Rico	Toxic Substances Control Act (TSCA) Inventory	Yes

*A "Yes" indicates that all components of this product comply with the inventory requirements administered by the governing country(s)

A "No" indicates that one or more components of the product are not listed or exempt from listing on the inventory administered by the governing country(s).

16. Other information, including date of preparation or last revision

Issue date	09-29-2021
Prepared by	Danica Fulmer
Version #	01
Disclaimer	The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. This information is accurate to the best of CRC's knowledge or obtained from sources believed by CRC to be accurate. Before using any product, read all warnings and directions on the label. For further clarification of any information contained on this (M)SDS consult your supervisor, a health & safety professional, or CRC Industries, Inc..
Revision information	This document has undergone significant changes and should be reviewed in its entirety.

CHICO® A SEALING COMPOUND/CHICO A3/CHICO A4/ CHICO A05/CHICO A200/CHICO A19PX/CHICO A39PX

SAFETY DATA SHEET

IF 1365

SAVE THESE INSTRUCTIONS FOR FUTURE REFERENCE

SECTION 1: PRODUCT & COMPANY IDENTIFICATION

Chemical Product Name: Chico A/Chico A3/Chico A4/Chico A05/
Chico A200/Chico A19PX/Chico A39PX
Product Description: Sealing compound
CAS Number: Mixture
Synonyms: NA
Recommended Use(s): Sealing compound
Company Information: Eaton's Crouse-Hinds Division
1201 Wolf Street
Syracuse, NY 13208 USA
(866) 764-5454
Telephone:
Emergency Phone: CHEMTREC (800) 424-9300

SECTION 2: HAZARDS IDENTIFICATION

OSHA HCS status: This product is a hazardous chemical, as defined by OSHA at 29 CFR 1910.1200. Hazards identified are based on hazards of the ingredients. This product has not been fully tested.

Relevant route of exposure/target organs: Dermal and inhalation.

OSHA/GHS signal word and hazard statements: **DANGER:** Causes serious eye damage. Causes skin irritation.

OSHA/GHS classification and pictograms:

Skin corrosion/irritation
Serious eye damage, eye irritation

Category 2
Category 1



OSHA/GHS precautionary statements:

Prevention: Wash hands and exposed skin thoroughly after handling. Wear protective gloves/protective clothing/eye protection/face protection. See Section 8 for recommendations on type of protective equipment to be worn.

Response: If on skin: Wash with plenty of water. Specific treatment: see first aid instructions on label. If skin irritation occurs: Get medical advice/attention. Take off contaminated clothing and wash it before reuse. If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a poison center/doctor.

GHS hazard and precautionary statement codes: See Section 16.

SECTION 3: COMPOSITION & INFORMATION ON INGREDIENTS

Component	CAS #	%
Cement, alumina, chemicals	65997-16-2	60-70
Bassanite (calcium sulfate hemihydrate)	10034-76-1	30-35
Haturite (tricalcium silicate)	12168-85-3	<4

SECTION 4: FIRST AID MEASURES

Eye contact: Holding eyelids away from the eyeballs, flush eyes thoroughly with lukewarm water for 15 minutes. Do not rub. If irritation persists, seek medical attention.

Skin contact: Remove contaminated clothing and wash skin thoroughly with soap and water. Do not rub or scratch skin. Use cream or lotion after washing. If irritation persists, seek medical attention.

Inhalation: If inhalation of dusts results in coughing, sneezing or nasal irritation, remove to fresh air until symptoms subside. Give oxygen or artificial respiration, if indicated. Seek medical attention.

Ingestion: Product can harden inside the body. If ingested, seek immediate medical attention.

Notes to physician: Ingestion of sufficient quantities can result in blockage or obstruction especially in the pyloric region of the digestive tract.

Most important symptoms/effects: Causes severe eye damage. Inhalation of dusts and fibers may cause upper respiratory irritation with coughing, sneezing and nasal irritation. Repeated exposure over time may affect the lungs (see below). Dusts may cause general skin irritation. Fibers may cause mechanical irritation and itching. Dusts may cause general eye irritation. Fibers may cause irritation and scratch the outer surface of the eye.

Indication of immediate medical attention and special treatment needed: Get medical attention immediately if product comes into contact with eyes or skin, or if it is inhaled. If ingested, get medical attention, if needed.

SECTION 5: FIRE FIGHTING MEASURES

Special fire fighting procedures: No unusual fire hazards.

Extinguishing media: Use media appropriate for surrounding fire.

Protective equipment: Firefighters should wear a NIOSH approved, full face piece self-contained breathing apparatus (SCBA) operated in positive pressure mode and full turnout gear.

Unusual fire or explosion hazards: Non-flammable and non-combustible.

Hazardous combustion products: Thermal decomposition may produce oxides of carbon.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal protection: Wear protective equipment appropriate for the level of exposure. If dust is present, wear NIOSH type N95 or N100 filter during clean-up. Avoid prolonged skin contact.

Spill procedures: Isolate the hazard and deny entry to unnecessary and unprotected personnel. Do not walk through or otherwise scatter spilled material. Clean dusts promptly to prevent dispersion. Do not inhale dusts.

Environmental precautions and clean-up methods: Use dry clean-up methods or a vacuum equipped with a filter sufficient to prevent re-circulation of dust into the workplace. Do not use compressed air to remove dusts from work and storage areas.

SECTION 7: HANDLING & STORAGE

Precautions: Periodically clean storage and work areas where this product is used or stored to minimize dust accumulation. Do not inhale dusts. Store in well-ventilated area in closed containers. Use dust collectors and local exhaust ventilation when cutting or trimming with power tools. Do not use compressed air or dry sweeping to remove dust from work area. Vacuum dusty clothing before removal. Launder work clothing separately and rinse washer after use. Avoid skin contact. Do not attempt to make a cast enclosing any part of the body using this material, as heat may cause severe burns and expansion may result in decreased circulation that may require surgical removal of affected tissue or amputation of limb.

Storage: Store in a cool, well-ventilated, non-combustible location, away from all sources of ignition. Keep away from heat, steam pipes and sunlight. Keep containers tightly closed.

SECTION 8: EXPOSURE CONTROLS & PERSONAL PROTECTION

Engineering controls/ventilation: Local exhaust ventilation used in combination with general ventilation as necessary to control air contaminants to at or below acceptable exposure guidelines.

Eye protection: Wear eye and face protection. Wear safety goggles that meet ANSI Z87 standards and/or are tested and approved under appropriate government standards.

Respiratory protection: Under normal working conditions with airborne exposures below acceptable exposure guidelines, none required. Where dust is present and for airborne exposures above acceptable limits, wear NIOSH approved respiratory protection, such as N95 or N100 respirator, in accordance with OSHA 29 CFR 1910.134.

Skin protection: Protective gloves and long sleeved clothing or coveralls with loose fitting cuffs and collars.

Component	CAS #	OSHA/PEL	ACGIH/TLV
Cement, alumina, chemicals	65997-16-2	Not established	Not established
Bassanite (calcium sulfate hemihydrate)*	10034-76-1	Not established	Not established
Haturite (tricalcium silicate)	12168-85-3	Not established	Not established
Total dust		15 mg/m ³	10 mg/m ³
Respirable dust		5 mg/m ³	3 mg/m ³

*Spain TLV 10 mg/m³.

SECTION 9: PHYSICAL & CHEMICAL PROPERTIES

- **Color:** Light gray
- **Physical form:** Powder
- **Odor:** Odorless
- **Odor characteristics:** NA
- **Odor threshold:** NA
- **pH (undiluted):** Not applicable
- **Flash point:** Not applicable
- **Flammability (solid, gas):** Non-flammable
- **Boiling point:** Not applicable
- **Evaporation rate:** Not applicable
- **Melting point:** 1300-1400°C
- **Lower explosive limit:** Not applicable
- **Upper explosive limit:** Not applicable
- **Vapor pressure:** Not applicable (at 70°F)
- **Vapor density:** Not applicable
- **Specific gravity:** 3
- **Solubility:** Slight
- **Auto-ignition temperature:** NA
- **Decomposition temperature:** >1450°C

SECTION 10: STABILITY & REACTIVITY

Stability: Stable under normal use and storage conditions.

Hazardous polymerization: Will not occur.

Oxidizing properties: None known for product.

Hazardous decomposition products: Thermal decomposition (above 1450°F) will produce toxic sulfur dioxide, metal oxides, calcium oxide and other oxidation products.

Incompatibilities: None known.

Conditions to avoid: When mixed with water, an exothermic reaction takes place. If large quantities of this product are mixed with sufficient quantities of water, steam can be formed. The heat from the steam can cause burns.

SECTION 11: TOXICOLOGICAL INFORMATION

Delayed and immediate effects: Skin and eyes: Causes severe burns.

Numerical measures of toxicity:

Oral LD50 (rat): No data is available for this material.

Inhalation LC50 (rat): No data is available for this material.

Dermal LD50: No data is available for this material.

Chronic effects: None known.

Carcinogenicity:

IARC: No

NTP: No

OSHA: No

Mutagenicity: No data is available for this material.

Reproductive toxicity: No data is available for this material.

Sensitization: No data is available for this material.

Signs and symptoms of overexposure:

If inhaled: Coughing, nasal congestion, laryngitis, respiratory irritation.

If ingested: Product will harden inside the body. Ingestion of sufficient quantities can result in blockage or obstruction, especially in the pyloric region of the digestive tract.

If on skin or eyes: Irritation, dryness, burns.

SECTION 12: ECOLOGICAL INFORMATION

This product is not expected to have an adverse effect on the environment. Avoid exposure to environment whenever possible.

Toxicity to fish, crustaceans and algae:

Cement, alumina, chemicals:

Oral LC50 (Oncorhynchus mykiss): > 100 mg/l (96 hour)

NOEC (Oncorhynchus mykiss): > 100 mg/l (96 hour)

EC50 (Daphnia magna) 6.6 mg/l (48 hour)

NOEC (Daphnia magna) 1.8 mg/l

EC50 (Pseudokirchnerella subcapitata) > 5.6 mg/l (72 hour)

NOEC (Pseudokirchnerella subcapitata) 3.2 mg/l (72 hour)

No data is available for other components of this material.

Ecotoxicological information: NA

Chemical fate information: NA

SECTION 13: DISPOSAL CONSIDERATIONS

Recycle, reclaim or dispose of contents/container to an approved landfill in accordance with local, regional, national, international regulations. Do not discard into any sewers, on the ground or into any body of water. It is the responsibility of the waste generator to determine the proper waste identification and disposal methods.

SECTION 14: TRANSPORT INFORMATION

Proper shipping name: Not classified as hazardous by DOT, IATA/ICAO and IMO.

Hazard class: Not classified as hazardous by DOT, IATA/ICAO and IMO.

Packing group: Not classified as hazardous by DOT, IATA/ICAO and IMO.

UN number: Not classified as hazardous by DOT, IATA/ICAO and IMO.

SECTION 15: REGULATORY INFORMATION

TSCA inventory status: All ingredients are listed on the TSCA inventory.

SARA Section 311/312 hazard categories: Immediate (acute) hazard.

Section 313 toxic chemicals: This product does not contain ingredients subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-To-Know Act (EPCRA) of 1986 and 40 CFR 372.

CERCLA RQ: This product does not contain ingredients subject to the report requirements of SARA 304 (CERCLA) and 302 (EHS).

California proposition 65: Not listed.

Canadian regulations: All components of this product are included in the Canadian Domestic Substances List (DSL) or the Canadian Non-domestic Substances List (NDSL).

WHMIS classification: D2A.

SECTION 16: OTHER INFORMATION

Revision number: Revision 6 (removed CAS numbers from Section 1)

Revision Date: April, 2024

Explanation of EU directive 1272/2009 codes

- P264 Wash hands thoroughly after handling.
- P280 Wear protective gloves/protective clothing/eye protection/face protection.
- P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
- P303 + P361 + P353 IF ON SKIN: Remove/take off immediately all contaminated clothing. Rinse skin with water/shower.
- P363 Wash contaminated clothing before reuse.
- P308 + P313 If exposed or concerned: Get medical advice/attention.
- P310 Immediately call a POISON CENTER or doctor/physician.
- P321 Specific treatment (see ... on this label).
- P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- P314 Get medical advice/attention if you feel unwell.

Abbreviations

CAS	Chemical Abstracts Service
CERCLA	Comprehensive Environmental Response Compensation and Liability Act
CFR	US Code of Federal Regulations
EC50	Concentration that will affect 50% of the sample aquatic population
HSIS	Australia Hazardous Substance Information System
IARC	International Agency for Research on Cancer
LD50	Lethal dose to 50% of exposed laboratory animals
NA	Not available
NIOSH	US National Institute of Occupational Safety and Health
NOEC	No observed effect concentration
NTP	US National Toxicology Program
OSHA	US Occupational Safety Health Administration
PEL	Permissible exposure limit
RQ	Reportable quantity
SARA	Superfund Amendments and Reauthorization Act
STEL	Short term exposure limit
TSCA	Toxic Substances Control Act
TWA	Time weighted average
UN	United Nations
WHMIS	Canada Workplace Hazardous Material Information System

DISCLAIMER

The information in this SAFETY DATA SHEET should be provided to all who will use, handle, store, transport or otherwise be exposed to this material. This information has been prepared for the guidance of plant engineering, operations and management, and for persons working with or handling this material. Eaton's Crouse-Hinds Division believes this information to be reliable and up-to-date as of the date of publication, but makes no warranty that it is.

All statements, technical information and recommendations contained herein are based on information and tests we believe to be reliable. The accuracy or completeness thereof are not guaranteed. In accordance with Eaton's Crouse-Hinds Division's "Terms and Conditions of Sale," and since conditions of use are outside our control, the purchaser should determine the suitability of the product for his intended use and assumes all risk and liability whatsoever in connection therewith.

SAVE THESE INSTRUCTIONS FOR FUTURE REFERENCE

SECTION 1: PRODUCT & COMPANY IDENTIFICATION

Chemical Product Name: Chico X Fiber/Chico X4/Chico X6/Chico X7
Product Description: Alumino-silicate Wool (ASW), Refractory ceramic fiber (RCF)
CAS Number: 142844-00-6
Synonyms: NA
Recommended Use(s): Refractory ceramic fiber used to create a dam or plug for sealing compound
Uses Advised Against: Not recommended for any use other than as advised above
Company Information: Eaton's Crouse-Hinds Division
1201 Wolf Street
Syracuse, NY 13208
Telephone: (866) 764-5454
Emergency Phone: CHEMTREC (800) 424-9300

SECTION 2: HAZARDS IDENTIFICATION

This product is classified as an "article" in accordance with 29 CFR 1910.1200(b)(6)(v) in the United States, and Section 12 of Canada's Hazardous Products Act. Although safety data sheets are not required for articles, Eaton is providing you with these safe use recommendations so that you can understand the conditions for the safe use of this product.

Relevant Route of Exposure/Target Organs: None

OSHA/GHS Signal Word and Hazard Statements: Warning; Suspected of causing cancer by inhalation

GHS Classification and Pictograms:

GHS Precautionary Statements:



Prevention: Do not handle until all safety instructions have been read and understood. Use respiratory protection as required; see section 8 of the Safety Data Sheet. If concerned about exposure, get medical advice.

Storage: Store in a manner to minimize airborne dust.

Disposal: Dispose of waste in accordance with local, state and federal regulations

SECTION 3: COMPOSITION AND INFORMATION ON INGREDIENTS

COMPONENT	CAS #	%
Refractories, Fibers, Aluminosilicate	142844-00-6	100

SECTION 4: FIRST AID MEASURES

When used as recommended in Section 1, exposures that require first aid are not anticipated for this article. The following guidance is provided if unanticipated use or exposures do occur.

Eye Contact: In case of eye contact flush abundantly with water; have eye bath available. Do not rub eyes.

Skin Contact: Handling of this material may generate mild mechanical temporary skin irritation. If this occurs, rinse affected areas with water and wash gently. Do not rub or scratch exposed skin.

Inhalation Nose and Throat: If these become irritated move to a dust free area, drink water and blow nose. If symptoms persist, seek medical advice.

Notes to Physician: Skin and respiratory effects are the result of temporary, mild mechanical irritation; fiber exposure does not result in allergic manifestations.

SECTION 5: FIRE FIGHTING MEASURES

Special Fire Fighting Procedures: No unusual fire hazards.

Extinguishing Media: Use media appropriate for surrounding fire.

Protective Equipment: Firefighters should wear a NIOSH approved, full face piece self-contained breathing apparatus (SCBA) operated in positive pressure mode and full turnout gear.

Unusual Fire or Explosion Hazards: Non-flammable and non-combustible.

SECTION 6: ACCIDENTAL RELEASE MEASURES

When used as recommended in Section 1, releases during use nor accidental releases are not anticipated. The following response measures are provided in the event that unanticipated use or releases occur.

Emergency Procedures: Minimize airborne dust. Compressed air or dry sweeping should not be used for cleaning. See Section 8 "Exposure Controls / Personal Protection" for exposure guidelines.

Methods and Materials for Containment and Cleaning Up: Frequently clean the work area with an appropriately filtered vacuum or wet sweeping to minimize the accumulation of debris. Do not use compressed air for clean-up.

SECTION 7: HANDLING & STORAGE

Precautions for Safe Handling: Handle fiber carefully to minimize airborne dust. Limit use of power tools unless in conjunction with local exhaust ventilation. Use hand tools whenever possible.

Conditions for Safe Storage: Store in a manner to minimize airborne dust.

Empty Containers: Product packaging may contain residue. Do not reuse.

SECTION 8: EXPOSURE CONTROLS & PERSONAL PROTECTION

Although this article is not anticipated to release quantities of hazardous chemicals that would produce a physical or health hazard, the following general exposure controls should be used as they would in any industrial environment.

Engineering Controls/Ventilation: Use engineering controls such as local exhaust ventilation, point of generation dust collection, down draft work stations, emission controlling tool designs, and materials handling equipment designed to minimize airborne fiber emissions.

Eye Protection: As necessary, wear goggles or safety glasses that seal to the face.

Respiratory Protection: When engineering and/or administrative controls are insufficient to maintain workplace concentrations below the 0.5 f/cc REG or a regulatory OEL, the use of appropriate respiratory protection, pursuant to the requirements of 5 OSHA Standards 29 CFR 1910.134 and 29 CFR 1926.103, is recommended. A NIOSH certified respirator with a filter efficiency of at least 95% should be used. The 95% filter efficiency recommendation is based on NIOSH respirator selection logic sequence for exposure to manmade mineral fibers. Pursuant to NIOSH recommendations, N-95 respirators are appropriate for exposures up to 10 times the NIOSH Recommended Exposure Limit (REL). With respect to RCF, both the NIOSH REL and the industry REG have been set at 0.5 fibers per cubic centimeter of air (f/cm³). Accordingly, N-95 would provide the necessary protection for exposures up to 5 f/cm³. Further, the Respirator Selection Guide published by 3M Corporation, the primary respirator manufacturer, specifically recommends use of N-95 respirators for RCF exposures. In cases where exposures are known to be above 5.0 f/cm³, 8 hour TWA, a filter efficiency of 100% should be used. Other factors to consider are the NIOSH filter series N, R or P – (N) Not resistant to oil, (R) Resistant to oil and (P) oil Proof. These recommendations are not designed to limit informed choices, provided that respiratory protection decisions comply with 29 CFR 1910.134.

Skin Protection: Wear gloves, head coverings and full body clothing as necessary to prevent skin irritation. Washable or disposable clothing may be used. If possible, do not take unwashed clothing home. If soiled work clothing must be taken home, employers should ensure employees are thoroughly trained on the best practices to minimize non-work dust exposure (e.g., vacuum clothes before leaving the work area, wash work clothing separately, rinse washer before washing other household clothes, etc.).

OSHA permissible exposure limit (PEL), American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Value (TLV), and any other exposure limit used or recommended by the chemical manufacturer, importer, or employer preparing the safety data sheet, where available

COMPONENT	OSHA PEL	ACGIH TLV	MANUFACTURER
Refractory Ceramic Fiber (RCF)	None established*	0.2 f/cc TLV, 8-hr. TWA	0.5 f/cc, 8-hr. TWA**

* Except for the state of California, where the PEL for RCF is 0.2 f/cc 8-hr TWA, there is no specific regulatory standard for RCF in the U.S. OSHA's "Particulate Not Otherwise Regulated (PNOR)" standard [29 CFR 1910.1000, Subpart Z, Air Contaminants] applies generally - Total Dust Total Dust 15 mg/m³; Respirable Fraction 5 mg/m³.

**In the absence of an OSHA PEL, HTIW Coalition has adopted a recommended exposure guideline (REG), as measured under NIOSH Method 7400 B. For further information on the history and development of the REG see "Rationale for the Recommended Exposure Guideline" at page 34 of the HTIW Coalition Product Stewardship Program http://www.htiwcoalition.org/documents/PSP_2012.pdf

OTHER OCCUPATIONAL EXPOSURE LEVELS (OEL): RCF-related occupational exposure limits vary internationally. Regulatory OEL examples include: California, 0.2 f/cc; Canadian provincial OELs ranging from 0.2 to 1.0 f/cc. The objectives and criteria underlying each of these OEL decisions also vary. The evaluation of occupational exposure limits and determining their relative applicability to the workplace is best performed, on a case-by-case basis, by a qualified Industrial Hygienist.

SECTION 9: PHYSICAL & CHEMICAL PROPERTIES

- **Color:** Gray or off white
- **Physical Form:** Fibrous wool
- **Odor:** Odorless
- **Odor Characteristics:** Not applicable
- **Odor Threshold:** Not applicable
- **pH (Undiluted):** Not applicable
- **Flash Point:** Not applicable
- **Flammability (Solid, Gas):** Not applicable
- **Boiling Point:** Not applicable
- **Evaporation Rate:** Not applicable
- **Melting Point:** 1760° C (3200° F)
- **Lower Explosive Limit:** Not applicable
- **Upper Explosive Limit:** Not applicable
- **Vapor Pressure:** Not applicable
- **Vapor Density:** Not applicable
- **Relative Density:** 2.50-2.75
- **Solubility:** Insoluble
- **Auto-ignition Temperature:** Not applicable
- **Decomposition Temperature:** Not applicable

SECTION 10: STABILITY & REACTIVITY

- Reactivity: RCF is non-reactive.
- Chemical stability As supplied: RCF is stable and inert.
- Possibility of hazardous reactions: None
- Conditions to avoid: Please refer to handling and storage advice in Section 7
- Incompatible materials: None
- Hazardous decomposition products: None

SECTION 11: TOXICOLOGY INFORMATION

Epidemiological studies of RCF production workers have indicated no increased incidence of respiratory disease or other significant health effects in occupationally exposed workers. In animal studies, long-term, high-dose inhalation exposure resulted in the development of respiratory disease in rats and hamsters.

Acute Toxicity and Immediate Effects: No data is available for this material.

Oral: No data available: Repeated dose studies have been carried out using gavage. No effect was found.

Inhalation: No data is available for this material.

Dermal: No data is available for this material.

Delayed and Chronic Effects: Repeated fiber inhalation over time may increase risk of developing lung cancer.

Carcinogenicity:

IARC: Group 2B (Possibly carcinogenic to humans)

NTP: Yes*

REACH: Yes

* NTP classifies ceramic fibers and glass wool fibers as substances, which are "reasonably anticipated to be human carcinogens."

Mutagenicity: Method: In vitro micronucleus test Species: Hamster (CHO) Dose: 1-35 mg/ml. Routes of administration: In suspension Results: Negative.

Reproductive Toxicity: There are no reports of any reproductive toxic effects of mineral fibers. Exposure to these fibers is via inhalation and effects seen are in the lung. Clearance of fibers is via the gut and the feces, so exposure of the reproductive organs is extremely unlikely.

Sensitization: No data is available for this material.

SECTION 12: ECOLOGICAL INFORMATION

- Ecotoxicity (aquatic and terrestrial, where available): No known aquatic toxicity.
- Persistence and degradability: These products are insoluble materials that remain stable over time and are chemically identical to inorganic compounds found in the soil and sediment; they remain inert in the natural environment.
- Bioaccumulative potential: No bioaccumulative potential.
- Mobility in soil: No mobility in soil.
- Other adverse effects (such as hazardous to the ozone layer): No adverse effects of this material on the environment are anticipated.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Management: To prevent waste materials from becoming airborne during waste storage, transportation and disposal, a covered container or plastic bagging is recommended.

Disposal: This product, as manufactured, is not classified as a hazardous waste according to Federal regulations (40 CFR 261). Any processing, use, alteration or chemical additions to the product, as purchased, may alter the disposal requirements. Under Federal regulations, it is the waste generator's responsibility to properly characterize a waste material, to determine if it is a "hazardous" waste. Check local, regional, state or provincial regulations to identify all applicable disposal requirements.

SECTION 14: TRANSPORT INFORMATION

- UN number UN proper shipping name: Not Applicable
- Transport hazard class(es): Not Applicable
- Packing group, if applicable: Not Applicable
- Environmental hazards (e.g., Marine pollutant (Yes/No)): Not Applicable
- Transport in bulk (according to Annex II of MARPOL 73/78 and the IBC: Not Applicable
- Special precautions which a user needs to be aware of, or needs to comply with, in connection with transport or conveyance either within or outside their premises: Not Applicable

SECTION 15: REGULATORY INFORMATION

UNITED STATES REGULATIONS:

EPA: Superfund Amendments and Reauthorization Act (SARA) Title III

- This product contains aluminum oxide (fibrous forms) which is reportable under Section 313 (40 CFR 372).

Sections 311 and 312 (40 CFR 370) apply (delayed hazard).

Toxic Substances Control Act (TSCA) - RCF has been assigned a CAS number; however, it is an "article" under TSCA and therefore exempt from listing on the TSCA inventory.

Comprehensive Environmental Response, Compensation and Liability Act (CERCLA) and the Clean Air Act (CAA) - This product contains fibers with an average diameter greater than one micron and thus is not considered a hazardous air pollutant.

OSHA: Comply with **Hazard Communication Standards** 29 CFR 1910.1200 and 29 CFR 1926.59 and the **Respiratory Protection Standards** 29 CFR 1910.134 and 29 CFR 1926.103.

California: WARNING: This product can expose you to "Ceramic fibers (airborne particles of respirable size)," which are known to the State of California to cause cancer. See "Labeling of HTIW Products Under California Proposition 65" at htiwcoalition.org for more information.

Other States: RCF products are not known to be regulated by states other than California; however, state and local OSHA and EPA regulations may apply to these products. If in doubt, contact your local regulatory agency.

INTERNATIONAL REGULATIONS:

Canada: Canadian Environmental Protection Act (CEPA) - All substances in this product are listed, as required, on the Domestic Substance List (DSL)

Europe: Integration of RCF into ANNEX XV of the REACH Regulation

RCF is classified under the CLP (classification, labelling and packaging of substances and mixtures) regulation as a category 1B carcinogen. On January 13, 2010 the European Chemicals Agency (ECHA) updated the candidate list for authorization (Annex XV of the REACH regulation) and added 14 new substances in this list including aluminosilicate refractory ceramic fibers. As a consequence, EU (European Union) or EEA (European Economic Area) suppliers of articles which contain aluminosilicate refractory ceramic fibers in a concentration above 0.1% (w/w) have to provide sufficient information, available to them, to their customers or upon requests to a consumer within 45 days of the receipt of the request. This information must ensure safe use of the article, and as minimum contains the name of the substance..

DISCLAIMER

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SECTION 16: OTHER INFORMATION

Revision Number: Revision 4

Revision Date: April 5, 2024

Abbreviations

CAS	Chemical Abstracts Service
GHS	Globally Harmonized System
HCS	Hazard Communication Standard
IARC	International Agency for Research on Cancer
LD50	Lethal dose to 50% of exposed laboratory animals
NA	Not available
NIOSH	U.S. National Institute of Occupational Safety and Health
NOEC	No observed effect concentration
NTP	U.S. National Toxicology Program
PEL	Permissible exposure limit
REACH	European Regulation on the Registration, Evaluation, Authorization and Restriction of Chemicals
STEL	Short term exposure limit
TWA	Time weighted average
UN	United Nations

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Product name: nVent® ERICO® Cadweld® Plus
Supersedes date:
SDS-ID:

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Version number: AU-EN/1.0

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product identifier

Product name: nVent® ERICO® Cadweld® Plus
Inclusive of material types: F20, F80, F33, XF19, F76
Applicable prefixes: ACB, ACC, SCC, SB, PB, CA, XF

Relevant identified uses of the substance or mixture and uses advised against

Application: Exothermic Welding material

Details of the supplier of the safety data sheet

Manufacturer: nVent
ERICO International Corporation
34600 Solon Road
Solon, Ohio 44139
Tel:(440) 248-0100

Supplier: nVent
ERICO Products Australia
323 Parramatta Rd
AUBURN NSW 2144

Further information can be obtained from: erico.compliance@nvent.com

Emergency telephone number

Emergency telephone: +01-813-248-0585 International

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SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture

WHS:
Acute toxicity – oral – Category 4
Acute toxicity - inhalation - Category 4
Serious eye damage/eye irritation - Category 1

Label elements



DANGER

Contains: Dicopper oxide (25-85%)
H302 Harmful if swallowed.
H332 Harmful if inhaled.
H318 Causes serious eye damage.

P260 Do not breathe dust/fume
P280 Wear eye protection and gloves.
P301 + P312 IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell.
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310 Immediately call a POISON CENTER or doctor/physician.
P273 Avoid release to the environment.

Additional information is listed in the safety data sheet.

Other hazards

PBT/vPvB: This product does not contain any PBT or vPvB substances.

Other: Dust may form explosive mixture with air. Improper use of the product or inadequate preparation of the conductors, moulds or surroundings can result in aggressive reactions. Self-propagating high temperature reaction will occur if heated above ignition temperature. Generates molten metal in excess of 2000°C, slag and dense, dusty smoke. The molten product can cause serious burns. Inhalation of powder or fumes may cause metal fume fever. Exposure to reaction by-products: See point 8.
Very toxic to aquatic life with long lasting effects.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Mixtures

Only classified substances above threshold limits or substances with an exposure limit are shown.

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WHS:

<u>%:</u>	<u>CAS-No.:</u>	<u>EC No.:</u>	<u>REACH Reg. No.:</u>	<u>Chemical name:</u>	<u>Hazard classification:</u>	<u>Notes:</u>
25-85	1317-39-1	215-270-7	-	Dicopper oxide	Acute Tox. 4;H302 Acute Tox. 4;H332 Eye Dam. 1;H318 Aquatic Acute 1;H400 Aquatic Chronic 1;H410	
1-30	1317-38-0	215-269-1	-	Copper oxide	Aquatic Acute 1;H400 Aquatic Chronic 1;H410	
1-30	7440-50-8	231-159-6	-	Copper	Aquatic Acute 1;H400 Aquatic Chronic 3;H412	
5-10	7429-90-5	231-072-3	-	Aluminium powder (stabilised)	Flam. Sol. 1;H228 Water-react. 2;H261 Skin Irrit. 2;H315 Eye Irrit. 2;H319 STOT SE 3;H335	
1-5	7789-75-5	232-188-7	-	Calcium fluoride	-	#
1-5	7440-31-5	231-141-8	-	Tin	-	#

Notes: #. The substance has been assigned an exposure limit.

References: The full text for all hazard statements is displayed in section 16.

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SECTION 4: FIRST AID MEASURES

Description of first aid measures

Molten product will cause skin burns and if in contact with eyes while in a molten state may cause serious damage.

Inhalation: Inhalation of welding fumes/Dust inhalation: Move into fresh air and keep at rest. In case of persistent throat irritation or coughing: Seek medical attention and bring these instructions.

Skin contact: Remove contaminated clothes and rinse skin thoroughly with water. If material is hot, treat for thermal burns and get immediate medical attention.

Eye contact: Dust in the eyes: Do not rub eye. Immediately flush with plenty of water for at least 15 minutes. Remove any contact lenses and open eyelids widely. If irritation persists: Seek medical attention and bring along these instructions.

Ingestion: Immediately rinse mouth and drink plenty of water. Keep person under observation. If person becomes uncomfortable seek hospital and bring these instructions.

Most important symptoms and effects, both acute and delayed

Symptoms/effects: Inhalation of powder or fumes may cause metal fume fever. Symptoms like headache, fatigue and nausea may appear. See section 11 for more detailed information on health effects and symptoms.

Indication of any immediate medical attention and special treatment needed

Medical attention/treatments: Burns (in contact with molten metal, slag or hot equipment): Flush with water immediately. While flushing, remove clothes which do not adhere to affected area. Call an ambulance. Continue flushing during transport to hospital.

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SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Extinguishing media: Extinguish with dry sand and/or flood with large amounts of water.

Extinguishing media which are not suitable: Hand water buckets or hand storage pumps. Molten metal contact with water can cause small pockets of superheated steam.

Use fire-extinguishing media appropriate for surrounding materials.

Special hazards arising from the substance or mixture

Specific hazards: During fire, gases hazardous to health may be formed.
Ignition temperature: >950°C

In the event that the packaging materials are ignited, the immediate and direct application of large quantities of water will effectively eliminate the spread of fire to the surrounding areas. The ignition of the packaging materials may, in rare cases, lead to ignition. Direct application of a continuous heavy stream of water is recommended.

Ignition of large quantities of exothermic materials may result in large volumes of dense smoke.

Advice for firefighters

Protective equipment for fire-fighters: Selection of respiratory protection for fire fighting: follow the general fire precautions indicated in the workplace.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Personal precautions: Avoid inhalation of dust. Do not breathe fumes. Avoid contact with skin and eyes. Follow precautions for safe handling described in this safety data sheet.

Remove sources of ignition. Ventilate well.

Environmental precautions

Environmental precautions: Precaution should be taken to prevent hot material and reaction byproducts from contact with combustible materials in surrounding areas. Avoid spreading dust or contaminated materials. Avoid discharge to the aquatic environment. Contact local authorities in case of spillage to drain/aquatic environment.

Methods and material for containment and cleaning up

Spill Cleanup Methods: Remove sources of ignition. Sweep up spilled substance and remove to safe place.
For large spills use natural fiber brush or broom with a conductive, non-sparking pan.

Reference to other sections

References: For personal protection, see section 8. For waste disposal, see section 13.

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SECTION 7: HANDLING AND STORAGE

Precautions for safe handling

Safe handling advice: Avoid inhalation of dust. Do not breathe fumes. Avoid contact with skin and eyes. Observe good chemical hygiene practices. nVent ERICO Cadweld Plus Welding and Filler Materials are designed for use in nVent ERICO Cadweld equipment only. Use of improper or damaged equipment can lead to exposure to molten metal and reaction byproducts, resulting in personal injury.

Technical measures: Do not smoke or use open fire or other sources of ignition. Work practice should minimise risk of contact.

Technical precautions: Confined space: Local exhaust is recommended.

Conditions for safe storage, including any incompatibilities

Technical measures for safe storage: nVent ERICO Cadweld Plus Welding and Filler Materials should be stored in a clean, dry and secure location. Storage should include provisions to minimize rough handling, excessive vibration and physical abuse. All outer packages must be stored in accordance with label markings.

Storage conditions: If evidence is present of damaged or contaminated products, these units should not be used.

If proper storage is maintained, nVent ERICO Cadweld Plus Materials do not exhibit any storage or shelf life.

Specific end use(s)

Specific use(s): Welding material

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SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

A detailed fume analysis was conducted on nVent ERICO Cadweld Plus Welding Materials. Reaction byproducts were tested for total dust, respirable dust, metals, acids, fluorides, various elements, and volatile organic compounds (VOC's). All sampling and analysis followed methodologies dictated by the National Institute of Occupational Safety and Health (NIOSH) and by the Occupational Safety and Health Administration (OSHA). A certified Industrial Hygienist did the sample collection and independent

Occupational exposure limits:

<u>CAS-No.:</u>	<u>Chemical name:</u>	<u>As:</u>	<u>Exposure limits:</u>	<u>Type:</u>	<u>Notes:</u>	<u>References</u>
7440-50-8	Copper, dusts & mists	Cu	1 mg/m3	TWA	-	WES
7440-50-8	Copper (fume)	-	0.2 mg/m3	TWA	-	WES
7440-31-5	Tin, metal	-	2 mg/m3	TWA	-	WES
7429-90-5	Aluminium, pyro powders	Al	5 mg/m3	TWA	-	WES
7429-90-5	Aluminium (welding fumes)	Al	5 mg/m3	TWA	-	WES
7429-90-5	Aluminium (metal dust)	-	10 mg/m3	TWA	-	WES
-	Fluorides	F	2.5 mg/m3	TWA	-	WES

Notes: WES: Workplace exposure standards for airborne contaminants.

Exposure controls

<u>Engineering measures:</u>	Provide adequate ventilation. Observe Occupational Exposure Limits and minimise the risk of inhalation of dust and fumes.
<u>Personal protection:</u>	Personal protection equipment should be chosen according to relevant AS/NZS standards and in discussion with the supplier of the personal protective equipment. Use special welding equipment for protection of eyes, skin and respiratory system.
<u>Respiratory equipment:</u>	Normal use precludes use of special protection as material is generally used out of doors, in small quantities and is of short duration. In case of inadequate ventilation and work of long duration or on large surface areas in confined rooms. Wear suitable respiratory equipment for dusts and metal fumes.
<u>Hand protection:</u>	Heat insulated protective gloves. Recommended for handling hot equipment.
<u>Eye protection:</u>	Wear goggles/face shield. Avoid direct eye contact with "flash" of light from reaction.
<u>Skin protection:</u>	Use protective clothing, which covers arms and legs.
<u>Hygiene measures:</u>	Wash hands after handling. Change contaminated clothing.

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SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Form: Granular.
Colour: Grey-black
Odour: Odourless.
pH: Not available.
Melting point / freezing point: 1093°C / 2000°F
Boiling point: Not available.
Flash point: Not relevant.
Evaporation rate: Not relevant.
Vapour pressure: Not relevant.
Vapour density: Not relevant.
Solubility: Insoluble in water
Partition coefficient (n-octanol/water): Not available.
Auto-ignition temperature (°C): > 950°C / 1742°F
Decomposition temperature (°C): Not available.
Viscosity: Not relevant.
Explosive properties: Not available.
Oxidising properties: Not available.

Other information

Other data: SPECIFIC GRAVITY (water=1): 5.5

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SECTION 10: STABILITY AND REACTIVITY

Reactivity

Reactivity: See hazardous reactions.

Chemical stability

Stability: Stable. Not sensitive to vibrations, shock or impact and is not subject to spontaneous ignition.

Possibility of hazardous reactions

Hazardous Reactions: Aggressive reactions are possible if excess moisture is present in the mould or on the conductors to be welded. Care should be taken to ensure proper preparation in accordance with instruction prints.

Conditions to avoid

Conditions/materials to avoid: Temperatures above ignition point. (950 °C)

Incompatible materials

Incompatible materials: Typical of problems associated with molten metals.

Hazardous decomposition products

Hazardous decomposition products: None under normal conditions. Polymerization will not occur.

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SECTION 11: TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute Toxicity (Oral): Harmful if swallowed.

Acute Toxicity (Dermal): Based on available data, the classification criteria are not met.

Acute Toxicity (Inhalation): Harmful if inhaled.

Skin Corrosion/Irritation: Based on available data, the classification criteria are not met.

Serious eye damage/irritation: Causes serious eye damage.

Respiratory or skin sensitisation: Based on available data, the classification criteria are not met.

Germ cell mutagenicity: Based on available data, the classification criteria are not met.

Carcinogenicity: Based on available data, the classification criteria are not met.

Reproductive Toxicity: Based on available data, the classification criteria are not met.

STOT - Single exposure: Based on available data, the classification criteria are not met.

STOT - Repeated exposure: Based on available data, the classification criteria are not met.

Aspiration hazard: Based on available data, the classification criteria are not met.

Inhalation: Dust may irritate throat and respiratory system and cause coughing. Heating above the melting point releases metallic oxides which may cause metal fume fever by inhalation. The symptoms are shivering, fever, malaise and muscular pain.

Skin contact: Dust has an irritating effect on moist skin. Prolonged and/or repeated contact: May cause eczema-like skin disorders (dermatitis). The molten product can cause serious burns.

Eye contact: Particles/fumes in the eyes may cause discomfort/irritation.

Ingestion: Ingestion may cause nausea, headache, dizziness and intoxication.

Specific effects: Frequent inhalation of dust over a long period of time increases the risk of developing lung diseases. Copper oxide may by repeated or prolonged inhalation occasionally cause ulceration and perforation of the nasal septum. Long term exposure to copper containing dusts may cause allergic dermatitis.

Toxicological data: LD50 (oral, rat): 1340 mg/kg (Dicopper oxide)

SAFETY DATA SHEET

Product name: nVent® ERICO® Cadweld® Plus
Supersedes date:
SDS-ID:

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SECTION 12: ECOLOGICAL INFORMATION

Toxicity

Ecotoxicity: Very toxic to aquatic life with long lasting effects.

Persistence and degradability

Degradability: The product solely consists of inorganic compounds which are not biodegradable.

Bioaccumulative potential

Bioaccumulative potential: No data available on bioaccumulation.

Mobility in soil

Mobility: The product is not volatile but may be spread by dust-raising handling.

Results of PBT and vPvB assessment

PBT/vPvB: This product does not contain any PBT or vPvB substances.

Other adverse effects

Other adverse effects: None known.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Dispose of waste and residues in accordance with local authority requirements. Waste is classified as hazardous waste.

SECTION 14: TRANSPORT INFORMATION

The product is covered by international regulation on the transport of dangerous goods (IMDG, IATA).

UN number

UN-No: 3077

UN proper shipping name

Proper Shipping Name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (Dicopper oxide)

Transport hazard class(es)

Class: 9

Packing group

PG: III

Environmental hazards

Marine pollutant: Yes.

Environmentally Hazardous substance: Yes.

Special precautions for user

Special precautions: -

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Transport in bulk: -

SAFETY DATA SHEET

Product name: nVent® ERICO® Cadweld® Plus

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SDS-ID:

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SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

All chemicals included in the product are AICS listed or exempted.

SUSMP: all ingredients are not listed or are excepted.

This product contains no ingredient listed in the model WHS Regulations 2011 (as amended), Schedule 10.

National regulation:

Work Health and Safety Regulations 2011, with amendments.

National model Code of Practice for the Labelling of Workplace Hazardous Chemicals (2018).

National model Code of Practice for the Preparation of Safety Data Sheets for Hazardous Chemicals (2018).

Workplace exposure standards for airborne contaminants (2019).

The Standard for the Uniform Scheduling of Medicines and Poisons (SUSMP).
Industrial Chemicals Act 2019.

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Product name: nVent® ERICO® Cadweld® Plus
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SECTION 16: OTHER INFORMATION

The user must be instructed in the proper work procedure and be familiar with the contents of these instructions.

Abbreviations and acronyms used in the safety data sheet: PBT = Persistent, Bioaccumulative and Toxic.
vPvB = very Persistent and very Bioaccumulative.
LD50 = Lethal Dose 50%.

Additional information: Classification according to WHS Regulations 2011: Calculation method.

Revision Date: 2020-11-03

Wording of H-statements:

H228	Flammable solid.
H261	In contact with water releases flammable gases.
H302	Harmful if swallowed.
H315	Causes skin irritation.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H332	Harmful if inhaled.
H335	May cause respiratory irritation.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.

The information on this data sheet represents our current data and is reliable provided that the product is used under the prescribed conditions and in accordance with the application specified on the packaging and/or in the technical guidance literature. Any other use of the product which involves using the product in combination with any other product or any other process is the responsibility of the user.



Revision Number: 002.0

Issue date: 03/31/2025

1. IDENTIFICATION

Product name: GE Advanced Silicone Window and Door Projects **IDH number:** 2810435

Product type/Recommended use: Joint sealant, silicone

Restriction of Use: None identified **Region:** United States

Company address: Henkel Corporation
One Henkel Way
Rocky Hill, Connecticut 06067

Contact information:
Telephone: +1 (860) 571-5100
MEDICAL EMERGENCY Phone: Poison Control Center
1-877-671-4608 (toll free) or 1-303-592-1711
TRANSPORT EMERGENCY Phone: CHEMTREC
1-800-424-9300 (toll free) or 1-703-527-3887
Internet: www.henkelna.com

2. HAZARDS IDENTIFICATION**EMERGENCY OVERVIEW**

WARNING: H317 - MAY CAUSE AN ALLERGIC SKIN REACTION.
H361 - SUSPECTED OF DAMAGING FERTILITY OR THE UNBORN CHILD.

HAZARD CLASS	HAZARD CATEGORY
SKIN SENSITIZATION	1
REPRODUCTIVE TOXICITY	2

PICTOGRAM(S)**Precautionary Statements**

Prevention: P201 - Obtain special instructions before use.
P202 - Do not handle until all safety precautions have been read and understood.
P261 - Avoid breathing dust or fumes.
P272 - Contaminated work clothing should not be allowed out of the workplace.

Response: P280 - Wear protective gloves, clothing, eye and face protection.
P302+P352 - IF ON SKIN: Wash with plenty of water.
P308+P313 - IF exposed or concerned: Get medical attention.
P333+P313 - If skin irritation or rash occurs: Get medical attention.
P362+P364 - Take off contaminated clothing and wash it before reuse.

Storage: P405 - Store locked up.

Disposal: P501 - Dispose of contents and/or container according to Federal, State/Provincial and local governmental regulations.

Other hazards Not available.

Classification complies with OSHA Hazard Communication Standard (29 CFR 1910.1200) and is consistent with the provisions of the United Nations Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

See Section 11 for additional toxicological information.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Hazardous Component(s)	CAS Number	Weight %*
Silica, amorphous, fumed, crystal-free	112945-52-5	5 - 10
1,1,1,3,3,3-Hexamethyldisilazane	999-97-3	1 - 5
Trimethoxy(methyl)silane	1185-55-3	1 - 5
octamethylcyclotetrasiloxane	556-67-2	0.1 - 1

* Exact percentages may vary or are trade secret. Concentration range is provided to assist users in providing appropriate protections.

4. FIRST AID MEASURES

First Aid Measures by likely routes of exposure

Inhalation:	If not breathing, give artificial respiration. If breathing is difficult, oxygen should be administered by qualified personnel. Move to fresh air. If symptoms persist, seek medical advice.
Skin contact:	Wipe off with paper towel or cloth. Rinse with running water and soap. Remove contaminated clothing and shoes. Wash contaminated clothing before reuse. Thoroughly clean shoes before reuse. Seek medical attention immediately.
Eye contact:	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Seek medical attention immediately.
Ingestion:	DO NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Seek medical attention immediately.
Most important symptoms and effects (acute and delayed):	The most important known symptoms and effects, both acute and delayed, are described in Section 11: Toxicological Information.
Indication of any immediate medical attention / special treatment needed:	Not available.

5. FIRE FIGHTING MEASURES

Extinguishing media:	Foam, dry chemical or carbon dioxide.
Improper extinguishing agents:	Not available.
Special firefighting procedures:	Wear self-contained breathing apparatus and full protective clothing, such as turn-out gear.
Unusual fire or explosion hazards:	Closed containers may rupture (due to build up of pressure) when exposed to extreme heat. In case of fire, keep containers cool with water spray.
Hazardous combustion products:	Oxides of carbon. Oxides of silicon. Formaldehyde. Toxic and irritating vapors.

6. ACCIDENTAL RELEASE MEASURES

Use personal protection recommended in Section 8, isolate the hazard area and deny entry to unnecessary and unprotected personnel.

Environmental precautions:	Do not allow product to enter sewer or waterways.
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Clean-up methods:

Ensure adequate ventilation. Wear appropriate personal protective equipment. Refer to Section 8 "Exposure Controls / Personal Protection" prior to clean up. Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Scrape up spilled material and place in a closed container for disposal.

7. HANDLING AND STORAGE

Handling:

Use only with adequate ventilation. Vapours should be extracted to avoid inhalation. Avoid contact with eyes, skin and clothing. Do not wear contact lenses. Do not handle contact lenses until all sealant has been removed from hands. Residual sealant may transfer to lenses and cause eye irritation. Wash thoroughly after handling. See Section 8 of the SDS for Personal Protective Equipment. Keep container closed.

Storage:

Keep in a cool, well ventilated area away from heat, sparks and open flame. Keep container tightly closed until ready for use.

For information on product shelf life, please review labels on container or check the Technical Data Sheet.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Employers should complete an assessment of all workplaces to determine the need for, and selection of, proper exposure controls and protective equipment for each task performed.

Hazardous Component(s)	ACGIH TLV	OSHA PEL	AIHA WEEL	OTHER
Silica, amorphous, fumed, crystal-free	10 mg/m3 TWA Inhalable dust. 3 mg/m3 TWA Respirable fraction. 3 mg/m3 TWA Respirable particles. 10 mg/m3 TWA Inhalable particles.	20 MPPCF TWA 0.8 mg/m3 TWA 50 MPPCF TWA Total dust. 5 mg/m3 TWA Respirable fraction. 15 mg/m3 TWA Total dust. 15 MPPCF TWA Respirable fraction.	None	None
1,1,1,3,3,3-Hexamethyldisilazane	None	None	10 ppm TWA 50 ppm STEL	None
Trimethoxy(methyl)silane	None	None	10 ppm (55 mg/m3) TWA	None
octamethylcyclotetrasiloxane	None	None	10 ppm TWA	None

Engineering controls:

Use local ventilation if general ventilation is insufficient to maintain vapor concentration below established exposure limits.

Respiratory protection:

Use NIOSH approved respirator if there is potential to exceed exposure limit(s).

Eye/face protection:

Safety goggles or safety glasses with side shields. Full face protection should be used if the potential for splashing or spraying of product exists. Safety showers and eye wash stations should be available.

Skin protection:

Use chemical resistant, impermeable clothing including gloves and either an apron or body suit to prevent skin contact.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state:

Solid

Color:

Clear

Odor:

Alcoholic

Odor threshold:

Not available.

pH:

Not applicable

Vapor pressure:

Not available.

Boiling point/range:

> 100 °C (> 212°F)

Melting point/ range:

Not available.

Density/Relative density:

1.01

Relative vapor density:

Not available.

Flash point:

Not available.

Flammable/Explosive limits - lower:

Not available.

Flammable/Explosive limits - upper:

Not available.

Autoignition temperature:

Not available.

Flammability:	Not classified as a flammability hazard
Evaporation rate:	Not available.
Solubility:	Not available.
Partition coefficient n-octanol/water (logarithmic value):	Not available.
VOC content:	3 %; 31 g/l (by weight, calculated using CARB method; g/L less water, less exempts calculated using SCAQMD method)
Dynamic viscosity:	Not available.
Kinematic viscosity:	Not available.
Particle characteristics:	Not available.
Decomposition temperature:	Not available.

10. STABILITY AND REACTIVITY

Stability:	Stable under normal conditions of storage and use.
Hazardous reactions:	None under normal processing.
Hazardous decomposition products:	Oxides of carbon. Oxides of silicon. Formaldehyde. Ammonia. Methanol.
Incompatible materials:	Acids and bases. Oxidizing agents. Amines.
Reactivity:	Not available.
Conditions to avoid:	Keep away from heat, ignition sources and incompatible materials. Protect from direct sunlight. Exposure to moisture.

11. TOXICOLOGICAL INFORMATION

Likely routes of exposure:	Skin, Inhalation, Eyes, Ingestion
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Potential Health Effects/Symptoms

Inhalation:	May cause respiratory tract irritation. When heated to temperatures exceeding 300° F (150° C) in the presence of air, silicones may form formaldehyde vapors. Formaldehyde is a potential cancer hazard and a known skin and respiratory sensitizer. Vapors irritate the eyes, nose and throat. Safe handling conditions may be maintained by keeping formaldehyde vapor concentrations below the OSHA permissible limit.
Skin contact:	Prolonged or repeated skin contact may cause skin irritation or allergic skin sensitization reaction.
Eye contact:	May cause eye irritation.
Ingestion:	Not expected under normal conditions of use. May cause gastrointestinal tract irritation if swallowed.

Hazardous Component(s)	LD50s and LC50s
Silica, amorphous, fumed, crystal-free	None
1,1,1,3,3,3-Hexamethyldisilazane	Oral LD50 (Rat) = 847 mg/kg Oral LD50 (Rabbit) = 1,100 mg/kg Oral LD50 (Mouse) = 850 mg/kg Inhalation LC50 (Rat, 4 h) = 8,700 mg/m3 Inhalation LC50 (Rat, 4 h) = 10.3 mg/l
Trimethoxy(methyl)silane	Inhalation LC50 (Rat, 4 h) = > 26000 ppm
octamethylcyclotetrasiloxane	Oral LD50 (Rat) = > 4,800 mg/kg Dermal LD50 (Rat) = > 2,000 mg/kg Dermal LD50 (Rabbit) = > 4,640 mg/kg Inhalation LC50 (Rat, 4 h) = 36 mg/l

Hazardous Component(s)	Immediate Health Effects	Delayed Health Effects	Chronic Health Effects
Silica, amorphous, fumed, crystal-free	Nuisance dust		
1,1,1,3,3,3-Hexamethyldisilazane	Irritant		
Trimethoxy(methyl)silane	Irritant	Allergen	
octamethylcyclotetrasiloxane	Irritant		

Hazardous Component(s)	NTP Carcinogen	IARC Carcinogen	OSHA Carcinogen (Specifically Regulated)
Silica, amorphous, fumed, crystal-free	No	No	No
1,1,1,3,3,3-Hexamethyldisilazane	No	No	No
Trimethoxy(methyl)silane	No	No	No
octamethylcyclotetrasiloxane	No	No	No

12. ECOLOGICAL INFORMATION

Ecological information: Not available.

13. DISPOSAL CONSIDERATIONS

Information provided is for unused product only.

Recommended method of disposal: Follow all local, state, federal and provincial regulations for disposal.

14. TRANSPORT INFORMATION

The transport information provided in this section only applies to the material/formulation itself, and is not specific to any packaging.

U.S. Department of Transportation Ground (49 CFR)

Proper shipping name:	Not regulated
Hazard class or division:	None
Identification number:	None
Packing group:	None

International Air Transportation (ICAO/IATA)

Proper shipping name:	Not regulated
Hazard class or division:	None
Identification number:	None
Packing group:	None

Water Transportation (IMO/IMDG)

Proper shipping name:	Not regulated
Hazard class or division:	None
Identification number:	None
Packing group:	None

15. REGULATORY INFORMATION

United States Regulatory Information

TSCA 8 (b) Inventory Status: All components are listed as active or are exempt from listing on the Toxic Substances Control Act (TSCA) inventory.

TSCA 12 (b) Export Notification: None above reporting de minimis

CERCLA/SARA Section 302 EHS: None above reporting de minimis.

CERCLA/SARA Section 311/312: Please refer to the GHS classification in Section 2

CERCLA/SARA Section 313: None above reporting de minimis.

California Proposition 65: This product contains a chemical known to the State of California to cause birth defects or other reproductive harm.

Canada Regulatory Information

CEPA DSL/NDSL Status: All components are listed on or are exempt from listing on the Canadian Domestic Substances List.

16. OTHER INFORMATION

This safety data sheet contains changes from the previous version in sections: This Safety Data Sheet contains changes from the previous version in Section(s): 11

Prepared by: Product Safety and Regulatory Affairs

Issue date: 03/31/2025

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CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Revision date: 10/27/2025

Issue date: 10/27/2025

Supersedes: 12/16/2024

Version: 5.0

SECTION 1: Identification

1.1. Identification

Product form	Mixture
Product name	CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25
Product code	BU Fire Protection

1.2. Other means of identification

No additional information available

1.3. Recommended use of the chemical and restrictions on use

Use of the substance/mixture	Firestop putty pad / stick / cable disc
Recommended use	For professional users only

1.4. Supplier's details

Supplier

Hilti, Inc.
Legacy Tower, Suite 1000
7250 Dallas Parkway
US TX 75024 Plano
USA
T +1 9724035800
1-800-879-8000 toll free, F +1 918 254 0522
us-sales@hilti.com

Department issuing data specification sheet

Hilti AG
Feldkircher Strasse
FL 9494 Schaan
Liechtenstein
T +423 234 2111
product.compliance-fire.protection@hilti.com

1.5. Emergency phone number

Emergency number	Emergency CONTACT (24-Hour-Number) GBK/Infotrac ID 101022 (USA domestic) 1 800 535 5053 or international (001) 352 323 3500
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SECTION 2: Hazard(s) identification

2.1. Classification of the substance or mixture

GHS-US classification

Not classified

2.2. GHS Label elements, including precautionary statements

GHS US labelling

No labelling applicable

2.3. Hazards associated with known or reasonably anticipated uses

No additional information available

2.4. Hazards not otherwise classified

No additional information available

2.5. Unknown acute toxicity (GHS US)

Not applicable

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

This mixture does not contain any substances to be mentioned according to the criteria for section 3.2 of HCS

SECTION 4: First-aid measures

4.1. Description of first aid measures

First-aid measures general	Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).
First-aid measures after inhalation	Remove person to fresh air and keep comfortable for breathing.
First-aid measures after skin contact	Wash skin with plenty of water.
First-aid measures after eye contact	Obtain medical attention if pain, blinking or redness persists. Rinse eyes with water as a precaution.
First-aid measures after ingestion	Rinse mouth. Do NOT induce vomiting. Obtain emergency medical attention. Call a poison center or a doctor if you feel unwell.

4.2. Most important symptoms and effects (acute and delayed)

Potential adverse human health effects and symptoms	Based on available data, the classification criteria are not met.
Symptoms/effects	Not expected to present a significant hazard under anticipated conditions of normal use.
Symptoms/effects after inhalation	Dust of the product, if present, may cause respiratory irritation after excessive inhalation exposure. Although no appropriate human or animal health effects data are known to exist, this material is expected to be an inhalation hazard.
Symptoms/effects after skin contact	None under normal conditions. Dust may cause irritation in skin folds or by contact in combination with tight clothing.
Symptoms/effects after eye contact	None under normal conditions. Dust from this product may cause eye irritation.
Symptoms/effects after ingestion	None under normal conditions.

4.3. Immediate medical attention and special treatment, if necessary

Treat symptomatically.

SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

Suitable extinguishing media	Foam. Dry powder. Carbon dioxide. Water spray. Sand.
Unsuitable extinguishing media	Do not use a heavy water stream.

5.2. Specific hazards arising from the chemical

Fire hazard	No fire hazard.
Explosion hazard	No direct explosion hazard.
Hazardous decomposition products in case of fire	Carbon dioxide. Carbon monoxide.

5.3. Special protective equipment and precautions for fire-fighters

Firefighting instructions	Use water spray or fog for cooling exposed containers. Exercise caution when fighting any chemical fire. Prevent fire fighting water from entering the environment. Do not enter fire area without proper protective equipment, including respiratory protection.
Protection during firefighting	Do not enter fire area without proper protective equipment, including respiratory protection. Do not attempt to take action without suitable protective equipment. Self-contained breathing apparatus. Complete protective clothing.

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

General measures Notify authorities if product enters sewers or public waters. Absorb spillage to prevent material damage.

For non-emergency personnel

Protective equipment Wear recommended personal protective equipment.
Emergency procedures Ventilate spillage area. Evacuate unnecessary personnel.

For emergency responders

Protective equipment Do not attempt to take action without suitable protective equipment. Equip cleanup crew with proper protection. For further information refer to section 8: "Exposure controls/personal protection".
Emergency procedures Ventilate area. Evacuate unnecessary personnel.

Environmental precautions Avoid release to the environment. Prevent entry to sewers and public waters.

6.2. Methods and materials for containment and cleaning up

For containment Using a clean shovel, put the material in a dry container and cover without compressing it.
Methods for cleaning up Mechanically recover the product. On land, sweep or shovel into suitable containers.
Other information Dispose of materials or solid residues at an authorized site.

See Section 8, Exposure controls and personal protection, For further information refer to section 13

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Additional hazards when processed Not expected to present a significant hazard under anticipated conditions of normal use.
Precautions for safe handling Ensure good ventilation of the work station. Wear personal protective equipment. Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work.
Hygiene measures Do not eat, drink or smoke when using this product. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures Keep in a cool, well-ventilated place away from heat.
Storage conditions Store at ambient temperature.
Incompatible products Strong bases. Strong acids.
Incompatible materials Sources of ignition. Direct sunlight.
Storage temperature 23 – 104 °F
Packaging materials Store always product in container of same material as original container.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

No additional information available

Additional information The product has a pasty consistency. Exposure limit values for respirable dusts are not relevant for this product.

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

8.2. Appropriate engineering controls

Appropriate engineering controls	Ensure good ventilation of the work station.
Environmental exposure controls	Avoid release to the environment.

8.3. Individual protection measures/Personal protective equipment

Personal protective equipment:

Avoid all unnecessary exposure. Protective goggles. Wear protective gloves. Protective clothing.

Hand protection:				
Wear suitable gloves tested to EN374. Suitable for short-term work or as a splash guard: Nitrile rubber gloves (> 0.1 mm). In case of permanent product contact:				
Type	Material	Permeation	Thickness (mm)	Penetration
Disposable gloves	Nitrile rubber (NBR)	6 (> 480 minutes)	>0,4	
Eye protection:				
Safety glasses				
Skin and body protection:				
Wear suitable protective clothing				
Respiratory protection:				
No respiratory protection needed under normal use conditions				

Personal protective equipment symbol(s):



Other information:

Do not eat, drink or smoke during use.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Solid
Appearance	Pasty.
Colour	red
Odour	characteristic
Odour threshold	Not determined
pH	Not relevant
Melting point	No data available
Freezing point	Not applicable
Boiling point	No data available
Flash point	Not applicable
Relative evaporation rate (butylacetate=1)	No data available
Flammability (solid, gas)	Non flammable.
Vapour pressure	No data available
Relative vapour density at 20°C	No data available
Relative density	No data available
Density	1.6 g/cm ³

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according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Molecular mass	Not determined
Solubility	Water: Insoluble
Partition coefficient n-octanol/water (Log Pow)	No data available
Auto-ignition temperature	Not applicable
Decomposition temperature	No data available
Viscosity, kinematic	Not applicable
Viscosity, dynamic	No data available
Explosive limits	Not applicable
Explosive properties	No data available
Oxidising properties	No data available

9.2. Other information

No additional information available

SECTION 10: Stability and reactivity

10.1. Reactivity

The product is non-reactive under normal conditions of use, storage and transport.

10.2. Chemical stability

Not established.

10.3. Possibility of hazardous reactions

Not established.

10.4. Conditions to avoid

Direct sunlight. Extremely high or low temperatures.

10.5. Incompatible materials

Strong acids. Strong bases.

10.6. Hazardous decomposition products

fume. Carbon monoxide. Carbon dioxide.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral)	Not classified
Acute toxicity (dermal)	Not classified
Acute toxicity (inhalation)	Not classified
Skin corrosion/irritation	Not classified
	pH: Not relevant
Serious eye damage/irritation	Not classified
	pH: Not relevant
Respiratory or skin sensitisation	Not classified
Germ cell mutagenicity	Not classified
Carcinogenicity	Not classified
Reproductive toxicity	Not classified
STOT-single exposure	Not classified
STOT-repeated exposure	Not classified
Aspiration hazard	Not classified
Viscosity, kinematic	Not applicable

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Potential adverse human health effects and symptoms	Based on available data, the classification criteria are not met.
Symptoms/effects	Not expected to present a significant hazard under anticipated conditions of normal use.
Symptoms/effects after inhalation	Dust of the product, if present, may cause respiratory irritation after excessive inhalation exposure. Although no appropriate human or animal health effects data are known to exist, this material is expected to be an inhalation hazard.
Symptoms/effects after skin contact	None under normal conditions. Dust may cause irritation in skin folds or by contact in combination with tight clothing.
Symptoms/effects after eye contact	None under normal conditions. Dust from this product may cause eye irritation.
Symptoms/effects after ingestion	None under normal conditions.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general	The product is not considered harmful to aquatic organisms nor to cause long-term adverse effects in the environment.
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12.2. Persistence and degradability

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25	
Persistence and degradability	Not established.

12.3. Bioaccumulative potential

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25	
Bioaccumulative potential	Not established.

12.4. Mobility in soil

No additional information available

12.5. Other adverse effects

Other information	Avoid release to the environment.
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SECTION 13: Disposal considerations

13.1. Disposal methods

Regional waste regulation	Disposal must be done according to official regulations.
Waste treatment methods	Dispose in a safe manner in accordance with local/national regulations. Dispose of contents/container in accordance with licensed collector's sorting instructions.
Sewage disposal recommendations	Disposal must be done according to official regulations.
Product/Packaging disposal recommendations	Dispose in a safe manner in accordance with local/national regulations. Disposal must be done according to official regulations.
Additional information	Do not re-use empty containers.
Ecological waste information	Avoid release to the environment.

SECTION 14: Transport information

In accordance with DOT / TDG / IMDG / IATA

DOT	TDG	IMDG	IATA
14.1. UN number			
Not regulated for transport			

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

DOT	TDG	IMDG	IATA
14.2. Proper Shipping Name			
Not regulated	Not regulated	Not regulated	Not regulated
14.3. Transport hazard class(es)			
Not regulated	Not regulated	Not regulated	Not regulated
14.4. Packing group			
Not regulated	Not regulated	Not regulated	Not regulated
14.5. Environmental hazards			
Not regulated	Not regulated	Not regulated	Not regulated
No supplementary information available			

14.6. Special precautions for user

DOT

Not regulated

TDG

Not regulated

IMDG

Not regulated

IATA

Not regulated

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable

SECTION 15: Regulatory information

15.1. US Federal regulations

All components of this product are present and listed as Active on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory

15.2. International regulations

No additional information available

15.3. US State regulations

California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm

SECTION 16: Other information

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Revision date

10/27/2025

Data sources

REGULATION (EC) No 1272/2008 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006.

Other information

None.

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

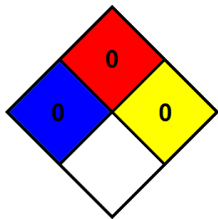
Abbreviations and acronyms	
ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute Toxicity Estimate
BCF	Bioconcentration factor
BLV	Biological limit value
BOD	Biochemical oxygen demand (BOD)
COD	Chemical oxygen demand (COD)
DMEL	Derived Minimal Effect level
DNEL	Derived-No Effect Level
EC-No.	European Community number
EC50	Median effective concentration
EN	European Standard
IARC	International Agency for Research on Cancer
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
LC50	Median lethal concentration
LD50	Median lethal dose
LOAEL	Lowest Observed Adverse Effect Level
NOAEC	No-Observed Adverse Effect Concentration
NOAEL	No-Observed Adverse Effect Level
NOEC	No-Observed Effect Concentration
OECD	Organisation for Economic Co-operation and Development
OEL	Occupational Exposure Limit
PBT	Persistent Bioaccumulative Toxic
PNEC	Predicted No-Effect Concentration
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
SDS	Safety Data Sheet
STP	Sewage treatment plant
ThOD	Theoretical oxygen demand (ThOD)
TLM	Median Tolerance Limit
VOC	Volatile Organic Compounds
CAS-No.	Chemical Abstract Service number
N.O.S.	Not Otherwise Specified

CFS-P BA / CP 618 / CFS-D 1" / CFS-D 25

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Abbreviations and acronyms	
vPvB	Very Persistent and Very Bioaccumulative
ED	Endocrine disruptor

NFPA health hazard	0 - Materials that, under emergency conditions, would offer no hazard beyond that of ordinary combustible materials.	
NFPA fire hazard	0 - Materials that will not burn under typical fire conditions, including intrinsically noncombustible materials such as concrete, stone, and sand.	
NFPA reactivity	0 - Material that in themselves are normally stable, even under fire conditions.	
Hazard Rating		
Health	0 Minimal Hazard - No significant risk to health	
Flammability	0 Minimal Hazard - Materials that will not burn	
Physical	0 Minimal Hazard - Materials that are normally stable, even under fire conditions, and will NOT react with water, polymerize, decompose, condense, or self-react. Non-Explosives.	
Personal protection	B - Safety glasses, Gloves	

Indication of changes:			
Section	Changed item	Change	Comments
			29 CFR § 1910.1200, Hazard Communication Standard (HCS)

SDS_US_Hilti

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

GC 22

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision date: 7/29/2024

Issue date: 7/29/2024 Supersedes: 8/16/2023

Version: 25.1

SECTION 1: Identification

1.1. Identification

Product form Mixture
Name GC 22
Product code BU Direct Fastening



1.2. Recommended use and restrictions on use

Recommended use For professional use only, Propellant for direct fastening tools.

1.3. Supplier

Supplier

Hilti, Inc.
Legacy Tower, Suite 1000
7250 Dallas Parkway
Plano, TX 75024
USA
T +1 9724035800
1-800-879-8000 toll free - F +1 918 254 0522

Department issuing data specification sheet

Hilti AG
Feldkircherstraße 100
Schaan, 9494
Liechtenstein
T +423 234 2111
product.compliance-direct.fastening@hilti.com

1.4. Emergency telephone number

Emergency number Emergency CONTACT (24-Hour-Number)
GBK/Infotrac ID 101022
(USA domestic) 1 800 535 5053
or international (001) 352 323 3500

SECTION 2: Hazard(s) identification

2.1. Classification of the substance or mixture

GHS-US classification

Flammable gases, Category 1	H220	Extremely flammable gas.
Gases under pressure : Compressed gas	H280	Contains gas under pressure; may explode if heated.
Full text of H-statements: see section 16		

2.2. GHS Label elements, including precautionary statements

GHS US labelling

Hazard pictograms (GHS US)



Signal word (GHS US) Danger
Hazard statements (GHS US)
H220 - Extremely flammable gas.
H280 - Contains gas under pressure; may explode if heated.

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Precautionary statements (GHS US)

P102 - Keep out of reach of children.
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P211 - Do not spray on an open flame or other ignition source.
P251 - Do not pierce or burn, even after use.
P381 - Eliminate all ignition sources if safe to do so.
P403 - Store in a well-ventilated place.
P410+P412 - Protect from sunlight. Do not expose to temperatures exceeding 50°C/122°F.

2.3. Other hazards which do not result in classification

No additional information available

2.4. Unknown acute toxicity (GHS US)

Not applicable

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Name	Product identifier	%	GHS-US classification
isobutane	CAS-No.: 75-28-5	55 - <65	Flam. Gas 1, H220 Press. Gas (Comp.), H280
propene	CAS-No.: 115-07-1	20 - <30	Flam. Gas 1, H220 Press. Gas (Comp.), H280
Propane	CAS-No.: 74-98-6	5 - <15	Flam. Gas 1, H220 Press. Gas (Comp.), H280

Full text of hazard classes and H-statements : see section 16

SECTION 4: First-aid measures

4.1. Description of first aid measures

First-aid measures general
First-aid measures after inhalation
First-aid measures after skin contact
First-aid measures after eye contact
First-aid measures after ingestion

Take off immediately all contaminated clothing.
Remove person to fresh air and keep comfortable for breathing.
Gently wash with plenty of soap and water.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical advice/attention.
Immediately consult a doctor/medical service.

4.2. Most important symptoms and effects (acute and delayed)

Potential adverse human health effects and symptoms

No additional information available. No harmful effects are to be expected if used properly.
The contained ingredients can be harmful, but they are hermetically enclosed in the article and can not be released.
The dismantling of the article is prohibited.

4.3. Immediate medical attention and special treatment, if necessary

Treat symptomatically.

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SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

Suitable extinguishing media	Carbon dioxide. Water spray. Dry powder. Alcohol resistant foam.
Unsuitable extinguishing media	Do not use a heavy water stream.

5.2. Specific hazards arising from the chemical

Explosion hazard	Heat may build pressure, rupturing closed containers, spreading fire and increasing risk of burns and injuries.
Hazardous decomposition products in case of fire	On burning: release of (highly) toxic gases/vapours. Thermal decomposition generates : Carbon dioxide. Carbon monoxide.

5.3. Special protective equipment and precautions for fire-fighters

Precautionary measures fire	Fight fire remotely due to the risk of explosion.
Firefighting instructions	DO NOT fight fire when fire reaches explosives. Evacuate area.
Protection during firefighting	Do not enter fire area without proper protective equipment, including respiratory protection. Self-contained breathing apparatus.
Other information	EN 12942. EN 12941.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

General measures	Evacuate area. Remove ignition sources.
------------------	---

6.1.1. For non-emergency personnel

Emergency procedures	Ventilate spillage area. Evacuate area. No open flames, no sparks, and no smoking.
----------------------	--

6.1.2. For emergency responders

Protective equipment	Do not attempt to take action without suitable protective equipment. Breathing apparatus.
----------------------	---

6.2. Environmental precautions

Avoid release to the environment. Prevent entry to sewers and public waters.

6.3. Methods and material for containment and cleaning up

Methods for cleaning up	Do not flush with water.
Other information	For further information refer to section 8: "Exposure controls/personal protection". For further information refer to section 13.

6.4. Reference to other sections

For further information refer to section 13. For further information refer to section 8: "Exposure controls/personal protection".

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Additional hazards when processed	Flammable gas. Do not pierce or burn, even after use. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
Precautions for safe handling	Do not spray on an open flame or other ignition source. Avoid contact with skin, eyes and clothing. Do not breathe vapours. Prevent the build-up of electrostatic charge.
Hygiene measures	Do not eat, drink or smoke when using this product. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

Technical measures	Proper grounding procedures to avoid static electricity should be followed.
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Storage conditions	Keep cool. Protect from sunlight. Keep in fireproof place. Store in dry protected location to prevent any moisture contact.
Incompatible materials	Heat sources. Direct sunlight. Sources of ignition.
Storage temperature	5 – 25 °C
Heat and ignition sources	Keep away from heat and direct sunlight. Keep away from ignition sources.
Information on mixed storage	Do not store with DX powder cartridges.

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

GC 22	
No additional information available	
isobutane (75-28-5)	
USA - ACGIH - Occupational Exposure Limits	
Local name	Isobutane
ACGIH OEL STEL	1000 ppm
Remark (ACGIH)	TLV® Basis: CNS impair
Regulatory reference	ACGIH 2022
propene (115-07-1)	
USA - ACGIH - Occupational Exposure Limits	
Local name	Propylene
ACGIH OEL TWA	500 ppm
Remark (ACGIH)	Asphyxia; URT irr
Regulatory reference	ACGIH 2023
Propane (74-98-6)	
USA - ACGIH - Occupational Exposure Limits	
Local name	Propane
Remark (ACGIH)	TLV® Basis: Simple Asphyxiant
Regulatory reference	ACGIH 2023
USA - OSHA - Occupational Exposure Limits	
Local name	Propane
OSHA PEL TWA	1800 mg/m³
	1000 ppm
Regulatory reference (US-OSHA)	OSHA Annotated Table Z-1

8.2. Appropriate engineering controls

Appropriate engineering controls	Ensure good ventilation of the work station.
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8.3. Individual protection measures/Personal protective equipment

Hand protection:				
In case of repeated or prolonged contact wear gloves				
Type	Material	Permeation	Thickness (mm)	Penetration
Disposable gloves	Nitrile rubber (NBR)	3 (> 60 minutes)	0,12	
Eye protection:				
Chemical goggles or safety glasses. ISO 16321-1. EN 170				
Skin and body protection:				
When using cartridge operated tools, sufficient ear protection must be worn.				
Respiratory protection:				
No respiratory protection needed under normal use conditions				

Personal protective equipment symbol(s):



SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Gas
Colour	Colourless
Odour	Sweet
Odour threshold	No data available
pH	No data available
Melting point	No data available
Freezing point	No data available
Boiling point	No data available
Flash point	-90 °C
Relative evaporation rate (butylacetate=1)	No data available
Flammability (solid, gas)	Extremely flammable gas.
Vapour pressure	8300 hPa
Relative vapour density at 20°C	No data available
Relative density	No data available
Density	0.6 g/cm³ (DIN 51757)
Solubility	insoluble in water.
Partition coefficient n-octanol/water (Log Pow)	No data available
Auto-ignition temperature	No data available
Decomposition temperature	No data available
Viscosity, kinematic	No data available
Viscosity, dynamic	No data available
Explosive limits	Lower explosion limit: 1.7 vol % Upper explosion limit: 11.1 vol %
Explosive properties	Product is not explosive. In use, may form flammable/explosive vapour-air mixture.
Oxidising properties	No data available

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9.2. Other information

Heat of combustion > 30 kJ/g NFPA 30B, Aerosol Classification Level: 3
Gas group Gases under pressure : Compressed gas

SECTION 10: Stability and reactivity

10.1. Reactivity

No additional information available

10.2. Chemical stability

Extremely flammable aerosol. Contains gas under pressure; may explode if heated. Extreme risk of explosion by shock, friction, fire or other sources of ignition.

10.3. Possibility of hazardous reactions

No additional information available

10.4. Conditions to avoid

Heat. Sparks. Open flame. Direct sunlight. Overheating.

10.5. Incompatible materials

No additional information available

10.6. Hazardous decomposition products

No additional information available

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral) Not classified (Based on available data, the classification criteria are not met)
Acute toxicity (dermal) Not classified (Based on available data, the classification criteria are not met)
Acute toxicity (inhalation) Not classified (Based on available data, the classification criteria are not met)

isobutane (75-28-5)	
LC50 Inhalation - Rat [ppm]	> 18000 ppm

propene (115-07-1)	
LC50 Inhalation - Rat	> 688 mg/m ³

Propane (74-98-6)	
LC50 Inhalation - Rat [ppm]	> 280000 ppm (literature)

Skin corrosion/irritation Not classified (Based on available data, the classification criteria are not met)
Serious eye damage/irritation Not classified (Based on available data, the classification criteria are not met)
Respiratory or skin sensitisation Not classified (Based on available data, the classification criteria are not met)
Germ cell mutagenicity Not classified (Based on available data, the classification criteria are not met)
Carcinogenicity Not classified (Based on available data, the classification criteria are not met)

propene (115-07-1)	
IARC group	3 - Not classifiable

Reproductive toxicity Not classified (Based on available data, the classification criteria are not met)
STOT-single exposure Not classified (Based on available data, the classification criteria are not met)
STOT-repeated exposure Not classified (Based on available data, the classification criteria are not met)

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Aspiration hazard	Not applicable
Viscosity, kinematic	No data available
Potential adverse human health effects and symptoms	No additional information available. No harmful effects are to be expected if used properly. The contained ingredients can be harmful, but they are hermetically enclosed in the article and can not be released. The dismantling of the article is prohibited.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general	Due to the consistency along with the low water solubility of the product a bioavailability is unlikely.
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isobutane (75-28-5)	
LC50 - Fish [1]	24.11 – 147.54 mg/l (Quantitative structure-activity relationship (QSAR))
EC50 - Crustacea [1]	7.02 – 69.43 mg/l (Quantitative structure-activity relationship (QSAR))
ErC50 algae	7.71 – 16.5 mg/l (Quantitative structure-activity relationship (QSAR))
propene (115-07-1)	
LC50 - Fish [1]	43.3 mg/l (72 h; Oncorhynchus mykiss (Rainbow trout); Quantitative structure-activity relationship (QSAR))
EC50 - Crustacea [1]	28.2 mg/l (48 h; daphnia; Quantitative structure-activity relationship (QSAR))
EC50 96h - Algae [1]	12.1 mg/l (algae; Quantitative structure-activity relationship (QSAR))

12.2. Persistence and degradability

isobutane (75-28-5)	
Persistence and degradability	Readily biodegradable.
propene (115-07-1)	
Persistence and degradability	Readily biodegradable in water.
Propane (74-98-6)	
Persistence and degradability	Readily biodegradable in water.

12.3. Bioaccumulative potential

isobutane (75-28-5)	
Partition coefficient n-octanol/water (Log Pow)	1.09 – 2.8 (20 °C)
Bioaccumulative potential	Bioaccumulation unlikely.
propene (115-07-1)	
Partition coefficient n-octanol/water (Log Kow)	1.77 (20 °C)
Bioaccumulative potential	Low bioaccumulation potential (Log Kow < 4).
Propane (74-98-6)	
Bioaccumulative potential	Low bioaccumulation potential (Log Kow < 4).

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12.4. Mobility in soil

No additional information available

12.5. Other adverse effects

Effect on global warming	No known effects from this product.
Other information	Avoid release to the environment.

SECTION 13: Disposal considerations

13.1. Disposal methods

Regional waste regulation	Disposal must be done according to official regulations.
Waste treatment methods	Dispose of contents/container in accordance with licensed collector's sorting instructions.
Product/Packaging disposal recommendations	Container under pressure. Do not drill or burn even after use.
Additional information	Flammable vapours may accumulate in the container.

SECTION 14: Transport information

In accordance with ADR / IMDG / IATA / RID

ADR	IMDG	IATA	RID
14.1. UN number or ID number			
UN 3150	UN 3150	UN 3150	UN 3150
14.2. UN proper shipping name			
HYDROCARBON GAS REFILLS FOR SMALL DEVICES	HYDROCARBON GAS REFILLS FOR SMALL DEVICES	Hydrocarbon gas Refills for small devices	HYDROCARBON GAS REFILLS FOR SMALL DEVICES
Transport document description			
UN 3150 HYDROCARBON GAS REFILLS FOR SMALL DEVICES, 2.1, (D)	UN 3150 HYDROCARBON GAS REFILLS FOR SMALL DEVICES, 2.1	UN 3150 Hydrocarbon gas Refills for small devices, 2.1	UN 3150 HYDROCARBON GAS REFILLS FOR SMALL DEVICES, 2.1
14.3. Transport hazard class(es)			
2.1	2.1	2.1	2.1
			
14.4. Packing group			
Not applicable	Not applicable	Not applicable	Not applicable
14.5. Environmental hazards			
Dangerous for the environment: No	Dangerous for the environment: No Marine pollutant: No	Dangerous for the environment: No	Dangerous for the environment: No
No supplementary information available			

14.6. Special precautions for user

Overland transport

Classification code (ADR) 6F

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Limited quantities (ADR)	0
Excepted quantities (ADR)	E0
Packing instructions (ADR)	P209
Mixed packing provisions (ADR)	MP9
Transport category (ADR)	2
Special provisions for carriage - Loading, unloading and handling (ADR)	CV9
Special provisions for carriage - Operation (ADR)	S2
Tunnel restriction code (ADR)	D

Transport by sea

Limited quantities (IMDG)	0
Excepted quantities (IMDG)	E0
Packing instructions (IMDG)	P003
EmS-No. (Fire)	F-D
EmS-No. (Spillage)	S-U
Stowage category (IMDG)	B
Stowage and handling (IMDG)	SW2
MFAG-No	115

Air transport

PCA Excepted quantities (IATA)	E0
PCA Limited quantities (IATA)	Forbidden
PCA limited quantity max net quantity (IATA)	Forbidden
PCA packing instructions (IATA)	201
PCA max net quantity (IATA)	1kg
CAO packing instructions (IATA)	201
CAO max net quantity (IATA)	15kg
Special provisions (IATA)	A802
ERG code (IATA)	10L

Rail transport

Classification code (RID)	6F
Limited quantities (RID)	0
Excepted quantities (RID)	E0
Packing instructions (RID)	P209
Mixed packing provisions (RID)	MP9
Transport category (RID)	2
Special provisions for carriage - Loading, unloading and handling (RID)	CW9
Colis express (express parcels) (RID)	CE2
Hazard identification number (RID)	23

14.7. Maritime transport in bulk according to IMO instruments

Not applicable

SECTION 15: Regulatory information

15.1. US Federal regulations

All components of this product are present and listed as Active on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory

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Chemical(s) subject to the reporting requirements of Section 313 or Title III of the Superfund Amendments and Reauthorization Act (SARA) of 1986 and 40 CFR Part 372.		
propene	CAS-No. 115-07-1	20 - <30%

15.2. International regulations

CANADA

isobutane (75-28-5)
Listed on the Canadian DSL (Domestic Substances List)

propene (115-07-1)
Listed on the Canadian DSL (Domestic Substances List)

Propane (74-98-6)
Listed on the Canadian DSL (Domestic Substances List)

EU-Regulations

No additional information available

National regulations

No additional information available

15.3. US State regulations

California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm

Component	State or local regulations
isobutane(75-28-5)	U.S. - Delaware - Pollutant Discharge Requirements - Reportable Quantities; U.S. - Massachusetts - Right To Know List; U.S. - New Jersey - Right to Know Hazardous Substance List; U.S. - New York City - Right to Know Hazardous Substances List; U.S. - Pennsylvania - RTK (Right to Know) List
propene(115-07-1)	U.S. - Delaware - Pollutant Discharge Requirements - Reportable Quantities; U.S. - Massachusetts - Right To Know List; U.S. - New Jersey - Right to Know Hazardous Substance List; U.S. - New York City - Right to Know Hazardous Substances List; U.S. - Pennsylvania - RTK (Right to Know) List
Propane(74-98-6)	U.S. - Delaware - Pollutant Discharge Requirements - Reportable Quantities; U.S. - Massachusetts - Right To Know List; U.S. - New Jersey - Right to Know Hazardous Substance List; U.S. - New York City - Right to Know Hazardous Substances List; U.S. - Pennsylvania - RTK (Right to Know) List

SECTION 16: Other information

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision date

07/29/2024

Data sources

European Chemicals Agency, <http://echa.europa.eu/>. manufacturer.

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Training advice	Department issuing data specification sheet.
Other information	NFPA 30B.

Full text of H-statements	
H220	Extremely flammable gas.
H280	Contains gas under pressure; may explode if heated.

Abbreviations and acronyms	
CAS-No.	Chemical Abstract Service number
ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute Toxicity Estimate
CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
DNEL	Derived-No Effect Level
EC50	Median effective concentration
ED	Endocrine disrupting properties
EC-No.	European Community number
EN	European Standard
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
IOELV	Indicative Occupational Exposure Limit Value
LC50	Median lethal concentration
LD50	Median lethal dose
NOEC	No-Observed Effect Concentration
OECD	Organisation for Economic Co-operation and Development
N.O.S.	Not Otherwise Specified
OEL	Occupational Exposure Limit
PBT	Persistent Bioaccumulative Toxic
PNEC	Predicted No-Effect Concentration
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
SDS	Safety Data Sheet
STP	Sewage treatment plant
TLM	Median Tolerance Limit
TRGS	Technical Rules for Hazardous Substances

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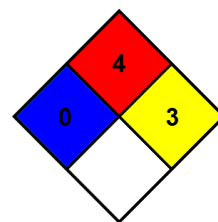
Abbreviations and acronyms	
VOC	Volatile Organic Compounds
WGK	Water Hazard Class
vPvB	Very Persistent and Very Bioaccumulative
NOAEL	No-Observed Adverse Effect Level
NOAEC	No-Observed Adverse Effect Concentration
LOAEL	Lowest Observed Adverse Effect Level

NFPA health hazard

NFPA fire hazard

NFPA reactivity

0 - Materials that, under emergency conditions, would offer no hazard beyond that of ordinary combustible materials.
 4 - Materials that rapidly or completely vaporize at atmospheric pressure and normal ambient temperature or that are readily dispersed in air and burn readily.
 3 - Materials that in themselves are capable of detonation or explosive decomposition or explosive reaction but that require a strong initiating source or must be heated under confinement before initiation.



Indication of changes:			
Section	Changed item	Change	Comments
1	Department issuing data specification sheet:	Modified	E-mail address of competent person responsible for the SDS
9	Physical and chemical properties	Added	NFPA 30B
16	Other data	Added	NFPA 30B

SDS US HILTI

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Revision date: 10/28/2025

Issue date: 10/28/2025

Supersedes: 10/30/2024

Version: 2.0

SECTION 1: Identification

1.1. Identification

Product form	Mixture
Trade name	FS-ONE MAX / CFS-FIL
Product code	BU Fire Protection



1.2. Other means of identification

No additional information available

1.3. Recommended use of the chemical and restrictions on use

Use of the substance/mixture	Firestop intumescent sealant
Recommended use	Adhesives, sealants

1.4. Supplier's details

Supplier

Hilti, Inc.
Legacy Tower, Suite 1000
7250 Dallas Parkway
US TX 75024 Plano
USA
T +1 9724035800
1-800-879-8000 toll free, F +1 918 254 0522
us-sales@hilti.com

Department issuing data specification sheet

Hilti AG
Feldkircher Strasse 100
FL 9494 Schaan
Liechtenstein
T +423 234 2111
product.compliance-fire.protection@hilti.com

1.5. Emergency phone number

Emergency number	Emergency CONTACT (24-Hour-Number) GBK/Infotrac ID 101022 (USA domestic) 1 800 535 5053 or international (001) 352 323 3500
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SECTION 2: Hazard(s) identification

2.1. Classification of the substance or mixture

GHS-US classification

Not classified

2.2. GHS Label elements, including precautionary statements

GHS US labelling

No labelling applicable

2.3. Hazards associated with known or reasonably anticipated uses

No additional information available

2.4. Hazards not otherwise classified

No additional information available

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

2.5. Unknown acute toxicity (GHS US)

Not applicable

SECTION 3: Composition/information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Name	Product identifier	%	GHS-US classification
Kaolin	CAS-No.: 1332-58-7	10 – 25	Not classified
fiberglass	CAS-No.: 65997-17-3	10 – 25	Not classified
Quartz (SiO ₂)	CAS-No.: 14808-60-7	2.5 – 5	Carc. 1A, H350 STOT RE 1, H372

Full text of hazard classes and H-statements : see section 16

SECTION 4: First-aid measures

4.1. Description of first aid measures

First-aid measures general	Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).
First-aid measures after inhalation	Get medical advice/attention if you feel unwell. Allow affected person to breathe fresh air. Allow the victim to rest.
First-aid measures after skin contact	Wash skin with plenty of water. If skin irritation occurs: Get medical advice/attention. Remove affected clothing and wash all exposed skin area with mild soap and water, followed by warm water rinse.
First-aid measures after eye contact	Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. Rinse immediately with plenty of water. Obtain medical attention if pain, blinking or redness persists.
First-aid measures after ingestion	Get medical advice/attention if you feel unwell. Rinse mouth. Do NOT induce vomiting. Obtain emergency medical attention.

4.2. Most important symptoms and effects (acute and delayed)

Potential adverse human health effects and symptoms	Based on available data, the classification criteria are not met.
Symptoms/effects	Not expected to present a significant hazard under anticipated conditions of normal use.

4.3. Immediate medical attention and special treatment, if necessary

No additional information available

SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

Suitable extinguishing media	Water spray. Dry powder. Foam. Carbon dioxide. Sand.
Unsuitable extinguishing media	Do not use a heavy water stream.

5.2. Specific hazards arising from the chemical

Hazardous decomposition products in case of fire	Carbon dioxide. Carbon monoxide.
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FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

5.3. Special protective equipment and precautions for fire-fighters

Firefighting instructions	Use water spray or fog for cooling exposed containers. Exercise caution when fighting any chemical fire. Prevent fire fighting water from entering the environment.
Protection during firefighting	Self-contained breathing apparatus. Complete protective clothing. Do not enter fire area without proper protective equipment, including respiratory protection.

SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

For non-emergency personnel

Emergency procedures	Evacuate unnecessary personnel.
----------------------	---------------------------------

For emergency responders

Protective equipment	For further information refer to section 8: "Exposure controls/personal protection". Equip cleanup crew with proper protection.
Emergency procedures	Ventilate area.
Environmental precautions	Prevent entry to sewers and public waters. Notify authorities if liquid enters sewers or public waters.

6.2. Methods and materials for containment and cleaning up

Methods for cleaning up	Mechanically recover the product. On land, sweep or shovel into suitable containers. Minimise generation of dust. Store away from other materials.
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For further information refer to section 13, See Section 8, Exposure controls and personal protection

SECTION 7: Handling and storage

7.1. Precautions for safe handling

Precautions for safe handling	Wear personal protective equipment. Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work. Provide good ventilation in process area to prevent formation of vapour.
Hygiene measures	Wash contaminated clothing before reuse. Do not eat, drink or smoke when using this product. Always wash hands after handling the product.

7.2. Conditions for safe storage, including any incompatibilities

Storage conditions	Keep cool. Store in a dry place. Keep only in the original container in a cool, well ventilated place away from : Keep container closed when not in use.
Incompatible products	Strong bases. Strong acids.
Incompatible materials	Sources of ignition. Direct sunlight.
Storage temperature	41 – 77 °F

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

FS-ONE MAX / CFS-FIL
No additional information available

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Quartz (SiO ₂) (14808-60-7)	
No additional information available	
USA - ACGIH - Occupational Exposure Limits	
Local name	Silica, crystalline, quartz
ACGIH® TLV® TWA	0.025 mg/m ³ (R - Respirable particulate matter)
Remark (ACGIH®)	TLV® Basis: Pulm fibrosis; lung cancer. Notations: A2 (Suspected Human Carcinogen)
Regulatory reference	ACGIH 2025
USA - OSHA - Occupational Exposure Limits	
Local name	Silica, crystalline quartz, respirable dust
Remark (OSHA)	(3) See Table Z-3.
Regulatory reference (US-OSHA)	OSHA Annotated Table Z-3 Mineral Dusts
fiberglass (65997-17-3)	
No additional information available	
USA - ACGIH - Occupational Exposure Limits	
ACGIH® TLV® TWA	1 fibers/cm ³ (Respirable fibers: length > 5 µm; aspect ratio ≥ 3:1, as determined by the membrane filter method at 400-450X magnification (4-mm objective), using phase-contrast illumination) 1 fibers/cm ³ (Respirable fibers: length > 5 µm; aspect ratio ≥ 3:1, as determined by the membrane filter method at 400-450X magnification (4-mm objective), using phase-contrast illumination) 5 mg/m ³ (Inhalable fraction)
Kaolin (1332-58-7)	
No additional information available	
USA - ACGIH - Occupational Exposure Limits	
Local name	Kaolin
ACGIH® TLV® TWA	2 mg/m ³ (Respirable fraction. The value is for particulate matter containing no asbestos and < 1% crystalline silica)
Remark (ACGIH®)	TLV® Basis: Pneumoconiosis. Notations: A4 (Not classifiable as a Human Carcinogen)
Regulatory reference	ACGIH 2025
USA - OSHA - Occupational Exposure Limits	
Local name	Kaolin
OSHA PEL TWA	15 mg/m ³ (Total dust) 5 mg/m ³ (Respirable fraction)
Regulatory reference (US-OSHA)	OSHA Annotated Table Z-1

Additional information The product has a pasty consistency. Exposure limit values for respirable dusts are not relevant for this product.

8.2. Appropriate engineering controls

No additional information available

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

8.3. Individual protection measures/Personal protective equipment

Personal protective equipment:

Protective clothing. Safety glasses. Gloves. Avoid all unnecessary exposure.

Hand protection:				
Wear suitable gloves tested to EN374. Suitable for short-term work or as a splash guard: Nitrile rubber gloves (> 0.1 mm). In case of permanent product contact:				
Type	Material	Permeation	Thickness (mm)	Penetration
Disposable gloves	Nitrile rubber (NBR)	6 (> 480 minutes)	>0,4	
Eye protection:				
Chemical goggles or safety glasses				
Skin and body protection:				
Wear suitable protective clothing				
Respiratory protection:				
When used in accordance with the instructions for use, no dust formation is expected. In case of dust formation use respirator with filter: P2				

Personal protective equipment symbol(s):



Other information:

Do not eat, drink or smoke during use.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	Solid
Appearance	Pasty.
Colour	red
Odour	characteristic
Odour threshold	Not determined
pH	≈ 7.85
Melting point	Not applicable
Freezing point	No data available
Boiling point	No data available
Flash point	Not applicable
Relative evaporation rate (butylacetate=1)	No data available
Flammability (solid, gas)	Not applicable. Non flammable.
Vapour pressure	No data available
Relative vapour density at 20°C	No data available
Relative density	No data available
Density	≈ 1.35 g/cm ³
Molecular mass	Not determined
Solubility	No data available
Partition coefficient n-octanol/water (Log Pow)	No data available
Auto-ignition temperature	No data available

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Decomposition temperature	No data available
Viscosity, kinematic	No data available
Viscosity, dynamic	No data available
Explosive limits	No data available
Explosive properties	No data available
Oxidising properties	No data available

9.2. Other information

VOC content	< 1 g/l ASTM D 2369 – 20, SCAQMD 1168 / All Other Architectural Sealants (Ilimit 50g/L)
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SECTION 10: Stability and reactivity

10.1. Reactivity

The product is non-reactive under normal conditions of use, storage and transport.

10.2. Chemical stability

Stable under normal conditions. Not established.

10.3. Possibility of hazardous reactions

No dangerous reactions known under normal conditions of use. Not established.

10.4. Conditions to avoid

None under recommended storage and handling conditions (see section 7). Direct sunlight. Extremely high or low temperatures.

10.5. Incompatible materials

Strong acids. Strong bases.

10.6. Hazardous decomposition products

Under normal conditions of storage and use, hazardous decomposition products should not be produced. fume. Carbon monoxide. Carbon dioxide.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral)	Not classified
Acute toxicity (dermal)	Not classified
Acute toxicity (inhalation)	Not classified
Skin corrosion/irritation	Not classified pH: ≈ 7.85
Serious eye damage/irritation	Not classified pH: ≈ 7.85
Respiratory or skin sensitisation	Not classified
Germ cell mutagenicity	Not classified
Carcinogenicity	Not classified

Quartz (SiO ₂) (14808-60-7)	
IARC group	1 - Carcinogenic to humans
National Toxicology Program (NTP) Status	Known Human Carcinogens

Reproductive toxicity	Not classified
STOT-single exposure	Not classified
STOT-repeated exposure	Not classified

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Quartz (SiO ₂) (14808-60-7)	
STOT-repeated exposure	Causes damage to organs through prolonged or repeated exposure.
Aspiration hazard	Not classified
Viscosity, kinematic	No data available
Potential adverse human health effects and symptoms	Based on available data, the classification criteria are not met.
Symptoms/effects	Not expected to present a significant hazard under anticipated conditions of normal use.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general	The product is not considered harmful to aquatic organisms nor to cause long-term adverse effects in the environment.
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12.2. Persistence and degradability

FS-ONE MAX / CFS-FIL	
Persistence and degradability	Not established.
Quartz (SiO ₂) (14808-60-7)	
Not rapidly degradable	
Persistence and degradability	Biodegradability: not applicable.
Chemical oxygen demand (COD)	Not applicable (inorganic)
ThOD	Not applicable (inorganic)
fiberglass (65997-17-3)	
Not rapidly degradable	
Persistence and degradability	Biodegradability: not applicable.
Chemical oxygen demand (COD)	Not applicable
ThOD	Not applicable
BOD (% of ThOD)	Not applicable
Kaolin (1332-58-7)	
Persistence and degradability	Biodegradability: not applicable.
Chemical oxygen demand (COD)	Not applicable (inorganic)
ThOD	Not applicable (inorganic)

12.3. Bioaccumulative potential

FS-ONE MAX / CFS-FIL	
Bioaccumulative potential	Not established.
Quartz (SiO ₂) (14808-60-7)	
Bioaccumulative potential	No bioaccumulation data available.

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

fiberglass (65997-17-3)	
Bioaccumulative potential	No bioaccumulation data available.

Kaolin (1332-58-7)	
Bioaccumulative potential	No bioaccumulation data available.

12.4. Mobility in soil

Quartz (SiO ₂) (14808-60-7)	
Surface tension	No data available in the literature
Ecology - soil	Low potential for mobility in soil.

fiberglass (65997-17-3)	
Ecology - soil	No (test)data on mobility of the substance available.

Kaolin (1332-58-7)	
Ecology - soil	No (test)data on mobility of the substance available.

12.5. Other adverse effects

Other information Avoid release to the environment.

SECTION 13: Disposal considerations

13.1. Disposal methods

Waste treatment methods	Dispose in a safe manner in accordance with local/national regulations.
Product/Packaging disposal recommendations	Dispose in a safe manner in accordance with local/national regulations.
Ecological waste information	Avoid release to the environment.

SECTION 14: Transport information

In accordance with DOT / TDG / IMDG / IATA

DOT	TDG	IMDG	IATA
14.1. UN number			
Not regulated for transport			
14.2. Proper Shipping Name			
Not applicable	Not applicable	Not applicable	Not applicable
14.3. Transport hazard class(es)			
Not applicable	Not applicable	Not applicable	Not applicable
14.4. Packing group			
Not applicable	Not applicable	Not applicable	Not applicable
14.5. Environmental hazards			
Dangerous for the environment: No	Dangerous for the environment: No	Dangerous for the environment: No Marine pollutant: No	Dangerous for the environment: No
No supplementary information available			

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

14.6. Special precautions for user

DOT

No data available

TDG

No data available

IMDG

No data available

IATA

No data available

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Not applicable

SECTION 15: Regulatory information

15.1. US Federal regulations

All components of this product are present and listed as Active on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory

15.2. International regulations

No additional information available

15.3. US State regulations

California Proposition 65 - This product does not contain any substances known to the state of California to cause cancer, developmental and/or reproductive harm

SECTION 16: Other information

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Revision date

10/28/2025

Data sources

REGULATION (EC) No 1272/2008 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006.

Other information

None.

Full text of hazard classes and H-statements	
H350	May cause cancer.
H372	Causes damage to organs through prolonged or repeated exposure

Abbreviations and acronyms	
CAS-No.	Chemical Abstract Service number
ADN	European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
ADR	European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE	Acute Toxicity Estimate

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Abbreviations and acronyms	
BCF	Bioconcentration factor
BLV	Biological limit value
BOD	Biochemical oxygen demand (BOD)
CLP	Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008
DMEL	Derived Minimal Effect level
DNEL	Derived-No Effect Level
EC-No.	European Community number
EC50	Median effective concentration
ED	Endocrine disruptor
EN	European Standard
IARC	International Agency for Research on Cancer
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
IOELV	Indicative Occupational Exposure Limit Value
LC50	Median lethal concentration
LD50	Median lethal dose
LOAEL	Lowest Observed Adverse Effect Level
N.O.S.	Not Otherwise Specified
NOAEC	No-Observed Adverse Effect Concentration
NOAEL	No-Observed Adverse Effect Level
NOEC	No-Observed Effect Concentration
vPvB	Very Persistent and Very Bioaccumulative
WGK	Water Hazard Class
VOC	Volatile Organic Compounds
SDS	Safety Data Sheet
RID	Regulations concerning the International Carriage of Dangerous Goods by Rail
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation (EC) No 1907/2006
PNEC	Predicted No-Effect Concentration
PBT	Persistent Bioaccumulative Toxic
OEL	Occupational Exposure Limit
OECD	Organisation for Economic Co-operation and Development
COD	Chemical oxygen demand (COD)
ThOD	Theoretical oxygen demand (ThOD)

FS-ONE MAX / CFS-FIL

Safety Data Sheet

according to 29 CFR § 1910.1200, Hazard Communication Standard (HCS)

Abbreviations and acronyms	
TRGS	Technical Rules for Hazardous Substances
TLM	Median Tolerance Limit
STP	Sewage treatment plant
ACGIH	American Conference of Government Industrial Hygienists
CSA	Chemical safety assessment
EWC	European waste catalogue
Log Kow	Partition coefficient n-octanol/water (Log Kow)
Log Pow	Partition coefficient n-octanol/water (Log Pow)
MAK	maximum workplace concentration
OSHA	Occupational Safety Health Administration
PPE	Personal protection equipment
TF	Technical function
TWA	Time Weighted Average
UFI	Unique Formula Identifier

NFPA health hazard

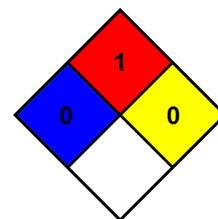
NFPA fire hazard

NFPA reactivity

0 - Materials that, under emergency conditions, would offer no hazard beyond that of ordinary combustible materials.

1 - Materials that must be preheated before ignition can occur.

0 - Material that in themselves are normally stable, even under fire conditions.



Hazard Rating

Health

Flammability

Physical

0 Minimal Hazard - No significant risk to health

0 Minimal Hazard - Materials that will not burn

0 Minimal Hazard - Materials that are normally stable, even under fire conditions, and will NOT react with water, polymerize, decompose, condense, or self-react. Non-Explosives.

Personal protection

B - Safety glasses, Gloves

Indication of changes:			
Section	Changed item	Change	Comments
			29 CFR § 1910.1200, Hazard Communication Standard (HCS)

SDS_US_Hilti

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.



Safety Data Sheet
OSHA Hazard Communication Standard
29 CFR 1910.1200. Prepared to GHS Rev 3.

Date of issue: 04.09.2023

Page: 1/7

Trade name: Clear Glide Wire Pulling Lubricant

SECTION 1: Identification

Product identifier used on the label:

Product Name: Clear Glide Wire Pulling Lubricant

Other means of identification:

Product Code Number: 31-388(-6), 31-381, 31-385(G), 31-2143

Recommended use of the chemical and restrictions on use:

Recommended use: Wire Pulling Lubricant

Recommended restrictions: Uses other than those described above.

Name, address, and telephone number of the chemical manufacturer, importer, or other responsible party:

Company Name: IDEAL INDUSTRIES, INC.

Company Address: Becker Place,
Sycamore, IL 60178

Company Telephone: Office hours (Mon – Fri)
7AM - 5 PM (CDT)
(815)895-5181

Company Contact Name: Darryl Docter.

Company Contact Email: IDEAL@IDEALINDUSTRIES.COM

Emergency phone number: 24 HOUR EMERGENCY NUMBER:
(815)895-5181.

SECTION 2: Hazard(s) identification

Classification of the chemical in accordance with paragraph (d) of §1910.1200:

Not classified as hazardous under OSHA HSC 2012

GHS Signal word: None Required

GHS Hazard statement(s): None required

GHS Hazard symbol(s): None required

GHS Precautionary statement(s): None required

Hazard(s) not otherwise

Classified (HNOC): None known

Percentage of ingredient(s) of unknown acute toxicity:

Not applicable

Clear Glide Wire Pulling Lubricant

SECTION 3: Composition/information on ingredients

None of chemical raw materials contained in this formulation are considered hazardous under the OSHA Hazard Communication Standard and therefore do not need reporting in this section.

SECTION 4: First-aid measures

Description of necessary measures, subdivided according to the different routes of exposure, i.e., inhalation, skin and eye contact, and ingestion:

Inhalation: Move affected person to fresh air and keep warm and at rest in a position comfortable for breathing. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Oxygen should only be administered by qualified personnel. Seek medical advice.

Skin contact: Wash with water and soap and rinse thoroughly. Seek medical advice if irritation or pain develops.

Eye contact: In case of contact with eyes, flush with water for at least 15 minutes. Assure adequate flushing by separating the eyelids with fingers. Call a physician if symptoms develop.

Ingestion: Do NOT induce vomiting. If swallowed, wash mouth out with water provided the person is conscious. NEVER GIVE LIQUIDS TO AN UNCONCIOUS PERSON. Call a physician if symptoms develop.

Most important symptoms/effects, acute and delayed:

None known

Indication of immediate medical attention and special treatment needed:

If any symptoms are observed, contact a physician and give them this SDS sheet. Treat symptomatically.

SECTION 5: Fire-fighting measures

Suitable (and unsuitable) extinguishing media:

Suitable extinguishing media: Dry chemical, foam, water spray, carbon dioxide.

Unsuitable extinguishing media: None known

Specific hazards arising from the chemical (e.g., nature of any hazardous combustion products):

None expected.

Hazardous combustion products may include the following substances: Carbon oxides

Special protective equipment and precautions for fire-fighters:

Use water spray or fog for cooling exposed containers. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Evacuate all non-emergency personnel from area. In addition, wear other appropriate protective equipment as conditions warrant (see Section 8).

Clear Glide Wire Pulling Lubricant

SECTION 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures:

No action shall be taken involving any personal risk or without suitable training. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through the spilled material. Minimize contact with skin or eyes. Provide adequate ventilation. Wear appropriate protective equipment, as conditions warrant (see Section 8). See Sections 2 and 7 for additional information on hazards and precautionary measures.

Environmental Precautions:

Stop spill/release if it can be done safely. Prevent spilled material from entering sewers, storm drains, other unauthorized drainage systems, and natural waterways.

Methods and material for containment and cleaning up:

Collect spillage and transfer to a lidded container for disposal or recovery. Dispose of via a licensed waste disposal contractor. See Section 1 for emergency contact information and Section 13 for waste disposal.

SECTION 7: Handling and storage

Precautions for safe handling:

Wear recommended personal protective equipment (See Section 8). Provide adequate ventilation in process areas. Avoid contact with eyes, skin and clothing. Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work.

Conditions for safe storage, including any incompatibles:

Keep only in original container. Store in a dry, well-ventilated place. Keep container closed when not in use. Make sure containers are properly labeled. Store away from incompatible materials. Store at 40-180 °F
Incompatible materials: Oxidizing agents.

SECTION 8: Exposure controls/personal protection

OSHA permissible exposure limit (PEL), American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Value (TLV), and any other exposure limit used or recommended by the chemical manufacturer, importer, or employer preparing the safety data sheet, where available.

None known.

Appropriate engineering controls:

Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits.

Provide eyewash station. Eye wash fountain and emergency showers are recommended. Concentrations should be monitored hazardous substances in the workplace in accordance with recognized test methods. Mode, method, type and frequency of testing and measurement of harmful factors in the working environment should meet the requirements of local/regional/national laws.

Clear Glide Wire Pulling Lubricant

Individual protection measures, such as personal protective equipment:

Eye/face protection: None required but the use of safety glasses is recommended. Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US).

Skin and Hand protection: None required but use of chemical resistant (rubber, nitrile) gloves is recommended. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

Respiratory protection: None required. Use supplied-air respiratory protection in enclosed spaces, if needed.

General hygiene considerations: Avoid contact with skin, eyes and clothing. When using, do not eat, drink or smoke. Wash hands before breaks and immediately after handling the product.

SECTION 9: Physical and chemical properties

Appearance (physical state, color, etc.):

Physical state: Gel.
Color: Clear, colorless.

Odor: Slight odor.

Odor threshold: Not determined.

pH: 7 - 8

Melting point/freezing point: Not determined.

Initial boiling point and boiling range: 212 °F (100 °C)

Flash point: None.

Evaporation rate: Not determined

Flammability (solid, gas): Not applicable

Upper/lower flammability or explosive limits

Flammability limit – lower (%): Not determined

Flammability limit – upper (%): Not determined

Explosive limit – lower (%): Not determined

Explosive limit – upper (%): Not determined

Vapor pressure: Not determined

Vapor density: Not determined

Relative density: 0.97 – 0.99

Solubility (ies): Not determined.

Partition coefficient (n-octanol/water): Not determined.

Auto-ignition temperature: Not determined

Decomposition temperature: Not determined

Viscosity: 25300-40300 cps

Clear Glide Wire Pulling Lubricant

SECTION 10: Stability and reactivity

Reactivity:	Not reactive under recommended storage and handling conditions.
Chemical stability:	Stable under recommended storage and handling conditions.
Possibility of hazardous reactions:	None known.
Conditions to avoid:	None known.
Incompatible materials:	Oxidizing agents
Hazardous decomposition Products:	None expected, although in a fire, Carbon oxides will be formed.

SECTION 11: Toxicological information

Information on likely routes of exposure:

Inhalation:	Expected to be a route of exposure
Ingestion:	Expected to be a route of exposure
Skin:	Expected to be a route of exposure
Eyes:	Expected to be a route of exposure

Target Organs: Eyes, Skin, Gastrointestinal system

Symptoms related to the physical, chemical, and toxicological characteristics:

None expected.

Delayed and immediate effects and chronic effects from short or long-term exposure:

None known.

Numerical measures of toxicity (such as acute toxicity estimates): None known.

Acute Toxicity:	Does not meet the criteria for classification
Skin corrosion/irritation:	Does not meet the criteria for classification
Serious eye damage/eye irritation:	Does not meet the criteria for classification
Respiratory sensitization:	Does not meet the criteria for classification
Skin sensitization:	Does not meet the criteria for classification
Germ cell mutagenicity:	Does not meet the criteria for classification
Carcinogenicity:	Does not meet the criteria for classification.
Reproductive toxicity:	Does not meet the criteria for classification
Specific target organ toxicity- Single exposure:	Does not meet the criteria for classification
Specific target organ toxicity- Repeat exposure:	Does not meet the criteria for classification,
Aspiration hazard:	Does not meet the criteria for classification

Whether the hazardous chemical is listed in the National Toxicology Program (NTP) Report on Carcinogens (latest edition) or has been found to be a potential carcinogen in the International Agency for Research on Cancer (IARC) Monographs (latest edition), or by OSHA: None listed.

Clear Glide Wire Pulling Lubricant

SECTION 12: Ecological information

Ecotoxicity (aquatic and terrestrial, where available):

Not expected to be toxic to the aquatic environment.

Persistence and Degradability:

No information available

Bioaccumulative Potential:

No information available.

Mobility in Soil:

No information available.

Other adverse effects (such as hazardous to the ozone layer):

No information available.

SECTION 13: Disposal considerations

Description of waste residues and information on their safe handling and methods of disposal, including the disposal of any contaminated packaging.

Product

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. Dispose via a licensed waste disposal contractor. Observe all federal, state/provincial and municipal regulations.

Contaminated packaging

Since emptied containers retain product residue, follow label warnings even after container is emptied. Empty containers should be properly labeled to supplier or everywhere there is a recovery program.

SECTION 14: Transport Information

US Department of Transportation Classification (49CFR)

Not regulated under DOT.

IMDG (Transport by sea)

Not regulated under IMDG.

IATA (Country variations may apply)

Not regulated under IATA.

Environmental hazards

Marine pollutant: No

Transport in bulk (according to Annex II of MARPOL 73/78 and the IBC Code)

No further relevant information available.

Special precautions which a user needs to be aware of, or needs to comply with, in connection with transport or conveyance either within or outside their premises.

None known

Clear Glide Wire Pulling Lubricant

SECTION 15: Regulatory Information

USA:

United States Federal Regulations: This SDS complies with the OSHA, 29 CFR 1910.1200. The product is not classified as hazardous under OSHA.

Toxic Substances Control Act (TSCA) - All substances in this product are listed, as required or are exempt from the TSCA inventory.

CERCLA RQ (lbs) Ingredients (> 0.1%):

None listed.

SARA Superfund and Reauthorization Act of 1986 Title III sections 302, 311, 312 and 313:

Section 302 Extremely Hazardous Substance (40 CFR 355, Appendix A) (> 0.1%):

No chemicals in this material are subject to the reporting requirements of SARA Title III, Section 302.

Section 311/312 (40 CFR 370) (> 0.1%):

Not applicable.

Section 313 Toxic Release Inventory (40 CFR 372) (> 0.1%):

None listed.

STATE REGULATIONS:

This SDS contains specific health and safety data is applicable for state requirements. For details on your regulatory requirements, you should contact the appropriate agency in your state.

California Proposition 65 (California Safe Drinking Water and Toxic Enforcement Act of 1986: None listed.

Massachusetts Right to Know:

No components are listed on the Massachusetts Right to Know list

New Jersey Right to Know:

No components are listed on the New Jersey Right to Know list

Pennsylvania Right to Know:

No components are listed on the Pennsylvania Right to Know list

SECTION 16: Other Information

Revision Date: April 9, 2023

To the best of our knowledge, the information contained herein is accurate. However IDEAL INDUSTRIES INC. does not assume any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards which exist.



Safety Data Sheet
OSHA Hazard Communication Standard
29 CFR 1910.1200. Prepared to GHS Rev 3.

Date of issue: 06.06.2023

Page: 1/8

Trade name: IDEAL Wipes Multi-Purpose Wipes

SECTION 1: Identification

Product identifier used on the label:

Product Name: IDEAL Wipes Multi-Purpose Wipes

Other means of identification:

Product Code Number: 38-500

Recommended use of the chemical and restrictions on use:

Recommended use: Cleaner.

Recommended restrictions: Uses other than those described above.

Name, address, and telephone number of the chemical manufacturer, importer, or other responsible party:

Company Name: IDEAL INDUSTRIES, INC.

Company Address: Becker Place,
Sycamore, IL 60178

Company Telephone: Office hours (Mon – Fri)
7AM - 5 PM (CDT)
(815)895-5181

Company Contact Name: Darryl Docter.

Company Contact Email: IDEAL@IDEALINDUSTRIES.COM

Emergency phone number: 24 HOUR EMERGENCY NUMBER:
(815)895-5181.

SECTION 2: Hazard(s) identification

Classification of the chemical in accordance with paragraph (d) of §1910.1200:

Not classified as hazardous under OSHA HSC 2012

GHS Signal word: None Required

GHS Hazard statement(s): None required

GHS Hazard symbol(s): None required

GHS Precautionary statement(s): None required

Hazard(s) not otherwise

Classified (HNO): None known

Percentage of ingredient(s) of unknown acute toxicity:

Not applicable

IDEAL Wipes Multi-Purpose Wipes

SECTION 3: Composition/information on ingredients

Chemical name	CAS#	Concentration (weight %)
Alcohols, C12-15, ethoxylated	68131-39-5	< 5%
Alcohols, C9-11, ethoxylated	68439-46-3	< 5%

NOTE: The balance of the ingredients is not classified as hazardous or are below the concentration limit to be classified as hazardous, under the criteria of the Federal OSHA Hazard Communication Standard 29CFR 1910.1200.

SECTION 4: First-aid measures

Description of necessary measures, subdivided according to the different routes of exposure, i.e., inhalation, skin and eye contact, and ingestion:

Inhalation: Move affected person to fresh air and keep warm and at rest in a position comfortable for breathing. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Oxygen should only be administered by qualified personnel. Call a physician if symptoms develop.

Skin contact: Wash immediately with plenty of water and soap for 15 minutes and rinse thoroughly. Remove clothing while washing. Call a physician if symptoms develop.

Eye contact: In case of contact with eyes, flush with water for at least 15 minutes. Assure adequate flushing by separating the eyelids with fingers. Do not apply neutralizing agents. Call a physician if symptoms develop.

Ingestion: Do NOT induce vomiting. If swallowed, wash mouth out with water provided the person is conscious. NEVER GIVE LIQUIDS TO AN UNCONCIOUS PERSON. Call a physician if symptoms develop.

Most important symptoms/effects, acute and delayed:
May cause eye irritation.

Indication of immediate medical attention and special treatment needed:
If any symptoms are observed, contact a physician and give them this SDS sheet. Treat symptomatically.

SECTION 5: Fire-fighting measures

Suitable (and unsuitable) extinguishing media:

Suitable extinguishing media: Alcohol-resistant foam, water spray or fog. Use fire-extinguishing media appropriate for surrounding materials.

Unsuitable extinguishing media: None known.

Specific hazards arising from the chemical (e.g., nature of any hazardous combustion products):

Harmful vapors may be released in a fire.

IDEAL Wipes Multi-Purpose Wipes

Hazardous combustion products may include the following substances: Carbon monoxide, Carbon dioxide.

Special protective equipment and precautions for fire-fighters:

Use water spray or fog for cooling exposed containers. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Evacuate all non-emergency personnel from area. In addition, wear other appropriate protective equipment as conditions warrant (see Section 8).

SECTION 6: Accidental release measures

Personal precautions, protective equipment and emergency procedures:

No action shall be taken involving any personal risk or without suitable training. Isolate the area. Evacuate personnel to safe areas. Approach from upwind. Ventilate the area. Keep away from incompatible products. Prevent further leakage or spillage if safe to do so.

Wear chemical resistant personal protective equipment, as conditions warrant (see Section 8). See Sections 2 and 7 for additional information on hazards and precautionary measures.

Environmental Precautions:

Stop spill/release if it can be done safely. Prevent spilled material from entering sewers, storm drains, other unauthorized drainage systems, and natural waterways.

Methods and material for containment and cleaning up:

Take up liquid spill with inert absorbent material. Scoop absorbed substance into containers. Flush contaminated areas with large amounts of water and direct wash waters to chemical sewer or collect for treatment. Dispose of via a licensed waste disposal contractor. See Section 1 for emergency contact information and Section 13 for waste disposal.

SECTION 7: Handling and storage

Precautions for safe handling:

Take precautionary measures against static discharges. Wear recommended personal protective equipment (See Section 8). Avoid eye and skin contact. Remove contaminated clothing. Keep away from incompatible products. Wash hands and other exposed areas with mild soap and water before eating, drinking or smoking and when leaving work.

Conditions for safe storage, including any incompatibles:

Store in a well-ventilated area. Store at ambient temperature. Keep container closed when not in use. Make sure containers are properly labeled.

Incompatible materials: Strong oxidizing agents.

SECTION 8: Exposure controls/personal protection

OSHA permissible exposure limit (PEL), American Conference of Governmental Industrial Hygienists (ACGIH) Threshold Limit Value (TLV), and any other exposure limit used or recommended by the chemical manufacturer, importer, or employer preparing the safety data sheet, where available.

IDEAL Wipes Multi-Purpose Wipes

Substance	OSHA PEL	ACGIH TLV	NIOSH IDLH
Alcohols, C12-15, ethoxylated	None known	None known	None known
Alcohols, C9-11, ethoxylated	None known	None known	None known

Appropriate engineering controls:

Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits.

Concentrations should be monitored hazardous substances in the workplace in accordance with recognized test methods. Mode, method, type and frequency of testing and measurement of harmful factors in the working environment should meet the requirements of local/regional/national laws.

Individual protection measures, such as personal protective equipment:

Eye/face protection: Wear safety glasses or chemical safety goggles. Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US).

Skin and Hand protection: Wear impervious gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

Respiratory protection: None normally required. In the case of dust or aerosol formation use respirator with an approved filter.

General hygiene considerations: Eye wash fountains should be located in the work areas. Take off contaminated clothing and shoes immediately. When using, do not eat, drink or smoke. Wash hands before breaks and immediately after handling the product.

SECTION 9: Physical and chemical properties

Appearance (physical state, color, etc.):

Physical state: Liquid.

Color: Clear.

Odor: Citrus.

Odor threshold: Not determined.

pH: 5.0 - 8.0

Melting point/freezing point: Not determined.

Initial boiling point and boiling range: 212 F.

Flash point: > 300 F.

Evaporation rate: Not determined.

Flammability (solid, gas): Not applicable

Upper/lower flammability or explosive limits

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Flammability limit – lower (%):	Not determined.
Flammability limit – upper (%):	Not determined.
Explosive limit – lower (%):	Not determined.
Explosive limit – upper (%):	Not determined.
Vapor pressure:	Not determined.
Vapor density:	Not determined.
Relative density:	1.
Solubility (ies):	Dispersible in water.
Partition coefficient (n-octanol/water):	Not determined.
Auto-ignition temperature:	Not determined.
Decomposition temperature:	Not determined.
Viscosity:	Not determined.

SECTION 10: Stability and reactivity

Reactivity:	Not expected to be chemically reactive.
Chemical stability:	Stable under recommended storage and handling conditions.
Possibility of hazardous reactions:	Hazardous reactions not anticipated.
Conditions to avoid:	None known.
Incompatible materials:	Avoid contact with strong oxidizers.
Hazardous decomposition Products:	None expected, however in case of fire, carbon oxides may be released.

SECTION 11: Toxicological information

Information on likely routes of exposure:

Inhalation:	Expected to be a route of exposure
Ingestion:	Expected to be a route of exposure
Skin:	Expected to be a route of exposure
Eyes:	Expected to be a route of exposure
Target Organs:	Eyes

Symptoms related to the physical, chemical, and toxicological characteristics:

May cause eye irritation.

Delayed and immediate effects and chronic effects from short or long-term exposure:

No additional information available.

Numerical measures of toxicity (such as acute toxicity estimates):

Substance	Test Type (species)	Value
Alcohols, C12-15, ethoxylated	LD ₅₀ Oral (Rat)	> 2000 mg/kg
	LD ₅₀ Dermal (Rabbit)	2500 mg/kg
	LC ₅₀ Inhalation (Rat)	None known

IDEAL Wipes Multi-Purpose Wipes

Alcohols, C9-11, ethoxylated	LD ₅₀ Oral (Rat)	3488 mg/kg
	LD ₅₀ Dermal (Rabbit)	> 2000 mg/kg
	LC ₅₀ Inhalation (Rat)	> 1600mg/m ³

Acute Toxicity:	Does not meet the criteria for classification.
Skin corrosion/irritation:	Does not meet the criteria for classification.
Serious eye damage/eye irritation:	Does not meet the criteria for classification.
Respiratory sensitization:	Does not meet the criteria for classification.
Skin sensitization:	Does not meet the criteria for classification.
Germ cell mutagenicity:	Does not meet the criteria for classification.
Carcinogenicity:	Does not meet the criteria for classification.
Reproductive toxicity:	Does not meet the criteria for classification.
Specific target organ toxicity-Single exposure:	Does not meet the criteria for classification.
Specific target organ toxicity-Repeat exposure:	Does not meet the criteria for classification.
Aspiration hazard:	Does not meet the criteria for classification.

Whether the hazardous chemical is listed in the National Toxicology Program (NTP) Report on Carcinogens (latest edition) or has been found to be a potential carcinogen in the International Agency for Research on Cancer (IARC) Monographs (latest edition), or by OSHA:

Component	IARC	NTP	ACGIH	OSHA
Alcohols, C12-15, ethoxylated	Not listed	Not listed	Not listed	Not listed
Alcohols, C9-11, ethoxylated	Not listed	Not listed	Not listed	Not listed

SECTION 12: Ecological information

Ecotoxicity (aquatic and terrestrial, where available):

Harmful to aquatic life with long lasting effects

Substance	Test Type	Species	Value
Alcohols, C12-15, ethoxylated	LC ₅₀	Fish - Pimephales promelas	0.628 mg/L 96h
	EC ₅₀	Aquatic Invertebrates - Daphnia magna	0.143 mg/L 48h
	EC ₅₀	Algae - Raphidocelis subcapitata	0.031 mg/L 72h

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Alcohols, C9-11, ethoxylated	LC ₅₀	Fish – Oncorhynchus mykiss	5 - 7 mg/L 96h
	EC ₅₀	Aquatic Invertebrates – Daphnia magna	2.5 mg/L 48h
	ErC ₅₀	Algae - Desmodesmus subspicatus	1.978 mg/L 72h

Persistence and Degradability:

Expected to be readily biodegradable

Bioaccumulative Potential:

Bioaccumulation in organisms is negligible, due to biotransformation and excretion of alcohol ethoxylates.

Mobility in Soil:

No data available

Other adverse effects (such as hazardous to the ozone layer):

No data available

SECTION 13: Disposal considerations

Description of waste residues and information on their safe handling and methods of disposal, including the disposal of any contaminated packaging.

Product

Dispose of waste materials in accordance with applicable local and national laws and regulations. Where possible, recycling is preferred to disposal or incineration. Contact the proper local authorities.

Contaminated packaging

Since emptied containers retain product residue, follow label warnings even after container is emptied. Dispose of as unused product.

SECTION 14: Transport Information

US Department of Transportation Classification (49CFR)

Not regulated under DOT.

IMDG (Transport by sea)

Not regulated under IMDG.

IATA (Country variations may apply)

Not regulated under IATA.

Environmental hazards

Marine pollutant: No

Transport in bulk (according to Annex II of MARPOL 73/78 and the IBC Code)

No further relevant information is available.

Special precautions which a user needs to be aware of, or needs to comply with, in connection with transport or conveyance either within or outside their premises.

IDEAL Wipes Multi-Purpose Wipes

None known.

SECTION 15: Regulatory Information

USA:

United States Federal Regulations: This SDS complies with the OSHA, 29 CFR 1910.1200. The product is NOT classified as hazardous under OSHA.

Toxic Substances Control Act (TSCA) - All components are listed on the TSCA inventory or are exempted.

CERCLA RQ (lbs) Ingredients (> 0.1%):

None of the components are listed.

SARA Superfund and Reauthorization Act of 1986 Title III sections 302, 311, 312 and 313:

Section 302 Extremely Hazardous Substance (40 CFR 355, Appendix A) (> 0.1%):

None of the components are listed.

Section 311/312 (40 CFR 370) (> 0.1%):

Not applicable

Section 313 Toxic Release Inventory (40 CFR 372) (> 0.1%):

None of the components are listed.

STATE REGULATIONS:

This SDS contains specific health and safety data that is applicable for state requirements. For details on your regulatory requirements, you should contact the appropriate agency in your state.

California Proposition 65 (California Safe Drinking Water and Toxic Enforcement Act of 1986: None of the components are listed.

Massachusetts Right to Know:

None of the components are listed on the Massachusetts Right to Know list.

New Jersey Right to Know:

None of the components are listed on the New Jersey Right to Know list.

Pennsylvania Right to Know:

None of the components are listed on the Pennsylvania Right to Know list.

SECTION 16: Other Information

Revision Date: June 6, 2023

To the best of our knowledge, the information contained herein is accurate. However IDEAL INDUSTRIES INC. does not assume any liability whatsoever for the accuracy or completeness of the information contained herein. Final determination of suitability of any material is the sole responsibility of the user. All materials may present unknown hazards and should be used with caution. Although certain hazards are described herein, we cannot guarantee that these are the only hazards which exist.



Product Information Sheet

May be used to comply with OSHA's Hazard Communication Standard 29 CFR 1910.1200.

This standard must be consulted for specific requirements.

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

Product Name: Lithium-ion Batteries - Rechargeable

Drawing Number: 58-97-0500

Issue Date: January 2018

Supersedes Date: April 2016

Milwaukee Electric Tool Corporation

13135 West Lisbon Road

Brookfield, Wisconsin USA 53005-2550

www.milwaukeeetool.com

Company Phone Number: 262-781-3600 or

1-800-729-3878

Emergency Contact Number: 1-800-424-9300

Chemtrec: United States only

For International: +1-703-741-5970

SECTION 2: HAZARDS IDENTIFICATION

Health	Environmental	Physical
Eye Irritation: No classified hazards	Acute Toxicity: No classified hazards	Flammable liquid: No classified hazards
Skin Irritation: No classified hazards	Chronic Toxicity: No classified hazards	
Acute Toxicity, Oral: No classified hazards		
Acute Toxicity, Inhalation: No classified hazards		

GHS Label

No applicable labeling

Hazard Statements	Precautionary Statements
No exposure during routine handling of product	

CLASSIFIED HAZARDS

This material is not hazardous under the criteria of the Federal OSHA Hazard Communication Standard 29CFR 1910.1200. This SDS contains valuable information for the safe handling and proper use of this product. Save this SDS for future reference.

OTHER HAZARDS

Flammable:

Organic components will burn if cell is incinerated. Combustion of cell contents may cause evolution of Hydrogen Fluoride.

Potential Health Effects:

Fluoride interferes with nerve impulse conduction causing severe pain or absence of sensations

WARNING:

No exposure during routine handling of product. Hydrofluoric Acid exposure during firefighting: This information is given for the use of professional fire fighters responding to a warehouse fire where fire from other materials may incinerate batteries. This section is provided solely in case of exposure, during firefighting, to the combustion by-products.

SECTION 3: COMPOSITION /INFORMATION OF INGREDIENTS

Chemical Name	CAS #	Concentration
Aluminum Foil	7429-90-5	0.1 - 10
Biphenyl (BP)	92-52-4	0.1 - 0.3
Copper Foil	7440-50-8	0.1 - 10
Linear & Cyclic Carbonate solvents	N/A	0 - 17
Graphite Powder/Carbon	7440-44-0	10 - 30
Metal Oxide or other Electrolyte (proprietary)	Confidential	10 - 50
Lithium Hexafluorophosphate (LiPF ₆)	21324-40-3	0 - 5
Polyvinylidene Flouride (PVDF)	24937-79-9	0.1 - 5
Styrene Butadiene Rubber (SBR)	N/A	<5
Aluminum, Steel, Nickel and other inert materials	N/A	Remainder

SECTION 4: FIRST AID MEASURES

No exposure during routine handling of product. Risk of exposure occurs only if the battery is mechanically or electrically abused.

No effect under routine handling and use to eyes, skin or if inhaled. Ingestion is not likely, given the physical size and state of the cell. If swallowed, seek medical attention immediately.

If exposure to internal materials within cell due to damaged outer casing the following actions are recommended:

EYE CONTACT:

Flush with water for 15 minutes without rubbing and immediately seek medical attention.

SKIN CONTACT:

Wash area immediately with soap and water. If irritation continues see medical attention.

INHALATION:

Leave area immediately and move to fresh air and seek medical attention.

INGESTION:

If swallowed, contact POISON CONTROL CENTER immediately.

SECTION 5: FIRE FIGHTING MEASURES

NFPA 704 Hazard Class



HMIS



0 (Minimal)
1 (Slight)
2 (Moderate)
3 (Serious)
4 (Severe)

SUITABLE EXTINGUISHING MEDIA:

Water spray, carbon dioxide, dry chemical powder or appropriate foam. Use agent appropriate for surrounding materials.

UNSUITABLE EXTINGUISHING MEDIA:

None.

PRODUCTS OF COMBUSTION:

Organic components will burn if incinerated. Combustion of cell contents may cause evolution of Hydrogen Fluoride. In case of fire in an adjacent area, use water, CO₂, or dry chemical extinguishers if cells are packed in their original containers since the fuel of the fire is basically paper products.

PROTECTION OF FIREFIGHTERS:

Hydrofluoric Acid exposure during firefighting: This information is given for the use of professional fire fighters responding to a warehouse fire where fire from other materials may incinerate batteries. This section is provided solely in case of exposure, during firefighting, to the combustion by-products.

SECTION 6: ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS:

Use standard industrial clothing in normal use. If handling large containers of cells wear steel-toed footwear.

ENVIRONMENTAL PRECAUTIONS:

No special precautions necessary.

METHODS FOR CONTAINMENT:

Transport container outdoors. Hold burned cells and fire cleanup solids for disposal as potential hazardous waste. Unburned cells are not hazardous waste. A fire with over 100 kg of cells burnt will likely require reporting to environmental officials. Always consult and obey all international, federal and local environmental laws.

METHODS FOR CLEAN-UP:

No data available

OTHER INFORMATION:

No data available

SECTION 7: HANDLING AND STORAGE

HANDLING:

Use only approved charging equipment. Do not disassemble battery or battery pack. Do not puncture, crush or dispose of in fire.

STORAGE:

Store in a cool, dry place away from sparks and flame. Keep below 125°C. Keep above -60°C. Charge between 0°C and 45°C.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Chemical Name	OSHA PEL	ACGIH TLV	California Prop 65 Reg. Y/N	IARC/NTP Y/N
Aluminum Foil	TWA 5mg/m ³ *	TWA 5mg/m ³ *	N	N
Biphenyl (BP)	NA	NA	N	N
Copper Foil	NA	NA	N	N
Linear & Cyclic Carbonate solvents	NA	NA	N	N
Graphite Powder/Carbon	NA	NA	N	N
Metal Oxide or other Electrolyte (proprietary)	NA	NA	N	N
Lithium Hexafluorophosphate (LiPF ₆)	NA	NA	N	N

Polyvinylidene Fluoride (PVDF)	NA	NA	N	N
Styrene Butadiene Rubber (SBR)	NA	NA	N	N
Aluminum, Steel, Nickel and other inert materials	NA	NA	N	N

EYE PROTECTION:

Not necessary under conditions of normal use

SKIN PROTECTION:

Not necessary under conditions of normal use

RESPIRATORY PROTECTION:

Not necessary under conditions of normal use

ENGINEERING CONTROLS:

Not necessary under conditions of normal use

GENERAL HYGIENE CONSIDERATIONS:

Not necessary under conditions of normal use

EXPOSURE GUIDELINES:

Not necessary under conditions of normal use

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Data represent typical values and are not intended to be specifications. NA=Not Applicable; ND=Not Determined

Physical state: Solid	Viscosity: NA
Colour: NA	Upper Explosive Limits (vol % in air): NA
Odor: Odorless	Lower Explosive Limits (vol % in air): NA
Odor Threshold: NA	Vapor pressure: NA
pH: NA	Vapor density: NA
Melting/Freezing Point: NA	Relative density: NA
VOC Content: NA	Solubility: NA
Boiling Point: NA	Partition Coefficient: NA
Flash Point: NA	Auto-ignition Temperature: NA
Evaporation Rate: NA	Decomposition Temperature: NA
Specific Gravity: NA	Flammability (solid, gas): Organic components will burn if cell is incinerated

SECTION 10: STABILITY AND REACTIVITY

INCOMPATIBLE MATERIALS:

Water, heat and strong acids.

DECOMPOSITION PRODUCTS MAY INCLUDE:

Hydrogen Fluoride, Phosphorus Oxides, Carbon Monoxide, Carbon Dioxide, Lithium Hydroxide, Manganese Oxides, Aluminum Oxide, possible fluoro-compounds, Carbon soot.

CONDITIONS TO AVOID:

Do not crush, puncture, incinerate, immerse in water or heat over 212°F (100°C). Steel casing slowly dissolves in strong mineral acids.

POLYMERIZATION:

Hazardous polymerization will not occur. Spontaneous decomposition will not occur at normal temperature.

CHEMICAL STABILITY:

This product is stable.

REACTIVITY:

Hazardous polymerization will not occur. Spontaneous decomposition will not occur at normal temperature.

SECTION 11: TOXICOLOGY INFORMATION

LIKELY ROUTES OF EXPOSURE: Inhalation, Eye and Skin contact

Eye contact, skin contact, skin absorption, inhalation only if burned. Hydrofluoric acid is extremely corrosive. Contact with hydrogen fluoride fumes is to be avoided. Permissible exposure limit is 3ppm. In case of contact with hydrogen fluoride fumes, immediately leave the area and seek first aid and emergency medical attention. Symptoms may have delayed onset. Fluoride ions penetrate skin readily causing destruction of deep tissue layers even bone. Fluoride interferes with nerve impulse conduction causing severe pain or absence of sensations. Immediately flush eyes or skin with water for at least 20 minutes to neutralize the acidity and remove some fluoride. Remove and destroy all contaminated clothing and permeable personal possessions. Before re-use, impermeable possessions should be soaked in benzalkonium chloride after washing. Following flushing of the affected areas, an iced aqueous solution of benzalkonium chloride or 2.5% calcium gluconate gel should be applied to react with the fluoride ion. Compresses and wraps may be used for areas where immersion is not practical. Medicated dressing should be changed every 2 minutes. Exposure to hydrofluoric acid fumes sufficient to cause pain requires immediate hospitalization for monitoring for pulmonary edema.

ACUTE SYMPTOMS AND EFFECTS:

Inhalation:	No further toxicological data known
Eye contact:	No further toxicological data known
Skin contact:	No further toxicological data known
Ingestion:	No further toxicological data known

OTHER:

No further data known.

SECTION 12: ECOLOGICAL INFORMATION

ECOTOXICOLOGICAL INFORMATION:

None in routine handling of product.

TOXICITY:

No data available

PERSISTENCE AND DEGRADABILITY (BIOPERSISTENCY & BIODEGRADABILITY):

None in routine handling of product.

POTENTIAL OF BIOACCUMULATION:

None in routine handling of product.

MOBILITY IN SOIL:

None in routine handling of product.

OTHER ADVERSE EFFECTS:

No data available

SECTION 13: DISPOSAL CONSIDERATIONS

DISPOSAL:

Dispose in accordance with appropriate regulations. Always consult and obey all international, federal, provincial/state and local hazardous waste disposal laws. Some jurisdictions require recycling of this spent product. Battery recycling is encouraged. Lithium ion batteries are safe for disposal in the normal municipal waste stream since they are not defined by the federal government as hazardous waste. However, Lithium ion batteries are recyclable.

This product does not contain mercury, cadmium or Lithium (metal).

DO NOT INCINERATE or subject battery cells to temperatures in excess of 212°F (100°C).

SECTION 14: TRANSPORTATION INFORMATION

U.S. DOT HAZARDOUS MATERIAL REGULATIONS (RE: GROUND TRANSPORT)

Proper Shipping Description:

UN3480 Lithium-ion batteries; UN3481 Lithium-ion batteries packed with or contained in equipment; Class 9.

Milwaukee Lithium-ion batteries are to be shipped in compliance with relevant requirements of HMR “49 CFR173.185”.

CANADA TRANSPORT DANGEROUS GOODS (RE: GROUND TRANSPORT)

Proper Shipping Description:

UN3480 Lithium-ion batteries; UN3481 Lithium-ion batteries packed with or contained in equipment; Class 9.

Milwaukee Lithium-ion batteries are to be shipped in compliance with relevant requirements of TDG “Part 2” (Section 2.43), or TDG “Schedule 2” (Special Provision 34), as applicable.

INTERNATIONAL DANGEROUS GOODS REGULATIONS (RE: AIR, SEA, GROUND TRANSPORT)

Proper Shipping Description:

UN3480 Lithium-ion batteries; UN3481 Lithium-ion batteries packed with or contained in equipment; Class 9.

Milwaukee Lithium-ion batteries are to be shipped in compliance with relevant requirements of the following DG Regulations:

- ICAO Technical Instructions or IATA Dangerous Goods Regulations (59th Edition): Packing Instructions 965; 966; 967 (Section I, or Section II, as applicable).
- IMDG Code: Packing Instruction P903, or Special Provision 188, as applicable.
- UN Model Regulations on the Transport of Dangerous Goods: Packing Instruction P903, or Special Provision 188, as applicable.
- UN European Agreements (ADR/RID/ADN): Packing Instruction P903, or Special Provision 188, as applicable.
- Australian Dangerous Goods (ADG): Packing Instruction P903, or Special Provision 188, as applicable.

IMPORTANT: The proper classification, packaging, labeling, marking, and documentation requirements for shipping Lithium-ion batteries is dependent upon whether the particular batteries are:

- a.) Rated at 100 Watt-hours (Wh) or less; or
- b.) Rated at greater than 100Wh.

Generally, Lithium-ion batteries rated 100Wh or less are “excepted” from certain Class 9 DG requirements. Always check compliance of Lithium-ion battery consignments against the current regulations governing the chosen mode of transport. When in doubt, contact the carrier or other trained Dangerous Goods professional to confirm acceptability.

UN 38.3 BATTERY TRANSPORTATION TESTING:

Milwaukee rechargeable Lithium-ion batteries listed in Section 1 have passed the relevant transportation test requirements as described in the UN *Manual of Tests and Criteria*, Part III, section 38.3.

UN 38.3 Test Reports are maintained on file at the corporate headquarters of Milwaukee Electric Tool Corporation located at 13135 W. Lisbon Rd., Brookfield, WI, USA 53005.

SECTION 15: REGULATORY INFORMATION

GLOBAL INVENTORIES

TSCA: United States	See Sec. 14. Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
DSL: Canada	See Sec. 14. Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
ECL: Korea	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
PICCS: Philippines	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
ENCS: Japan	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
AICS: Australia	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
IECS: China	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.
EINECS: European Union	Compliant with, relevant transportation test requirements as described in the UN Manual of Tests & Criteria, Part III, Sub-section 38.3.

SARA 313 Information:

SARA Title III Section 313: This product does not contain regulated levels of any toxic chemical subject to the reporting requirements of Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA) and 40 CFR part 372.

California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65)

This product does not contain regulated levels of any toxic chemical subject to the reporting requirements of California Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65)

WHMIS: Canadian Workplace

This product does not contain regulated levels of any toxic chemical subject to the reporting requirements

SECTION 16: OTHER INFORMATION

ABBREVIATIONS:

TSCA	Toxic Substance Control Act
ICAO	International Civil Aviation Organization
IMDG	International Maritime Dangerous
OSHA	Occupational Safety and Health
IARC/NTP	International Agency for Research on Cancer/National Toxicology Program
SARA	Superfund Amendments and Reauthorization Act of 1986
ACGIH	American Conference of Governmental Industrial Hygienists

NIOSH/MSHA National Institute for Occupational Safety Health/
Mine Safety and Health Administration
WHMIS Workplace Hazardous Materials Information System

Prepared by: Milwaukee Electric Tool Corporation

The batteries referenced herein are considered exempt articles and are not subject to the OSHA Hazard Communication Standard; therefore a SDS is not required. This sheet is being provided as a service to our customers.

The information and recommendations set forth are made in good faith and believed to be accurate as of the date of preparation. **MILWAUKEE ELECTRIC TOOL CORPORATION** makes no warranty, expressed or implied, regarding the accuracy of this data or the results to be obtained from the use thereto.



SAFE USE INSTRUCTION SHEET

Creation Date 28-Jul-2016

Revision Date 05-Jun-2023

Version 8

0. GENERAL INFORMATION

This Safe Use Instruction Sheet is the document provided by Owens Corning to communicate recommended safe handling and use instructions for manufactured articles neither regulated by OSHA Hazard Communication Standard, 29 CFR 1910.1200 nor by the Canada Hazardous Products Regulation SOR/2015-17 (WHMIS 2015)

1. IDENTIFICATION

Product Name	Bonded Mineral Wool (Unfaced)
Synonyms	Thermafiber® FireSpan® (40, 90 & 120), Thermafiber® Safing™, Thermafiber® FireLedge®, Thermafiber® RainBarrier®, HD, Thermafiber® RainBarrier® 45, Thermafiber® SAFB™ (Sound Attenuation Fire Blanket), Thermafiber® VersaBoard®, Thermafiber® TopStop®, Thermafiber® FixtureShield®, Thermafiber® U.S. Coast Guard Felt, Thermafiber® Industrial Felt, Thermafiber® Industrial Board, Thermafiber® Industrial Blanket, Thermafiber® Metal Mesh Blankets, Thermafiber® Industrial Fabrication Board, Thermafiber® Fire & Sound Guard® Plus, RainBarrier® ci HC 80, RainBarrier® ci HC Plus 110, RainBarrier® ci HC Max
Product code	OCMW00004
Recommended Use	Curtain Wall Insulation, Fire Containment Insulation, Thermafiber® Impasse® No Backer Bar™ System, Continuous Insulation, Sound Control Insulation, Commercial Insulation, Residential & Light Commercial Insulation, Head-of-Wall Insulation, Light Fixture Insulation, Maritime Insulation, Industrial Felt, Industrial Insulation, Semi-Refractory Felt, Horticulture
Manufacturer Address	Owens Corning Mineral Wool, LLC 3711 Mill St, Wabash, IN, 46992
Company Phone Number	1-800-GET-PINK or 1-800-438-7465
E-mail address	safetydatasheet@owenscorning.com
Company Website	http://owenscorning.com/

2. HAZARDS IDENTIFICATION

Regulatory Status	This product is considered an article. 29 CFR 1910.1200(c) definition of an article is as follows: "Article" means a manufactured item other than a fluid or particle: (i) which is formed to a specific shape or design during manufacture; (ii) which has end use function(s) dependent in whole or in part upon its shape or design during end use; and (iii) which under normal conditions of use does not release more than very small quantities, e.g., minute or trace amounts of a hazardous chemical (as determined under paragraph (d) of this section), and does not pose a physical hazard or health risk to employees This product is considered an article per the Canadian Hazardous Products Regulation SOR/2015-17 Manufactured articles which meet the definition of the Canadian Hazardous Products Act (any article that is formed to a specific shape or design during manufacture, the intended use of which when in that form is dependent in whole or in part on its shape or design, and that, when being installed, if the intended use of the article requires it to be installed, and under normal conditions of use, will not release or otherwise cause an individual to be exposed to a hazardous product) are not regulated by the Canadian Hazardous Products Regulation SOR/2015-17
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3. COMPOSITION/INFORMATION ON INGREDIENTS

The product contains no substances which at their given concentration, are considered to be hazardous to health

Mineral wool may cause temporary skin and mucous membranes itching due to the mechanical abrasion effects of fibers.

Chemical name	CAS No.	Weight-%	Trade Secret
Mineral Wool	65997-17-3	90-100	*

4. FIRST AID MEASURES

Description of First Aid Measures

- | | |
|---------------------|---|
| Eye contact | <ul style="list-style-type: none">• DO NOT rub or scratch eyes• Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes• If eye irritation persists: Get medical advice/attention |
| Skin contact | <ul style="list-style-type: none">• Wash off immediately with soap and plenty of cold water• DO NOT use warm water because this will open up the pores of the skin, which will cause further penetration of fibers and dust• Use a wash cloth to help remove fibers and dust• DO NOT rub or scratch affected area• Remove contaminated clothing and shoes• If fibers are seen penetrating from the skin, the fibers can be removed by applying and removing adhesive tape so that the fibers adhere to the tape and are pulled out of the skin• If skin irritation persists, call a physician• Never use compressed air to remove fibers from skin |
| Inhalation | <ul style="list-style-type: none">• Remove to fresh air• If symptoms persist, call a physician |
| Ingestion | <ul style="list-style-type: none">• Accidental ingestion of this product is unlikely• Rinse mouth with water and drink water to remove fibers from the throat• If this does occur watch person for several days to make sure intestinal blockage does not occur• If symptoms persist, call a physician |

5. FIRE-FIGHTING MEASURES

- | | |
|--|---|
| Suitable extinguishing media | <ul style="list-style-type: none">• Use extinguishing measures that are appropriate to local circumstances and the surrounding environment |
| Protective equipment and precautions for firefighters | <ul style="list-style-type: none">• As in any fire, wear self-contained breathing apparatus (positive-pressure), MSHA/NIOSH (approved or equivalent) and full protective gear |

6. ACCIDENTAL RELEASE MEASURES

- | | |
|--------------------------------|--|
| Personal precautions | <ul style="list-style-type: none">• Avoid contact with eyes and skin |
| Methods for cleaning up | <ul style="list-style-type: none">• Use personal protective equipment as required• Avoid creating dust• Clean contaminated surface thoroughly• Cover powder spill with plastic sheet or tarp to minimize spreading and keep powder dry• Take up mechanically, placing in appropriate containers for disposal |

7. HANDLING AND STORAGE

- | | |
|--------------------------------------|--|
| Precautions for safe handling | <ul style="list-style-type: none">• Prevent and/or minimize dust formation• Do not breathe dust• Wear appropriate personal protective equipment in case of direct contact with the product |
| Storage Conditions | <ul style="list-style-type: none">• Keep product in packaging until use to minimize potential dust generation |

- Product should be kept dry and undercover

Incompatible materials

- None known based on information supplied

8. EXPOSURE CONTROLS/PERSONAL PROTECTION**Exposure Guidelines****Engineering Controls**

Provide local exhaust and/or general ventilation to maintain exposure below regulatory and recommended limits

Dust collection system must be used in transferring operations, cutting or other dust generating processes, such as using power tools

Vacuum or wet clean-up methods should be used

Individual protection measures, such as personal protective equipment**Eye/face protection**

- Wear safety glasses with side shields (or goggles)

Skin and body protection

- Wear protective gloves
- Wear long-sleeved shirt and long pants

Respiratory protection

- When workers are facing airborne particulates/dust concentrations above the exposure limits, they must use an appropriate certified respirator
- A properly fitted NIOSH approved disposable N 95 type dust respirator or better is recommended

General Hygiene Considerations

- Wash hands before breaks and immediately after handling products
- Remove and wash contaminated clothing before re-use

9. PHYSICAL AND CHEMICAL PROPERTIES**Physical state**

Solid

Appearance

Fibrous

Odor

No information available

Color

No information available

Water solubility

Insoluble in water

10. STABILITY AND REACTIVITY

Possibility of Hazardous Reactions • None under normal processing conditions

Hazardous Decomposition Products • None known based on information supplied

11. TOXICOLOGICAL INFORMATION**Product Information**

Dusts may cause mechanical irritation to eyes and skin. Ingestion may cause transient irritation of throat, stomach and gastrointestinal tract. Inhalation may cause coughing, nose and throat irritation, and sneezing. High exposures may cause difficulty breathing, congestion, and chest tightness

Carcinogenicity

This product does not contain any carcinogens or potential carcinogens as listed by OSHA, IARC or NTP

12. ECOLOGICAL INFORMATION

This product is not expected to be hazardous for the environment.

13. DISPOSAL CONSIDERATIONS

Disposal should be in accordance with applicable regional, national and local laws and regulations

14. TRANSPORT INFORMATION

This material is not subject to regulation as a hazardous material for shipping

15. REGULATORY INFORMATION**International Inventories**

This product is classified as an article. Articles are exempted from registration or listing under chemicals inventories like TSCA (USA), DSL/NDL (CAN), REACH (EU), ENCS (JP), IECSC (CN), KECL (KR), PICCS (PH), AICS (AUS)

California Proposition 65

This product can expose you to chemicals including those listed below, which is [are] known to the State of California to cause cancer, birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov

Chemical name	California Proposition 65
Formaldehyde 50-00-0	Carcinogen

Note: The substance is only relevant in trace amount of release due to the binder composition. The substance that is subject to California Proposition 65 in this product is in a trace amount that is not subject to the occupational exposure limit.

16. OTHER INFORMATION

Creation Date 28-Jul-2016
Revision Date 05-Jun-2023
Revision Note SDS sections updated 1, components review

Disclaimer

Reasonable care has been taken in the preparation of this information, but the manufacturer makes no warranty of merchantability or any other warranty, expressed or implied, with respect to this information. The manufacturer makes no representations and assumes no liability for any direct, incidental or consequential damages resulting from its use.

End of Safe Use Instruction Sheet



SAFETY DATA SHEET

Section 1 – Product & Company Identification

Product Name: RIDGID Nu-Clear Thread Cutting Oil (United States)

Product Catalog No.: 11461, 11481, 41575, 41585, 42513, 70835

Recommended Use: Thread Cutting

Restrictions on Use: Industrial use only

Company Information:

<u>North America</u> Ridge Tool Company 400 Clark Street Elyria, Ohio 44035-6001 1-800-519-3456 (8:00 am – 5:00 pm EST, M-F) Emergency Telephone call 9-1-1 or local emergency number www.RIDGID.com	<u>Australia</u> Ridge Tool Australia 127 Metrolink Circuit Campbellfield, VIC 3061 1-800-743-443 (8:30 am – 5:00 pm AEST, M-F) Emergency Telephone call 000 or local emergency number www.RIDGID.com.au
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Operating Standard: 6-101
Revision: N
EC Number: 005080
Issue Date: January 14, 2026
Last Revision Date: April 13, 2023

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- Castellano – pág. 25



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 2 – Hazards Identification

Hazard Classification Not classified as hazardous under GHS

Label Elements

Hazard Symbol: No symbol
Signal Word: No signal word
Hazard Statement: Not applicable
Precautionary Statements Not applicable

Other hazards which do not result in GHS classification: None

Section 3 – Composition / Information On Ingredients

General information: This product does not contain silicone or chlorinated additives.

Hazardous Component(s):

Chemical name	CAS-No.	Concentration
Distillates (petroleum), hydrotreated heavy paraffinic	64742-54-7	20 - <50%
Paraffin oils (petroleum), catalytic dewaxed heavy	64742-70-7	20 - <50%
Soybean oil	8001-22-7	1 - <5%

Specific chemical identities and/or exact percentages have been withheld as trade secrets.



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 4 – First Aid Measures

Ingestion:	Rinse mouth thoroughly. Call a POISON CENTER/doctor if you feel unwell. Do NOT induce vomiting.
Inhalation:	Move to fresh air. Call a POISON CENTER/doctor if you feel unwell.
Skin Contact:	Remove contaminated clothing and shoes. Wash contact areas with soap and water. If skin irritation occurs: Get medical advice/attention.
Eye contact:	Flush thoroughly with water. If irritation occurs, get medical assistance. Continue to rinse for at least 15 minutes.

Most important symptoms/effects, acute and delayed

Symptoms: No data available.

Indication of immediate medical attention and special treatment needed

Treatment: Get medical attention if symptoms occur.

Section 5 – Fire Fighting Measures

General Fire Hazards: No unusual fire or explosion hazards noted.

Suitable (and unsuitable) extinguishing media

Suitable extinguishing media: Water spray, fog, CO₂, dry chemical, or regular foam. Use fire-extinguishing media appropriate for surrounding materials.

Unsuitable extinguishing media: Do not use water jet as an extinguisher, as this will spread the fire.

Specific hazards arising from the chemical: Heat may cause the containers to explode. During fire, gases hazardous to health may be formed.



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Special protective equipment and precautions for fire-fighters

Special fire-fighting procedures: No data available.

Special protective equipment for fire-fighters: Firefighters must use standard protective equipment including flame retardant coat, helmet with face shield, gloves, rubber boots, and in enclosed spaces, SCBA.

Section 6 – Accidental Release Measures
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Personal precautions, protective equipment and emergency procedures:	See Section 8 of the SDS for Personal Protective Equipment. Do not touch damaged containers or spilled material unless wearing appropriate protective clothing. Keep unauthorized personnel away. Ensure adequate ventilation.
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Methods and material for containment and cleaning up:	Absorb with sand or other inert absorbent. Stop the flow of material, if this is without risk.
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Environmental Precautions:	Avoid release to the environment. Do not contaminate water sources or sewer. Prevent further leakage or spillage if safe to do so.
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Section 7 – Handling And Storage

Precautions for safe handling:	Observe good industrial hygiene practices. Wear appropriate personal protective equipment. Do not expose to intense heat as product may expand and pressurize container.
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Conditions for safe storage, including any incompatibilities:	Shelf Life: 720 Days Store in original tightly closed container. Avoid contact with oxidizing agents. Store away from incompatible materials.
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Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 8 – Exposure Controls / Personal Protection

Exposure Limits

Chemical name	Type	Exposure Limit Values	Source
Distillates (petroleum), hydrotreated heavy paraffinic - Mist.	PEL	5 mg/m ³	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended (01 2017)
Distillates (petroleum), hydrotreated heavy paraffinic - Mist.	TWA	5 mg/m ³	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended (1989)
Paraffin oils (petroleum), catalytic dewaxed heavy - Inhalable fraction.	TWA	5 mg/m ³	US. ACGIH Threshold Limit Values, as amended (03 2014)
Paraffin oils (petroleum), catalytic dewaxed heavy - Mist.	PEL	5 mg/m ³	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended (02 2006)
Paraffin oils (petroleum), catalytic dewaxed heavy - Mist.	TWA	5 mg/m ³	US. OSHA Table Z-1-A (29 CFR 1910.1000), as amended (1989)
Soybean oil - Total dust.	PEL	15 mg/m ³	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended (02 2006)
Soybean oil - Respirable fraction.	PEL	5 mg/m ³	US. OSHA Table Z-1 Limits for Air Contaminants (29 CFR 1910.1000), as amended (02 2006)

Protective Measures: Use personal protective equipment as required.

Respiratory Protection: In case of inadequate ventilation use suitable respirator. Seek advice from supervisor on the company's respiratory protection standards.

Eye Protection: Wear safety glasses with side shields (or goggles).

Skin and Body Protection: Wear protective clothing appropriate for the risk of exposure. Be aware of other hazards such as rotating parts. Contact health and safety professional or manufacturer for specific information. Wear chemical-resistant gloves, footwear, and protective clothing appropriate for the risk of exposure. Contact health and safety professional or manufacturer for specific information.

Hygiene measures: Always observe good personal hygiene measures, such as washing after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing to remove contaminants. Discard contaminated footwear that



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

cannot be cleaned.

Section 9 – Physical And Chemical Properties

Appearance:

Physical state:	liquid
Form:	No data available
Color:	Yellow
Odor:	Mild petroleum/solvent
Odor threshold:	No data available
pH:	No data available
Melting point/freezing point:	No data available
Initial boiling point and boiling range:	No data available
Flash Point:	196.11 °C (385.00 °F)
Evaporation rate:	No data available
Flammability (solid, gas):	No data available
Upper/lower limit on flammability or explosive limits:	
Flammability limit - upper (%):	No data available
Flammability limit - lower (%):	No data available
Explosive limit - upper:	No data available
Explosive limit - lower:	No data available
Vapor pressure:	No data available
Vapor density:	No data available
Relative density:	0.878
Solubility(ies):	
Solubility in water:	Insoluble
Solubility (other):	No data available
Partition coefficient (n-octanol/water):	No data available
Auto-ignition temperature:	No data available
Decomposition temperature:	No data available
Viscosity:	43 mm ² /s (40 °C, Measured)
Other information :	
VOC:	1.1 % (Method 24) 9.4 g/l (ASTM E 1868-10)



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 10 – Stability And Reactivity

Reactivity:	Not reactive during normal use.
Chemical Stability:	Material is stable under normal conditions.
Possibility of hazardous reactions:	None under normal conditions.
Conditions to avoid:	Avoid heat or contamination.
Incompatible Materials:	No data available.
Hazardous Decomposition Products:	Thermal decomposition or combustion may liberate carbon oxides and other toxic gases or vapors. Contains a component which may release flammable substances, including trimethylpentene, by distillation in systems with solvent recovery. This may lead to accumulation in the solvent circuit.

Section 11 – Toxicological Information

Information on likely routes of exposure

Ingestion:	May be ingested by accident. Ingestion may cause irritation and malaise.
Inhalation:	Inhalation is the primary route of exposure. In high concentrations, vapors, fumes or mists may irritate nose, throat and mucus membranes.
Skin Contact:	Prolonged skin contact may cause redness and irritation.



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Eye contact:	Eye contact is possible and should be avoided.
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Symptoms related to the physical, chemical and toxicological characteristics

Ingestion:	No data available.
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Inhalation:	No data available.
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Skin Contact:	No data available.
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Eye contact:	No data available.
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Information on toxicological effects

Acute toxicity (list all possible routes of exposure)

Oral Product:	ATEmix (): 2000 - 5000 mg/kg
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Dermal Product:	ATEmix (): 2000 - 5000 mg/kg
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Inhalation Product:	No data available.
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Repeated dose toxicity Product:	No data available.
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Skin Corrosion/Irritation Product:	No data available.
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Serious Eye Damage/Eye Irritation Product:	No data available.
---	--------------------

Respiratory or Skin Sensitization Product:	No data available.
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Carcinogenicity Product:	No data available.
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IARC Monographs on the Evaluation of Carcinogenic Risks to Humans:	No carcinogenic components identified
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Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

US. National Toxicology Program (NTP) Report on Carcinogens:

No carcinogenic components
identified

**US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050), as
amended:**

No carcinogenic components
identified

Germ Cell Mutagenicity:

**In vitro
Product:**

No data available.

**In vivo
Product:**

No data available.

**Reproductive toxicity
Product:**

No data available.

**Specific Target Organ Toxicity - Single
Exposure Product:**

No data available.

**Specific Target Organ Toxicity - Repeated
Exposure
Product:**

No data available.

**Aspiration Hazard
Product:**

No data available.

Other effects:

No data available.

Section 12 – Ecological Information
--

General information:

This product has not been evaluated
for ecological toxicity or other
environmental effects.



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 13 – Disposal Consideration

Disposal instructions:

Discharge, treatment, or disposal may be subject to national, state, or local laws. Dispose of waste at an appropriate treatment and disposal facility in accordance with applicable laws and regulations, and product characteristics at time of disposal. It is the responsibility of the product user or owner to determine at the time of disposal, which waste regulations must be applied.

Contaminated Packaging:

Empty containers should be taken to an approved waste handling site for recycling or disposal.

Section 14 – Transportation Information

DOT

Not regulated.

IMDG

Not regulated.

IATA

Not regulated.

Section 15 – Regulatory Information

US Federal Regulations

US. OSHA Specifically Regulated Substances (29 CFR 1910.1001-1050), as amended

None present or none present in regulated quantities.

Superfund Amendments and Reauthorization Act of 1986 (SARA)



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Hazard categories

Not classified as hazardous under GHS

SARA 313 (TRI Reporting)

None present or none present in regulated quantities.

US State Regulations

US. California Proposition 65

No ingredient requiring a warning under CA Prop 65.



Product Name : RIDGID Nu-Clear Thread Cutting Oil (United States)

Section 16 – Other Information

Prepared by:..... Ridge Tool Company

OPSTD 6-101

Revision N

EC Number 005080

Issue Date: January 14, 2026

Last Revision Date: April 13, 2023

RIDGE TOOL BELIEVES THE STATEMENTS, TECHNICAL INFORMATION AND RECOMMENDATIONS CONTAINED HEREIN ARE RELIABLE BUT THEY ARE GIVEN WITHOUT WARRANTY OR GUARANTEE OF ANY KIND, EXPRESSED OR IMPLIED, AND WE ASSUME NO RESPONSIBILITY FOR ANY LOSS, DAMAGE OR EXPENSE, DIRECT OR CONSEQUENTIAL, ARISING OUT OF THEIR USE.



FICHE SANTÉ/SÉCURITÉ

1 – Identification du produit et du fournisseur

Produit: RIDGID Nu-Clear Thread Cutting Oil (Etats-Unis)

Réf. catalogue: 11461, 11481, 41575, 41585, 42513, 70835

Emploi recommandé: Filetage mécanique

Restrictions d'utilisation: Usage industriel seulement

Fournisseur:

North America
Ridge Tool Company
400 Clark Street
Elyria, Ohio 44035-6001
1-800-519-3456
(Etats-Unis) (du lundi au vendredi de 8h à 17h EST)
Téléphone d'urgence:
composer le 9-1-1 ou appeler les services d'urgences appropriés
www.RIDGID.com

Norme De Fonctionnement: 6-101

Révision: N

ECN: 002150

Date de publication: 14 janvier 2026

Dernière revision: 13 avril 2023



Produit: RIDGID Nu-Clear Thread Cutting Oil (Etats-Unis)

2 – Identification des risques

Classe de Danger Non classé comme dangereux selon le SGH

Éléments d'étiquetage :

Symbole de Danger:	Aucun symbole
Mention d'Avertissement:	Aucun mot indicateur.
Mention de Danger:	Non applicable
Conseils de Prudence	Non applicable

Autres dangers ne donnant pas lieu à classement selon le SGH: Aucun(e).

3 – Composition du produit et renseignements sur ses ingrédients

Informations générales: Ce produit ne contient pas de silicone ou d'additifs chlorés.

Composant(s) dangereux:

Désignation chimique	N° CAS	Concentration
distillats paraffiniques lourds (pétrole), hydrotraités	64742-54-7	20 - <50%
huiles de paraffine lourdes (pétrole)	64742-70-7	20 - <50%
l'huile de soja	8001-22-7	1 - <5%

Les identités chimiques spécifiques et/ou les pourcentages exacts ont été refusées comme les secrets commerciaux.



Produit: RIDGID Nu-Clear Thread Cutting Oil (Etats-Unis)

4 – Premiers soins

- Ingestion:** Rincer soigneusement la bouche. Appeler un CENTRE ANTIPOISON/un médecin en cas de malaise. NE PAS faire vomir.
- Inhalation:** Transporter à l'air frais. Appeler un CENTRE ANTIPOISON/un médecin en cas de malaise.
- Contact avec la Peau:** Enlever les vêtements et les chaussures contaminés. Laver les zones de contact à l'eau et au savon. En cas d'irritation cutanée: consulter un médecin.
- Contact oculaire:** Rincer avec soin à l'eau. En cas d'irritation, consulter un médecin. Continuer à rincer pendant au moins 15 minutes.

Symptômes/effets les plus importants, aigus et différés

Symptômes: Aucune information disponible.

Indication d'un besoin médical immédiat et traitement spécial requis

Traitement: Consulter un médecin en cas de symptômes.

5 – Lutte contre les incendies

Dangers d'Incendie Généraux: Aucun risque exceptionnel d'incendie et d'explosion.

Moyens d'extinction appropriés (et inappropriés)

Moyens d'extinction appropriés: Eau pulvérisée, brouillard, CO₂, agent chimique sec ou mousse standard. Choisir le moyen d'extinction de l'incendie en tenant compte d'autres produits chimiques éventuels.

Moyens d'extinction inappropriés: Ne pas lutter contre l'incendie au jet d'eau pour ne pas propager les flammes.



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**Dangers spécifiques
du produit
chimique:**

La chaleur peut provoquer l'explosion des récipients. En cas d'incendie, des gaz dangereux pour la santé peuvent se former.

Équipement de protection spécial et précautions pour les pompiers

**Procédures
spéciales de lutte
contre l'incendie:**

Aucune information disponible.

**Équipement de
protection
spécial pour le
personnel
préposé à la lutte
contre le feu:**

Les pompiers doivent porter un équipement de protection standard, notamment vêtement ignifuge, casque à masque facial, gants, bottes en caoutchouc et, dans les espaces clos, un appareil respiratoire autonome.

6 – Lutte contre les déversements accidentels
--

**Précautions
individuelles,
équipement de
protection et
procédures d'urgence:**

Voir l'équipement de protection individuelle à la Section 8. Ne pas toucher les récipients endommagés ou le produit déversé à moins de porter les vêtements de protection appropriés. Maintenir à distance le personnel non autorisé. Assurer une ventilation adéquate.

**Méthodes et matériel de
confinement et de
nettoyage:**

Absorber le produit avec du sable ou un autre absorbant inerte. Arrêter le débit de matière, si ceci est sans risque.

**Précautions pour la
protection de
l'environnement:**

Éviter le rejet dans l'environnement. Ne pas contaminer les sources d'eau ou les égouts. Endiguer la fuite ou le déversement si cela peut être fait sans danger.

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7 – Manipulation et stockage

Précautions à prendre pour une manipulation sans danger:

Se conformer aux bonnes pratiques d'hygiène industrielle. Porter un équipement de protection personnelle approprié. N'exposez pas à la chaleur intense comme le produit peut développer et pressuriser le récipient.

Conditions d'un stockage sûr, y compris les éventuelles incompatibilités:

Durée de conservation: 720 jours Conserver dans le récipient d'origine hermétiquement fermé. Éviter tout contact avec des agents comburants. Conserver à l'écart des matières incompatibles.

8 – Risques d'exposition et protection individuelle

Limites d'Exposition

Désignation chimique	Type	Valeurs Limites d'Exposition	Source
distillats paraffiniques lourds (pétrole), hydrotraités - Brouillard	PEL	5 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (01 2017)
distillats paraffiniques lourds (pétrole), hydrotraités - Brouillard	TWA	5 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (1989)
huiles de paraffine lourdes (pétrole) - Fraction inhalable.	TWA	5 mg/m ³	Etats-Unis. ACGIH, valeurs limites d'exposition, dans leur version modifiée (03 2014)
huiles de paraffine lourdes (pétrole) - Brouillard	PEL	5 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (02 2006)
huiles de paraffine lourdes (pétrole) - Brouillard	TWA	5 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (1989)
l'huile de soja - poussière totales	PEL	15 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (02 2006)
l'huile de soja - Fraction alvéolaire.	PEL	5 mg/m ³	Les Etats-Unis. La Table d'OSHA z-1 les Limites pour les Polluants Aériens (29 CFR 1910.1000) (02 2006)

Mesures de protection: Utiliser l'équipement de protection individuel requis.

Protection respiratoire: En cas de ventilation insuffisante, porter un appareil respiratoire approprié. Demander l'avis du superviseur sur



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les normes de protection respiratoire de la société.

Protection des Yeux: Porter des lunettes de sécurité à écrans latéraux ou des lunettes étanches.

Protection de la peau et du corps: Porter des vêtements de protection appropriés au risque d'exposition. Soyez conscient des autres dangers tels que les pièces en rotation. Contacter un professionnel de la santé et de la sécurité ou un fabricant pour obtenir des informations spécifiques. Porter des gants, des chaussures et des vêtements de protection résistant aux produits chimiques, et correspondant au risque d'exposition. Contacter un professionnel de l'hygiène et sécurité ou le fabricant pour tout détail.

Mesures d'hygiène: Toujours adopter de bonnes pratiques d'hygiène personnelle, telles que lavage après manipulation de la substance et avant de manger, de boire ou de fumer. Laver régulièrement la tenue de travail pour éliminer les contaminants. Mettre au rebut les chaussures qui ne peuvent pas être lavées.

9 – Caractéristiques physiques et chimiques

Aspect

État:	liquide
Forme:	Aucune information disponible.
Couleur:	Jaune
Odeur:	Légère, Pétrole/solvant
Seuil de perception de l'odeur:	Aucune information disponible.
pH:	Aucune information disponible.
Point de fusion/point de congélation:	Aucune information disponible.
Température d'ébullition initiale et intervalle d'ébullition:	Aucune information disponible.
Point d'éclair:	196.11 °C (385.00 °F)
Taux d'évaporation:	Aucune information disponible.
Inflammabilité (solide, gaz):	Aucune information disponible.
Limites supérieures/inférieures d'inflammabilité ou d'explosivité	
Limites d'inflammabilité - supérieure (%):	Aucune information disponible.
Limites d'inflammabilité - inférieure (%):	Aucune information disponible.

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Limites d'explosivité - supérieure:	Aucune information disponible.
Limites d'explosivité - inférieure:	Aucune information disponible.
Pression de vapeur:	Aucune information disponible.
Densité de vapeur:	Aucune information disponible.
Densité relative:	0.878
Solubilités	
Solubilité dans l'eau:	Insoluble
Solubilité (autre):	Aucune information disponible.
Coefficient de partition (n-octanol/eau):	Aucune information disponible.
Température d'auto-inflammation:	Aucune information disponible.
Température de décomposition:	Aucune information disponible.
Viscosité:	43 mm ² /s (40 °C, Mesurée)
Autres informations	
VOC:	1.1 % (Method 24) 9.4 g/l (ASTM E 1868-10)

10 – Stabilité et réactivité

Réactivité:	Non réactif pendant l'utilisation normale.
Stabilité chimique:	Ce produit est stable dans des conditions normales.
Possibilité de réactions dangereuses:	Aucun(e)(s) dans les conditions normales.
Conditions à éviter:	Éviter tout chauffage ou contamination.
Matières incompatibles:	Aucune information disponible.
Produits de décomposition dangereux:	La décomposition thermique ou la combustion peut libérer des oxydes de carbone et d'autres gaz ou vapeurs toxiques. Contient un composant qui peut libérer substances inflammables, y compris triméthylpentène, par distillation dans les systèmes de récupération de solvant. Cela peut entraîner une accumulation dans le solvant du circuit

11 – Données toxicologiques

Informations sur les voies d'exposition probables

Ingestion:	Peut être ingéré par accident. L'ingestion peut provoquer irritation et malaises.
Inhalation:	L'inhalation est la principale voie d'exposition. À concentration élevée, les vapeurs, émanations ou brouillards peuvent être irritants pour le nez, la gorge et les muqueuses.
Contact avec la Peau:	Le contact prolongé avec la peau peut entraîner des rougeurs et de l'irritation.
Contact oculaire:	Le contact oculaire est possible ; il doit être évité.

Symptômes liés aux caractéristiques physiques, chimiques et toxicologiques

Ingestion:	Aucune information disponible.
Inhalation:	Aucune information disponible.
Contact avec la Peau:	Aucune information disponible.
Contact oculaire:	Aucune information disponible.

Informations sur les effets toxicologiques

Toxicité aiguë (répertorier toutes les voies d'exposition possibles)

Ingestion Produit:	ETAmél (): 2000 - 5000 mg/kg
Contact avec la peau Produit:	ETAmél (): 2000 - 5000 mg/kg
Inhalation Produit:	Aucune information disponible.
Toxicité à dose	Aucune information disponible.



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répétée Produit:

Corrosion ou Irritation de la Peau Produit: Aucune information disponible.

Blessure ou Irritation Grave des Yeux Produit: Aucune information disponible.

Sensibilisation Respiratoire ou Cutanée Produit: Aucune information disponible.

Cancérogénicité Produit: Aucune information disponible.

Monographies du CIRC sur l'évaluation des risques de cancérogénicité pour l'homme:

Aucun composant cancérigène identifié

États-Unis. Rapport du NTP (National Toxicology Program) sur les cancérogènes :

Aucun composant cancérigène identifié

ÉTATS-UNIS. Substances spécialement réglementées par l'OSHA (29 CFR 1910.1001-1050), dans sa version modifiée:

Aucun composant cancérigène identifié

Mutagénicité des Cellules Germinales

In vitro Produit: Aucune information disponible.

In vivo Produit: Aucune information disponible.

Toxicité pour la reproduction Produit: Aucune information disponible.

Toxicité Spécifique au Niveau de l'Organe Cible- Exposition Unique Produit: Aucune information disponible.

Toxicité Spécifique au Niveau de l'Organe Aucune information disponible.



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**Cible- Expositions
répétées Produit:**

**Risque
d'AspirationnnProduit:** Aucune information disponible.

Autres effets: Aucune information disponible.

12 – Données écologiques

Informations générales: Ce produit n'a pas été évalué pour la toxicité écologique ou d'autres effets de l'environnement.

13 – Recyclage

**Instructions pour
l'élimination:** Le rejet, le traitement et l'élimination peuvent être soumis à des lois nationales, régionales ou locales. Éliminer les déchets dans une installation de traitement et d'élimination des déchets appropriée conformément aux lois et aux réglementations en vigueur et en fonction des caractéristiques du produit au moment de l'élimination. C'est la responsabilité de l'utilisateur de produit ou du propriétaire pour déterminer au moment de la disposition, qui se perdent les règlements doivent être appliqués.

**Emballages
Contaminés:** Les conteneurs vides doivent être acheminés vers un site agréé pour le traitement des déchets à des fins de recyclage ou d'élimination.



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14 – Transport

Ministère des
transports des États-
Unis (Department of
Transportation, DOT)

Non réglementé.

IMDG

Non réglementé.

IATA

Non réglementé.

15 – Réglementation

Réglementations Fédérales des Etats-Unis

ÉTATS-UNIS. Substances spécialement réglementées par l'OSHA (29 CFR 1910.1001-1050), dans sa version modifiée

Aucun présent ou aucun présent dans des quantités réglementées.

Superfund Amendments and Reauthorization Act de 1986 (SARA)

Catégories de
danger

Non classé comme dangereux selon le SGH

**SARA 313
(Déclaration au
TRI)**

Aucun présent ou aucun présent dans des quantités réglementées.

États-Unis - Réglementation des États

**États-Unis -
Proposition 65 de
la Californie**

Aucun composant réglementé par la Proposition 65 de la Californie n'est présent.



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16 – Renseignements divers

Rédaction Ridge Tool Company

Norme De Fonctionnement 6-101

Révision N

ECN..... 005080

Date de publication..... 14 janvier 2026

Dernière révision 13 avril 2023

Quoi que la société Ridge Tool estime que les affirmations, informations techniques et recommandations ci-présentes sont dignes de confiance, celles-ci ne sont données qu'à titre indicatif, sans aucune garantie expresse ou implicite, et ne sauraient engager la responsabilité civile de la société en cas de pertes, dommages et intérêts, voire frais directs ou indirects relevant de leur application.



HOJA DE DATOS DE SEGURIDAD

Sección 1 – Identificación del producto y la compañía

Nombre del producto: RIDGID Nu-Clear Thread Cutting Oil (Estados Unidos)

No. de catálogo: 11461, 11481, 41575, 41585, 42513, 70835

Uso recomendado: Para cortar roscas

Restricciones de utilización: Uso industria seulement

Nombre de la compañía:

North America
Ridge Tool Company
400 Clark Street
Elyria, Ohio 44035-6001, EE. UU.
Teléfono 1-800-519-3456 (EE. UU.) (8:00 a 17:00 hora estándar del este, lunes a viernes)
Teléfono de emergencia: Llame al 9-1-1 o al teléfono de emergencia local
www.RIDGID.com

Estándar De Funcionamiento: 6-101

Révision: N

ECN: 005080

Fecha de publicación: 14 de enero de 2026

Fecha de la última revisión: 13 de abril de 2023



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Sección 2 – Identificación de peligros

Clasificación de Peligro No clasificado como peligroso bajo GHS

Elementos de la etiqueta

Símbolo de Peligro	No hay símbolo
Palabra de Advertencia	No hay palabra de advertencia.
Indicación de Peligro:	No aplicable
Consejos de Prudencia	No aplicable

Otros peligros que no dan lugar a clasificación SGA: Ninguno

Sección 3 – Composición e información sobre ingredientes

Información general: Este producto no contiene silicona o aditivos clorados.

Componente(s) peligroso(s):

Determinación química	No. CAS	Concentración
destilados (petróleo), fracción parafínica pesada tratada con hidrógeno	64742-54-7	20 - <50%
aceites de parafina (petróleo)	64742-70-7	20 - <50%
aceite de soja	8001-22-7	1 - <5%

Las identidades químicas específicas y/o los porcentajes exactos han sido retenidos como secretos de fabricación.



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Sección 4 – Primeros auxilios

- Ingestión:** Enjuagar a fondo la boca. Llamar a un CENTRO DE TOXICOLOGÍA / médico si la persona se encuentra mal. NO provocar el vómito.
- Inhalación:** Trasladar al aire libre. Llamar a un CENTRO DE TOXICOLOGÍA / médico si la persona se encuentra mal.
- Contacto con la Piel:** Quitar ropa y zapatos contaminados. Lave las áreas de contacto con agua y jabón. En caso de irritación cutánea: Consultar a un médico.
- Contacto con los ojos:** Lave con abundante agua. Si aparece irritación, busque asistencia médica. Continuar enjuagando durante al menos 15 minutos.

Los síntomas y efectos más importantes, tanto los agudos como los retardados

Síntomas: No hay datos disponibles.

Indicación de asistencia médica inmediata y tratamiento especial necesario

Tratamiento: Obtenga atención médica en caso de síntomas.

Sección 5 – Medidas contra incendios

Riesgos Generales de Incendio: Ningún riesgo excepcional de incendio o explosión señalado.

Medios de extinción adecuados (y no adecuados)

Medios de extinción apropiados: Agua pulverizada, neblina, CO₂, polvos químicos, o espuma normal. Seleccione el medio de extinción más apropiado, teniendo en cuenta la posible presencia de otros productos químicos.

Medios de extinción no No utilice chorro de agua, pues extendería el fuego.



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apropiados:

**Peligros específicos
derivados de la
sustancia química:**

El calor puede ocasionar explosión de los recipientes. En caso de incendio se pueden formar gases nocivos.

Equipo especial de protección y medias de precaución para los bomberos

**Medidas
especiales de
lucha contra
incendios:**

No hay datos disponibles.

**Equipos de
protección
especial que
debe llevar el
personal de
lucha contra
incendios:**

Los bomberos deben utilizar un equipo de protección estándar incluyendo chaqueta ignífuga, casco con careta, guantes, botas de goma, y, en espacios cerrados, equipo de respiración autónomo (SCBA, según sus siglas en inglés).

Sección 6 – Medidas en caso de liberación accidental

**Precauciones
personales, equipo de
protección y
procedimientos de
emergencia:**

Consulte la sección 8 de la FDS sobre equipo de protección personal. No toque los recipientes dañados o el material derramado a menos que esté usando ropa protectora adecuada. Mantener alejado al personal no autorizado. Asegúrese una ventilación apropiada.

**Métodos y material de
contención y de
limpieza:**

Absorber con arena u otro absorbente inerte. Detenga el flujo del material, si esto no representa un riesgo.

**Precauciones relativas
al medio ambiente:**

Evitar su liberación al medio ambiente. No contamine el drenaje o el alcantarillado. Impedir nuevos escapes o derrames de forma segura.



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Sección 7 – Manipulación y almacenamiento

Precauciones para una manipulación segura:

Respete las normas para una manipulación correcta de productos químicos. Use equipo protector personal adecuado. No exponga al calor intenso cuando el producto puede ampliar y presurizar el contenedor.

Condiciones de almacenamiento seguro, incluidas posibles incompatibilidades:

Vida útil: 720 días Guárdese en el recipiente original bien cerrado. Evite el contacto con agentes reductores. Consérvese alejado de materiales incompatibles.

Sección 8 – Controles contra la exposición: protección personal

Valores Límite

Determinación química	Tipo	Valores Límite de Exposición	Fuente
destilados (petróleo), fracción parafínica pesada tratada con hidrógeno - Niebla	PEL	5 mg/m ³	NOS. OSHA la tabla Z-1 límites para contaminantes del aire (29 CFR 1910.1000) (01 2017)
destilados (petróleo), fracción parafínica pesada tratada con hidrógeno - Niebla	TWA	5 mg/m ³	NOS. OSHA la Tabla Z-1-A (29 CFR 1910.1000) (1989)
aceites de parafina (petróleo) - Fracción inhalable	TWA	5 mg/m ³	US. Valores límite de umbral de la ACGIH, en su forma enmendada (03 2014)
aceites de parafina (petróleo) - Niebla	PEL	5 mg/m ³	NOS. OSHA la tabla Z-1 límites para contaminantes del aire (29 CFR 1910.1000) (02 2006)
aceites de parafina (petróleo) - Niebla	TWA	5 mg/m ³	NOS. OSHA la Tabla Z-1-A (29 CFR 1910.1000) (1989)
aceite de soja - Polvo total	PEL	15 mg/m ³	NOS. OSHA la tabla Z-1 límites para contaminantes del aire (29 CFR 1910.1000) (02 2006)
aceite de soja - Fracción respirable	PEL	5 mg/m ³	NOS. OSHA la tabla Z-1 límites para contaminantes del aire (29 CFR 1910.1000) (02 2006)

Medidas de protección: Utilizar los equipos de protección individual según las necesidades.



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- Protección respiratoria:** En caso de ventilación insuficiente, utilice un equipo respiratorio adecuado. Consulte al supervisor sobre la norma de la compañía de protección respiratoria.
- Protección de los Ojos:** Use gafas de seguridad con protectores laterales (o gafas estancas).
- Protección de la Piel y del Cuerpo:** Use ropa protectora apropiada para el riesgo de exposición. Tenga en cuenta otros peligros, como las piezas giratorias. Comuníquese con el profesional o fabricante de salud y seguridad para obtener información específica. Lleve guantes resistentes a los productos químicos, zapatos y traje protectores adecuados para el riesgo de exposición. Contacte con un especialista en salud y seguridad profesional o con el fabricante para obtener información específica.
- Medidas de higiene:** Seguir siempre buenas medidas de higiene personal, como lavarse después de manipular el material y antes de comer, beber y/o fumar. Lave rutinariamente la ropa de trabajo para eliminar los contaminantes. Deseche el calzado contaminado que no se pueda limpiar.

Sección 9 – Propiedades físicas y químicas

Aspecto

Forma/estado:	líquido
Forma/Figura:	No hay datos disponibles.
Color:	Amarillo
Olor:	Ligero, petróleo/solvente
Umbral de olor:	No hay datos disponibles.
pH:	No hay datos disponibles.
Punto de fusión / Punto de congelación:	No hay datos disponibles.
Punto inicial de ebullición e intervalo de ebullición:	No hay datos disponibles.
Punto de inflamación:	196.11 °C (385.00 °F)
Tasa de evaporación:	No hay datos disponibles.
Inflamabilidad (sólido, gas):	No hay datos disponibles.
Límites superior/inferior de inflamabilidad o de explosividad	
Límite superior de inflamabilidad (LSI) (%):	No hay datos disponibles.
Límite inferior de inflamabilidad (LII) (%):	No hay datos disponibles.
Límite superior de explosividad:	No hay datos disponibles.



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Límite inferior de explosividad:	No hay datos disponibles.
Presión de vapor:	No hay datos disponibles.
Densidad del vapor:	No hay datos disponibles.
Densidad relativa:	0.878
Solubilidad(es)	
Solubilidad en agua:	Insoluble
Solubilidad (otra):	No hay datos disponibles.
Coeficiente de reparto (n-octanol/agua):	No hay datos disponibles.
Temperatura de autoignición:	No hay datos disponibles.
Temperatura de descomposición:	No hay datos disponibles.
Viscosidad:	43 mm ² /s (40 °C, medido)
Otros datos	
VOC:	1.1 % (Method 24) 9.4 g/l (ASTM E 1868-10)

Sección 10 – Estabilidad y reactividad

Reactividad:	No reactivo durante uso normal.
Estabilidad química:	El material es estable bajo condiciones normales.
Posibilidad de reacciones peligrosas:	Ningunos en circunstancias normales.
Condiciones que deben evitarse:	Evite el calor o la contaminación.
Materiales incompatibles:	No hay datos disponibles.
Productos de descomposición peligrosos:	La descomposición térmica o la combustión pueden liberar óxido de carbono u otros gases o vapores tóxicos. Contiene un componente que puede liberar sustancias inflamables, incluyendo trimetilpenteno, por destilación en sistemas con recuperación de disolvente. Esto puede conducir a la acumulación en el circuito de disolvente.



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Sección 11 – Información toxicológica

Información sobre posibles vías de exposición

Ingestión:	Puede ingerirse accidentalmente. La ingestión puede causar irritación y malestar.
Inhalación:	La inhalación es la principal vía de exposición. En concentraciones altas, los vapores, humos o neblinas pueden irritar la nariz, la garganta y las membranas mucosas.
Contacto con la Piel:	El contacto prolongado con la piel puede causar rubor e irritación.
Contacto con los ojos:	El contacto con los ojos es posible y debe evitarse.

Síntomas relacionados a las características físicas, químicas y toxicológicas

Ingestión:	No hay datos disponibles.
Inhalación:	No hay datos disponibles.
Contacto con la Piel:	No hay datos disponibles.
Contacto con los ojos:	No hay datos disponibles.

Información sobre los efectos toxicológicos

Toxicidad aguda (listar todas las vías de exposición posibles)

Ingestión Producto	ETAmézcla (): 2000 - 5000 mg/kg
Contacto dermal Producto	ETAmézcla (): 2000 - 5000 mg/kg
Inhalación Producto	No hay datos disponibles.



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Toxicidad por dosis repetidas Producto No hay datos disponibles.

Corrosión/Irritación Cutáneas Producto No hay datos disponibles.

Lesiones Oculares Graves/Irritación Ocular Producto: No hay datos disponibles.

Sensibilización de la Piel o Respiratoria Producto: No hay datos disponibles.

Carcinogenicidad Producto: No hay datos disponibles.

Monografías de IARC sobre la evaluación de los riesgos carcinogénicos para los humanos:

No se identificaron componentes carcinogénicos

Programa Nacional de Toxicología de EUA (NTP). Reporte sobre carcinógenos:

No se identificaron componentes carcinogénicos

EEUU. Sustancias específicamente reguladas por la OSHA (29 CFR 1910.1001-1050), en su forma enmendada:

No se identificaron componentes carcinogénicos

Mutagenicidad en Células Germinales

En vitro Producto: No hay datos disponibles.

En vivo Producto: No hay datos disponibles.

Toxicidad para la reproducción Producto: No hay datos disponibles.

Toxicidad Sistémica No hay datos disponibles.



Producto: RIDGID Nu-Clear Thread Cutting Oil (Estados Unidos)

**Específica de Órganos
Diana- Exposición
Única Producto:**

**Toxicidad Sistémica
Específica de Órganos
Diana- Exposiciones
Repetidas Producto:** No hay datos disponibles.

**Peligro por Aspiración
Producto:** No hay datos disponibles.

Otros síntomas: No hay datos disponibles.

Sección 12 –Información ecológica

Información general: Este producto no ha sido evaluado para la toxicidad ecológica u otros efectos ambientales.

Sección 13 – Consideraciones relativas a la eliminación

Instrucciones para la eliminación: Las actividades de descarga, tratamiento o eliminación pueden estar sujetos a leyes nacionales, estatales o locales. Elimine el residuo en una instalación adecuada de tratamiento y eliminación de acuerdo con las leyes y reglamentos correspondientes y características del producto en el momento de la eliminación. Es responsabilidad del usuario del producto o propietario para determinar en el momento de la disposición, que las regulaciones de residuos debe ser aplicado.

Envases Contaminados: Los contenedores vacíos deben ser llevados a un sitio de manejo aprobado para desechos, para el reciclado o eliminación.



Producto: RIDGID Nu-Clear Thread Cutting Oil (Estados Unidos)

Sección 14 – Información de transporte

DOT	No reglamentado.
IMDG	No reglamentado.
IATA	No reglamentado.

Sección 15 – Información sobre reglamentos

Reglamentos Federales de EE.UU.

EEUU. Sustancias específicamente reguladas por la OSHA (29 CFR 1910.1001-1050), en su forma enmendada

No están presentes, o no están presentes en las cantidades reguladas.

Ley de Enmiendas y Reautorización del Superfondo de 1986 (SARA)

Categorías de peligro	No clasificado como peligroso bajo GHS
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SARA 313 (Reporte TRI, Acerca del Inventario de Liberación de Sustancias Tóxicas)	No están presentes, o no están presentes en las cantidades reguladas.
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Regulaciones de un Estado de EUA

Proposición 65 del Estado de California, EUA	No hay presencia de ningún ingrediente regulado por CA Prop 65.
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Producto: RIDGID Nu-Clear Thread Cutting Oil (Estados Unidos)

Sección 16 – Información adicional

Preparado por Ridge Tool Company

Estándar De Funcionamiento. 6-101

Revision N

ECN..... 005080

Fecha de emisión 14 de enero de 2026

Fecha de la última revisión..... 13 de abril de 2023

RIDGE TOOL CONSIDERA QUE TODAS LAS DECLARACIONES, INFORMACIÓN TÉCNICA Y RECOMENDACIONES EN EL PRESENTE DOCUMENTO SON CONFIABLES, PERO SE PRESENTAN SIN GARANTÍA ALGUNA, SEA EXPRESA O IMPLÍCITA, Y NO ASUMIMOS RESPONSABILIDAD ALGUNA POR PÉRDIDAS, DAÑOS O GASTOS, DIRECTOS O CONSECUENTES, QUE SURJAN DE SU USO.



Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.
Issue date: 5/5/2016 Revision date: 2/18/2022 Version: 2.0

SECTION 1: Identification

1.1. Identification

Product form : Mixture
Product name : Precision Grout Non Shrink
Product code : Not available

1.2. Recommended use and restrictions on use

Recommended use : Various

1.3. Supplier

Manufacturer

Sakrete of North America
625 Griffith Rd., Ste 100
Charlotte, NC 28217
T 866-725-7383

1.4. Emergency telephone number

Emergency number : CHEMTREC (800) 424-9300

SECTION 2: Hazard(s) identification

2.1. Classification of the substance or mixture

GHS US classification

Skin Irrit. 2	Causes skin irritation
Eye Dam. 1	Causes serious eye damage
Skin Sens. 1B	May cause an allergic skin reaction
Carc. 1A	May cause cancer
Repr. 1A	May damage fertility or the unborn child
STOT SE 3	May cause respiratory irritation
STOT RE 1	Causes damage to organs (lungs) through prolonged or repeated exposure

2.2. GHS Label elements, including precautionary statements

GHS US labeling

Hazard pictograms (GHS US) :



Signal word (GHS US) :

Danger

Hazard statements (GHS US) :

Causes skin irritation
May cause an allergic skin reaction
Causes serious eye damage
May cause respiratory irritation
May cause cancer
May damage fertility or the unborn child
Causes damage to organs (lungs) through prolonged or repeated exposure
Obtain special instructions before use.
Do not handle until all safety precautions have been read and understood.
Do not breathe dust, fume, gas, mist, spray, vapors.

Precautionary statements (GHS US) :

Precision Grout Non Shrink

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according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Wash hands thoroughly after handling.
Do not eat, drink or smoke when using this product.
Use only outdoors or in a well-ventilated area. If not in a well-ventilated area, wear a NIOSH-approved respirator or other dust mask when using the product to avoid or minimize exposure to dust.
Contaminated work clothing must not be allowed out of the workplace.
Wear protective gloves/protective clothing/eye protection/face protection.
If on skin: Wash with plenty of water.
Take off contaminated clothing and wash it before reuse.
If skin irritation or rash occurs: Get medical advice/attention.
If inhaled: Remove person to fresh air and keep comfortable for breathing.
Call a poison center or doctor if you feel unwell.
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
Immediately call a poison center or doctor.
If exposed or concerned: Get medical advice/attention.
Store in a well-ventilated place. Keep container tightly closed.
Store locked up.
Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

2.3. Other hazards which do not result in classification

No additional information available

2.4. Unknown acute toxicity (GHS US)

Not applicable

SECTION 3: Composition/Information on ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Name	Product identifier	%
Quartz	CAS-No.: 14808-60-7	10 – 30
Cement, portland, chemicals	CAS-No.: 65997-15-1	10 – 30
Sulfuric acid, calcium salt (1:1)	CAS-No.: 7778-18-9	1 – 5
Limestone	CAS-No.: 1317-65-3	1 – 5
Flue dust, zinc-refining	CAS-No.: 69012-63-1	0.5 – 1.5
Formaldehyde	CAS-No.: 50-00-0	< 0.1
Ethylene oxide	CAS-No.: 75-21-8	< 0.1
Chromium, ion (Cr 6+)*	CAS-No.: 18540-29-9	< 0.1

*Hexavalent chromium is included due to dermal sensitivity associated with the component.

Chemical name, CAS number and/or exact concentration have been withheld as a trade secret

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

SECTION 4: First-aid measures

4.1. Description of first aid measures

First-aid measures after inhalation	: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician if you feel unwell.
First-aid measures after skin contact	: IF ON SKIN: Brush off loose particles from skin. Immerse in cool water. Take off contaminated clothing and wash it before reuse. If skin irritation or rash occurs: Get medical advice/attention.
First-aid measures after eye contact	: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a poison center or doctor/physician.
First-aid measures after ingestion	: Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical advice/attention if you feel unwell.

4.2. Most important symptoms and effects (acute and delayed)

Symptoms/effects after inhalation	: May cause respiratory irritation.
Symptoms/effects after skin contact	: Causes skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin. May cause an allergic skin reaction. May cause burns in the presence of moisture. Skin contact during hydration may slowly develop sufficient heat that may cause severe burns possibly resulting in permanent injury. Do not allow product to harden around any body part or allow continuous, prolonged contact with skin. Handling can cause dry skin.
Symptoms/effects after eye contact	: Causes serious eye damage. Symptoms may include discomfort or pain, excess blinking and tear production, with marked redness and swelling of the conjunctiva. May cause burns.
Symptoms/effects after ingestion	: May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.
Chronic symptoms	: May cause cancer through inhalation of dust. May damage fertility or the unborn child. Causes damage to organs (lungs) through prolonged or repeated exposure.

4.3. Immediate medical attention and special treatment, if necessary

Symptoms may be delayed. In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

SECTION 5: Fire-fighting measures

5.1. Suitable (and unsuitable) extinguishing media

Suitable extinguishing media	: Water spray. Dry powder. Foam.
Unsuitable extinguishing media	: Do not use water jet.

5.2. Specific hazards arising from the chemical

Fire hazard	: Products of combustion may include, and are not limited to: oxides of carbon. irritating vapors.
Hazardous decomposition products in case of fire	: Toxic fumes may be released.

5.3. Special protective equipment and precautions for fire-fighters

Protection during firefighting	: Keep upwind of fire. Wear full fire fighting turn-out gear (full Bunker gear) and respiratory protection (SCBA).
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SECTION 6: Accidental release measures

6.1. Personal precautions, protective equipment and emergency procedures

General measures	: Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.
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6.1.1. For non-emergency personnel

No additional information available

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

6.1.2. For emergency responders

No additional information available

6.2. Environmental precautions

Prevent entry to sewers and public waters.

6.3. Methods and material for containment and cleaning up

- | | |
|-------------------------|--|
| For containment | : Contain spill, then place in a suitable container. Minimize dust generation. Do not flush to sewer or allow to enter waterways. Use appropriate Personal Protective Equipment (PPE). |
| Methods for cleaning up | : Vacuum or sweep material and place in a disposal container. Provide ventilation. |

6.4. Reference to other sections

For further information refer to section 8: "Exposure controls/personal protection".

SECTION 7: Handling and storage

7.1. Precautions for safe handling

- | | |
|-------------------------------|--|
| Precautions for safe handling | : Obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Avoid contact with skin and eyes. Do not breathe dust/fume/gas/mist/vapors/spray. Do not swallow. When using do not eat, drink or smoke. Handle and open container with care. Avoid generating dust. The use of compressed air for cleaning clothing, equipment, etc, is not recommended. Good housekeeping is important to prevent accumulation of dust. Use only outdoors or in a well-ventilated area. If not in a well-ventilated area, wear a NIOSH-approved respirator or other dust mask when using the product to avoid or minimize exposure to dust.
Formaldehyde, Ethylene oxide, and Chromium, ion (Cr 6+) are subject to the standard 29 CFR 1910.1048, 29 CFR 1910.1047, and 29 CFR 1910.1026 which may contain specific requirements for handling including protective equipment, regulated areas, monitoring and medical surveillance. The employer should review the standard and assure compliance with applicable requirements. |
| Hygiene measures | : Take off contaminated clothing and wash it before reuse. Contaminated work clothing should not be allowed out of the workplace. Wash hands, forearms and face thoroughly after handling. |

7.2. Conditions for safe storage, including any incompatibilities

- | | |
|--------------------|---|
| Storage conditions | : Keep out of the reach of children. Store locked up. Keep away from food, drink and animal feedingstuffs. Store in dust-tight, dry, labelled containers. Avoid any dust buildup by frequent cleaning and suitable construction of the storage area. Do not store in an area equipped with emergency water sprinklers. Keep container tightly closed when not in use. Store in a cool, well-ventilated place. |
|--------------------|---|

SECTION 8: Exposure controls/personal protection

8.1. Control parameters

Precision Grout Non Shrink	
No additional information available	
Quartz (14808-60-7)	
USA - ACGIH - Occupational Exposure Limits	
ACGIH OEL TWA	0.025 mg/m³ (respirable particulate matter)
ACGIH chemical category	Suspected Human Carcinogen

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Quartz (14808-60-7)	
USA - OSHA - Occupational Exposure Limits	
Local name	Quartz (Total Dust) (Silica: Crystalline)
OSHA PEL (TWA) [1]	50 µg/m³ (Respirable crystalline silica)
Remark (OSHA)	Table Z-3. For OSHA PEL (TWA) use formula: (30 mg/m³ / (%SiO₂+2)) for mg/m³. CAS No. source: eCFR Table Z-1.
Regulatory reference (US-OSHA)	OSHA Annotated Table Z-3 Mineral Dusts
Cement, portland, chemicals (65997-15-1)	
USA - ACGIH - Occupational Exposure Limits	
Local name	Portland cement
ACGIH OEL TWA	1 mg/m³ (particulate matter containing no asbestos and <1% crystalline silica, respirable particulate matter)
Remark (ACGIH)	TLV® Basis: Pulm func; resp symptoms; asthma. Notations: A4 (Not classifiable as a Human Carcinogen)
ACGIH chemical category	Not Classifiable as a Human Carcinogen
Regulatory reference	ACGIH 2020
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [1]	15 mg/m³ (total dust) 5 mg/m³ (respirable fraction)
Sulfuric acid, calcium salt (1:1) (7778-18-9)	
USA - ACGIH - Occupational Exposure Limits	
ACGIH OEL TWA	10 mg/m³ (inhalable particulate matter)
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [1]	15 mg/m³ (total dust) 5 mg/m³ (respirable fraction)
Limestone (1317-65-3)	
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [1]	15 mg/m³ (total dust) 5 mg/m³ (respirable fraction)
Flue dust, zinc-refining (69012-63-1)	
No additional information available	
Formaldehyde (50-00-0)	
USA - ACGIH - Occupational Exposure Limits	
ACGIH OEL TWA [ppm]	0.1 ppm
ACGIH OEL STEL [ppm]	0.3 ppm
ACGIH chemical category	Confirmed Human Carcinogen, dermal sensitizer
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [2]	0.75 ppm
OSHA PEL (STEL) [2]	2 ppm (see 29 CFR 1910.1048)

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Ethylene oxide (75-21-8)	
USA - ACGIH - Occupational Exposure Limits	
ACGIH OEL TWA [ppm]	1 ppm
ACGIH chemical category	Suspected Human Carcinogen
USA - ACGIH - Biological Exposure Indices	
BEI (BLV)	Parameter: N-(2-Hydroxyethyl)valine (HEV) hemoglobin adducts - Medium: blood - Sampling time: not critical (nonspecific) Parameter: S-(2-Hydroxyethyl)mercapturic acid (HEMA) - Medium: urine - Sampling time: end of shift (nonspecific, population based)
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [2]	1 ppm
OSHA PEL (STEL) [2]	5 ppm (see 29 CFR 1910.1047)
Chromium, ion (Cr 6+) (18540-29-9)	
USA - OSHA - Occupational Exposure Limits	
OSHA PEL (TWA) [1]	5 µg/m³

8.2. Appropriate engineering controls

Appropriate engineering controls	: Ensure good ventilation of the work station. Provide readily accessible eye wash stations and safety showers.
Environmental exposure controls	: Avoid release to the environment.

8.3. Individual protection measures/Personal protective equipment

Hand protection:
Wear suitable waterproof gloves
Eye protection:
Wear approved eye protection (properly fitted dust- or splash-proof chemical safety goggles) and face protection (face shield).
Skin and body protection:
Wear suitable waterproof protective clothing
Respiratory protection:
A NIOSH approved dust mask or filtering facepiece is recommended in poorly ventilated areas or when permissible exposure limits may be exceeded. Respirators should be selected by and used under the direction of a trained health and safety professional following requirements found in OSHA's respirator standard (29 CFR 1910.134) and ANSI's standard for respiratory protection (Z88.2).

Other information:

Do not eat, smoke or drink where material is handled, processed or stored. Wash hands carefully before eating or smoking. Handle according to established industrial hygiene and safety practices.

SECTION 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical state	: Solid
Appearance	: Powder
Color	: No data available
Odor	: No data available

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Odor threshold	: No data available
pH	: No data available
pH solution	: 12 – 13
Melting point	: No data available
Freezing point	: Not applicable
Boiling point	: No data available
Flash point	: Not applicable
Relative evaporation rate (butyl acetate=1)	: No data available
Flammability (solid, gas)	: Not flammable.
Vapor pressure	: No data available
Relative vapor density at 20 °C	: No data available
Relative density	: No data available
Solubility	: No data available
Partition coefficient n-octanol/water	: No data available
Auto-ignition temperature	: Not applicable
Decomposition temperature	: No data available
Viscosity, kinematic	: Not applicable
Viscosity, dynamic	: No data available
Explosion limits	: Not applicable
Explosive properties	: No data available
Oxidizing properties	: No data available

9.2. Other information

VOC content : 0%, Not applicable; 0 wt, Not applicable.

SECTION 10: Stability and reactivity

10.1. Reactivity

No dangerous reactions known under normal conditions of use.

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

No dangerous reactions known under normal conditions of use.

10.4. Conditions to avoid

Heat. Incompatible materials. Moisture.

10.5. Incompatible materials

Wet cement is alkaline and incompatible with acid, ammonium salts and aluminum metal.

10.6. Hazardous decomposition products

May include, and are not limited to: oxides of carbon. irritating vapors.

SECTION 11: Toxicological information

11.1. Information on toxicological effects

Acute toxicity (oral)	: Not classified
Acute toxicity (dermal)	: Not classified
Acute toxicity (inhalation)	: Not classified

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Sulfuric acid, calcium salt (1:1) (7778-18-9)	
LD50 oral rat	> 3000 mg/kg
LC50 inhalation rat	> 3.26 mg/l air Animal: rat, Guideline: OECD Guideline 403 (Acute Inhalation Toxicity)
Flue dust, zinc-refining (69012-63-1)	
LD50 dermal rat	> 2000 mg/kg
LC50 inhalation rat	> 5.371 mg/l/4h
Formaldehyde (50-00-0)	
LD50 oral rat	100 mg/kg
LD50 dermal rat	> 2000 mg/kg
LC50 inhalation rat	480 ppm
Ethylene oxide (75-21-8)	
LD50 oral rat	72 mg/kg
LC50 inhalation rat	800 ppm
Skin corrosion/irritation	: Causes skin irritation.
Serious eye damage/irritation	: Causes serious eye damage.
Respiratory or skin sensitization	: May cause an allergic skin reaction.
Germ cell mutagenicity	: Not classified
Carcinogenicity	: May cause cancer through inhalation of dust.
Quartz (14808-60-7)	
IARC group	1 - Carcinogenic to humans
National Toxicology Program (NTP) Status	Known Human Carcinogens
In OSHA Hazard Communication Carcinogen list	Yes
Sulfuric acid, calcium salt (1:1) (7778-18-9)	
NOAEL (chronic,oral,animal/male,2 years)	256 mg/kg body weight Animal: rat, Animal sex: male, Guideline: other:No data, Remarks on results: other:Effect type: carcinogenicity (migrated information)
NOAEL (chronic,oral,animal/female,2 years)	284 mg/kg body weight Animal: rat, Animal sex: female, Guideline: other:No data, Remarks on results: other:Effect type: carcinogenicity (migrated information)
Formaldehyde (50-00-0)	
IARC group	1 - Carcinogenic to humans
National Toxicology Program (NTP) Status	Known Human Carcinogens
In OSHA Hazard Communication Carcinogen list	Yes
In OSHA Specifically Regulated Carcinogen list	Yes
Ethylene oxide (75-21-8)	
IARC group	1 - Carcinogenic to humans
National Toxicology Program (NTP) Status	Known Human Carcinogens
In OSHA Hazard Communication Carcinogen list	Yes
In OSHA Specifically Regulated Carcinogen list	Yes
Chromium, ion (Cr 6+) (18540-29-9)	
IARC group	1 - Carcinogenic to humans

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Chromium, ion (Cr 6+) (18540-29-9)	
In OSHA Hazard Communication Carcinogen list	Yes
In OSHA Specifically Regulated Carcinogen list	Yes
Reproductive toxicity	: May damage fertility or the unborn child.
STOT-single exposure	: May cause respiratory irritation.
Cement, portland, chemicals (65997-15-1)	
STOT-single exposure	May cause respiratory irritation.
Formaldehyde (50-00-0)	
STOT-single exposure	May cause damage to organs. May cause drowsiness or dizziness. May cause respiratory irritation.
Ethylene oxide (75-21-8)	
STOT-single exposure	May cause respiratory irritation.
STOT-repeated exposure	: Causes damage to organs (lungs) through prolonged or repeated exposure. Respirable crystalline silica in the form of quartz or cristobalite from occupational sources is listed by the International Agency for Research on Cancer (IARC) and National Toxicology Program (NTP) as a lung carcinogen. Prolonged exposure to respirable crystalline silica has been known to cause silicosis, a lung disease, which may be disabling. While there may be a factor of individual susceptibility to a given exposure to respirable silica dust, the risk of contracting silicosis and the severity of the disease is clearly related to the amount of dust exposure and the length of time (usually years) of exposure.
Quartz (14808-60-7)	
STOT-repeated exposure	Causes damage to organs through prolonged or repeated exposure.
Sulfuric acid, calcium salt (1:1) (7778-18-9)	
LOAEL (oral,rat,90 days)	237 mg/kg body weight Animal: rat, Animal sex: male, Guideline: OECD Guideline 422 (Combined Repeated Dose Toxicity Study with the Reproduction / Developmental Toxicity Screening Test)
NOAEL (oral,rat,90 days)	79 mg/kg body weight Animal: rat, Animal sex: male, Guideline: OECD Guideline 422 (Combined Repeated Dose Toxicity Study with the Reproduction / Developmental Toxicity Screening Test)
Limestone (1317-65-3)	
STOT-repeated exposure	Causes damage to organs through prolonged or repeated exposure.
Flue dust, zinc-refining (69012-63-1)	
STOT-repeated exposure	May cause damage to organs through prolonged or repeated exposure.
Ethylene oxide (75-21-8)	
STOT-repeated exposure	Causes damage to organs through prolonged or repeated exposure.
Aspiration hazard	: Not classified
Viscosity, kinematic	: Not applicable
Symptoms/effects after inhalation	: May cause respiratory irritation.
Symptoms/effects after skin contact	: Causes skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin. May cause an allergic skin reaction. May cause burns in the presence of moisture. Skin contact during hydration may slowly develop sufficient heat that may cause severe burns possibly resulting in permanent injury. Do not allow product to harden around any body part or allow continuous, prolonged contact with skin. Handling can cause dry skin.
Symptoms/effects after eye contact	: Causes serious eye damage. Symptoms may include discomfort or pain, excess blinking and tear production, with marked redness and swelling of the conjunctiva. May cause burns.

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Symptoms/effects after ingestion	: May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.
Chronic symptoms	: May cause cancer through inhalation of dust. May damage fertility or the unborn child. Causes damage to organs (lungs) through prolonged or repeated exposure.
Other information	: Likely routes of exposure: ingestion, inhalation, skin and eye.

SECTION 12: Ecological information

12.1. Toxicity

Ecology - general : May cause long-term adverse effects in the aquatic environment.

Sulfuric acid, calcium salt (1:1) (7778-18-9)	
LC50 - Fish [1]	2980 mg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
LC50 - Fish [2]	> 1970 mg/l (Exposure time: 96 h - Species: Pimephales promelas [static])
Flue dust, zinc-refining (69012-63-1)	
LC50 - Fish [1]	> 100 mg/l (Exposure time: 96 h - Species: Danio rerio [static])
Formaldehyde (50-00-0)	
LC50 - Fish [1]	1.8 mg/l
EC50 - Crustacea [1]	2 mg/l (Exposure time: 48 h - Species: Daphnia magna)
LC50 - Fish [2]	1510 µg/l (Exposure time: 96 h - Species: Lepomis macrochirus [static])
EC50 - Crustacea [2]	11.3 – 18 mg/l (Exposure time: 48 h - Species: Daphnia magna [Static])
Ethylene oxide (75-21-8)	
LC50 - Fish [1]	84 mg/l
EC50 - Crustacea [1]	137 – 300 mg/l (Exposure time: 48 h - Species: Daphnia magna)
Chromium, ion (Cr 6+) (18540-29-9)	
LC50 - Fish [1]	36.2 mg/l (Exposure time: 96 h - Species: Pimephales promelas)
LC50 - Fish [2]	7.6 mg/l (Exposure time: 96 h - Species: Oncorhynchus mykiss)

12.2. Persistence and degradability

Precision Grout Non Shrink

Persistence and degradability	Not established.
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12.3. Bioaccumulative potential

Precision Grout Non Shrink

Bioaccumulative potential	Not established.
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Formaldehyde (50-00-0)

Partition coefficient n-octanol/water	0.35 (at 25 °C)
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Ethylene oxide (75-21-8)

Partition coefficient n-octanol/water	-0.3 (at 25 °C)
---------------------------------------	-----------------

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

12.4. Mobility in soil

Precision Grout Non Shrink

Ecology - soil	No data available.
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12.5. Other adverse effects

Other adverse effects : No data available.
Other information : No other effects known.

SECTION 13: Disposal considerations

13.1. Disposal methods

Product/Packaging disposal recommendations : Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation. The generation of waste should be avoided or minimized wherever possible.

SECTION 14: Transport information

In accordance with DOT

14.1. UN number

Not regulated for transport

14.2. UN proper shipping name

Proper Shipping Name (DOT) : Not applicable

14.3. Transport hazard class(es)

DOT

Transport hazard class(es) (DOT) : Not applicable

14.4. Packing group

Packing group (DOT) : Not applicable

14.5. Environmental hazards

Other information : No supplementary information available.

14.6. Special precautions for user

Special transport precautions : Do not handle until all safety precautions have been read and understood.

14.7. Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable

SECTION 15: Regulatory information

15.1. US Federal regulations

All components of this product are listed, or excluded from listing, on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory.

Precision Grout Non Shrink

Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

15.2. International regulations

No additional information available

15.3. US State regulations

⚠ WARNING: This product can expose you to chemicals including Quartz, Formaldehyde, and Propylene Oxide, which are known to the State of California to cause cancer, and Chromium ion (Cr 6+) and Ethylene oxide, which are known to the State of California to cause birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov.

SECTION 16: Other information

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Issue date : 05/05/2016
Revision date : 02/18/2022
Other information : None.
Prepared by : Nexreg Compliance Inc.
www.Nexreg.com



Full text of H-phrases

Carc. 1A	Carcinogenicity Category 1A
Eye Dam. 1	Serious eye damage/eye irritation Category 1
Repr. 1A	Reproductive toxicity Category 1A
Skin Irrit. 2	Skin corrosion/irritation Category 2
Skin Sens. 1B	Skin sensitization, category 1B
STOT RE 1	Specific target organ toxicity (repeated exposure) Category 1
STOT SE 3	Specific target organ toxicity — Single exposure, Category 3, Respiratory tract irritation

Indication of changes:

SDS update . GHS classification.

Safety Data Sheet (SDS), USA

Disclaimer: We believe the statements, technical information and recommendations contained herein are reliable, but they are given without warranty or guarantee of any kind. The information contained in this document applies to this specific material as supplied. It may not be valid for this material if it is used in combination with any other materials. It is the user's responsibility to satisfy oneself as to the suitability and completeness of this information for the user's own particular use.



SAFETY DATA SHEET

Prepared to U.S. OSHA, CMA, ANSI, Canadian WHMIS, the Korean ISHA (Notice 2009-68), the Japanese Industrial Standard JIS Z 7250: 2000, Mexican NOM018-STPS 2000, SPRING Singapore, and the Global Harmonization Standard

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY UNDERTAKING

IDENTIFICATION OF THE MIXTURE

TRADE/MATERIAL NAME:
RELEVANT USE of the SUBSTANCE:
USES ADVISED AGAINST:
SUPPLIER/MANUFACTURER'S NAME:
Address:

Business Phone:
Emergency Phone:

SpecSeal® Smoke 'N' Sound Acoustical Sealant
Firestop and Sound Transmission
None
Specified Technologies Inc.
210 Evans Way,
Somerville, New Jersey 08876
(908) 526-8000 (8:00am to 5:00pm Eastern Standard Time)
U.S., Canada: 1-800-255-3924 (24 hrs)
International: +1-813-248-0585 (collect-24 hrs)

EMAIL of Competent Person for Information on SDS: techserv@stifiestop.com

NOTE: ALL United States Occupational Safety and Health Administration Standard (29 CFR 1910.1200), U.S. State equivalent Standards, Canadian WHMIS [Controlled Products Regulations], Mexican NOM018-STPS 2000, SPRING Singapore, and Japanese JIS Z7250 required information is included in appropriate sections based on the U.S. ANSI Z400.1-2010 format. This product has been classified in accordance with the hazard criteria of the countries listed above.

2. HAZARD IDENTIFICATION

GLOBAL HARMONIZATION AND JAPANESE JIS Z7253 LABELING AND CLASSIFICATION: This product has been classified per UN GHS Standards under U.S., Japanese and other applicable regulations that require Global Harmonization compliance.

Classification: Carcinogenic Category 1, Eye Irritation Category 2A, Specific Target Organ Toxicity (Inhalation-Respiratory Irritation) Single Exposure Category 3

Signal Word: Warning

Hazard Statements: H351: This product contains trace amounts of a suspected human carcinogen by inhalation: however, this hazard is not expected to be significant due to viscosity and consistency of the mixture.

H319: Causes serious eye irritation. H335: May cause respiratory irritation.

Precautionary Statements:

Prevention: P201: Obtain special instructions before use. P202: Do not handle until all safety precautions have been read and understood. P261: Avoid breathing vapors, fumes. P271: Use only in a well-ventilated area. P280: Wear protective gloves, clothing, eye protection and face protection. P284: Wear respiratory protection.

Response: P308 + P313: IF exposed or concerned: Get medical advice/attention. P305 + P351 + P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. P337 + P313: If eye irritation persists: Get medical advice/attention. P304 + P340: If inhaled, remove victim to fresh air and keep at rest in a position comfortable for breathing. P312: Call a POISON CENTER or doctor if you feel unwell. P321: Specific treatment (remove from exposure and treat symptoms).

Storage: P403 + P233 + P405: Store in a well-ventilated place. Keep container tightly closed. Store locked up.

Disposal: P501: Dispose of contents/containers in accordance with all local, regional, national and international regulations.

Hazard Symbols: GHS07, GHS08



3. COMPOSITION and INFORMATION ON INGREDIENTS

Hazardous Components:

Chemical Name	CAS #	Chinese IECSC Inventory	Japanese ENCS #	Korean ECL #	Taiwan NESCI ECS	WT%	LABEL ELEMENTS GHS & Japanese JIS Z7253 Classification Korean ISHA Classification GHS Hazard Codes
Titanium Dioxide	13463-67-7	Listed	1-558	KE-33390		1-2%	SELF CLASSIFICATION GHS & JAPANESE JIS Z7253, KOREAN ISHA: Classification: Carcinogenic Cat. 2 Hazard Codes: H351i
Crystalline Silica	14808-60-7	Listed	1-548	KE-29983		0.05-0.5%	SELF CLASSIFICATION GHS & JAPANESE JIS Z7253, KOREAN ISHA: Classification: Carcinogenic Cat. 1, STOT (Inhalation-Lungs) RE Cat. 2 Hazard Statement Codes: H350, H373

4. FIRST-AID MEASURES

DESCRIPTION OF FIRST AID MEASURES: Contaminated individuals must be taken for medical attention if any adverse effects occur. Remove contaminated clothing and shoes. Take a copy of this SDS to health professional with victim. Wash clothing and thoroughly clean shoes before reuse. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Take a copy of label and SDS to physician or health professional with the contaminated individual.

Skin Exposure: If adverse skin effects occur, discontinue use and flush contaminated area. Seek medical attention if adverse effect occurs after flushing.

Inhalation: If fumes or vapors are inhaled, remove victim to fresh air. Seek medical attention if adverse effect continues after removal to fresh air.

Eye Exposure: If this product contaminates the eyes, rinse eyes under gently running water. Use sufficient force to open eyelids and then "roll" eyes while flushing. Minimum flushing is for 20 minutes. **Ingestion:** If this product is swallowed, CALL PHYSICIAN OR POISON CONTROL CENTER FOR MOST CURRENT INFORMATION. If professional advice is not available, DO NOT INDUCE VOMITING. Never induce vomiting or give diluents (milk or water) to someone who is unconscious, having convulsions, or unable to swallow.

MEDICAL CONDITIONS AGGRAVATED BY EXPOSURE: Pre-existing respiratory disorders may be aggravated by overexposures to this product.

INDICATION OF IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT IF NEEDED: Treat symptoms and eliminate exposure.

5. FIRE-FIGHTING MEASURES

FLASH POINT: 320 °C (608 °F)

AUTOIGNITION TEMPERATURE: Not available.

FLAMMABLE LIMITS (in air by volume, %): Not applicable.

FIRE EXTINGUISHING MEDIA: Use extinguishing materials suitable for the surrounding area.

UNSUITABLE FIRE EXTINGUISHING MEDIA: None known.

UNUSUAL FIRE AND EXPLOSION HAZARDS: This product is formulated to be non-flammable and non-combustible.

When involved in a fire, this material may decompose and produce irritating vapors and toxic gases

Explosion Sensitivity to Mechanical Impact: Not sensitive.

Explosion Sensitivity to Static Discharge: Not sensitive.

SPECIAL PROTECTIVE ACTIONS FOR FIRE-FIGHTERS: No Special protective actions for fire-fighters are anticipated.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS AND EMERGENCY PROCEDURES: Uncontrolled releases should be responded to by trained personnel using pre-planned procedures. Proper protective equipment should be used. Call CHEMTREC (1-800-424-9300) for emergency assistance. Or if in Canada, call CANUTEC (613-996-6666).

PERSONAL PROTECTIVE EQUIPMENT: Proper protective equipment should be used.

Small Spills: Wear rubber gloves, splash goggles, and appropriate body protection.

Large Spills: Minimum Personal Protective Equipment should be chemical resistant gloves

METHODS FOR CLEAN-UP AND CONTAINMENT: Spills of this product present minimal hazard.

Small Spills: Small releases can be carefully swept up or cleaned up using a damp sponge or polypads.

METHODS FOR CLEAN-UP AND CONTAINMENT (continued):

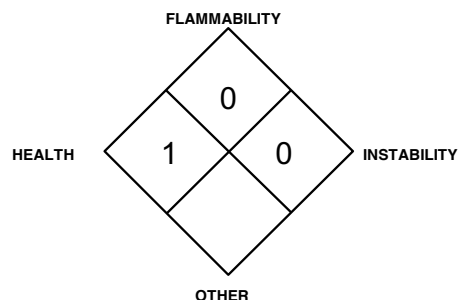
Large Spills: Access to the spill area should be restricted. For large spills, dike or otherwise contain spill and sweep-up or vacuum with non-sparking vacuum.

All Spills: Place all spill residue in a double plastic bag or other containment and seal.

Close off sewers and take other measures to protect human health and the environment as necessary. Rinse area with soap and water solution and follow with a water rinse. Do not mix with wastes from other materials. Dispose of in accordance with applicable Federal, State, and local procedures (see Section 13, Disposal Considerations). For spills on water, contain, minimize dispersion and collect. Dispose of recovered material and report spill per regulatory requirements.

ENVIRONMENTAL PRECAUTIONS: Avoid release to the environment. Run-off water may be contaminated by other materials and should be contained to prevent possible environmental damage.

NFPA RATING



Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate
3 = Serious 4 = Severe

7. HANDLING and USE

PRECAUTIONS FOR SAFE HANDLING: As with all chemicals, avoid getting this material ON YOU or IN YOU. Do not eat, drink, smoke, or apply cosmetics while handling this product. Wash hands thoroughly after handling this product or containers of this product. Avoid breathing fumes or vapors generated by this product. Use in a well-ventilated location.

CONDITIONS FOR SAFE STORAGE: Store containers in a cool, dry location, away from direct sunlight, sources of intense heat. Do not store above 55°C (131°F)

SPECIFIC END USE(S): This product is for use as a sealant. Follow all industry standards for use of this product.

PROTECTIVE PRACTICES DURING MAINTENANCE OF CONTAMINATED EQUIPMENT: Follow practices indicated in Section 6 (Accidental Release Measures). Collect all rinsates and dispose of according to applicable Federal, State, and local procedures.

8. EXPOSURE CONTROLS - PERSONAL PROTECTION

EXPOSURE LIMITS/CONTROL PARAMETERS:

Ventilation and Engineering Controls: Use with adequate ventilation to ensure exposure levels are maintained below the limits provided below (if applicable). Exhaust directly to the outside, taking necessary precautions for environmental protection.

Workplace Exposure Limits/Control Parameters:

CHEMICAL NAME	CAS #	EXPOSURE LIMITS IN AIR							
		ACGIH-TLVs		OSHA-PELs		NIOSH-RELs		NIOSH	OTHER
		TWA mg/m ³	STEL mg/m ³	TWA mg/m ³	STEL mg/m ³	TWA mg/m ³	STEL mg/m ³	IDLH mg/m ³	mg/m ³
Crystalline Silica (Quartz)	14808-60-7	0.025 (resp. fract.)	NE	0.05 mg/m ³ (resp. dust)	NE	0.05 (resp. dust)	NE	50	Carcinogen: IARC-1, MAK-1 (respirable fraction), NIOSH-Ca, NTP-K (respirable fraction), TLV-A2
Titanium Dioxide	13463-67-7	10	NE	15 (total dust); 10 (vacated 1989 PEL)	NE	See Pocket Guide App. A		5000 (Ca)	Carcinogen: IARC-2B, MAK-3A, NIOSH-Ca; TLV-A3

NE = Not Established. See Section 16 for Definitions of Other Terms Used

International Occupational Exposure Limits: Currently, the following additional exposure limit values have been established by various countries for the components of this mixture. More current limits may be available; individual countries should be consulted to determine if newer limits are available.

CRYSTALLINE SILICA:

Australia: TWA = 0.1 mg/m³, JUL 2008
Belgium: TWA = 0.1 mg/m³ (resp. dust), MAR 2002
Denmark: TWA = 0.1 mg/m³ (respirable), carc, MAY 2011
Denmark: TWA = 0.1 mg/m³ (resp.), carc, MAY 2011
Denmark: TWA = 0.3 mg/m³ (total), MAY 2011
Finland: TWA = 0.05 mg/m³, resp. dust, SEP 2009
France: VME = 0.1 mg/m³, (resp), FEB 2006
Iceland: TWA = 0.1 mg/m³ (resp. dust), NOV 2011
Japan: OEL-C = 0.03 mg/m³ (respirable), APR 2007
Korea: TWA = 0.1 mg/m³, 2006
Mexico: TWA = 0.1 mg/m³ (respirable), 2004
The Netherlands: MAC-TGG = 0.075 mg/m³, 2003
New Zealand: TWA = 0.2 mg/m³ (respirable dust), JAN 2002
Norway: TWA = 0.1 mg/m³ (resp. dust), JAN 1999
Norway: TWA = 0.3 mg/m³ (total dust), JAN 1999

Peru: TWA = 0.05 mg/m³, JUL 2005

Russia: TWA = 1 mg/m³, STEL = 3 mg/m³, JUN 2003

Sweden: TWA = 0.1 mg/m³ (resp. dust), JUN 2005

Switzerland: MAK-W = 0.15 mg/m³, DEC 2006

Thailand: TWA = 10 mg/m³ (resp. dust), JAN 1993

Thailand: TWA = 30 mg/m³ (total dust), JAN 1993

United Kingdom: TWA = 0.1 mg/m³ (resp. dust), OCT 2007

In Argentina, Bulgaria, Colombia, Jordan, Singapore, Vietnam check ACGIH TLV

GROUND LIMESTONE:

Belgium: TWA = 10 mg/m³, MAR 2002

Hungary: TWA = 10 mg/m³, SEP 2000

Japan: OEL = 2 mg/m³ (resp. dust), 84 mg/m³ (total dust), MAY 2012

Korea: TWA = 10 mg/m³, 2006

Mexico: TWA = 10 mg/m³; STEL = 20 mg/m³ (inhalable), 2004

The Netherlands: MAC-TGG = 10 mg/m³, 2003

New Zealand: TWA = 10 mg/m³ (inspirable dust), JAN 2002

Norway: TWA = 5 mg/m³, JAN 1999

Peru: TWA = 10 mg/m³, JUL 2005

Poland: MAC(TWA) = 10 mg(Ti)/m³, MAC(STEL) = 30 mg(Ti)/m³, JAN 1999

Russia: TWA = 10 mg/m³, JUN 2003

Sweden: TWA = 5 mg/m³ (total dust), JUN 2005

Switzerland: MAK-W = 3 mg/m³, DEC 2006

Turkey: TWA = 15 mg/m³, JAN 1993

United Kingdom: TWA = 10 mg/m³ (inhal. dust), OCT 2007

United Kingdom: TWA = 4 mg/m³ (resp. dust), OCT 2007

In Argentina, Bulgaria, Colombia, Jordan, Singapore, Vietnam check ACGIH TLV

PROTECTIVE EQUIPMENT: The following information on appropriate Personal Protective Equipment is provided to assist employers in complying with OSHA regulations found in 29 CFR Subpart I (beginning at 1910.132, including U.S. Federal OSHA Respiratory Protection (29 CFR 1910.134), OSHA Eye Protection 29 CFR 1910.133, OSHA Hard Protection 29 CFR 1910.138, OSHA Foot Protection 29 CFR 1910.136 and OSHA Body Protection 29 CFR 1910.132), equivalent standards of Canada (including CSA Respiratory Standard Z94.4-02, Z94.3-M1982, Industrial Eye and Face Protectors and CSA Standard Z195-02, Protective Footwear), or standards of Japan (including JIS T 8116:2005 for glove selection, JIS T 8150:2006 for respiratory PPE, JIS T 8147:2003 for eye protectors, and JIS T 8030:2005 for protective clothing). Please reference applicable regulations and standards for relevant details.

Respiratory Protection: Maintain airborne contaminant concentrations below exposure limits listed above. For materials without listed

8. EXPOSURE CONTROLS - PERSONAL PROTECTION (Continued)

exposure limits, minimize respiratory exposure. If necessary, use only respiratory protection authorized under appropriate regulations.

Eye Protection: Wear splash goggles or safety glasses as appropriate for the task.

Hand Protection: During manufacture or other similar operations, wear the appropriate hand protection for the process.

Skin Protection: Use appropriate protective clothing. If necessary, refer to the U.S. OSHA Technical Manual (Section VII: Personal Protective Equipment) or other appropriate regulations. Full-body chemical protective clothing is recommended for emergency response procedures.

9. PHYSICAL and CHEMICAL PROPERTIES

FORM: Paste.

MOLECULAR FORMULA: Mixture.

ODOR: Mild acrylic.

FLAMMABLE LIMITS (in air by volume, %): Not applicable.

DECOMPOSITION TEMPERATURE: Not available.

AUTOIGNITION TEMPERATURE: Not available.

FREEZING/MELTING POINT: Not available.

VAPOR PRESSURE: Not available.

VAPOR DENSITY (air = 1): Not available.

EVAPORATION RATE (*n*-BuAc = 1): > 1

SOLUBILITY IN WATER: Dissolves when wet; insoluble when cured.

COEFFICIENT WATER/OIL DISTRIBUTION: Not established.

HOW TO DETECT THIS SUBSTANCE (warning properties in event of accidental release): The appearance may be characteristics to distinguish a release of this product.

COLOR: White.

MOLECULAR WEIGHT: Mixture.

ODOR THRESHOLD: Not available.

OXIDIZING PROPERTIES: Not applicable.

PERCENT VOLATILE: 24

FLASH POINT: 320 C 608 F

BOILING POINT: 100-105°C (212-221°F)

SPECIFIC GRAVITY (water = 1): 1.3 gm/L

CARB VOC: 0.28 wt % (calc.)

VOC (U.S. EPA Method 24): 20 g/L

SOLUBILITY IN SOLVENTS: Not available.

pH: Not available.

10. STABILITY and REACTIVITY

CHEMICAL STABILITY: This product is stable when properly stored at normal temperature and pressures (see Section 7, Handling and Storage).

DECOMPOSITION PRODUCTS: Combustion: If exposed to extremely high temperatures, thermal decomposition may generate irritating fumes and toxic Hydrolysis: None known.

MATERIALS WITH WHICH SUBSTANCE IS INCOMPATIBLE: This product is incompatible with strong oxidizers.

POSSIBILITY OF HAZARDOUS POLYMERIZATION OR REACTION: Will not occur.

CONDITIONS TO AVOID: Avoid exposure to or contact with extreme temperatures and incompatible chemicals.

11. TOXICOLOGICAL INFORMATION

SYMPTOMS OF EXPOSURE BY ROUTE OF EXPOSURE: The health hazard information provided below is pertinent to employees using this product in an occupational setting. The following paragraphs describe the symptoms of exposure by route of exposure.

Inhalation: Inhalation of fumes or vapors may cause irritation of the nose, throat, and lungs and cause coughing. Removal to fresh air should relieve symptoms. This product contains trace amounts of a suspected human carcinogen by inhalation; however, this hazard is not expected to be significant due to viscosity and consistency of the mixture.

Contact with Skin or Eyes: Direct eye contact may cause irritation, redness, and tearing from mechanical irritation. Prolonged or repeated skin exposures may cause dermatitis (dry red skin).

Skin Absorption: Components are not known to be absorbed through intact skin.

Ingestion: Ingestion is not a significant route of occupational exposure and is unlikely to occur.

Injection: Accidental injection of this product, via laceration or puncture by a contaminated object can cause redness at the site of injection. Animal data for the Crystalline Silica component indicate that it may cause carcinogenic effects by this route of exposure.

HEALTH EFFECTS OR RISKS FROM EXPOSURE: Exposure to this product may cause the following health effects:

Acute: Inhalation of fumes or vapors may cause irritation of respiratory system. Eye contact may cause mechanical irritation.

Chronic: Prolonged or repeated skin exposure may cause dermatitis (dry red skin). This product contains trace amounts of a suspected human carcinogen by

HAZARDOUS MATERIAL IDENTIFICATION SYSTEM

HEALTH HAZARD	(BLUE)	1
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FLAMMABILITY HAZARD	(RED)	0
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PHYSICAL HAZARD	(YELLOW)	0
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PROTECTIVE EQUIPMENT

EYES	RESPIRATORY	HANDS	BODY
	SEE SECTION 8		SEE SECTION 8

For Routine Industrial Use and Handling Applications

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate
3 = Serious 4 = Severe * = Chronic hazard

11. TOXICOLOGICAL INFORMATION (Suite)

inhalation: however, this hazard is not expected to be significant due to viscosity and consistency of the mixture.

TARGET ORGANS: Acute: Skin, eyes, respiratory system. Chronic: Skin.

TOXICITY DATA: Currently, the following toxicological data are available for components of 1% or more concentration.

Due to the large amount of data available for the Titanium Dioxide and Crystalline Silica components, only human data, LD50 Oral Rat and Mouse, LD50 Skin Rabbit and Rat, LC50 Inhalation Rat and Mouse, carcinogenic and mutation data are provided. Contact STI for information on additional data for these components.

PROPRIETARY BENZOATE ESTERS:

LD₅₀ (Skin-Rat) > 2000 mg/kg

LC₅₀ (Inhalation-Rat) 4 hours = > 220 mg/L

TITANIUM DIOXIDE:

Standard Draize Test (Skin-Human) 300 µg/3 days-intermittent: Mild

TC (Inhalation-Rat) 10 mg/m³/18 hours/2 years-intermittent: Tumorigenic: carcinogenic by RTECS criteria; Lungs, Thorax, or Respiration: tumors

TDLo (Intratracheal-Mouse) 100 mg/kg: Tumorigenic: increased incidence of tumors in susceptible strains

TITANIUM DIOXIDE (continued):

TD (Intramuscular-Rat) 260 mg/kg/84 weeks-intermittent: Tumorigenic: equivocal tumorigenic agent by RTECS criteria; Blood: lymphoma, including Hodgkin's disease; Tumorigenic: tumors at site of application

TDLo (Intramuscular-Rat) 360 mg/kg/2 years-intermittent: Tumorigenic: neoplastic by RTECS criteria; Blood: lymphoma, including Hodgkin's disease; Tumorigenic: tumors at site of application

TCLo (Inhalation-Rat) 250 mg/m³/6 hours/2 years-intermittent: Tumorigenic: carcinogenic by RTECS criteria; Lungs, Thorax, or Respiration: tumors

DNA Damage (Human Lung) 100 µg/plate

DNA Damage (Human Lung) 20 µg/disk/4 hours

Sister Chromatid Exchange (Human Lymphocyte) 2 µmol/L/72 hours

Micronucleus Test (Human Lymphocyte) 5 µmol/L/72 hours

Micronucleus Test (Intraperitoneal-Mouse) 3 gm/kg/3 days-continuous

Micronucleus Test (Hamster Ovary) 5 µmol/L

DNA Inhibition (Hamster Lung) 500 mg/L

Sister Chromatid Exchange (Hamster Ovary) 1 µmol/L

IRRITANCY OF PRODUCT: Inhalation of fumes or vapors may cause respiratory irritation. Eye contact may cause irritation. Prolonged skin contact may cause irritation.

SENSITIZATION OF PRODUCT: This product is not currently known to cause allergic skin or respiratory reaction.

CARCINOGENIC POTENTIAL OF COMPONENTS: Components of this product are listed by agencies tracking the carcinogenic potential of chemical compounds, as follows:

CRYSTALLINE SILICA: ACGIH-TLV-A2 (Suspected Human Carcinogen); IARC-1 (Carcinogenic to Humans); MAK-1 (Substances that Cause Cancer in Man and Can Be Assumed to Make a Significant Contribution to Cancer Risk); NIOSH-Ca (Potential Occupational Carcinogen with No Further Categorization); NTP-K (Known to Be a Human Carcinogen)

TITANIUM DIOXIDE: ACGIH TLV-A3 (Confirmed Animal Carcinogen); IARC-3 (Unclassifiable as to Carcinogenicity in Humans); NIOSH-Ca (Potential Occupational Carcinogen, with No Further Categorization)

CARCINOGENIC POTENTIAL OF COMPONENTS (continued): The remaining components are not found on the following lists: U.S. EPA, U.S. NTP, U.S. OSHA, U.S. NIOSH, GERMAN MAK, IARC, or ACGIH and therefore is neither considered to be nor suspected to be a cancer-causing agent by these agencies.

REPRODUCTIVE TOXICITY INFORMATION: Components of this product have no reported mutagenic, embryotoxic, teratogenic or reproductive toxicity.

ACGIH BIOLOGICAL EXPOSURE INDICES (BEIs): Currently, there are no ACGIH Biological Exposure Indices (BEIs) determined for this material.

DEGREE OF EFFECT TO THE HEALTH OF THE POLLUTING AGENT OF ENVIRONMENT OF WORK (per Mexican NOM-010 STPS-1999): 0

12. ECOLOGICAL INFORMATION

ALL WORK PRACTICES MUST BE AIMED AT ELIMINATING ENVIRONMENTAL CONTAMINATION.

MOBILITY: This product has not been tested for mobility in soil.

PERSISTENCE AND BIODEGRADABILITY: This product has not been tested for persistence or biodegradability. The mineral components are not expected to biodegrade to great extent.

BIO-ACCUMULATION POTENTIAL: This product has not been tested for bio-accumulation potential.

ECOTOXICITY: This product has not been tested for aquatic or animal toxicity. All releases to terrestrial, atmospheric and aquatic environments should be avoided.

OTHER ADVERSE EFFECTS: This material is not listed as having ozone depletion potential.

ENVIRONMENTAL EXPOSURE CONTROLS: Controls

should be engineered to prevent release to the environment, including procedures to prevent spills, atmospheric release and release to waterways.

13. DISPOSAL CONSIDERATIONS

DISPOSAL METHODS: It is the responsibility of the generator to determine at the time of disposal whether the product meets the criteria of a hazardous waste per regulations of the area in which the waste is generated and/or disposed of. Waste disposal must be in accordance with appropriate Federal, State, and local regulations. This product, if unaltered by use, may be disposed of by treatment at a permitted facility or as advised by your local hazardous waste regulatory authority. Shipment of wastes must be done with appropriately permitted and registered transporters.

DISPOSAL CONTAINERS: Waste materials must be placed in and shipped in appropriate 5-gallon or 55-gallon poly or metal waste pails or drums. Permeable cardboard containers are not appropriate and should not be used. Ensure that any required marking or labeling of the containers be done to all applicable regulations.

PRECAUTIONS TO BE FOLLOWED DURING WASTE HANDLING: Wear proper protective equipment when handling waste materials.

U.S. EPA WASTE NUMBER: Not applicable.

14. TRANSPORTATION INFORMATION

U.S. DEPARTMENT OF TRANSPORTATION REGULATIONS: This product is not classified as dangerous goods, per U.S. DOT regulations, under 49 CFR 172.101.

TRANSPORT CANADA TRANSPORTATION OF DANGEROUS GOODS REGULATIONS: This product is not classified as Dangerous Goods, per regulations of Transport Canada.

INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA): This product is not classified as dangerous goods under rules of IATA.

INTERNATIONAL MARITIME ORGANIZATION (IMO) DESIGNATION: This product is not classified as Dangerous Goods by the International Maritime Organization.

OFFICIAL MEXICAN STANDARD; REGULATION FOR THE TRANSPORT OF DANGEROUS GOODS AND RESIDUES: This product is not classified as Dangerous Goods, per transport regulations of Mexico.

SINGAPORE STANDARD 286: PART A: This product has no requirements under the Specification for Caution Labeling for Hazardous Substances, Part 4: Marking of Packages, Containers and Vehicles, as it does not meet the criteria for any hazard class under this regulation.

TRANSPORT IN BULK ACCORDING TO THE IBC CODE: See the information under the individual jurisdiction listings for IBC information.

ENVIRONMENTAL HAZARDS: This material does not meet the criteria of environmentally hazardous according to the criteria of the UN Model Regulations (as reflected in the IMDG Code, ADR, RID, and ADN) and is not listed in Annex III under MARPOL 73/78.

15. REGULATORY INFORMATION

UNITED STATES REGULATIONS:

U.S. SARA Reporting Requirements: This product is not subject to the reporting requirements of Sections 302, 304, and 313 of Title III of the Superfund Amendments and Reauthorization Act.

U.S. SARA Hazard Categories (Section 311/312, 40 CFR 370-21): ACUTE: Yes; CHRONIC: Yes; FIRE: No; REACTIVE: No; SUDDEN RELEASE: No

UNITED STATES REGULATIONS (continued):

U.S. SARA Threshold Planning Quantity (TPQ): There are no specific Threshold Planning Quantities for components. The default Federal SDS submission and inventory requirement filing threshold of 10,000 lb (4,540 kg) may apply, per 40 CFR 370.20.

U.S. CERCLA Reportable Quantity (RQ): Not applicable.

U.S. TSCA Inventory Status: Components of this product are listed on the TSCA Inventory.

California Safe Drinking Water and Toxic Enforcement Act (Proposition 65): The Crystalline Silica component (airborne, unbound particles of respirable size) and Titanium Dioxide are found on the Proposition 65 List of chemicals known to the state to cause cancer. However due to the form of the product, the Proposition 65 warning for these components is not applicable.

CANADIAN REGULATIONS:

Canadian DSL/NDSL Inventory Status: Components are on the DSL or NDSL Inventories.

Canadian Environmental Protection Act (CEPA) Priorities Substances Lists: Components are not on the CEPA Priorities Substances Lists.

Canadian WHMIS Classification and Symbols: This product would be categorized as a Controlled Product, D2B (Other Toxic Effects- Potential Carcinogenic Effect, Irritation) as per the Controlled Product Regulations.



CHINESE REGULATIONS:

Chinese Inventory of Existing Chemical Substances Status: Components listed by CAS# are listed on the Chinese Inventory of Existing Chemical Substances (IECSC).

JAPANESE REGULATIONS:

Japanese ENCS: Components listed by CAS# are on the ENCS Inventory or are excepted.

15. REGULATORY INFORMATION (Continued)

Japanese Ministry of Economy, Trade, and Industry (METI) Status: Components are not listed as Class I Specified Chemical Substances, Class II Specified Chemical Substances, or Designated Chemical Substances by the Japanese METI.

Poisonous and Deleterious Substances Control Law: Components are not listed as a Specified Poisonous Substance under the Poisonous and Deleterious Substances Control Law.

KOREAN REGULATIONS:

Korean Existing Chemicals List (ECL) Status: Components listed by CAS# are listed on the Korean ECL Inventory.

MEXICAN REGULATIONS:

Mexican Workplace Regulations (NOM-018-STPS-2000): This product is classified as hazardous.

SINGAPORE REGULATIONS:

List of Controlled Hazardous Substances: Components listed by CAS# are not listed on the Singapore List of Controlled Substances.

Code of Practice on Pollution Control Requirements: The components identified by CAS# in Section 2 (Composition and Information on Ingredients) NOT are subject to the requirements under the Singapore Code of Practice on Pollution Control.

TAIWANESE REGULATIONS:

Taiwan Existing Chemical Substances Inventory Status: Components listed by CAS# are listed on the Taiwan Existing Chemicals List.

16. OTHER INFORMATION

REFERENCES AND DATA SOURCES: Contact the supplier for information.

METHODS OF EVALUATING INFORMATION FOR THE PURPOSE OF CLASSIFICATION: Criteria of the GHS were used for classification.

PREPARED BY: Specified Technologies Inc, 210 Evans Way Somerville NJ 08876

DATE OF PRINTING: December 14, 2023

REVISED: June 5, 2018

January 14, 2020

January 17, 2023

REVISION DETAILS: Revised Proposition 65 statement.
Revised hazard identification profile.
Periodic document review.



SAFETY DATA SHEET

Prepared to U.S. OSHA, CMA, ANSI, Canadian WHMIS, the Korean ISHA (Notice 2009-68), the Japanese Industrial Standard JIS Z 7250: 2000, Mexican NOM018-STPS 2000, SPRING Singapore, and the Global Harmonization Standard

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY UNDERTAKING

IDENTIFICATION OF THE MIXTURE

TRADE/MATERIAL NAME:

RELEVANT USE of the SUBSTANCE:

USES ADVISED AGAINST:

SUPPLIER/MANUFACTURER'S NAME:

Address:

Business Phone:

Emergency Phone:

SpecSeal® Series FP Intumescent Firestop Plug

Firestop Device

None

Specified Technologies Inc.

210 Evans Way,

Somerville, New Jersey 08876

(908) 526-8000 (8:00am to 5:00pm Eastern Standard Time)

U.S., Canada: 1-800-255-3924 (24 hrs)

International: +1-813-248-0585 (Collect-24 hrs)

EMAIL of Competent Person for Information on SDS:

techserv@stifiirestop.com

NOTE: ALL United States Occupational Safety and Health Administration Standard (29 CFR 1910.1200), U.S. State equivalent Standards, Canadian WHMIS [Controlled Products Regulations], Mexican NOM018-STPS 2000, SPRING Singapore, and Japanese JIS Z7250 required information is included in appropriate sections based on the U.S. ANSI Z400.1-2010 format. This product has been classified in accordance with the hazard criteria of the countries listed above.

This product is defined as an "Article" under the U.S. Federal OSHA Hazard Communication Standard (29 CFR 1910.1200), EU Directives, and the Canadian Workplace Hazardous Materials Standard. Refer to Section 15 (Regulatory Information) for specific regulatory citations. As articles, this product presents negligible health and physical hazards under reasonably anticipated circumstances of use. Subsequently, a Material Safety Data Sheet is not required under Standards cited above. This document is prepared to provide persons using this product with additional safety information.

2. HAZARD IDENTIFICATION

GLOBAL HARMONIZATION AND EU CLP REGULATION (EC) 1272/2008 LABELING AND CLASSIFICATION: This product is an article and is not required to be classified under CLP Regulation (EC) 1272/2008.

EU 67/548/EEC LABELING AND CLASSIFICATION: This product is an article and is not required to be classified under European Community Council Directive 67/548/EEC or subsequent Directives.

KOREAN ISHA (Notice 2009-68) LABELING AND CLASSIFICATION: As an article, this product is not subject to ISHA Notice 2009-68.

3. COMPOSITION and INFORMATION ON INGREDIENTS

This product is an article and as such no components of this product pose a hazard; no component information is given in this SDS.

4. FIRST-AID MEASURES

Skin Exposure: As an article, no need for first aid is anticipated.

Inhalation: As an article, no need for first aid is anticipated.

Eye Exposure: As an article, no need for first aid is anticipated.

Ingestion: As an article, no need for first aid is anticipated.

5. FIRE-FIGHTING MEASURES

FIRE EXTINGUISHING MEDIA: Use extinguishing materials suitable for the surrounding area.

UNSUITABLE FIRE EXTINGUISHING MEDIA: None known.

UNUSUAL FIRE AND EXPLOSION HAZARDS: This product is formulated to be non-flammable and non-combustible.

SPECIAL PROTECTIVE ACTIONS FOR FIRE-FIGHTERS: No Special protective actions for fire-fighters are anticipated.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS AND EMERGENCY PROCEDURES: Not applicable.

PERSONAL PROTECTIVE EQUIPMENT: Not applicable.

METHODS FOR CLEAN-UP AND CONTAINMENT: Not applicable.

ENVIRONMENTAL PRECAUTIONS: Not applicable.

7. HANDLING and STORAGE

PRECAUTIONS FOR SAFE HANDLING: No special requirements.

CONDITIONS FOR SAFE STORAGE: No special requirements.

8. EXPOSURE CONTROLS - PERSONAL PROTECTION

EXPOSURE LIMITS/CONTROL PARAMETERS: As an article which does not release or otherwise result in exposure to hazardous chemicals under normal use, no personal protective equipment (PPE) is required

9. PHYSICAL and CHEMICAL PROPERTIES

FORM: Solid.

ODOR: Not available.

FLAMMABLE LIMITS (in air by volume, %): Not available.

DECOMPOSITION TEMPERATURE: Not available.

AUTOIGNITION TEMPERATURE: Not available.

FREEZING/MELTING POINT: Not available.

VAPOR PRESSURE: Not applicable.

VAPOR DENSITY (air = 1): Not applicable.

EVAPORATION RATE (*n*-BuAc = 1): Not applicable.

SOLUBILITY IN WATER: Insoluble.

COEFFICIENT WATER/OIL DISTRIBUTION: Not established.

COLOR: Red.

ODOR THRESHOLD: Not available.

OXIDIZING PROPERTIES: Not applicable.

PERCENT VOLATILE: 0

FLASH POINT: Not available.

BOILING POINT: Not applicable.

SPECIFIC GRAVITY (water = 1): Not applicable.

CARB VOC: Not applicable.

SCAQMD (U.S. EPA Method 24): Not applicable.

SOLUBILITY IN SOLVENTS: Not applicable.

pH: Not applicable.

10. STABILITY and REACTIVITY

CHEMICAL STABILITY: This product is stable when properly stored at normal temperatures.

11. TOXICOLOGICAL INFORMATION

SYMPTOMS OF EXPOSURE BY ROUTE OF EXPOSURE:

Inhalation: Due to the form of the product, inhalation is unlikely.

Contact with Skin or Eyes: Due to the form of the product, contact with the eyes is unlikely.

Skin Absorption: Due to form of product, skin absorption is not a likely route of exposure.

Ingestion: Ingestion is not a likely route of exposure, due to the form of the product.

Injection: Injection is not likely, due to the form of the product.

12. ECOLOGICAL INFORMATION

MOBILITY: As an article, this product will not be mobile in soil.

PERSISTENCE AND BIODEGRADABILITY: No specific information is available regarding persistence and biodegradability.

BIO-ACCUMULATION POTENTIAL: This product has not been tested for bio-accumulation potential.

ECOTOXICITY: This product has not been tested for aquatic or animal toxicity. All releases to terrestrial, atmospheric and aquatic environments should be avoided.

13. DISPOSAL CONSIDERATIONS

DISPOSAL METHODS: Waste disposal must be in accordance with appropriate Federal, State, and local regulations.
U.S. EPA WASTE NUMBER: Not applicable.

14. TRANSPORTATION INFORMATION

U.S. DEPARTMENT OF TRANSPORTATION REGULATIONS: This product is not classified as dangerous goods, per U.S. DOT regulations, under 49 CFR 172.101.
TRANSPORT CANADA TRANSPORTATION OF DANGEROUS GOODS REGULATIONS: This product is not classified as Dangerous Goods, per regulations of Transport Canada.
INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA): This product is not classified as dangerous goods under rules of IATA.
INTERNATIONAL MARITIME ORGANIZATION (IMO) DESIGNATION: This product is not classified as Dangerous Goods by the International Maritime Organization.
OFFICIAL MEXICAN STANDARD; REGULATION FOR THE TRANSPORT OF DANGEROUS GOODS AND RESIDUES: This product is not classified as Dangerous Goods, per transport regulations of Mexico.
SINGAPORE STANDARD 286: PART A: This product has no requirements under the Specification for Caution Labeling for Hazardous Substances, Part 4: Marking of Packages, Containers and Vehicles, as it does not meet the criteria for any hazard class under this regulation.
TRANSPORT IN BULK ACCORDING TO THE IBC CODE: See the information under the individual jurisdiction listings for IBC information.
ENVIRONMENTAL HAZARDS: This material does not meet the criteria of environmentally hazardous according to the criteria of the UN Model Regulations (as reflected in the IMDG Code, ADR, RID, and ADN) and is not listed in Annex III under MARPOL 73/78.

15. REGULATORY INFORMATION

UNITED STATES REGULATIONS:

U.S. SARA Reporting Requirements: As an article, this product is not subject to the reporting requirements of Sections 302, 304, and 313 of Title III of the Superfund Amendments and Reauthorization Act.
U.S. SARA Hazard Categories (Section 311/312, 40 CFR 370-21): ACUTE: No; CHRONIC: No; FIRE: No; REACTIVE: No; SUDDEN RELEASE: No
U.S. SARA Threshold Planning Quantity (TPQ): As an article, this product is not subject to Threshold Planning Quantities, per 40 CFR 370.20.
U.S. CERCLA Reportable Quantity (RQ): Not applicable.
U.S. TSCA Inventory Status: Components of this product are listed on the TSCA Inventory.
California Safe Drinking Water and Toxic Enforcement Act (Proposition 65): No component is on the California Proposition 65 lists.

CANADIAN REGULATIONS:

Canadian DSL/NDSL Inventory Status: Components are on the DSL or NDSL Inventories.
Canadian Environmental Protection Act (CEPA) Priorities Substances Lists: As a phthalate, the Undecyl Alcohol Phthalate component is regulated by the CEPA as a substance that may present, to individuals in Canada, the greatest potential for exposure; or is persistent or bioaccumulative in accordance with the regulations, and inherently toxic to human beings or to non-human organisms, as determined by laboratory or other studies..
Canadian WHMIS Classification and Symbols: As an article, this product is not subject the Controlled Product Regulations.

CHINESE REGULATIONS:

Chinese Inventory of Existing Chemical Substances Status: As an article, this product is not subjected to requirements under the Chinese Inventory of Existing Chemical Substances (IECSC).

JAPANESE REGULATIONS:

Japanese ENCS: As an article, this product is not subjected to requirements under ENCS Inventory.
Japanese Ministry of Economy, Trade, and Industry (METI) Status: As an article, this product is not subjected to requirements under the Japanese METI.
Poisonous and Deleterious Substances Control Law: As an article, this product is not subjected to requirements under the Poisonous and Deleterious Substances Control Law.

KOREAN REGULATIONS:

Korean Existing Chemicals List (ECL) Status: As an article, this product is not subjected to requirements under the Korean ECL Inventory.

MEXICAN REGULATIONS:

Mexican Workplace Regulations (NOM-018-STPS-2000): This product is not classified as hazardous.

SINGAPORE REGULATIONS:

List of Controlled Hazardous Substances: As an article, this product is not subjected to requirements under the Singapore List of Controlled Substances.
Code of Practice on Pollution Control Requirements: As an article, this product is not subjected to requirements under the Singapore Code of Practice on Pollution Control.

TAIWANESE REGULATIONS:

Taiwan Existing Chemical Substances Inventory Status: As an article, this product is not subjected to requirements under the Taiwan Existing Chemicals List.

16. OTHER INFORMATION

REFERENCES AND DATA SOURCES: Contact the supplier for information.

METHODS OF EVALUATING INFORMATION FOR THE PURPOSE OF CLASSIFICATION: Criteria of the GHS were used for classification.

PREPARED BY: Specified Technologies Inc.

DATE OF PRINTING: December 14, 2023

REVISED: January 23, 2017
January 16, 2023

REVISION DETAILS: Periodic document review.

SAFETY DATA SHEET



SDS-00062 [Carlton Low VOC "Quick-Set" Solvent Cement
for ENT and PVC Plastic Pipe]

Section 1. Identification

GHS product identifier : SDS-00062 [Carlton Low VOC "Quick-Set" Solvent Cement for ENT and PVC Plastic Pipe]
Product code : VC9992, VC9995C-RT
Other means of identification : Not available.
Product type : Liquid.

Relevant identified uses of the substance or mixture and uses advised against

Product use : Cements.
Area of application : Consumer applications.
Manufacturer : ABB Installation Products Inc.
860 Ridge Lake Blvd.
Memphis, TN 38120, US
Telephone no.: 1-888-862-3289

Emergency telephone number (with hours of operation) : INFOTRAC – 24 Hours 1-800-535-5053
+1 352-323-3500 (Outside USA)

Section 2. Hazards identification

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance or mixture : H225 FLAMMABLE LIQUIDS - Category 2
H332 ACUTE TOXICITY (inhalation) - Category 4
H319 EYE IRRITATION - Category 2A
H351 CARCINOGENICITY - Category 2
H335 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE)
(Respiratory tract irritation) - Category 3
H336 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE)
(Narcotic effects) - Category 3

GHS label elements

Hazard pictograms :



Signal word : Danger

Date of revision

: 26- March 2025

Date of previous issue

: 18-June 2024

Version : H

1/19

Section 2. Hazards identification

Hazard statements	: H225 - Highly flammable liquid and vapor. H319 - Causes serious eye irritation. H332 - Harmful if inhaled. H335 - May cause respiratory irritation. H336 - May cause drowsiness or dizziness. H351 - Suspected of causing cancer.
<u>Precautionary statements</u>	
Prevention	: P201 - Obtain special instructions before use. P202 - Do not handle until all safety precautions have been read and understood. P280 - Wear protective gloves, protective clothing and eye or face protection. P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P241 - Use explosion-proof electrical, ventilating or lighting equipment. P242 - Use non-sparking tools. P243 - Take action to prevent static discharges. P271 - Use only outdoors or in a well-ventilated area. P261 - Avoid breathing vapor. P264 - Wash thoroughly after handling.
Response	: P308 + P313 - IF exposed or concerned: Get medical advice or attention. P304 + P340, P312 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor if you feel unwell. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P337 + P313 - If eye irritation persists: Get medical advice or attention.
Storage	: P405 - Store locked up. P403 + P233 - Store in a well-ventilated place. Keep container tightly closed. P403 + P235 - Keep cool.
Disposal	: P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements	: Avoid contact with skin and clothing. Wash thoroughly after handling.
Hazards not otherwise classified	: Prolonged or repeated contact may dry skin and cause irritation.

Section 3. Composition/information on ingredients

Substance/mixture	: Mixture
Other means of identification	: Not available.

Ingredient name	Other names	%	Identifiers
tetrahydrofuran	-	≥40 - ≤60	CAS: 109-99-9
butanone	-	≥5 - ≤25	CAS: 78-93-3
cyclohexanone	-	≥5 - ≤25	CAS: 108-94-1
acetone	-	≥5 - ≤25	CAS: 67-64-1
Ethene, chloro-, homopolymer	-	≥5 - ≤25	CAS: 9002-86-2

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Date of revision	: 26- March 2025	Date of previous issue	: 18-June 2024	Version	: H	2/19
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Section 4. First aid measures

Description of necessary first aid measures

- | | |
|---------------------|---|
| Eye contact | : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention. |
| Inhalation | : Remove victim to fresh air and keep at rest in a position comfortable for breathing. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention. If necessary, call a poison center or physician. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. |
| Skin contact | : Wash skin thoroughly with soap and water or use recognized skin cleanser. Remove contaminated clothing and shoes. Continue to rinse for at least 10 minutes. Get medical attention. Wash clothing before reuse. Clean shoes thoroughly before reuse. |
| Ingestion | : Wash out mouth with water. Remove dentures if any. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention. If necessary, call a poison center or physician. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. |

Most important symptoms/effects, acute and delayed

Potential acute health effects

- | | |
|---------------------|---|
| Eye contact | : Causes serious eye irritation. |
| Inhalation | : Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation. |
| Skin contact | : Defatting to the skin. May cause skin dryness and irritation. |
| Ingestion | : Can cause central nervous system (CNS) depression. |

Over-exposure signs/symptoms

- | | |
|---------------------|---|
| Eye contact | : Adverse symptoms may include the following:
pain or irritation
watering
redness |
| Inhalation | : Adverse symptoms may include the following:
respiratory tract irritation
coughing
nausea or vomiting
headache
drowsiness/fatigue
dizziness/vertigo
unconsciousness |
| Skin contact | : Adverse symptoms may include the following:
irritation
dryness
cracking |
| Ingestion | : No specific data. |

Indication of immediate medical attention and special treatment needed, if necessary

Section 4. First aid measures

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Foam Carbon dioxide (CO₂), dry chemical powder
- Unsuitable extinguishing media** : Do not use water jet.

Specific hazards arising from the chemical : Class 1B Flammable liquid. Vapors can travel to a source of ignition and flashback.

- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
halogenated compounds

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

For non-emergency personnel : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Shut off all ignition sources. No flares, smoking or flames in hazard area. Avoid breathing vapor or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.

For emergency responders : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

Environmental precautions : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and materials for containment and cleaning up

Section 6. Accidental release measures

- Small spill** : Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
- Large spill** : Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations.

Section 7. Handling and storage

Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see Section 8). Avoid exposure - obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapor or mist. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Do not enter storage areas and confined spaces unless adequately ventilated. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Store and use away from heat, sparks, open flame or any other ignition source. Use explosion-proof electrical (ventilating, lighting and material handling) equipment. Use only non-sparking tools. Take precautionary measures against electrostatic discharges. Empty containers retain product residue and can be hazardous. Do not reuse container.
- Advice on general occupational hygiene** : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

- Conditions for safe storage, including any incompatibilities** : Store in accordance with local regulations. Store in a segregated and approved area. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Store locked up. Eliminate all ignition sources. Separate from oxidizing materials. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
tetrahydrofuran	ACGIH TLV (United States, 1/2024) A3. Absorbed through skin. TWA 8 hours: 50 ppm. STEL 15 minutes: 100 ppm. NIOSH REL (United States, 10/2020) TWA 10 hours: 200 ppm. TWA 10 hours: 590 mg/m³. STEL 15 minutes: 250 ppm. STEL 15 minutes: 735 mg/m³. OSHA PEL (United States, 5/2018)

Section 8. Exposure controls/personal protection

butanone

TWA 8 hours: 200 ppm.
TWA 8 hours: 590 mg/m³.
CAL OSHA PEL (United States, 5/2018)
STEL 15 minutes: 735 mg/m³.
STEL 15 minutes: 250 ppm.
TWA 8 hours: 590 mg/m³.
TWA 8 hours: 200 ppm.
ACGIH TLV (United States, 1/2024) Absorbed through skin.
TWA 8 hours: 75 ppm.
STEL 15 minutes: 150 ppm.
NIOSH REL (United States, 10/2020)
TWA 10 hours: 200 ppm.
TWA 10 hours: 590 mg/m³.
STEL 15 minutes: 300 ppm.
STEL 15 minutes: 885 mg/m³.
OSHA PEL (United States, 5/2018)
TWA 8 hours: 200 ppm.
TWA 8 hours: 590 mg/m³.
CAL OSHA PEL (United States, 5/2018)
STEL 15 minutes: 885 mg/m³.
STEL 15 minutes: 300 ppm.
TWA 8 hours: 590 mg/m³.
TWA 8 hours: 200 ppm.

cyclohexanone

ACGIH TLV (United States, 1/2024) A3. Absorbed through skin.
TWA 8 hours: 20 ppm.
STEL 15 minutes: 50 ppm.
NIOSH REL (United States, 10/2020) Absorbed through skin.
TWA 10 hours: 25 ppm.
TWA 10 hours: 100 mg/m³.
OSHA PEL (United States, 5/2018)
TWA 8 hours: 50 ppm.
TWA 8 hours: 200 mg/m³.
CAL OSHA PEL (United States, 5/2018) Absorbed through skin.

acetone

TWA 8 hours: 100 mg/m³.
TWA 8 hours: 25 ppm.
ACGIH TLV (United States, 1/2024) A4.
TWA 8 hours: 250 ppm.
STEL 15 minutes: 500 ppm.
NIOSH REL (United States, 10/2020)
TWA 10 hours: 250 ppm.
TWA 10 hours: 590 mg/m³.
OSHA PEL (United States, 5/2018)
TWA 8 hours: 1000 ppm.
TWA 8 hours: 2400 mg/m³.
CAL OSHA PEL (United States, 5/2018)
STEL 15 minutes: 1780 mg/m³.
STEL 15 minutes: 750 ppm.
C: 3000 ppm.
TWA 8 hours: 1200 mg/m³.
TWA 8 hours: 500 ppm.
ACGIH TLV (United States, 1/2024) A4.
TWA 8 hours: 1 mg/m³. Form: Respirable fraction.

Ethene, chloro-, homopolymer

Section 8. Exposure controls/personal protection

Biological exposure indices

Ingredient name	Exposure indices
tetrahydrofuran	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, tetrahydrofuran [in urine]. Sampling time: end of shift.
butanone	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, methyl ethyl ketone [in urine]. Sampling time: end of shift.
cyclohexanone	ACGIH BEI (United States, 1/2024) BEI: 80 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], 1,2-cyclohexanediol [in urine]. Sampling time: end of shift at end of workweek. BEI: 8 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], cyclohexanol [in urine]. Sampling time: end of shift.
acetone	ACGIH BEI (United States, 1/2024) BEI: 25 mg/l, acetone [in urine]. Sampling time: end of shift.

Appropriate engineering controls

- : Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.

Environmental exposure controls

- : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures

- : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Section 8. Exposure controls/personal protection

- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
- Skin protection**
- Hand protection** : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.
- Body protection** : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. When there is a risk of ignition from static electricity, wear anti-static protective clothing. For the greatest protection from static discharges, clothing should include anti-static overalls, boots and gloves. Recommended: Rubber gloves.
- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

- Physical state** : Liquid.
- Color** : Blue. Aqua
- Odor** : Ether-like.
- Odor threshold** : 0.88 ppm
- pH** : Not available.
- Melting point/freezing point** : -108.5°C (-163.3°F)
- Boiling point or initial boiling point and boiling range** : 56°C (132.8°F)
- Flash point** : Closed cup: -20°C (-4°F)
- Evaporation rate** : >1 (butyl acetate = 1)
- Flammability** : Not available.
- Lower and upper explosion limit/flammability limit** : Lower: 1.8%
Upper: 12.8%
- Vapor pressure** : 25.3 kPa (190 mm Hg)
- Relative vapor density** : 2.5 [Air = 1]
- Relative density** : 0.935
- Density** : Not available.
- Solubility(ies)** :

Media	Result
water	Not soluble

Section 9. Physical and chemical properties

Partition coefficient: n-octanol/water : Not applicable.

Auto-ignition temperature :	Ingredient name	°C	°F	Method
	tetrahydrofuran	215	419	DIN 51794

Decomposition temperature : Not available.

SADT : Not available.

Viscosity : Dynamic (room temperature): Not available.
Kinematic (room temperature): Not available.
Kinematic (40°C (104°F)): Not available.

Particle characteristics

Median particle size : Not applicable.

Other information

Physical/chemical properties comments : No additional information.

Section 10. Stability and reactivity

Reactivity : No specific test data related to reactivity available for this product or its ingredients.

Chemical stability : The product is stable.

Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.
Under normal conditions of storage and use, hazardous polymerization will not occur.

Conditions to avoid : Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition. Do not allow vapor to accumulate in low or confined areas.

Incompatible materials : Reactive or incompatible with the following materials:
oxidizing materials, acids, Bases.

Hazardous decomposition products : Hydrogen chloride, carbon dioxide, hydrocarbons.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result
tetrahydrofuran	Rat - Oral - LD50 1650 mg/kg Rat - Male, Female - Dermal - LD50 OECD [Acute Dermal Toxicity] >2000 mg/kg
butanone	Rabbit - Dermal - LD50 6480 mg/kg Rat - Oral - LD50 2737 mg/kg

Date of revision

: 26- March 2025

Date of previous issue

: 18-June 2024

Version : H

9/19

Section 11. Toxicological information

cyclohexanone	Rat - Inhalation - LC50 Vapor 33.36 mg/l [4 hours]	
	Rat - Oral - LD50 1800 mg/kg	
	Rabbit - Dermal - LD50 950 mg/kg	
acetone	Rat - Inhalation - LC50 Gas. 8000 ppm [4 hours]	<u>Toxic effects:</u> Behavioral - Altered sleep time (including change in righting reflex) Behavioral - Tremor
	Rat - Oral - LD50 5800 mg/kg	
	Rabbit - Dermal - LD50 20000 mg/kg	
	Rat - Female - Inhalation - LC50 Vapor 76 mg/l [4 hours]	

Conclusion/Summary [Product] : Not available.

Skin corrosion/irritation

Product/ingredient name	Result
butanone	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 14 mg
	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 402 mg
	Rabbit - Skin - Moderate irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg
cyclohexanone	Rabbit - Skin - Mild irritant Amount/concentration applied: 500 mg
acetone	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg
	Rabbit - Skin - Mild irritant Amount/concentration applied: 395 mg

Conclusion/Summary [Product] : Not available.

Serious eye damage/eye irritation

Product/ingredient name	Result
cyclohexanone	Rabbit - Eyes - Severe irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 250 ug
	Rabbit - Eyes - Severe irritant Amount/concentration applied: 20 mg
acetone	Rabbit - Eyes - Mild irritant

Section 11. Toxicological information

Amount/concentration applied: 10 uL

Rabbit - Eyes - Moderate irritant

Duration of treatment/exposure: 24 hours

Amount/concentration applied: 20 mg

Rabbit - Eyes - Severe irritant

Amount/concentration applied: 20 mg

Conclusion/Summary [Product] : Not available.

Respiratory corrosion/irritation

Conclusion/Summary [Product] : Not available.

Respiratory or skin sensitization

Skin

Conclusion/Summary [Product] : Not available.

Respiratory

Conclusion/Summary [Product] : Not available.

Germ cell mutagenicity

Conclusion/Summary [Product] : Not available.

Carcinogenicity

Conclusion/Summary [Product] : Not available.

Classification

Product/ingredient name	OSHA	IARC	NTP
tetrahydrofuran	-	2B	-
cyclohexanone	-	3	-
Ethene, chloro-, homopolymer	-	3	-

Reproductive toxicity

Conclusion/Summary [Product] : Not available.

Specific target organ toxicity (single exposure)

Product/ingredient name	Result
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Section 11. Toxicological information

tetrahydrofuran	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
butanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
cyclohexanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Product/ingredient name	Result
butanone	ASPIRATION HAZARD - Category 1
cyclohexanone	ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Eye contact	: Causes serious eye irritation.
Inhalation	: Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation.
Skin contact	: Defatting to the skin. May cause skin dryness and irritation.
Ingestion	: Can cause central nervous system (CNS) depression.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact	: Adverse symptoms may include the following: pain or irritation watering redness
Inhalation	: Adverse symptoms may include the following: respiratory tract irritation coughing nausea or vomiting headache drowsiness/fatigue dizziness/vertigo unconsciousness
Skin contact	: Adverse symptoms may include the following: irritation dryness cracking
Ingestion	: No specific data.

Section 11. Toxicological information

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Potential chronic health effects

Conclusion/Summary [Product] : Not available.

General : Prolonged or repeated contact can defat the skin and lead to irritation, cracking and/or dermatitis.

Carcinogenicity : Suspected of causing cancer. Risk of cancer depends on duration and level of exposure.

Mutagenicity : No known significant effects or critical hazards.

Reproductive toxicity : No known significant effects or critical hazards.

Numerical measures of toxicity

Acute toxicity estimates

Product/ingredient name	Oral(mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapors) (mg/l)	Inhalation (dusts and mists) (mg/l)
SDS-00062 [Carlton Low VOC "Quick-Set" Solvent Cement for ENT and PVC Plastic Pipe]	2273.2	2683.1	24000	N/A	1.5
tetrahydrofuran	1650	2500	N/A	N/A	N/A
butanone	2737	6480	N/A	33.36	N/A
cyclohexanone	1800	950	8000	N/A	N/A
acetone	5800	20000	N/A	76	N/A

Section 12. Ecological information

Toxicity

Product/ingredient name

tetrahydrofuran

Result

Acute - LC50 - Fresh water

Fish - Fathead minnow - *Pimephales promelas*

Age: 34 days; Size: 18.6 mm; Weight: 0.111 g

2160 mg/l [96 hours]

Chronic - NOEC - Fresh water

Fish - Fathead minnow - *Pimephales promelas* - Embryo

Age: <24 hours

367 mg/l [33 days]

Effect: Mortality

Effect: Growth

Section 12. Ecological information

butanone	Acute - EC50 - Fresh water	<u>Effect</u> : Intoxication
	Daphnia - Water flea - <i>Daphnia magna</i> - Larvae	
	<u>Age</u> : <24 hours	
	5091 mg/l [48 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Fathead minnow - <i>Pimephales promelas</i>	
	<u>Age</u> : 31 days; <u>Size</u> : 22 mm; <u>Weight</u> : 0.167 g	
	3220 mg/l [96 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Population
	Algae - Diatom - <i>Skeletonema costatum</i> >500 mg/l [96 hours]	
cyclohexanone	Acute - NOEC - Fresh water	OECD 203 [Fish, Acute Toxicity Test]
	Fish	
	1170 mg/l [96 hours]	
	Acute - NOEC - Fresh water	OECD 202 [Daphnia sp. Acute Immobilization Test and Reproduction Test]
	Daphnia - Daphnia	
	68 mg/l [48 hours]	
	Acute - NOEC - Fresh water	OECD 201 [Alga, Growth Inhibition Test]
	Algae - Algae	
	566 mg/l [72 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
acetone	Fish - Fathead minnow - <i>Pimephales promelas</i>	
	<u>Age</u> : 30 days; <u>Size</u> : 20.2 mm; <u>Weight</u> : 0.127 g	
	527 mg/l [96 hours]	
	Chronic - EC10	<u>Effect</u> : Population
	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase	
	<u>Age</u> : 7 days	
	3.56 mg/l [72 hours]	
	Acute - EC50	<u>Effect</u> : Population
	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase	
	<u>Age</u> : 7 days	
	32.9 mg/l [72 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Daphnia - Water flea - <i>Daphnia magna</i>	
	10 mg/l [48 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i>	
	20.565 mg/l [96 hours]	
	Chronic - NOEC - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i>	
	4.95 mg/l [96 hours]	
	Chronic - NOEC - Fresh water	<u>Effect</u> : Population
	Crustaceans - Daphnia - <i>Daphniidae</i>	
	0.016 ml/l [21 days]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Guppy - <i>Poecilia reticulata</i>	
	<u>Age</u> : 4 to 12 months; <u>Size</u> : 2 to 10 cm; <u>Weight</u> : 0.5 to 14 g	
	5600 ppm [96 hours]	

Section 12. Ecological information

Conclusion/Summary [Product] : Not available.

Persistence and degradability

Product/ingredient name	Result	
tetrahydrofuran	82% [28 days]	Ready Biodegradability - Manometric Respirometry Test
butanone	Aerobic 98% [28 days] - Readily	OECD [Ready Biodegradability - Closed Bottle Test]

Conclusion/Summary [Product] : Not available.

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
tetrahydrofuran	-	-	Inherent
butanone	-	-	Readily
cyclohexanone	-	-	Readily
acetone	-	-	Readily

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
tetrahydrofuran	0.45	-	Low
butanone	0.3	-	Low
cyclohexanone	0.86	-	Low
acetone	-0.23	-	Low

Mobility in soil

Soil/Water partition coefficient : Not available.

Other adverse effects

No known significant effects or critical hazards.

Section 13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

RCRA Toxic hazardous waste "U" List

Section 13. Disposal considerations

Ingredient	CAS #	Status	Reference number
Tetrahydrofuran (I)	109-99-9	Listed	U213
Methyl ethyl ketone (MEK) (I,T)	78-93-3	Listed	U159
Cyclohexanone (I)	108-94-1	Listed	U057
Acetone (I)	67-64-1	Listed	U002

Section 14. Transport information

	DOT Classification	IMDG	IATA
UN number	UN1133	UN1133	UN1133
UN proper shipping name	Adhesives	ADHESIVES	Adhesives
Transport hazard class(es)	3 	3 	3 
Packing group	II	II	II
Environmental hazards	No.	No.	No.

Additional information

DOT Classification	: Reportable quantity 1666.7 lbs / 756.67 kg [213.79 gal / 809.27 L]. Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements. Limited quantity Yes. Packaging instruction Exceptions: 150. Non-bulk: 173. Bulk: 242. Quantity limitation Passenger aircraft/rail: 5 L. Cargo aircraft: 60 L. Special provisions 149, B52, IB2, T4, TP1, TP8
IMDG	: Emergency schedules F-E, S-D
IATA	: Quantity limitation Passenger and Cargo Aircraft: 5 L. Packaging instructions: 353. Cargo Aircraft Only: 60 L. Packaging instructions: 364. Limited Quantities - Passenger Aircraft: 1 L. Packaging instructions: Y341. Special provisions A3

Special precautions for user : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to IMO instruments : Not available.

Section 15. Regulatory information

U.S. Federal regulations : TSCA 8(a) PAIR: tetrahydrofuran
 TSCA 8(a) CDR Exempt/Partial exemption: Not determined
 United States inventory (TSCA 8b): All components are active or exempted.

TSCA 12(b) - Chemical export notification

Not applicable.

Clean Air Act Section 112 (b) Hazardous Air Pollutants (HAPs) : Not listed

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304

Composition/information on ingredients

No products were found.

SARA 304 RQ : Not applicable.

SARA 311/312

Classification : FLAMMABLE LIQUIDS - Category 2
 ACUTE TOXICITY (inhalation) - Category 4
 EYE IRRITATION - Category 2A
 CARCINOGENICITY - Category 2
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
 HNOC - Defatting irritant

Composition/information on ingredients

Name	%	Classification
tetrahydrofuran	≥40 - ≤60	FLAMMABLE LIQUIDS - Category 2 ACUTE TOXICITY (oral) - Category 4 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 HNOC - Defatting irritant
butanone	≥5 - ≤25	FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1 HNOC - Defatting irritant

Section 15. Regulatory information

cyclohexanone	≥5 - ≤25	FLAMMABLE LIQUIDS - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 3 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1
acetone	≥5 - ≤25	FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A HNOC - Defatting irritant
Ethene, chloro-, homopolymer	≥5 - ≤25	COMBUSTIBLE DUSTS

SARA 313

Not applicable.

State regulations

Massachusetts	: The following components are listed: TETRAHYDROFURAN; METHYL ETHYL KETONE; CYCLOHEXANONE; ACETONE
New York	: The following components are listed: Tetrahydrofuran; Methyl ethyl ketone; Cyclohexanone; Acetone
New Jersey	: The following components are listed: TETRAHYDROFURAN; METHYL ETHYL KETONE; CYCLOHEXANONE; ACETONE; PVC
Pennsylvania	: The following components are listed: FURAN, TETRAHYDRO-; 2-BUTANONE; CYCLOHEXANONE; 2-PROPANONE

California Prop. 65

 **WARNING:** This product can expose you to tetrahydrofuran, which is known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

Ingredient name	No significant risk level	Maximum acceptable dosage level
tetrahydrofuran	-	-

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

Section 16. Other information

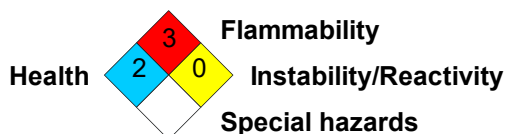
Hazardous Material Information System (U.S.A.)

Health	*	2
Flammability		3
Physical hazards		0

Caution: HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. Although HMIS® ratings and the associated label are not required on SDSs or products leaving a facility under 29 CFR 1910.1200, the preparer may choose to provide them. HMIS® ratings are to be used with a fully implemented HMIS® program. HMIS® is a registered trademark and service mark of the American Coatings Association, Inc.

The customer is responsible for determining the PPE code for this material. For more information on HMIS® Personal Protective Equipment (PPE) codes, consult the HMIS® Implementation Manual.

National Fire Protection Association (U.S.A.)



Procedure used to derive the classification

Classification	Justification
FLAMMABLE LIQUIDS - Category 2	On basis of test data
ACUTE TOXICITY (inhalation) - Category 4	Expert judgment
EYE IRRITATION - Category 2A	Calculation method
CARCINOGENICITY - Category 2	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3	Calculation method

History

Date of issue/Date of revision	: 03/25/2025
Date of previous issue	: No previous validation
Version	: 1
Prepared by	: Sphera Solutions
Key to abbreviations	: ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods IMO = International Maritime Organization LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) N/A = Not available SGG = Segregation Group TDG = Transportation of Dangerous Goods UN = United Nations

References : HCS (U.S.A.) - Hazard Communication Standard
International transport regulations

Indicates information that has changed from previously issued version.

Notice to reader

ABB Installation Products Inc. cannot anticipate all conditions under which this information and its product, or the products of other manufacturers in combination with its product, may be used. It is the user's responsibility to ensure safe conditions for handling, storage and disposal of the product, and to assume liability for loss, injury, damage or expense due to improper use. The information in the sheet was written based on the best knowledge and experience currently available.

SAFETY DATA SHEET



SDS-00060 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]

Section 1. Identification

GHS product identifier : SDS-00060 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]
Product code : VC9964, VC9963, VC9963C, VC9962, VC9961P, VC9924-24, VC9923, VC9922, VC9941P, VC9LV4-24, VC9LV2, VC9LV3
Other means of identification : Not available.
Product type : Liquid.

Relevant identified uses of the substance or mixture and uses advised against

Product use : Cements.
Area of application : Consumer applications.
Manufacturer : ABB Installation Products Inc.
860 Ridge Lake Blvd.
Memphis, TN 38120, US
Telephone no.: 1-888-862-3289

Emergency telephone number (with hours of operation) : INFOTRAC – 24 Hours 1-800-535-5053
+1 352-323-3500 (Outside USA)

Section 2. Hazards identification

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance or mixture : H225 FLAMMABLE LIQUIDS - Category 2
H332 ACUTE TOXICITY (inhalation) - Category 4
H319 EYE IRRITATION - Category 2A
H351 CARCINOGENICITY - Category 2
H335 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
H336 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3

GHS label elements

Hazard pictograms :



Signal word : Danger

Date of revision

: 26- March 2025

Date of previous issue

: 18- June 2024

Version : H

1/19

Section 2. Hazards identification

Hazard statements	: H225 - Highly flammable liquid and vapor. H319 - Causes serious eye irritation. H332 - Harmful if inhaled. H335 - May cause respiratory irritation. H336 - May cause drowsiness or dizziness. H351 - Suspected of causing cancer.
<u>Precautionary statements</u>	
Prevention	: P201 - Obtain special instructions before use. P202 - Do not handle until all safety precautions have been read and understood. P280 - Wear protective gloves, protective clothing and eye or face protection. P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P241 - Use explosion-proof electrical, ventilating or lighting equipment. P242 - Use non-sparking tools. P243 - Take action to prevent static discharges. P271 - Use only outdoors or in a well-ventilated area. P261 - Avoid breathing vapor. P264 - Wash thoroughly after handling.
Response	: P308 + P313 - IF exposed or concerned: Get medical advice or attention. P304 + P340, P312 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor if you feel unwell. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P337 + P313 - If eye irritation persists: Get medical advice or attention.
Storage	: P405 - Store locked up. P403 + P233 - Store in a well-ventilated place. Keep container tightly closed. P403 + P235 - Keep cool.
Disposal	: P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements	: Avoid contact with skin and clothing. Wash thoroughly after handling.
Hazards not otherwise classified	: Prolonged or repeated contact may dry skin and cause irritation.

Section 3. Composition/information on ingredients

Substance/mixture	: Mixture
Other means of identification	: Not available.

Ingredient name	Other names	%	Identifiers
tetrahydrofuran	-	≥20 - ≤40	CAS: 109-99-9
butanone	-	≥10 - ≤30	CAS: 78-93-3
cyclohexanone	-	≥10 - ≤30	CAS: 108-94-1
acetone	-	≥10 - ≤30	CAS: 67-64-1
Ethene, chloro-, homopolymer	-	≥10 - ≤30	CAS: 9002-86-2

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Date of revision	: 26- March 2025	Date of previous issue	: 18- June 2024	Version	: H	2/19
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Section 4. First aid measures

Description of necessary first aid measures

- | | |
|---------------------|---|
| Eye contact | : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention. |
| Inhalation | : Remove victim to fresh air and keep at rest in a position comfortable for breathing. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention. If necessary, call a poison center or physician. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. |
| Skin contact | : Wash skin thoroughly with soap and water or use recognized skin cleanser. Remove contaminated clothing and shoes. Continue to rinse for at least 10 minutes. Get medical attention. Wash clothing before reuse. Clean shoes thoroughly before reuse. |
| Ingestion | : Wash out mouth with water. Remove dentures if any. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention. If necessary, call a poison center or physician. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband. |

Most important symptoms/effects, acute and delayed

Potential acute health effects

- | | |
|---------------------|---|
| Eye contact | : Causes serious eye irritation. |
| Inhalation | : Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation. |
| Skin contact | : Defatting to the skin. May cause skin dryness and irritation. |
| Ingestion | : Can cause central nervous system (CNS) depression. |

Over-exposure signs/symptoms

- | | |
|---------------------|---|
| Eye contact | : Adverse symptoms may include the following:
pain or irritation
watering
redness |
| Inhalation | : Adverse symptoms may include the following:
respiratory tract irritation
coughing
nausea or vomiting
headache
drowsiness/fatigue
dizziness/vertigo
unconsciousness |
| Skin contact | : Adverse symptoms may include the following:
irritation
dryness
cracking |
| Ingestion | : No specific data. |

Indication of immediate medical attention and special treatment needed, if necessary

Section 4. First aid measures

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Foam
Carbon dioxide.
dry chemical powder

- Unsuitable extinguishing media** : Do not use water jet.

- Specific hazards arising from the chemical** : Class 1B Flammable liquid. Vapors can travel to a source of ignition and flashback.

- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
halogenated compounds

- Special protective actions for fire-fighters** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.

- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

- For non-emergency personnel** : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Shut off all ignition sources. No flares, smoking or flames in hazard area. Avoid breathing vapor or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.

- For emergency responders** : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

- Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Section 6. Accidental release measures

Methods and materials for containment and cleaning up

- Small spill

: Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
- Large spill

: Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations.

Section 7. Handling and storage

Precautions for safe handling

- Protective measures

: Put on appropriate personal protective equipment (see Section 8). Avoid exposure - obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapor or mist. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Do not enter storage areas and confined spaces unless adequately ventilated. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Store and use away from heat, sparks, open flame or any other ignition source. Use explosion-proof electrical (ventilating, lighting and material handling) equipment. Use only non-sparking tools. Take precautionary measures against electrostatic discharges. Empty containers retain product residue and can be hazardous. Do not reuse container.
- Advice on general occupational hygiene

: Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

- Conditions for safe storage, including any incompatibilities

: Store in accordance with local regulations. Store in a segregated and approved area. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Store locked up. Eliminate all ignition sources. Separate from oxidizing materials. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
tetrahydrofuran	ACGIH TLV (United States, 1/2024) A3. Absorbed through skin. TWA 8 hours: 50 ppm. STEL 15 minutes: 100 ppm. NIOSH REL (United States, 10/2020) TWA 10 hours: 200 ppm. TWA 10 hours: 590 mg/m³. STEL 15 minutes: 250 ppm.

Section 8. Exposure controls/personal protection

butanone

STEL 15 minutes: 735 mg/m³.
OSHA PEL (United States, 5/2018)
 TWA 8 hours: 200 ppm.
 TWA 8 hours: 590 mg/m³.
CAL OSHA PEL (United States, 5/2018)
 STEL 15 minutes: 735 mg/m³.
 STEL 15 minutes: 250 ppm.
 TWA 8 hours: 590 mg/m³.
 TWA 8 hours: 200 ppm.
ACGIH TLV (United States, 1/2024) Absorbed through skin.
 TWA 8 hours: 75 ppm.
 STEL 15 minutes: 150 ppm.
NIOSH REL (United States, 10/2020)
 TWA 10 hours: 200 ppm.
 TWA 10 hours: 590 mg/m³.
 STEL 15 minutes: 300 ppm.
 STEL 15 minutes: 885 mg/m³.
OSHA PEL (United States, 5/2018)
 TWA 8 hours: 200 ppm.
 TWA 8 hours: 590 mg/m³.
CAL OSHA PEL (United States, 5/2018)
 STEL 15 minutes: 885 mg/m³.
 STEL 15 minutes: 300 ppm.
 TWA 8 hours: 590 mg/m³.
 TWA 8 hours: 200 ppm.

cyclohexanone

ACGIH TLV (United States, 1/2024) A3. Absorbed through skin.
 TWA 8 hours: 20 ppm.
 STEL 15 minutes: 50 ppm.
NIOSH REL (United States, 10/2020) Absorbed through skin.
 TWA 10 hours: 25 ppm.
 TWA 10 hours: 100 mg/m³.
OSHA PEL (United States, 5/2018)
 TWA 8 hours: 50 ppm.
 TWA 8 hours: 200 mg/m³.

acetone

CAL OSHA PEL (United States, 5/2018) Absorbed through skin.
 TWA 8 hours: 100 mg/m³.
 TWA 8 hours: 25 ppm.
ACGIH TLV (United States, 1/2024) A4.
 TWA 8 hours: 250 ppm.
 STEL 15 minutes: 500 ppm.
NIOSH REL (United States, 10/2020)
 TWA 10 hours: 250 ppm.
 TWA 10 hours: 590 mg/m³.
OSHA PEL (United States, 5/2018)
 TWA 8 hours: 1000 ppm.
 TWA 8 hours: 2400 mg/m³.
CAL OSHA PEL (United States, 5/2018)
 STEL 15 minutes: 1780 mg/m³.
 STEL 15 minutes: 750 ppm.
 C: 3000 ppm.
 TWA 8 hours: 1200 mg/m³.
 TWA 8 hours: 500 ppm.
ACGIH TLV (United States, 1/2024) A4.

Ethene, chloro-, homopolymer

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Section 8. Exposure controls/personal protection

TWA 8 hours: 1 mg/m³. Form: Respirable fraction.

Biological exposure indices

Ingredient name	Exposure indices
tetrahydrofuran	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, tetrahydrofuran [in urine]. Sampling time: end of shift.
butanone	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, methyl ethyl ketone [in urine]. Sampling time: end of shift.
cyclohexanone	ACGIH BEI (United States, 1/2024) BEI: 80 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], 1,2-cyclohexanediol [in urine]. Sampling time: end of shift at end of workweek. BEI: 8 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], cyclohexanol [in urine]. Sampling time: end of shift.
acetone	ACGIH BEI (United States, 1/2024) BEI: 25 mg/l, acetone [in urine]. Sampling time: end of shift.

- Appropriate engineering controls

: Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.
- Environmental exposure controls

: Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

- Hygiene measures

: Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Section 8. Exposure controls/personal protection

- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
- Skin protection**
- Hand protection** : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.
- Body protection** : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. When there is a risk of ignition from static electricity, wear anti-static protective clothing. For the greatest protection from static discharges, clothing should include anti-static overalls, boots and gloves. Recommended: Rubber gloves.
- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

- Physical state** : Liquid.
- Color** : Clear. Gray. White.
- Odor** : Ether-like.
- Odor threshold** : 0.88 ppm
- pH** : Not available.
- Melting point/freezing point** : -108.5°C (-163.3°F)
- Boiling point or initial boiling point and boiling range** : 56°C (132.8°F)
- Flash point** : Closed cup: -20°C (-4°F)
- Evaporation rate** : >1 (butyl acetate = 1)
- Flammability** : Not available.
- Lower and upper explosion limit/flammability limit** : Lower: 1.8%
Upper: 12.8%
- Vapor pressure** : 25.3 kPa (190 mm Hg)
- Relative vapor density** : 2.5 [Air = 1]
- Relative density** : 0.925
- Density** : Not available.
- Solubility(ies)** : Not available.
- Partition coefficient: n-octanol/water** : Not applicable.
- Auto-ignition temperature** :

Section 9. Physical and chemical properties

Ingredient name	°C	°F	Method
tetrahydrofuran	215	419	DIN 51794

Decomposition temperature : Not available.
SADT : Not available.
Viscosity : Dynamic (room temperature): Not available.
 Kinematic (room temperature): Not available.
 Kinematic (40°C (104°F)): Not available.

Particle characteristics

Median particle size : Not applicable.

Other information

Physical/chemical properties comments : No additional information.

Section 10. Stability and reactivity

Reactivity : No specific test data related to reactivity available for this product or its ingredients.

Chemical stability : The product is stable.

Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.
 Under normal conditions of storage and use, hazardous polymerization will not occur.

Conditions to avoid : Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition. Do not allow vapor to accumulate in low or confined areas.

Incompatible materials : Reactive or incompatible with the following materials:
 oxidizing materials, acids, Bases.

Hazardous decomposition products : Hydrogen chloride, carbon oxides (CO, CO₂), hydrocarbons.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result
tetrahydrofuran	Rat - Oral - LD50 1650 mg/kg Rat - Male, Female - Dermal - LD50 OECD [Acute Dermal Toxicity] >2000 mg/kg
butanone	Rabbit - Dermal - LD50 6480 mg/kg Rat - Oral - LD50 2737 mg/kg Rat - Inhalation - LC50 Vapor 33.36 mg/l [4 hours]

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cyclohexanone	Rat - Oral - LD50 1800 mg/kg Rabbit - Dermal - LD50 950 mg/kg Rat - Inhalation - LC50 Gas. 8000 ppm [4 hours]	
acetone	Rat - Oral - LD50 5800 mg/kg Rabbit - Dermal - LD50 20000 mg/kg Rat - Female - Inhalation - LC50 Vapor 76 mg/l [4 hours]	<u>Toxic effects:</u> Behavioral - Altered sleep time (including change in righting reflex) Behavioral - Tremor

Conclusion/Summary [Product] : Not available.

Skin corrosion/irritation

Product/ingredient name	Result
butanone	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 14 mg Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 402 mg Rabbit - Skin - Moderate irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg
cyclohexanone	Rabbit - Skin - Mild irritant Amount/concentration applied: 500 mg
acetone	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg Rabbit - Skin - Mild irritant Amount/concentration applied: 395 mg

Conclusion/Summary [Product] : Not available.

Serious eye damage/eye irritation

Product/ingredient name	Result
cyclohexanone	Rabbit - Eyes - Severe irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 250 ug Rabbit - Eyes - Severe irritant Amount/concentration applied: 20 mg
acetone	Rabbit - Eyes - Mild irritant Amount/concentration applied: 10 uL Rabbit - Eyes - Moderate irritant

Section 11. Toxicological information

Duration of treatment/exposure: 24 hours
 Amount/concentration applied: 20 mg
Rabbit - Eyes - Severe irritant
 Amount/concentration applied: 20 mg

Conclusion/Summary [Product] : Not available.

Respiratory corrosion/irritation

Conclusion/Summary [Product] : Not available.

Respiratory or skin sensitization

Skin

Conclusion/Summary [Product] : Not available.

Respiratory

Conclusion/Summary [Product] : Not available.

Germ cell mutagenicity

Conclusion/Summary [Product] : Not available.

Carcinogenicity

Conclusion/Summary [Product] : Not available.

Classification

Product/ingredient name	OSHA	IARC	NTP
tetrahydrofuran	-	2B	-
cyclohexanone	-	3	-
Ethene, chloro-, homopolymer	-	3	-

Reproductive toxicity

Conclusion/Summary [Product] : Not available.

Specific target organ toxicity (single exposure)

Product/ingredient name	Result
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Section 11. Toxicological information

tetrahydrofuran	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
butanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
cyclohexanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Product/ingredient name	Result
butanone	ASPIRATION HAZARD - Category 1
cyclohexanone	ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Eye contact	: Causes serious eye irritation.
Inhalation	: Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation.
Skin contact	: Defatting to the skin. May cause skin dryness and irritation.
Ingestion	: Can cause central nervous system (CNS) depression.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact	: Adverse symptoms may include the following: pain or irritation watering redness
Inhalation	: Adverse symptoms may include the following: respiratory tract irritation coughing nausea or vomiting headache drowsiness/fatigue dizziness/vertigo unconsciousness
Skin contact	: Adverse symptoms may include the following: irritation dryness cracking
Ingestion	: No specific data.

Section 11. Toxicological information

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Potential chronic health effects

Conclusion/Summary [Product] : Not available.

General : Prolonged or repeated contact can defat the skin and lead to irritation, cracking and/or dermatitis.

Carcinogenicity : Suspected of causing cancer. Risk of cancer depends on duration and level of exposure.

Mutagenicity : No known significant effects or critical hazards.

Reproductive toxicity : No known significant effects or critical hazards.

Numerical measures of toxicity

Acute toxicity estimates

Product/ingredient name	Oral(mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapors) (mg/l)	Inhalation (dusts and mists) (mg/l)
SDS-00060 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]	2506.3	2732.3	24000	N/A	1.5
tetrahydrofuran	1650	2500	N/A	N/A	N/A
butanone	2737	6480	N/A	33.36	N/A
cyclohexanone	1800	950	8000	N/A	N/A
acetone	5800	20000	N/A	76	N/A

Section 12. Ecological information

Toxicity

Product/ingredient name

tetrahydrofuran

Result

Acute - LC50 - Fresh water

Fish - Fathead minnow - *Pimephales promelas*

Age: 34 days; Size: 18.6 mm; Weight: 0.111 g

2160 mg/l [96 hours]

Chronic - NOEC - Fresh water

Fish - Fathead minnow - *Pimephales promelas* - Embryo

Age: <24 hours

367 mg/l [33 days]

Effect: Mortality

Effect: Growth

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butanone	Acute - EC50 - Fresh water	<u>Effect</u> : Intoxication
	Daphnia - Water flea - <i>Daphnia magna</i> - Larvae	
	<u>Age</u> : <24 hours	
	5091 mg/l [48 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Fathead minnow - <i>Pimephales promelas</i>	
	<u>Age</u> : 31 days; <u>Size</u> : 22 mm; <u>Weight</u> : 0.167 g	
	3220 mg/l [96 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Population
	Algae - Diatom - <i>Skeletonema costatum</i> >500 mg/l [96 hours]	
cyclohexanone	Acute - NOEC - Fresh water	OECD 203 [Fish, Acute Toxicity Test]
	Fish	
	1170 mg/l [96 hours]	
	Acute - NOEC - Fresh water	OECD 202 [Daphnia sp. Acute Immobilization Test and Reproduction Test]
	Daphnia - Daphnia	
	68 mg/l [48 hours]	
	Acute - NOEC - Fresh water	OECD 201 [Alga, Growth Inhibition Test]
	Algae - Algae	
	566 mg/l [72 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
acetone	Fish - Fathead minnow - <i>Pimephales promelas</i>	
	<u>Age</u> : 30 days; <u>Size</u> : 20.2 mm; <u>Weight</u> : 0.127 g	
	527 mg/l [96 hours]	
	Chronic - EC10	<u>Effect</u> : Population
	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase	
	<u>Age</u> : 7 days	
	3.56 mg/l [72 hours]	
	Acute - EC50	<u>Effect</u> : Population
	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase	
	<u>Age</u> : 7 days	
acetone	32.9 mg/l [72 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Daphnia - Water flea - <i>Daphnia magna</i>	
	10 mg/l [48 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i>	
	20.565 mg/l [96 hours]	
	Chronic - NOEC - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i>	
	4.95 mg/l [96 hours]	
acetone	Chronic - NOEC - Fresh water	<u>Effect</u> : Population
	Crustaceans - Daphnia - <i>Daphniidae</i>	
	0.016 ml/l [21 days]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Guppy - <i>Poecilia reticulata</i>	
	<u>Age</u> : 4 to 12 months; <u>Size</u> : 2 to 10 cm; <u>Weight</u> : 0.5 to 14 g	
acetone	5600 ppm [96 hours]	

Section 12. Ecological information

Conclusion/Summary [Product] : Not available.

Persistence and degradability

Product/ingredient name	Result	
tetrahydrofuran	82% [28 days]	Ready Biodegradability - Manometric Respirometry Test
butanone	Aerobic 98% [28 days] - Readily	OECD [Ready Biodegradability - Closed Bottle Test]

Conclusion/Summary [Product] : Not available.

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
tetrahydrofuran	-	-	Inherent
butanone	-	-	Readily
cyclohexanone	-	-	Readily
acetone	-	-	Readily

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
tetrahydrofuran	0.45	-	Low
butanone	0.3	-	Low
cyclohexanone	0.86	-	Low
acetone	-0.23	-	Low

Mobility in soil

Soil/Water partition coefficient : Not available.

Other adverse effects

No known significant effects or critical hazards.

Section 13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

RCRA Toxic hazardous waste "U" List

Section 13. Disposal considerations

Ingredient	CAS #	Status	Reference number
Tetrahydrofuran (I)	109-99-9	Listed	U213
Methyl ethyl ketone (MEK) (I,T)	78-93-3	Listed	U159
Cyclohexanone (I)	108-94-1	Listed	U057
Acetone (I)	67-64-1	Listed	U002

Section 14. Transport information

	DOT Classification	IMDG	IATA
UN number	UN1133	UN1133	UN1133
UN proper shipping name	Adhesives	ADHESIVES	Adhesives
Transport hazard class(es)	3 	3 	3 
Packing group	II	II	II
Environmental hazards	No.	No.	No.

Additional information

DOT Classification	: Reportable quantity 2500 lbs / 1135 kg [324.15 gal / 1227 L]. Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements. Limited quantity Yes. Packaging instruction Exceptions: 150. Non-bulk: 173. Bulk: 242. Quantity limitation Passenger aircraft/rail: 5 L. Cargo aircraft: 60 L. Special provisions 149, B52, IB2, T4, TP1, TP8
IMDG	: Emergency schedules F-E, S-D
IATA	: Quantity limitation Passenger and Cargo Aircraft: 5 L. Packaging instructions: 353. Cargo Aircraft Only: 60 L. Packaging instructions: 364. Limited Quantities - Passenger Aircraft: 1 L. Packaging instructions: Y341. Special provisions A3
Special precautions for user	: Transport within user's premises: always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.
Transport in bulk according to IMO instruments	: Not available.

Section 15. Regulatory information

U.S. Federal regulations : TSCA 8(a) PAIR: tetrahydrofuran
 TSCA 8(a) CDR Exempt/Partial exemption: Not determined
 United States inventory (TSCA 8b): All components are active or exempted.

TSCA 12(b) - Chemical export notification

Not applicable.

Clean Air Act Section 112 (b) Hazardous Air Pollutants (HAPs) : Not listed

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304

Composition/information on ingredients

No products were found.

SARA 304 RQ : Not applicable.

SARA 311/312

Classification : FLAMMABLE LIQUIDS - Category 2
 ACUTE TOXICITY (inhalation) - Category 4
 EYE IRRITATION - Category 2A
 CARCINOGENICITY - Category 2
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
 HNOC - Defatting irritant

Composition/information on ingredients

Name	%	Classification
tetrahydrofuran	≥20 - ≤40	FLAMMABLE LIQUIDS - Category 2 ACUTE TOXICITY (oral) - Category 4 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 HNOC - Defatting irritant
butanone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1 HNOC - Defatting irritant

Section 15. Regulatory information

cyclohexanone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 3 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1
acetone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A HNOC - Defatting irritant
Ethene, chloro-, homopolymer	≥10 - ≤30	COMBUSTIBLE DUSTS

SARA 313

Not applicable.

State regulations

Massachusetts	: The following components are listed: TETRAHYDROFURAN; METHYL ETHYL KETONE; CYCLOHEXANONE; ACETONE
New York	: The following components are listed: Tetrahydrofuran; Methyl ethyl ketone; Cyclohexanone; Acetone
New Jersey	: The following components are listed: TETRAHYDROFURAN; METHYL ETHYL KETONE; CYCLOHEXANONE; ACETONE; PVC
Pennsylvania	: The following components are listed: FURAN, TETRAHYDRO-; 2-BUTANONE; CYCLOHEXANONE; 2-PROPANONE

California Prop. 65

 **WARNING:** This product can expose you to tetrahydrofuran, which is known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

Ingredient name	No significant risk level	Maximum acceptable dosage level
tetrahydrofuran	-	-

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

Section 16. Other information

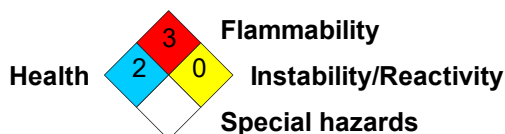
Hazardous Material Information System (U.S.A.)

Health	2
Flammability	3
Physical hazards	0

Caution: HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. Although HMIS® ratings and the associated label are not required on SDSs or products leaving a facility under 29 CFR 1910.1200, the preparer may choose to provide them. HMIS® ratings are to be used with a fully implemented HMIS® program. HMIS® is a registered trademark and service mark of the American Coatings Association, Inc.

The customer is responsible for determining the PPE code for this material. For more information on HMIS® Personal Protective Equipment (PPE) codes, consult the HMIS® Implementation Manual.

National Fire Protection Association (U.S.A.)



Procedure used to derive the classification

Classification	Justification
FLAMMABLE LIQUIDS - Category 2	On basis of test data
ACUTE TOXICITY (inhalation) - Category 4	Expert judgment
EYE IRRITATION - Category 2A	Calculation method
CARCINOGENICITY - Category 2	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3	Calculation method

History

Date of issue/Date of revision	: 03/25/2025
Date of previous issue	: No previous validation
Version	: 1
Prepared by	: Sphera Solutions
Key to abbreviations	: ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods IMO = International Maritime Organization LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) N/A = Not available SGG = Segregation Group TDG = Transportation of Dangerous Goods UN = United Nations

References : HCS (U.S.A.) - Hazard Communication Standard
International transport regulations

Indicates information that has changed from previously issued version.

Notice to reader

ABB Installation Products Inc. cannot anticipate all conditions under which this information and its product, or the products of other manufacturers in combination with its product, may be used. It is the user's responsibility to ensure safe conditions for handling, storage and disposal of the product, and to assume liability for loss, injury, damage or expense due to improper use. The information in the sheet was written based on the best knowledge and experience currently available.

SAFETY DATA SHEET



SDS-00061 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]

Section 1. Identification

GHS product identifier : SDS-00061 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]
Product code : VC9984, VC9983, VC9983C, VC9982, VC9981P
Other means of identification : Not available.
Product type : Liquid.

Relevant identified uses of the substance or mixture and uses advised against

Product use : Cements.
Area of application : Consumer applications.
Manufacturer : ABB Installation Products Inc.
860 Ridge Lake Blvd.
Memphis, TN 38120, US
Telephone no.: 1-888-862-3289

Emergency telephone number (with hours of operation) : INFOTRAC – 24 Hours 1-800-535-5053
+1 352-323-3500 (Outside USA)

Section 2. Hazards identification

OSHA/HCS status : This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Classification of the substance or mixture : H225 FLAMMABLE LIQUIDS - Category 2
H332 ACUTE TOXICITY (inhalation) - Category 4
H319 EYE IRRITATION - Category 2A
H351 CARCINOGENICITY - Category 2
H335 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
H336 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H225 - Highly flammable liquid and vapor.
H319 - Causes serious eye irritation.
H332 - Harmful if inhaled.
H335 - May cause respiratory irritation.
H336 - May cause drowsiness or dizziness.
H351 - Suspected of causing cancer.

Date of revision : 26- March 2025 **Date of previous issue** : 18-June 2025 **Version** : G

1/19

Section 2. Hazards identification

Precautionary statements

Prevention	: P201 - Obtain special instructions before use. P202 - Do not handle until all safety precautions have been read and understood. P280 - Wear protective gloves, protective clothing and eye or face protection. P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P241 - Use explosion-proof electrical, ventilating or lighting equipment. P242 - Use non-sparking tools. P243 - Take action to prevent static discharges. P271 - Use only outdoors or in a well-ventilated area. P261 - Avoid breathing vapor. P264 - Wash thoroughly after handling.
Response	: P308 + P313 - IF exposed or concerned: Get medical advice or attention. P304 + P340, P312 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor if you feel unwell. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P337 + P313 - If eye irritation persists: Get medical advice or attention.
Storage	: P405 - Store locked up. P403 + P233 - Store in a well-ventilated place. Keep container tightly closed. P403 + P235 - Keep cool.
Disposal	: P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.
Supplemental label elements	: Avoid contact with skin and clothing. Wash thoroughly after handling.
Hazards not otherwise classified	: Prolonged or repeated contact may dry skin and cause irritation.

Section 3. Composition/information on ingredients

Substance/mixture	: Mixture
Other means of identification	: Not available.

Ingredient name	Other names	%	Identifiers
tetrahydrofuran	-	≥25 - ≤50	CAS: 109-99-9
Ethene, chloro-, homopolymer	-	≥10 - ≤30	CAS: 9002-86-2
cyclohexanone	-	≥10 - ≤30	CAS: 108-94-1
acetone	-	≥10 - ≤30	CAS: 67-64-1
butanone	-	≥10 - ≤30	CAS: 78-93-3

Any concentration shown as a range is to protect confidentiality or is due to batch variation.

There are no additional ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified and hence require reporting in this section.

Occupational exposure limits, if available, are listed in Section 8.

Section 4. First aid measures

Description of necessary first aid measures

- Eye contact** : Immediately flush eyes with plenty of water, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Continue to rinse for at least 10 minutes. Get medical attention.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Get medical attention. If necessary, call a poison center or physician. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.
- Skin contact** : Wash skin thoroughly with soap and water or use recognized skin cleanser. Remove contaminated clothing and shoes. Continue to rinse for at least 10 minutes. Get medical attention. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Ingestion** : Wash out mouth with water. Remove dentures if any. If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Stop if the exposed person feels sick as vomiting may be dangerous. Do not induce vomiting unless directed to do so by medical personnel. If vomiting occurs, the head should be kept low so that vomit does not enter the lungs. Get medical attention. If necessary, call a poison center or physician. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Maintain an open airway. Loosen tight clothing such as a collar, tie, belt or waistband.

Most important symptoms/effects, acute and delayed

Potential acute health effects

- Eye contact** : Causes serious eye irritation.
- Inhalation** : Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation.
- Skin contact** : Defatting to the skin. May cause skin dryness and irritation.
- Ingestion** : Can cause central nervous system (CNS) depression.

Over-exposure signs/symptoms

- Eye contact** : Adverse symptoms may include the following:
pain or irritation
watering
redness
- Inhalation** : Adverse symptoms may include the following:
respiratory tract irritation
coughing
nausea or vomiting
headache
drowsiness/fatigue
dizziness/vertigo
unconsciousness
- Skin contact** : Adverse symptoms may include the following:
irritation
dryness
cracking
- Ingestion** : No specific data.

Indication of immediate medical attention and special treatment needed, if necessary

Section 4. First aid measures

- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.
- Specific treatments** : No specific treatment.
- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

See toxicological information (Section 11)

Section 5. Fire-fighting measures

Extinguishing media

- Suitable extinguishing media** : Foam, carbon dioxide, dry chemical.
- Unsuitable extinguishing media** : Do not use water jet.

Specific hazards arising from the chemical : Class 1B Flammable liquid. Vapors can travel to a source of ignition and flashback.

- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
halogenated compounds

Special protective actions for fire-fighters : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training. Move containers from fire area if this can be done without risk. Use water spray to keep fire-exposed containers cool.

Special protective equipment for fire-fighters : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

Personal precautions, protective equipment and emergency procedures

- For non-emergency personnel** : No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Shut off all ignition sources. No flares, smoking or flames in hazard area. Avoid breathing vapor or mist. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment.
- For emergency responders** : If specialized clothing is required to deal with the spillage, take note of any information in Section 8 on suitable and unsuitable materials. See also the information in "For non-emergency personnel".

Environmental precautions : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Methods and materials for containment and cleaning up

Section 6. Accidental release measures

- Small spill** : Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.
- Large spill** : Stop leak if without risk. Move containers from spill area. Use spark-proof tools and explosion-proof equipment. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Wash spillages into an effluent treatment plant or proceed as follows. Dispose of via a licensed waste disposal contractor. Contaminated absorbent material may pose the same hazard as the spilled product. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations.

Section 7. Handling and storage

Precautions for safe handling

- Protective measures** : Put on appropriate personal protective equipment (see Section 8). Avoid exposure - obtain special instructions before use. Do not handle until all safety precautions have been read and understood. Do not get in eyes or on skin or clothing. Do not ingest. Avoid breathing vapor or mist. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Do not enter storage areas and confined spaces unless adequately ventilated. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Store and use away from heat, sparks, open flame or any other ignition source. Use explosion-proof electrical (ventilating, lighting and material handling) equipment. Use only non-sparking tools. Take precautionary measures against electrostatic discharges. Empty containers retain product residue and can be hazardous. Do not reuse container.
- Advice on general occupational hygiene** : Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.

- Conditions for safe storage, including any incompatibilities** : Store in accordance with local regulations. Store in a segregated and approved area. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10) and food and drink. Store locked up. Eliminate all ignition sources. Separate from oxidizing materials. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination. See Section 10 for incompatible materials before handling or use.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
tetrahydrofuran	ACGIH TLV (United States, 1/2024) A3. Absorbed through skin. TWA 8 hours: 50 ppm. STEL 15 minutes: 100 ppm. NIOSH REL (United States, 10/2020) TWA 10 hours: 200 ppm. TWA 10 hours: 590 mg/m³. STEL 15 minutes: 250 ppm. STEL 15 minutes: 735 mg/m³. OSHA PEL (United States, 5/2018)

Section 8. Exposure controls/personal protection

Ethene, chloro-, homopolymer

cyclohexanone

acetone

butanone

TWA 8 hours: 200 ppm.

TWA 8 hours: 590 mg/m³.

CAL OSHA PEL (United States, 5/2018)

STEL 15 minutes: 735 mg/m³.

STEL 15 minutes: 250 ppm.

TWA 8 hours: 590 mg/m³.

TWA 8 hours: 200 ppm.

ACGIH TLV (United States, 1/2024) A4.

TWA 8 hours: 1 mg/m³. Form: Respirable fraction.

ACGIH TLV (United States, 1/2024) A3. Absorbed through skin.

TWA 8 hours: 20 ppm.

STEL 15 minutes: 50 ppm.

NIOSH REL (United States, 10/2020) Absorbed through skin.

TWA 10 hours: 25 ppm.

TWA 10 hours: 100 mg/m³.

OSHA PEL (United States, 5/2018)

TWA 8 hours: 50 ppm.

TWA 8 hours: 200 mg/m³.

CAL OSHA PEL (United States, 5/2018) Absorbed through skin.

TWA 8 hours: 100 mg/m³.

TWA 8 hours: 25 ppm.

ACGIH TLV (United States, 1/2024) A4.

TWA 8 hours: 250 ppm.

STEL 15 minutes: 500 ppm.

NIOSH REL (United States, 10/2020)

TWA 10 hours: 250 ppm.

TWA 10 hours: 590 mg/m³.

OSHA PEL (United States, 5/2018)

TWA 8 hours: 1000 ppm.

TWA 8 hours: 2400 mg/m³.

CAL OSHA PEL (United States, 5/2018)

STEL 15 minutes: 1780 mg/m³.

STEL 15 minutes: 750 ppm.

C: 3000 ppm.

TWA 8 hours: 1200 mg/m³.

TWA 8 hours: 500 ppm.

ACGIH TLV (United States, 1/2024) Absorbed through skin.

TWA 8 hours: 75 ppm.

STEL 15 minutes: 150 ppm.

NIOSH REL (United States, 10/2020)

TWA 10 hours: 200 ppm.

TWA 10 hours: 590 mg/m³.

STEL 15 minutes: 300 ppm.

STEL 15 minutes: 885 mg/m³.

OSHA PEL (United States, 5/2018)

TWA 8 hours: 200 ppm.

TWA 8 hours: 590 mg/m³.

CAL OSHA PEL (United States, 5/2018)

STEL 15 minutes: 885 mg/m³.

STEL 15 minutes: 300 ppm.

TWA 8 hours: 590 mg/m³.

TWA 8 hours: 200 ppm.

Section 8. Exposure controls/personal protection

Biological exposure indices

Ingredient name	Exposure indices
tetrahydrofuran	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, tetrahydrofuran [in urine]. Sampling time: end of shift.
cyclohexanone	ACGIH BEI (United States, 1/2024) BEI: 80 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], 1,2-cyclohexanediol [in urine]. Sampling time: end of shift at end of workweek. BEI: 8 mg/l [Semi-quantitative: The determinant is an indicator of exposure to the chemical, but the quantitative interpretation of the measurement is ambiguous. These determinants should be used as a screening test if a quantitative test is not practical or as a confirmatory test if the quantitative test is not specific and the origin of the determinant is in question.], cyclohexanol [in urine]. Sampling time: end of shift.
acetone	ACGIH BEI (United States, 1/2024) BEI: 25 mg/l, acetone [in urine]. Sampling time: end of shift.
butanone	ACGIH BEI (United States, 1/2024) BEI: 2 mg/l, methyl ethyl ketone [in urine]. Sampling time: end of shift.

Appropriate engineering controls

- : Use only with adequate ventilation. Use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. The engineering controls also need to keep gas, vapor or dust concentrations below any lower explosive limits. Use explosion-proof ventilation equipment.

Environmental exposure controls

- : Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

Individual protection measures

Hygiene measures

- : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Section 8. Exposure controls/personal protection

- Eye/face protection** : Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists, gases or dusts. If contact is possible, the following protection should be worn, unless the assessment indicates a higher degree of protection: chemical splash goggles.
- Skin protection**
- Hand protection** : Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary. Considering the parameters specified by the glove manufacturer, check during use that the gloves are still retaining their protective properties. It should be noted that the time to breakthrough for any glove material may be different for different glove manufacturers. In the case of mixtures, consisting of several substances, the protection time of the gloves cannot be accurately estimated.
- Body protection** : Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product. When there is a risk of ignition from static electricity, wear anti-static protective clothing. For the greatest protection from static discharges, clothing should include anti-static overalls, boots and gloves. Recommended: Rubber gloves.
- Other skin protection** : Appropriate footwear and any additional skin protection measures should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.
- Respiratory protection** : Based on the hazard and potential for exposure, select a respirator that meets the appropriate standard or certification. Respirators must be used according to a respiratory protection program to ensure proper fitting, training, and other important aspects of use.

Section 9. Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

Appearance

- Physical state** : Liquid. [Clear Liquid.]
- Color** : Clear.
- Odor** : Ether-like.
- Odor threshold** : 0.88 ppm
- pH** : Not available.
- Melting point/freezing point** : -108.5°C (-163.3°F)
- Boiling point or initial boiling point and boiling range** : 56°C (132.8°F)
- Flash point** : Closed cup: -20°C (-4°F)
- Evaporation rate** : >1 (butyl acetate = 1)
- Flammability** : Not available.
- Lower and upper explosion limit/flammability limit** : Lower: 1.8%
Upper: 12.8%
- Vapor pressure** : 25.3 kPa (190 mm Hg)
- Relative vapor density** : 2.5 [Air = 1]
- Relative density** : 0.93
- Density** : Not available.
- Solubility(ies)** :

Media	Result
water	Not soluble

Section 9. Physical and chemical properties

Partition coefficient: n-octanol/water : Not applicable.

Auto-ignition temperature :	Ingredient name	°C	°F	Method
	tetrahydrofuran	215	419	DIN 51794

Decomposition temperature : Not available.

SADT : Not available.

Viscosity : Dynamic (room temperature): Not available.
Kinematic (room temperature): Not available.
Kinematic (40°C (104°F)): Not available.

Particle characteristics

Median particle size : Not applicable.

Other information

Physical/chemical properties comments : No additional information.

Section 10. Stability and reactivity

Reactivity : No specific test data related to reactivity available for this product or its ingredients.

Chemical stability : The product is stable.

Possibility of hazardous reactions : Under normal conditions of storage and use, hazardous reactions will not occur.
Under normal conditions of storage and use, hazardous polymerization will not occur.

Conditions to avoid : Avoid all possible sources of ignition (spark or flame). Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition. Do not allow vapor to accumulate in low or confined areas.

Incompatible materials : Reactive or incompatible with the following materials:
oxidizing materials, acids, Bases.

Hazardous decomposition products : carbon oxides Hydrogen chloride Hydrocarbon.

Section 11. Toxicological information

Information on toxicological effects

Acute toxicity

Product/ingredient name	Result
tetrahydrofuran	Rat - Oral - LD50 1650 mg/kg Rat - Male, Female - Dermal - LD50 OECD [Acute Dermal Toxicity] >2000 mg/kg
cyclohexanone	Rat - Oral - LD50 1800 mg/kg Rabbit - Dermal - LD50 950 mg/kg

Section 11. Toxicological information

acetone	Rat - Inhalation - LC50 Gas. 8000 ppm [4 hours]	Toxic effects: Behavioral - Altered sleep time (including change in righting reflex) Behavioral - Tremor
	Rat - Oral - LD50 5800 mg/kg	
butanone	Rabbit - Dermal - LD50 20000 mg/kg	
	Rat - Female - Inhalation - LC50 Vapor 76 mg/l [4 hours]	
	Rabbit - Dermal - LD50 6480 mg/kg	
	Rat - Oral - LD50 2737 mg/kg	
	Rat - Inhalation - LC50 Vapor 33.36 mg/l [4 hours]	
Conclusion/Summary [Product] : Not available.		

Skin corrosion/irritation

Product/ingredient name	Result
cyclohexanone	Rabbit - Skin - Mild irritant Amount/concentration applied: 500 mg
acetone	Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg
butanone	Rabbit - Skin - Mild irritant Amount/concentration applied: 395 mg Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 14 mg Rabbit - Skin - Mild irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 402 mg Rabbit - Skin - Moderate irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 500 mg
Conclusion/Summary [Product] : Not available.	

Serious eye damage/eye irritation

Product/ingredient name	Result
cyclohexanone	Rabbit - Eyes - Severe irritant Duration of treatment/exposure: 24 hours Amount/concentration applied: 250 ug
acetone	Rabbit - Eyes - Severe irritant Amount/concentration applied: 20 mg Rabbit - Eyes - Mild irritant

Section 11. Toxicological information

Amount/concentration applied: 10 uL

Rabbit - Eyes - Moderate irritant

Duration of treatment/exposure: 24 hours

Amount/concentration applied: 20 mg

Rabbit - Eyes - Severe irritant

Amount/concentration applied: 20 mg

Conclusion/Summary [Product] : Not available.

Respiratory corrosion/irritation

Conclusion/Summary [Product] : Not available.

Respiratory or skin sensitization

Skin

Conclusion/Summary [Product] : Not available.

Respiratory

Conclusion/Summary [Product] : Not available.

Germ cell mutagenicity

Conclusion/Summary [Product] : Not available.

Carcinogenicity

Conclusion/Summary [Product] : Not available.

Classification

Product/ingredient name	OSHA	IARC	NTP
tetrahydrofuran	-	2B	-
Ethene, chloro-, homopolymer	-	3	-
cyclohexanone	-	3	-

Reproductive toxicity

Conclusion/Summary [Product] : Not available.

Specific target organ toxicity (single exposure)

Product/ingredient name	Result
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Section 11. Toxicological information

tetrahydrofuran	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
cyclohexanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
butanone	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3

Specific target organ toxicity (repeated exposure)

Not available.

Aspiration hazard

Product/ingredient name

Result

cyclohexanone

ASPIRATION HAZARD - Category 1

butanone

ASPIRATION HAZARD - Category 1

Information on the likely routes of exposure

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

Potential acute health effects

Eye contact	: Causes serious eye irritation.
Inhalation	: Harmful if inhaled. Can cause central nervous system (CNS) depression. May cause drowsiness or dizziness. May cause respiratory irritation.
Skin contact	: Defatting to the skin. May cause skin dryness and irritation.
Ingestion	: Can cause central nervous system (CNS) depression.

Symptoms related to the physical, chemical and toxicological characteristics

Eye contact	: Adverse symptoms may include the following: pain or irritation watering redness
Inhalation	: Adverse symptoms may include the following: respiratory tract irritation coughing nausea or vomiting headache drowsiness/fatigue dizziness/vertigo unconsciousness
Skin contact	: Adverse symptoms may include the following: irritation dryness cracking
Ingestion	: No specific data.

Section 11. Toxicological information

Delayed and immediate effects and also chronic effects from short and long term exposure

Short term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Long term exposure

Potential immediate effects : Not available.

Potential delayed effects : Not available.

Potential chronic health effects

Conclusion/Summary [Product] : Not available.

General : Prolonged or repeated contact can defat the skin and lead to irritation, cracking and/or dermatitis.

Carcinogenicity : Suspected of causing cancer. Risk of cancer depends on duration and level of exposure.

Mutagenicity : No known significant effects or critical hazards.

Reproductive toxicity : No known significant effects or critical hazards.

Numerical measures of toxicity

Acute toxicity estimates

Product/ingredient name	Oral(mg/kg)	Dermal (mg/kg)	Inhalation (gases) (ppm)	Inhalation (vapors) (mg/l)	Inhalation (dusts and mists) (mg/l)
SDS-00061 [Carlton Low-VOC Solvent Cement for PVC Plastic Pipe]	2416.7	2714.3	24000	N/A	1.5
tetrahydrofuran	1650	2500	N/A	N/A	N/A
cyclohexanone	1800	950	8000	N/A	N/A
acetone	5800	20000	N/A	76	N/A
butanone	2737	6480	N/A	33.36	N/A

Section 12. Ecological information

Toxicity

Product/ingredient name

tetrahydrofuran

Result

Acute - LC50 - Fresh water

Fish - Fathead minnow - *Pimephales promelas*

Age: 34 days; Size: 18.6 mm; Weight: 0.111 g

2160 mg/l [96 hours]

Chronic - NOEC - Fresh water

Fish - Fathead minnow - *Pimephales promelas* - Embryo

Age: <24 hours

367 mg/l [33 days]

Effect: Mortality

Effect: Growth

Section 12. Ecological information

cyclohexanone	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Fathead minnow - <i>Pimephales promelas</i> <u>Age</u> : 30 days; <u>Size</u> : 20.2 mm; <u>Weight</u> : 0.127 g 527 mg/l [96 hours]	
	Chronic - EC10	<u>Effect</u> : Population
acetone	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase <u>Age</u> : 7 days 3.56 mg/l [72 hours]	
	Acute - EC50	<u>Effect</u> : Population
	Algae - Green algae - <i>Chlamydomonas reinhardtii</i> - Exponential growth phase <u>Age</u> : 7 days 32.9 mg/l [72 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Daphnia - Water flea - <i>Daphnia magna</i> 10 mg/l [48 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i> 20.565 mg/l [96 hours]	
	Chronic - NOEC - Marine water	<u>Effect</u> : Reproduction
	Algae - Green algae - <i>Ulva pertusa</i> 4.95 mg/l [96 hours]	
	Chronic - NOEC - Fresh water	<u>Effect</u> : Population
	Crustaceans - Daphnia - <i>Daphniidae</i> 0.016 ml/l [21 days]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
butanone	Fish - Guppy - <i>Poecilia reticulata</i> <u>Age</u> : 4 to 12 months; <u>Size</u> : 2 to 10 cm; <u>Weight</u> : 0.5 to 14 g 5600 ppm [96 hours]	
	Acute - EC50 - Fresh water	<u>Effect</u> : Intoxication
	Daphnia - Water flea - <i>Daphnia magna</i> - Larvae <u>Age</u> : <24 hours 5091 mg/l [48 hours]	
	Acute - LC50 - Fresh water	<u>Effect</u> : Mortality
	Fish - Fathead minnow - <i>Pimephales promelas</i> <u>Age</u> : 31 days; <u>Size</u> : 22 mm; <u>Weight</u> : 0.167 g 3220 mg/l [96 hours]	
	Acute - EC50 - Marine water	<u>Effect</u> : Population
	Algae - Diatom - <i>Skeletonema costatum</i> >500 mg/l [96 hours]	
	Acute - NOEC - Fresh water	OECD 203 [Fish, Acute Toxicity Test]
	Fish 1170 mg/l [96 hours]	
	Acute - NOEC - Fresh water	OECD 202 [Daphnia sp. Acute Immobilization Test and Reproduction Test]
	Daphnia - Daphnia 68 mg/l [48 hours]	
	Acute - NOEC - Fresh water	OECD 201 [Alga, Growth Inhibition Test]
	Algae - Algae 566 mg/l [72 hours]	

Section 12. Ecological information

Conclusion/Summary [Product] : Not available.

Persistence and degradability

Product/ingredient name	Result	
tetrahydrofuran	82% [28 days]	Ready Biodegradability - Manometric Respirometry Test
butanone	Aerobic 98% [28 days] - Readily	OECD [Ready Biodegradability - Closed Bottle Test]

Conclusion/Summary [Product] : Not available.

Product/ingredient name	Aquatic half-life	Photolysis	Biodegradability
tetrahydrofuran	-	-	Inherent
cyclohexanone	-	-	Readily
acetone	-	-	Readily
butanone	-	-	Readily

Bioaccumulative potential

Product/ingredient name	LogP _{ow}	BCF	Potential
tetrahydrofuran	0.45	-	Low
cyclohexanone	0.86	-	Low
acetone	-0.23	-	Low
butanone	0.3	-	Low

Mobility in soil

Soil/Water partition coefficient : Not available.

Other adverse effects

No known significant effects or critical hazards.

Section 13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Vapor from product residues may create a highly flammable or explosive atmosphere inside the container. Do not cut, weld or grind used containers unless they have been cleaned thoroughly internally. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

RCRA Toxic hazardous waste "U" List

Section 13. Disposal considerations

Ingredient	CAS #	Status	Reference number
Tetrahydrofuran (I)	109-99-9	Listed	U213
Cyclohexanone (I)	108-94-1	Listed	U057
Acetone (I)	67-64-1	Listed	U002
Methyl ethyl ketone (MEK) (I,T)	78-93-3	Listed	U159

Section 14. Transport information

	DOT Classification	IMDG	IATA
UN number	UN1133	UN1133	UN1133
UN proper shipping name	Adhesives	ADHESIVES	Adhesives
Transport hazard class(es)	3 	3 	3 
Packing group	II	II	II
Environmental hazards	No.	No.	No.

Additional information

DOT Classification

: **Reportable quantity** 2000 lbs / 908 kg [257.92 gal / 976.34 L]. Package sizes shipped in quantities less than the product reportable quantity are not subject to the RQ (reportable quantity) transportation requirements.

Limited quantity Yes.

Packaging instruction Exceptions: 150. Non-bulk: 173. Bulk: 242.

Quantity limitation Passenger aircraft/rail: 5 L. Cargo aircraft: 60 L.

Special provisions 149, B52, IB2, T4, TP1, TP8

IMDG

: **Emergency schedules** F-E, S-D

IATA

: **Quantity limitation** Passenger and Cargo Aircraft: 5 L. Packaging instructions: 353. Cargo Aircraft Only: 60 L. Packaging instructions: 364. Limited Quantities - Passenger Aircraft: 1 L. Packaging instructions: Y341.

Special provisions A3

Special precautions for user : **Transport within user's premises:** always transport in closed containers that are upright and secure. Ensure that persons transporting the product know what to do in the event of an accident or spillage.

Transport in bulk according to IMO instruments : Not available.

Section 15. Regulatory information

U.S. Federal regulations : TSCA 8(a) PAIR: tetrahydrofuran
 TSCA 8(a) CDR Exempt/Partial exemption: Not determined
 United States inventory (TSCA 8b): All components are active or exempted.

TSCA 12(b) - Chemical export notification

Not applicable.

Clean Air Act Section 112 (b) Hazardous Air Pollutants (HAPs) : Not listed

Clean Air Act Section 602 Class I Substances : Not listed

Clean Air Act Section 602 Class II Substances : Not listed

DEA List I Chemicals (Precursor Chemicals) : Not listed

DEA List II Chemicals (Essential Chemicals) : Not listed

SARA 302/304

Composition/information on ingredients

No products were found.

SARA 304 RQ : Not applicable.

SARA 311/312

Classification : FLAMMABLE LIQUIDS - Category 2
 ACUTE TOXICITY (inhalation) - Category 4
 EYE IRRITATION - Category 2A
 CARCINOGENICITY - Category 2
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3
 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3
 HNOC - Defatting irritant

Composition/information on ingredients

Name	%	Classification
tetrahydrofuran	≥25 - ≤50	FLAMMABLE LIQUIDS - Category 2 ACUTE TOXICITY (oral) - Category 4 EYE IRRITATION - Category 2A CARCINOGENICITY - Category 2 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 HNOC - Defatting irritant
Ethene, chloro-, homopolymer	≥10 - ≤30	COMBUSTIBLE DUSTS
cyclohexanone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 3 ACUTE TOXICITY (oral) - Category 4 ACUTE TOXICITY (dermal) - Category 3 ACUTE TOXICITY (inhalation) - Category 4 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3

Section 15. Regulatory information

acetone	≥10 - ≤30	SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1 FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A HNOC - Defatting irritant
butanone	≥10 - ≤30	FLAMMABLE LIQUIDS - Category 2 EYE IRRITATION - Category 2A SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3 SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3 ASPIRATION HAZARD - Category 1 HNOC - Defatting irritant

SARA 313

Not applicable.

State regulations

Massachusetts

: The following components are listed: TETRAHYDROFURAN; CYCLOHEXANONE; ACETONE; METHYL ETHYL KETONE

New York

: The following components are listed: Tetrahydrofuran; Cyclohexanone; Acetone; Methyl ethyl ketone

New Jersey

: The following components are listed: TETRAHYDROFURAN; PVC; CYCLOHEXANONE; ACETONE; METHYL ETHYL KETONE

Pennsylvania

: The following components are listed: FURAN, TETRAHYDRO-; CYCLOHEXANONE; 2-PROPANONE; 2-BUTANONE

California Prop. 65

 **WARNING:** This product can expose you to tetrahydrofuran, which is known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

Ingredient name	No significant risk level	Maximum acceptable dosage level
tetrahydrofuran	-	-

International regulations

Chemical Weapon Convention List Schedules I, II & III Chemicals

Not listed.

Montreal Protocol

Not listed.

Stockholm Convention on Persistent Organic Pollutants

Not listed.

Rotterdam Convention on Prior Informed Consent (PIC)

Not listed.

UNECE Aarhus Protocol on POPs and Heavy Metals

Not listed.

Section 16. Other information

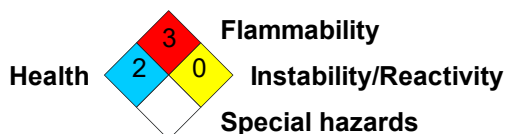
Hazardous Material Information System (U.S.A.)

Health	*	2
Flammability		3
Physical hazards		0

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National Fire Protection Association (U.S.A.)



Procedure used to derive the classification

Classification	Justification
FLAMMABLE LIQUIDS - Category 2	On basis of test data
ACUTE TOXICITY (inhalation) - Category 4	Expert judgment
EYE IRRITATION - Category 2A	Calculation method
CARCINOGENICITY - Category 2	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Respiratory tract irritation) - Category 3	Calculation method
SPECIFIC TARGET ORGAN TOXICITY (SINGLE EXPOSURE) (Narcotic effects) - Category 3	Calculation method

History

Date of issue/Date of revision	: 03/25/2025
Date of previous issue	: No previous validation
Version	: 1
Prepared by	: Sphera Solutions
Key to abbreviations	: ATE = Acute Toxicity Estimate BCF = Bioconcentration Factor GHS = Globally Harmonized System of Classification and Labelling of Chemicals IATA = International Air Transport Association IBC = Intermediate Bulk Container IMDG = International Maritime Dangerous Goods IMO = International Maritime Organization LogPow = logarithm of the octanol/water partition coefficient MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution) N/A = Not available SGG = Segregation Group TDG = Transportation of Dangerous Goods UN = United Nations

References : HCS (U.S.A.) - Hazard Communication Standard
International transport regulations

Indicates information that has changed from previously issued version.

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